

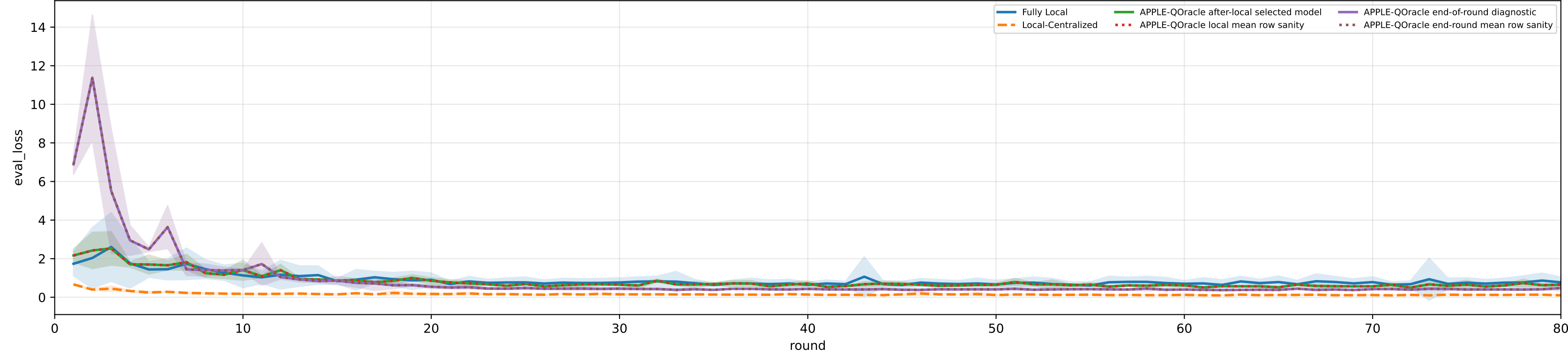
Controlled q-label heterogeneity: iid | APPLE-QOracle diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

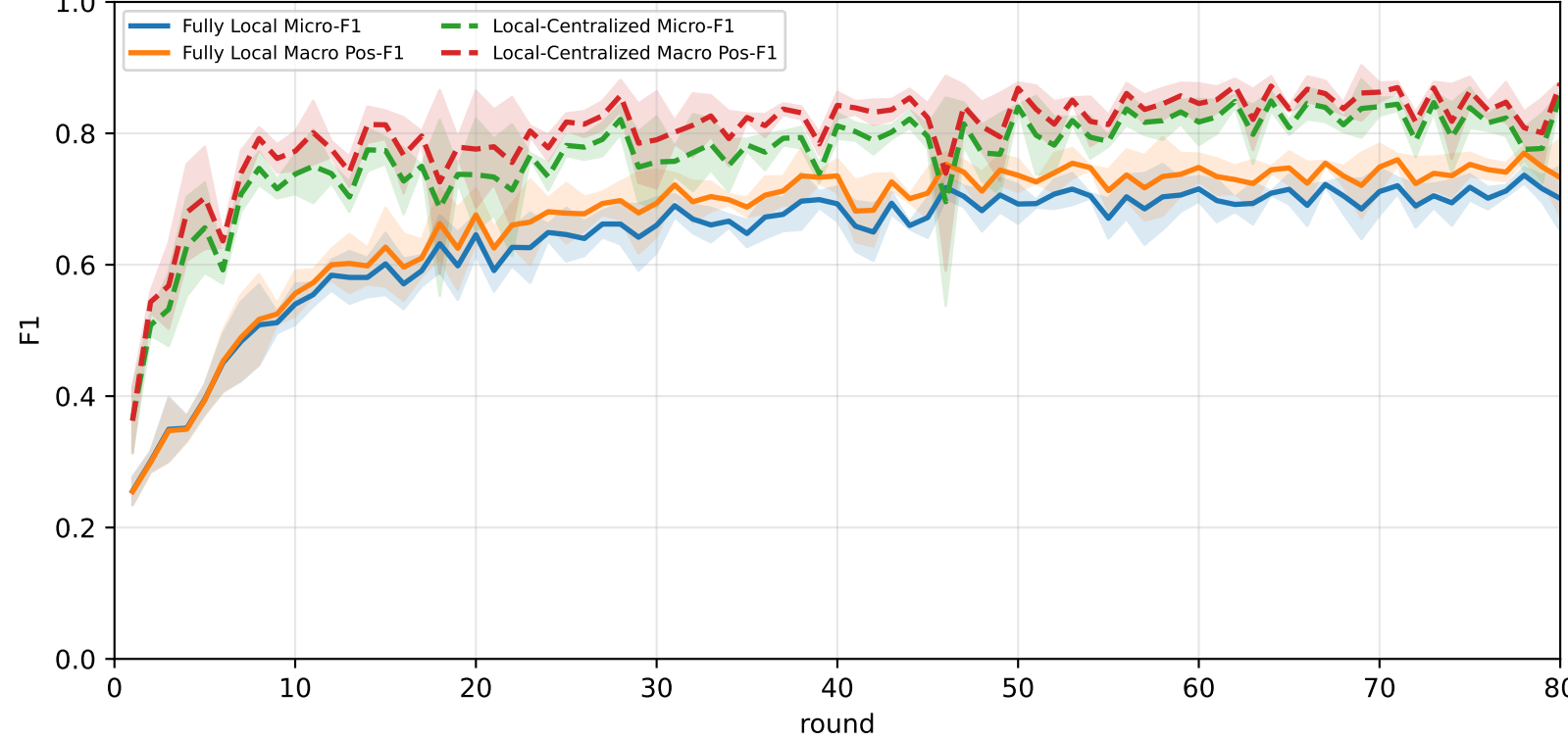
Best test performance (Fully Local vs Local-Centralized vs APPLE-QOracle, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	67.9% ± 1.6%	72.3% ± 1.5%	<u>90.0% ± 1.5%</u>	<u>85.0% ± 1.7%</u>	67.5% ± 3.5%	<u>66.8% ± 2.2%</u>	50.9% ± 1.2%
Local-Centralized	85.0% ± 0.7%	87.0% ± 0.6%	98.0% ± 1.1%	94.5% ± 1.4%	89.7% ± 1.8%	80.3% ± 0.5%	72.5% ± 1.3%
APPLE-QOracle	<u>70.3% ± 1.0%</u>	<u>73.3% ± 1.6%</u>	87.4% ± 5.3%	82.3% ± 3.0%	<u>72.8% ± 2.7%</u>	66.4% ± 1.0%	<u>56.4% ± 0.4%</u>

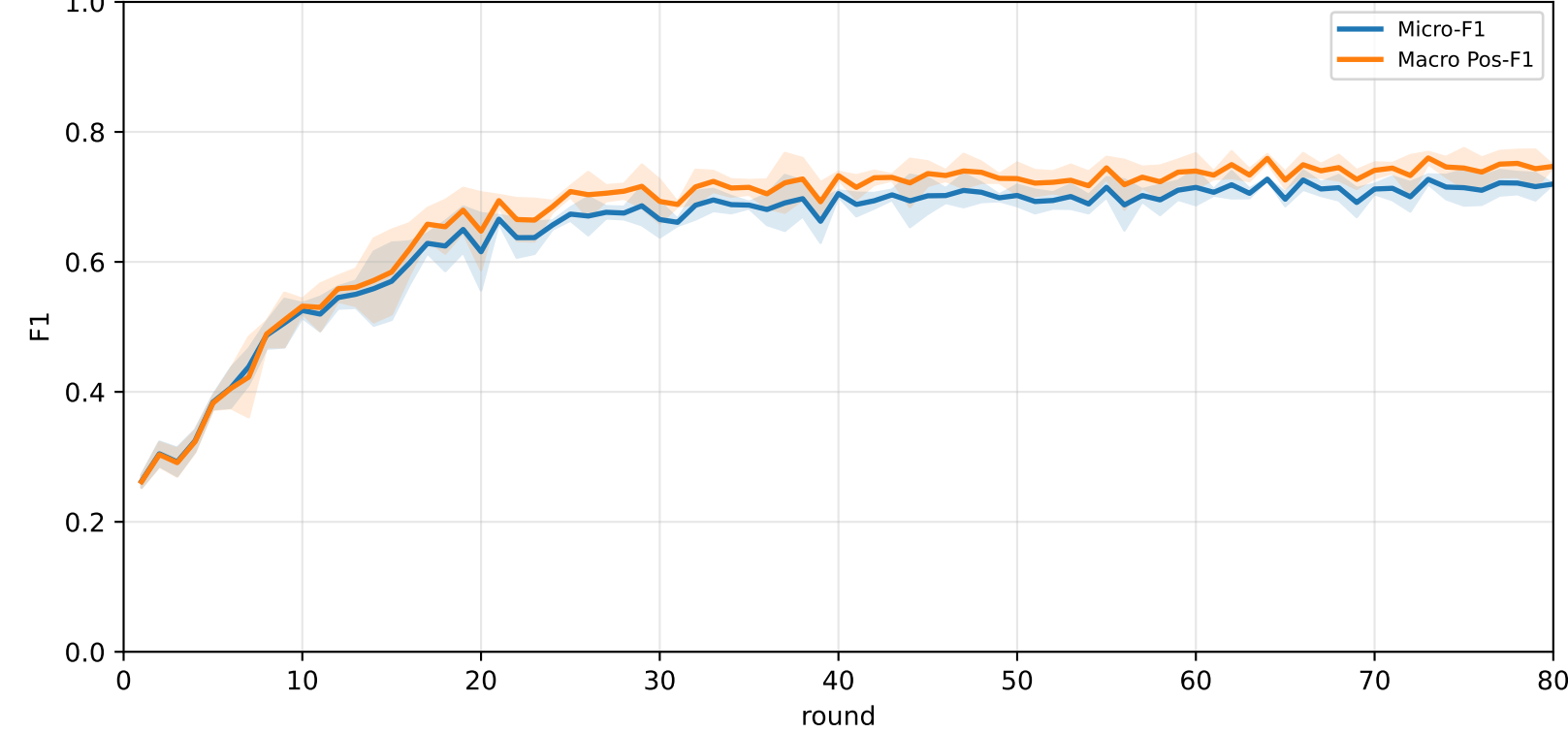
Validation loss: baselines vs selected APPLE-QOracle and end-of-round mismatch (rounds=80)



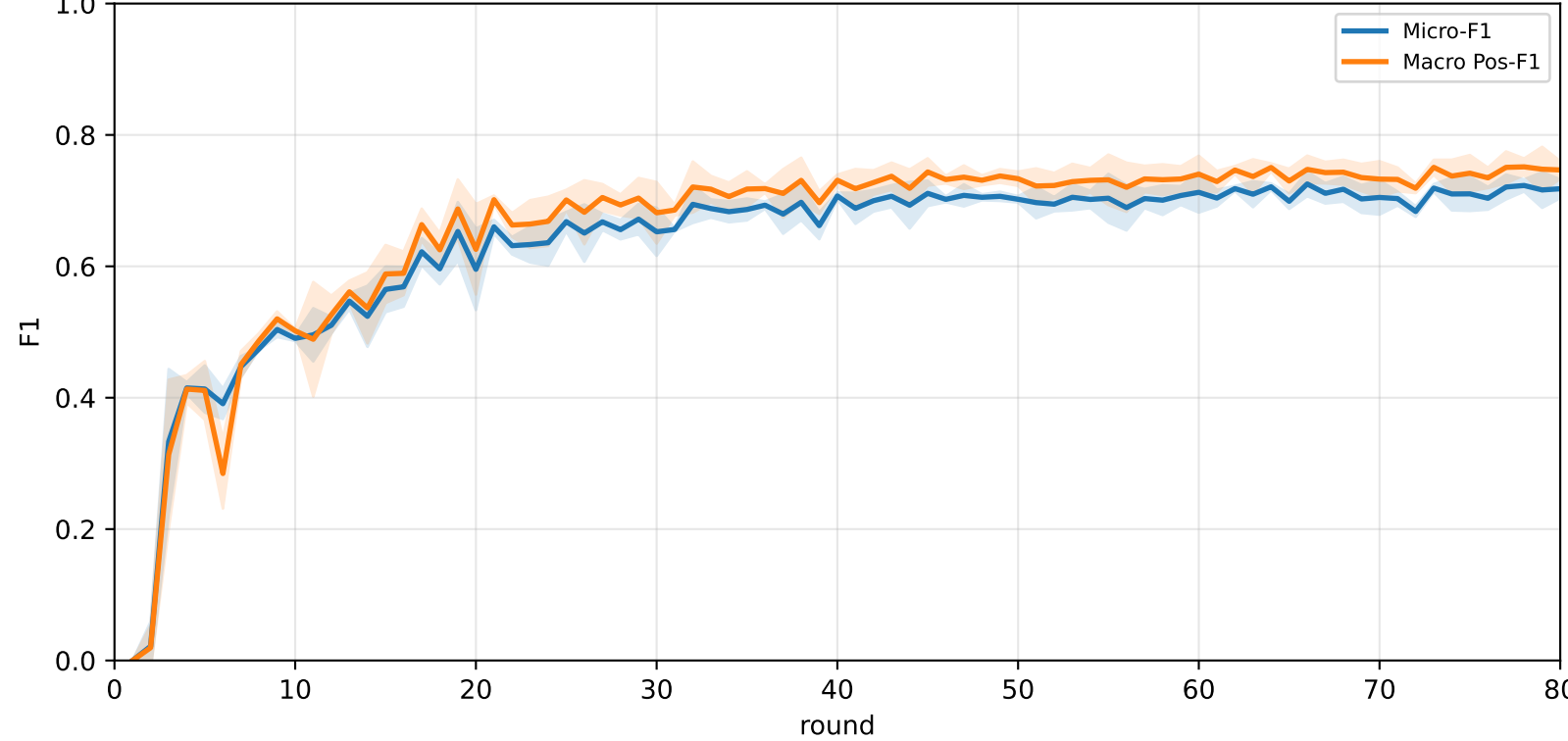
Pooled baseline micro / macro F1



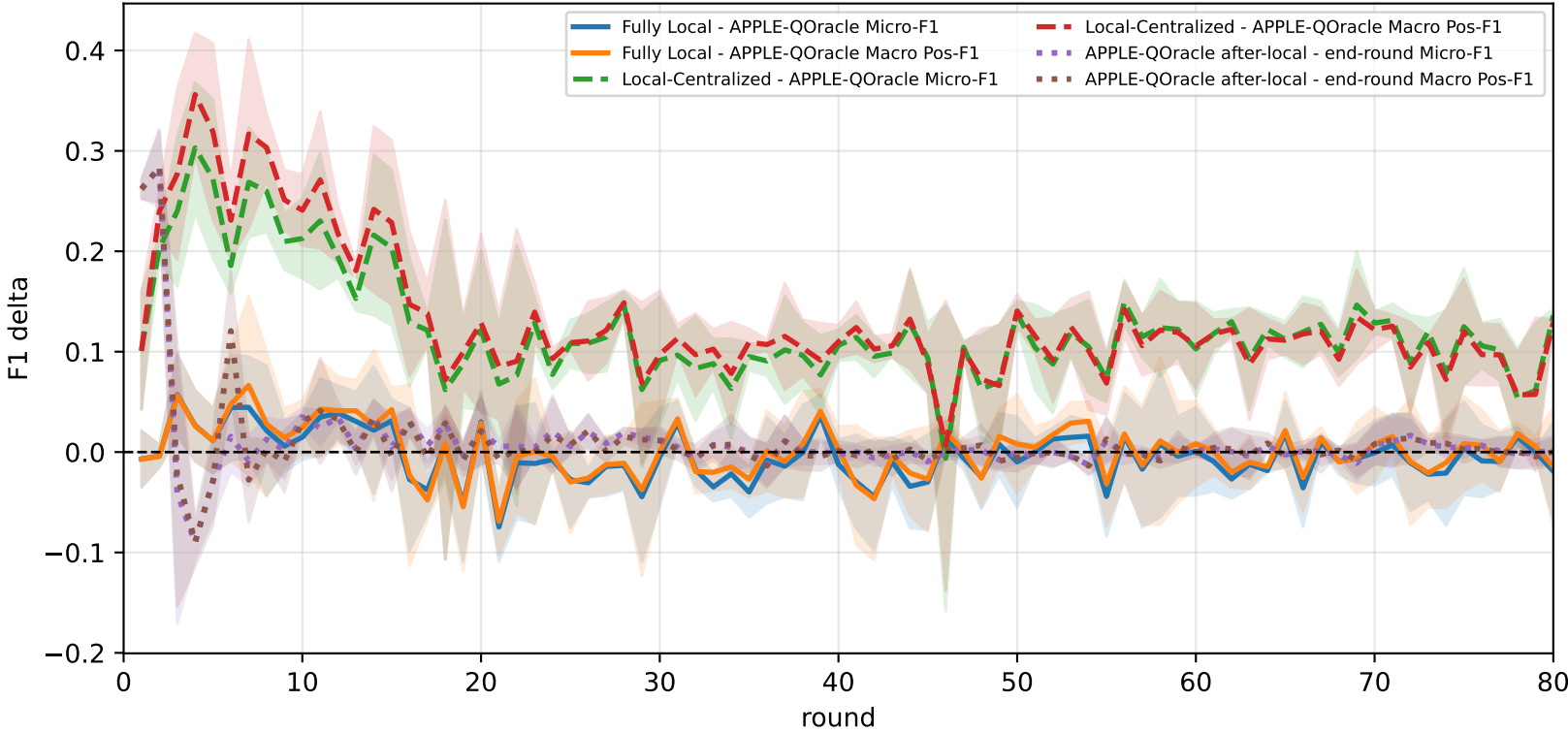
Pooled APPLE-QOracle micro / macro F1 (after-local selected model)



Pooled APPLE-QOracle micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QOracle mismatch gap



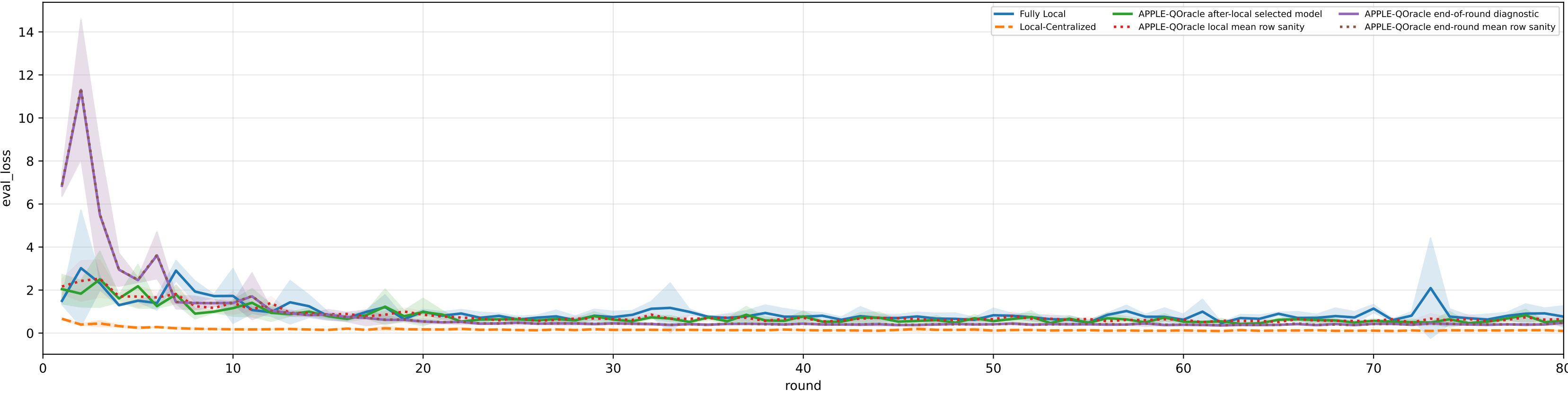
Controlled q-label heterogeneity: iid | APPLE-QOracle diagnostics | family=q_task_cycle2 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

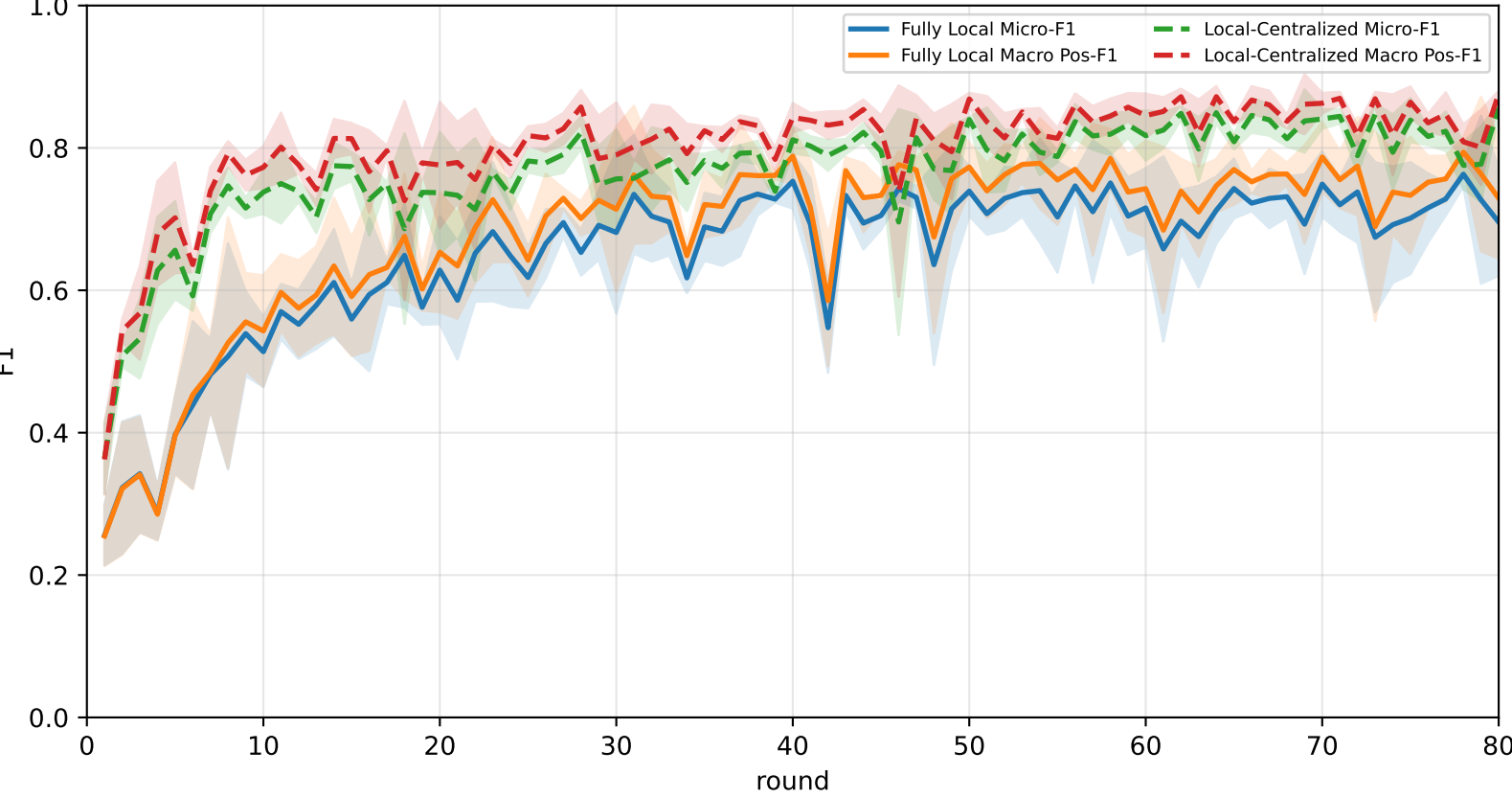
Best test performance (Fully Local vs Local-Centralized vs APPLE-QOracle, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	<u>70.1% ± 3.2%</u>	<u>74.3% ± 2.5%</u>	<u>93.1% ± 1.1%</u>	<u>86.7% ± 1.6%</u>	70.3% ± 4.6%	<u>67.5% ± 2.2%</u>	53.8% ± 5.0%
Local-Centralized	85.0% ± 0.7%	87.0% ± 0.6%	98.0% ± 1.1%	94.5% ± 1.4%	89.7% ± 1.8%	80.3% ± 0.5%	72.5% ± 1.3%
APPLE-QOracle	69.8% ± 0.8%	72.7% ± 2.1%	86.3% ± 9.4%	82.3% ± 4.1%	<u>72.2% ± 4.0%</u>	65.8% ± 3.3%	<u>56.8% ± 1.5%</u>

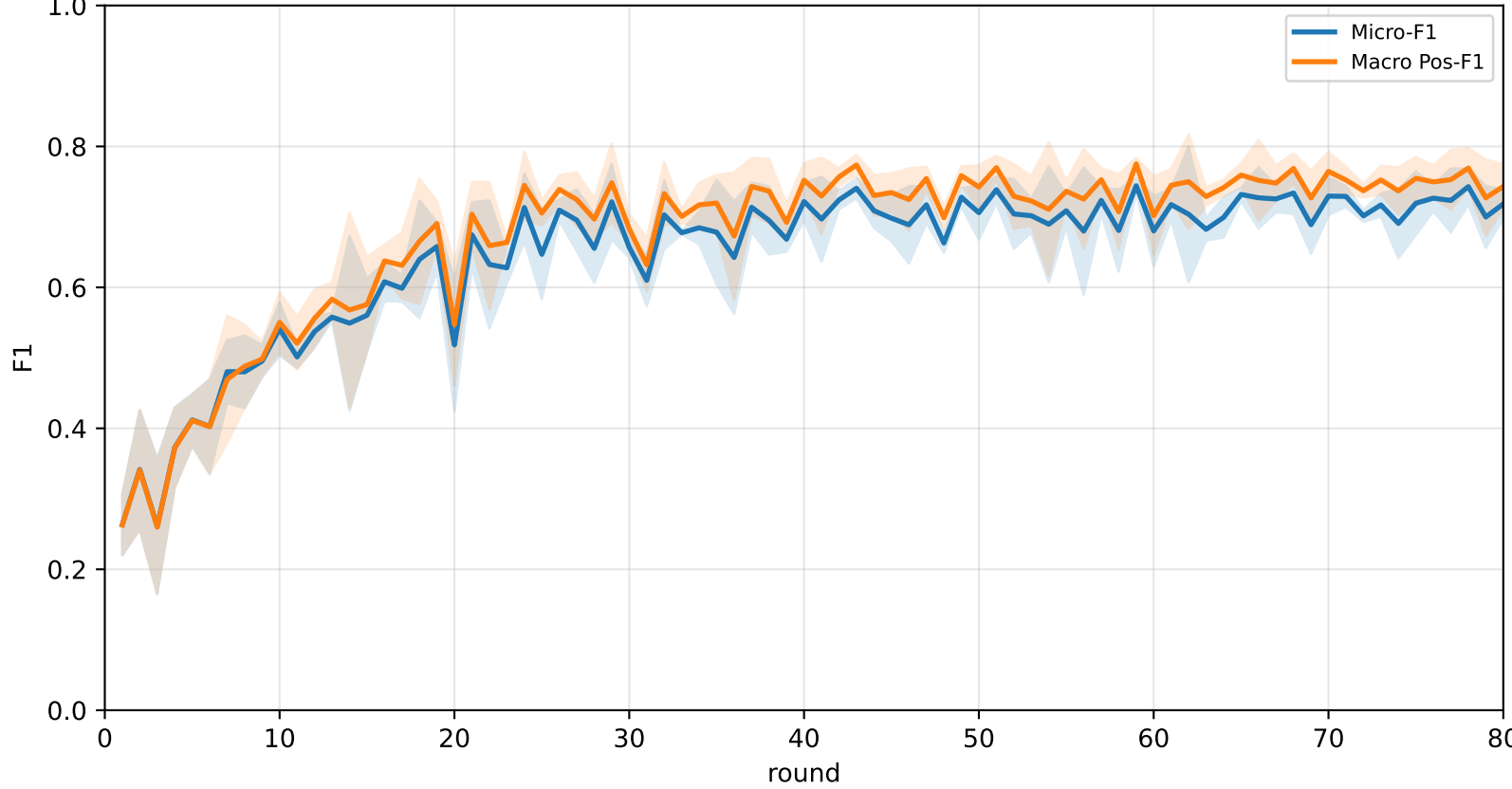
Validation loss: baselines vs selected APPLE-QOracle and end-of-round mismatch (rounds=80)



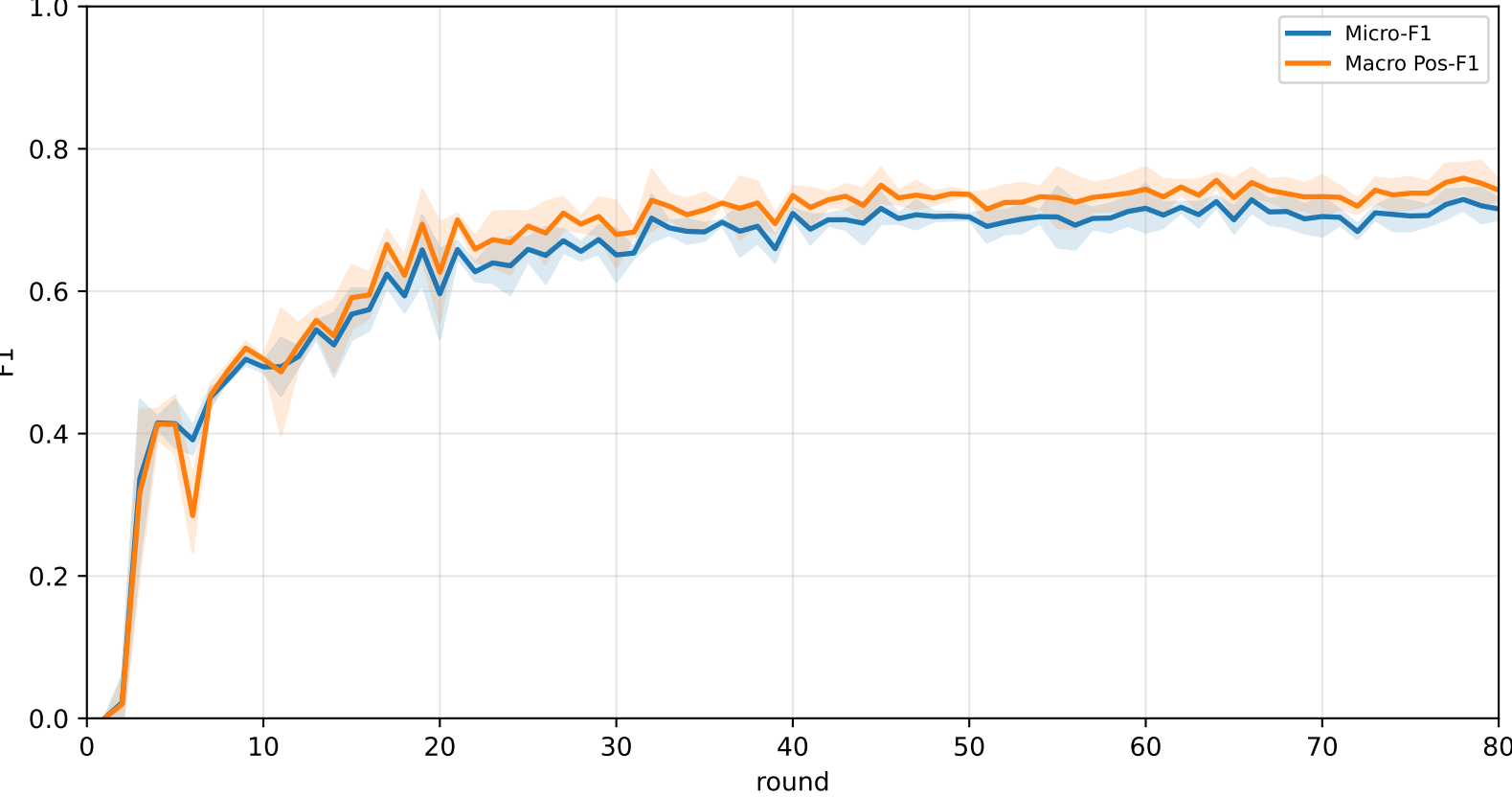
Pooled baseline micro / macro F1



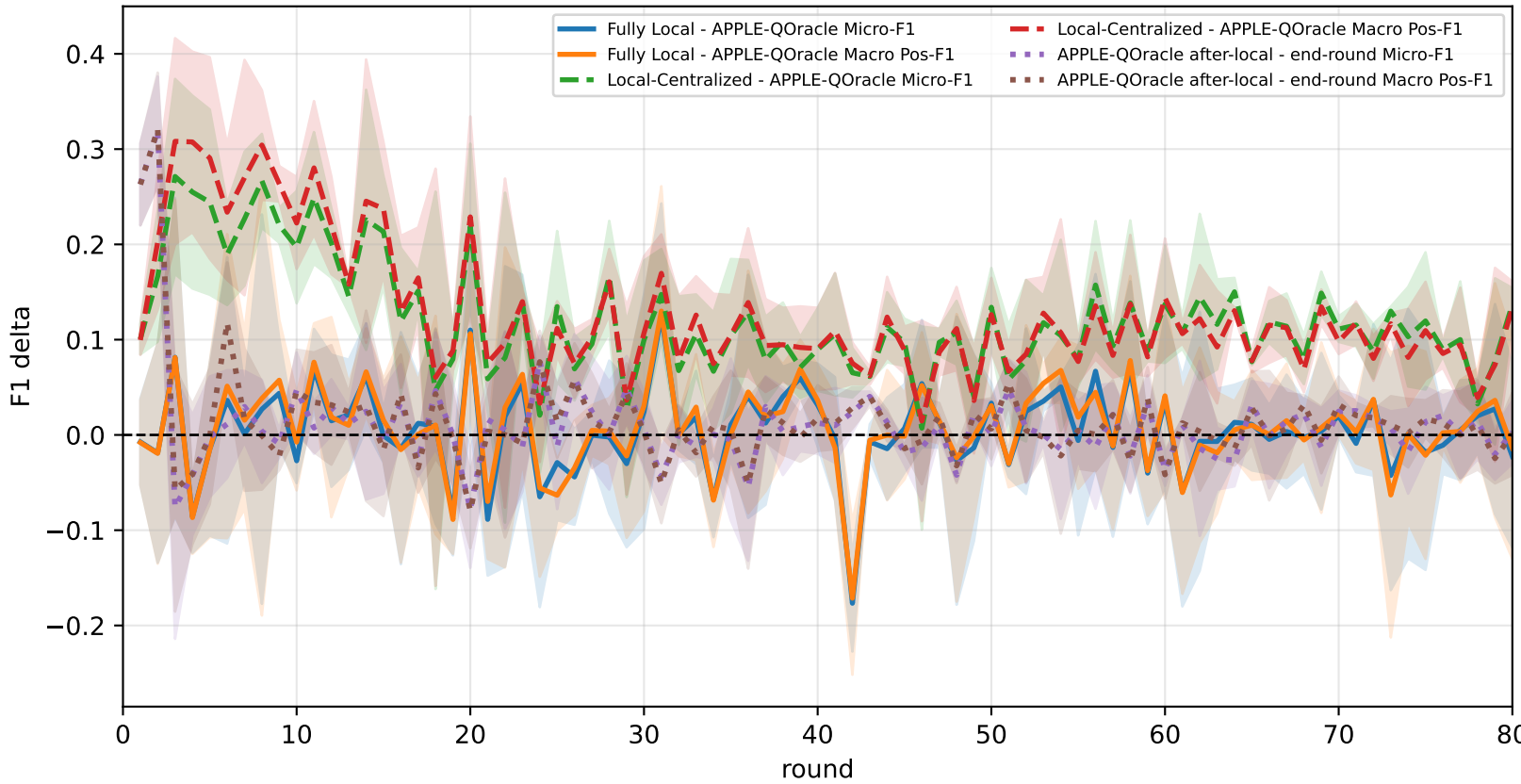
Pooled APPLE-QOracle micro / macro F1 (after-local selected model)



Pooled APPLE-QOracle micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QOracle mismatch gap



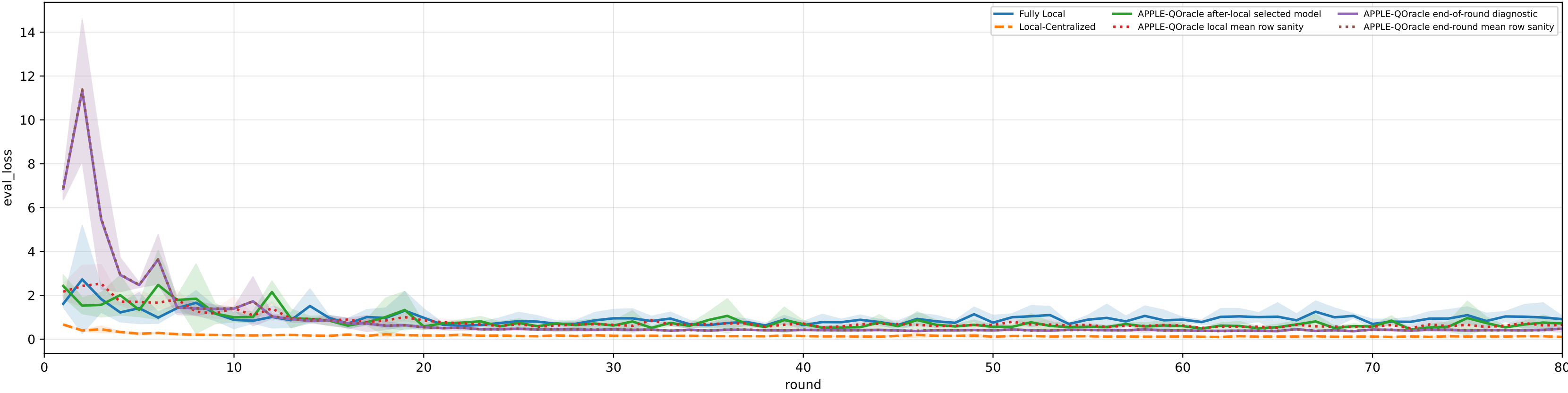
Controlled q-label heterogeneity: iid | APPLE-QOracle diagnostics | family=q_task_cycle3 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

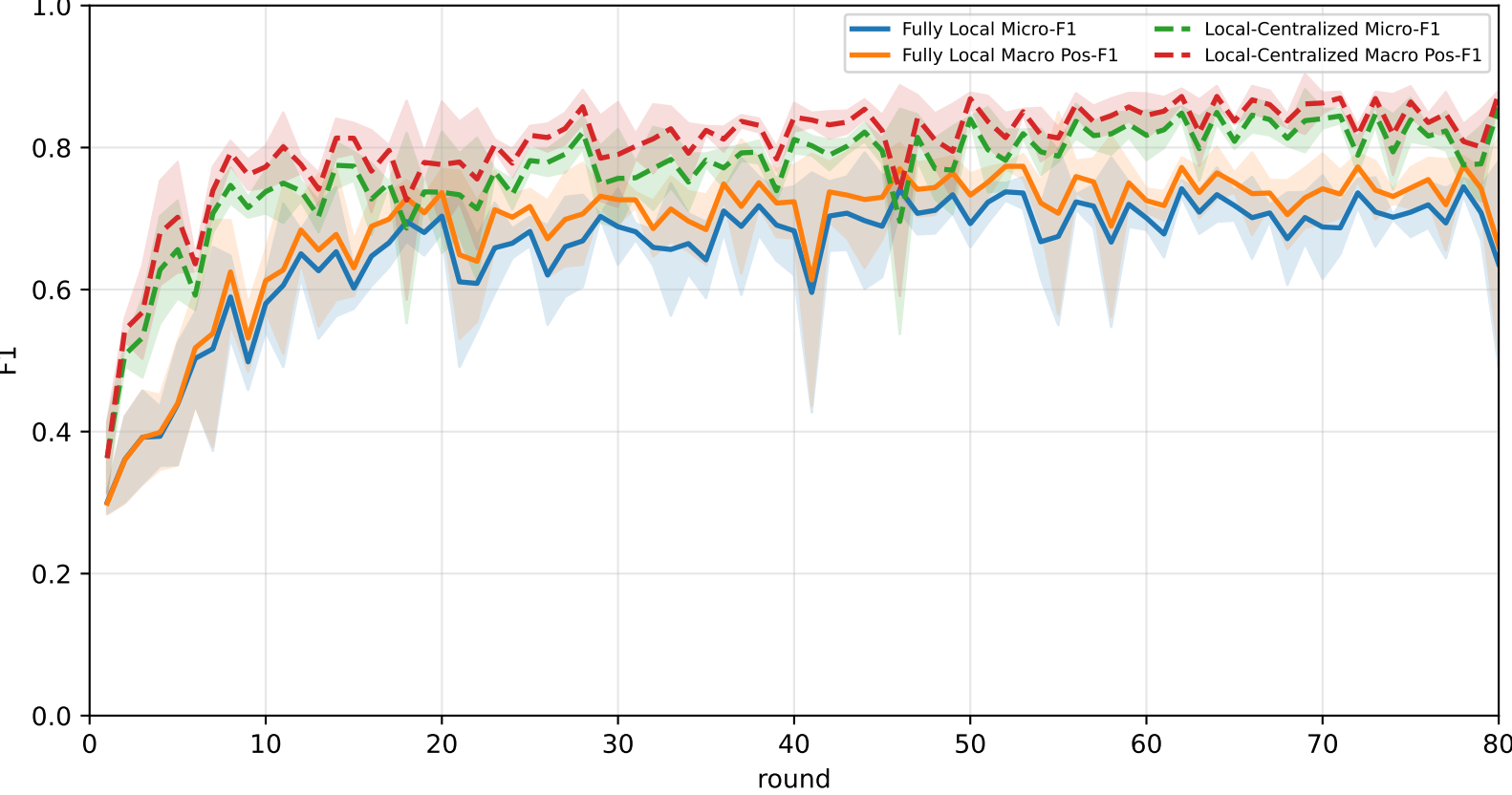
Best test performance (Fully Local vs Local-Centralized vs APPLE-QOracle, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	69.5% ± 0.8%	73.5% ± 0.6%	90.4% ± 0.9%	<u>85.2% ± 1.6%</u>	67.7% ± 0.6%	<u>71.2% ± 1.1%</u>	52.9% ± 2.1%
Local-Centralized	85.0% ± 0.7%	87.0% ± 0.6%	98.0% ± 1.1%	94.5% ± 1.4%	89.7% ± 1.8%	80.3% ± 0.5%	72.5% ± 1.3%
APPLE-QOracle	<u>71.2% ± 3.3%</u>	<u>74.6% ± 2.6%</u>	<u>93.1% ± 1.1%</u>	84.3% ± 1.6%	<u>70.8% ± 6.5%</u>	68.0% ± 6.2%	<u>57.1% ± 3.0%</u>

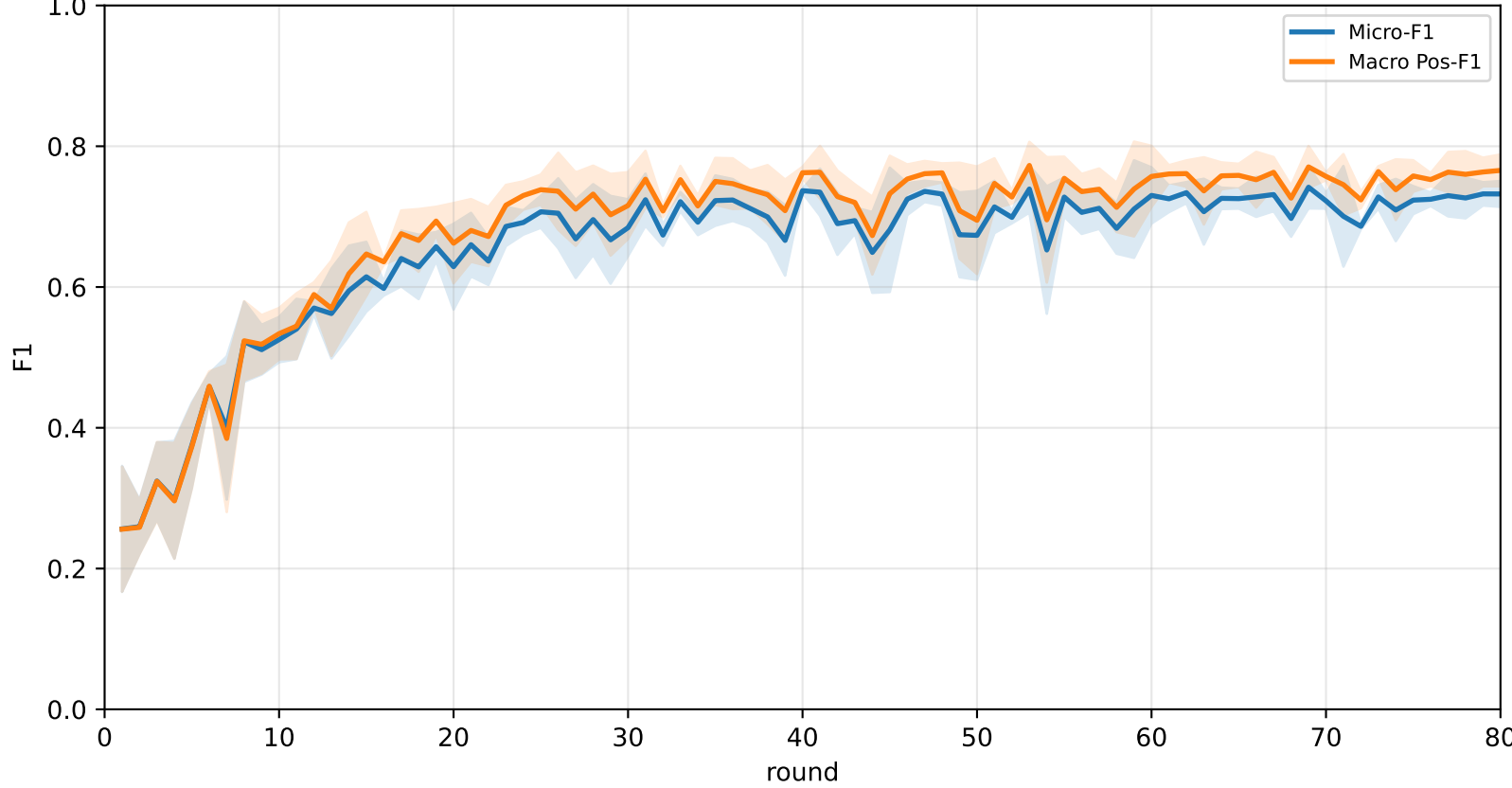
Validation loss: baselines vs selected APPLE-QOracle and end-of-round mismatch (rounds=80)



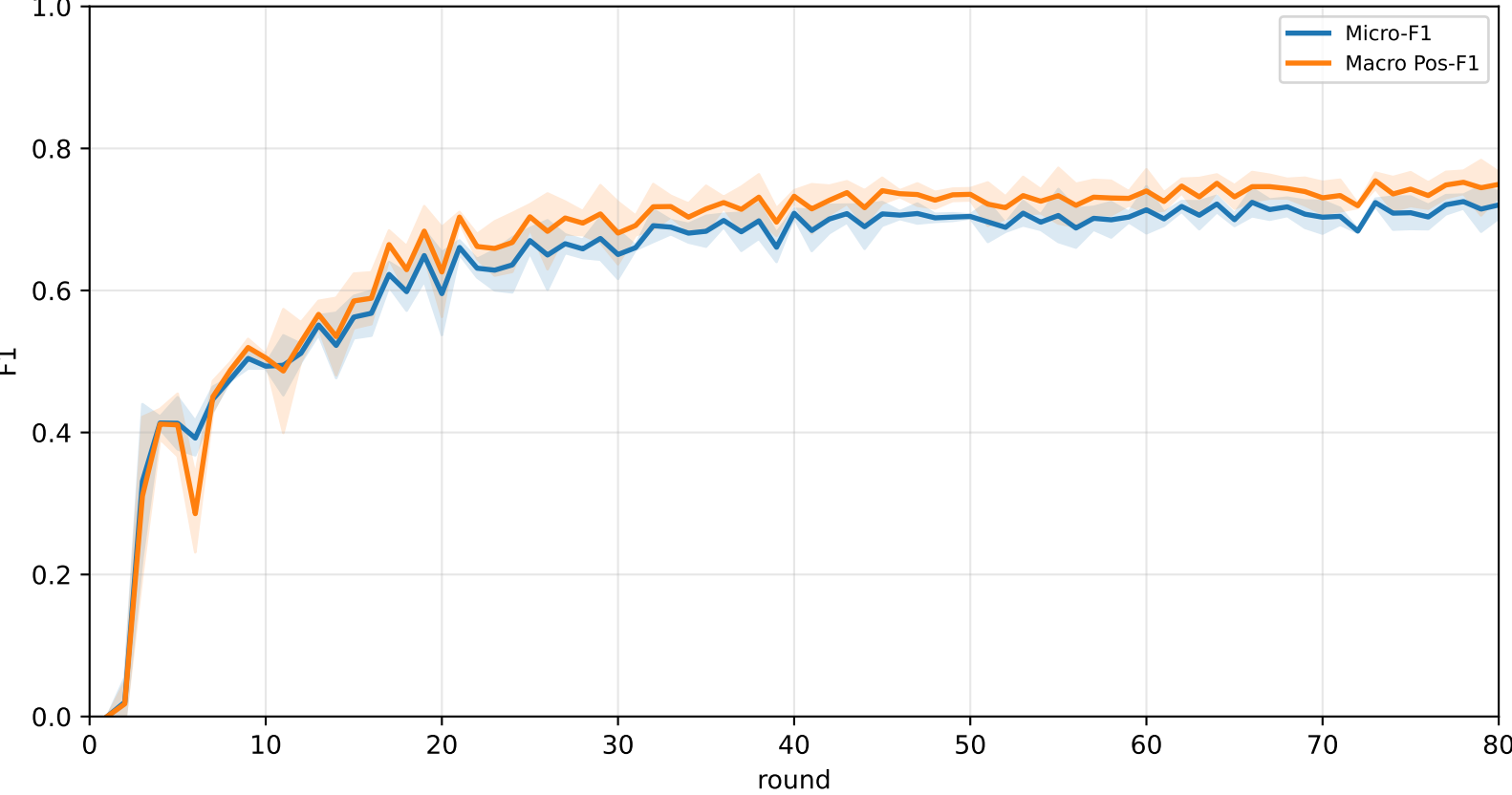
Pooled baseline micro / macro F1



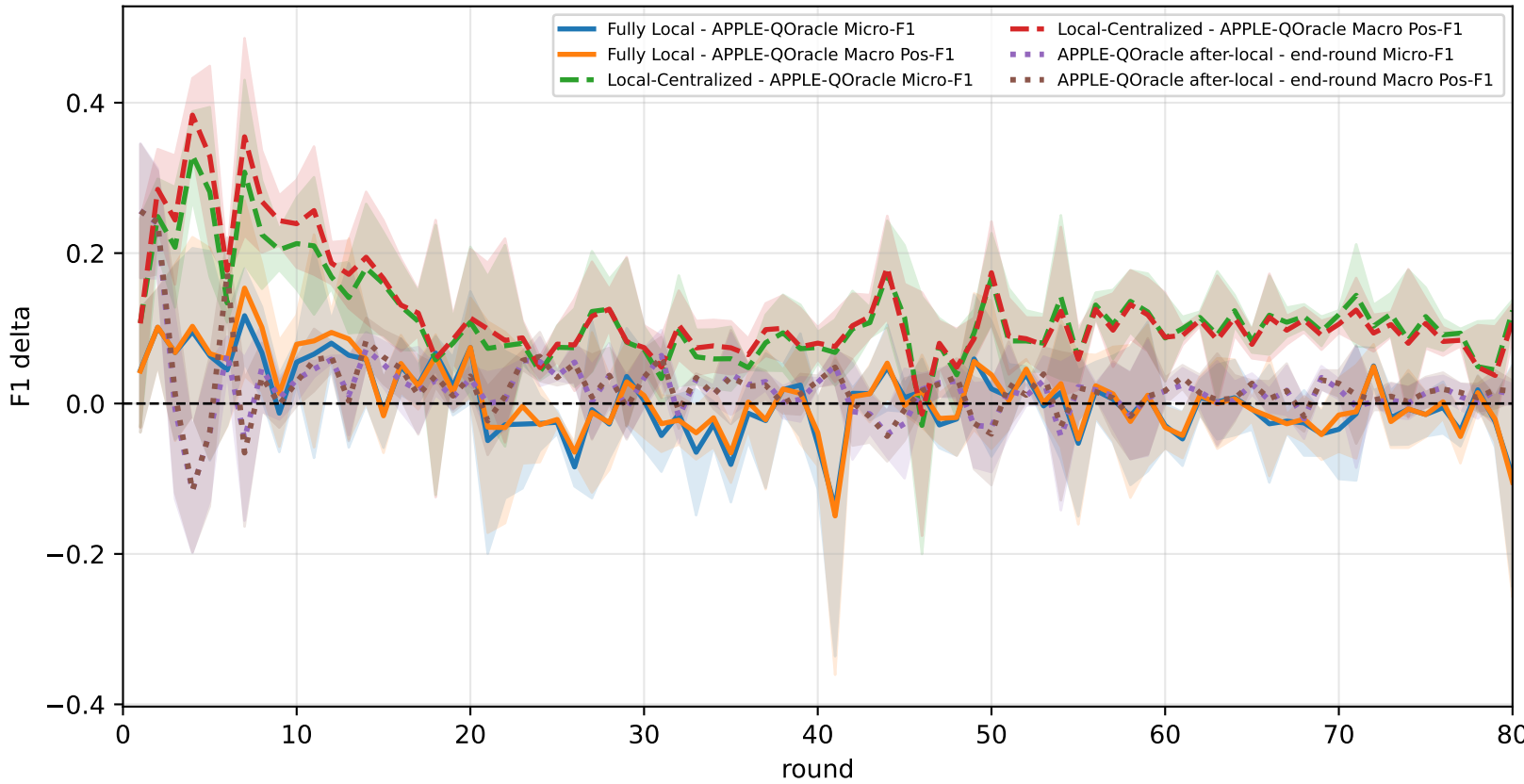
Pooled APPLE-QOracle micro / macro F1 (after-local selected model)



Pooled APPLE-QOracle micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QOracle mismatch gap

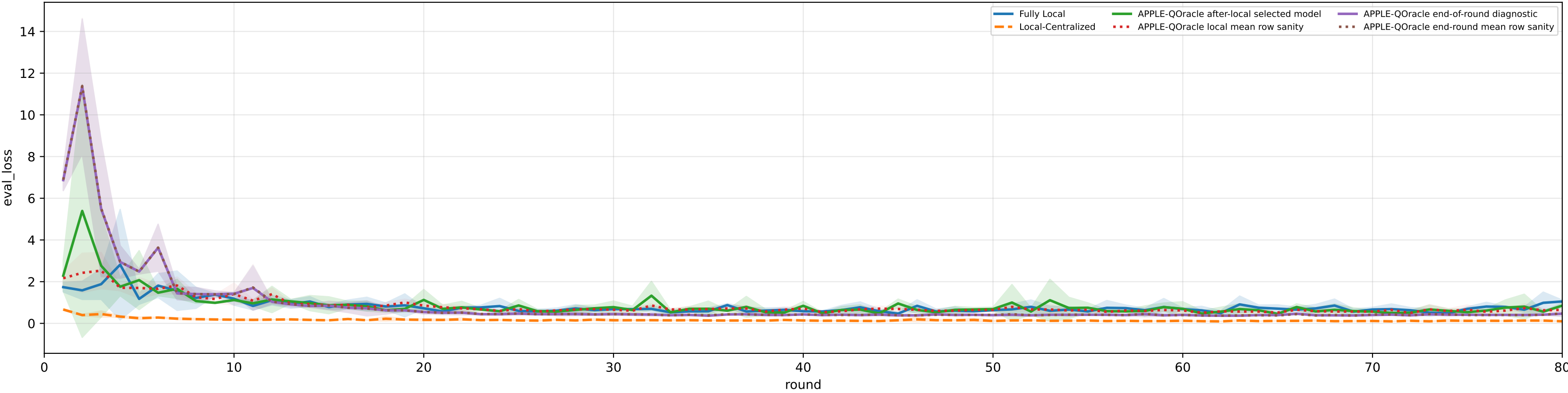


family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

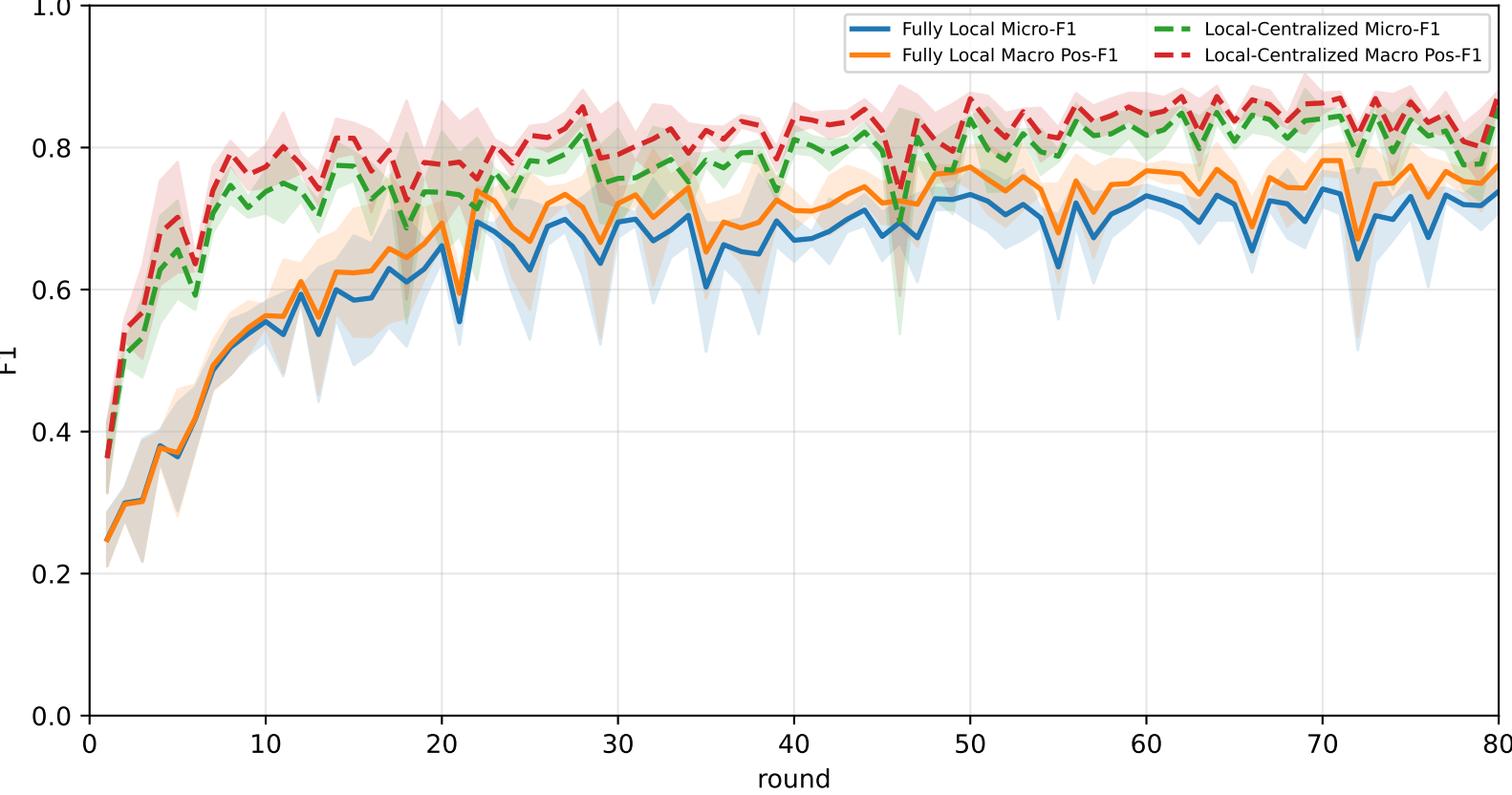
Best test performance (Fully Local vs Local-Centralized vs APPLE-QOracle, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	64.8% ± 3.3%	70.1% ± 2.7%	<u>92.2% ± 3.4%</u>	<u>82.7% ± 3.5%</u>	65.7% ± 8.3%	60.9% ± 7.0%	48.9% ± 2.7%
Local-Centralized	85.0% ± 0.7%	87.0% ± 0.6%	98.0% ± 1.1%	94.5% ± 1.4%	89.7% ± 1.8%	80.3% ± 0.5%	72.5% ± 1.3%
APPLE-QOracle	<u>72.4% ± 1.6%</u>	<u>74.8% ± 1.8%</u>	87.4% ± 1.1%	<u>82.7% ± 5.9%</u>	<u>76.2% ± 3.7%</u>	<u>68.5% ± 1.1%</u>	<u>59.3% ± 0.6%</u>

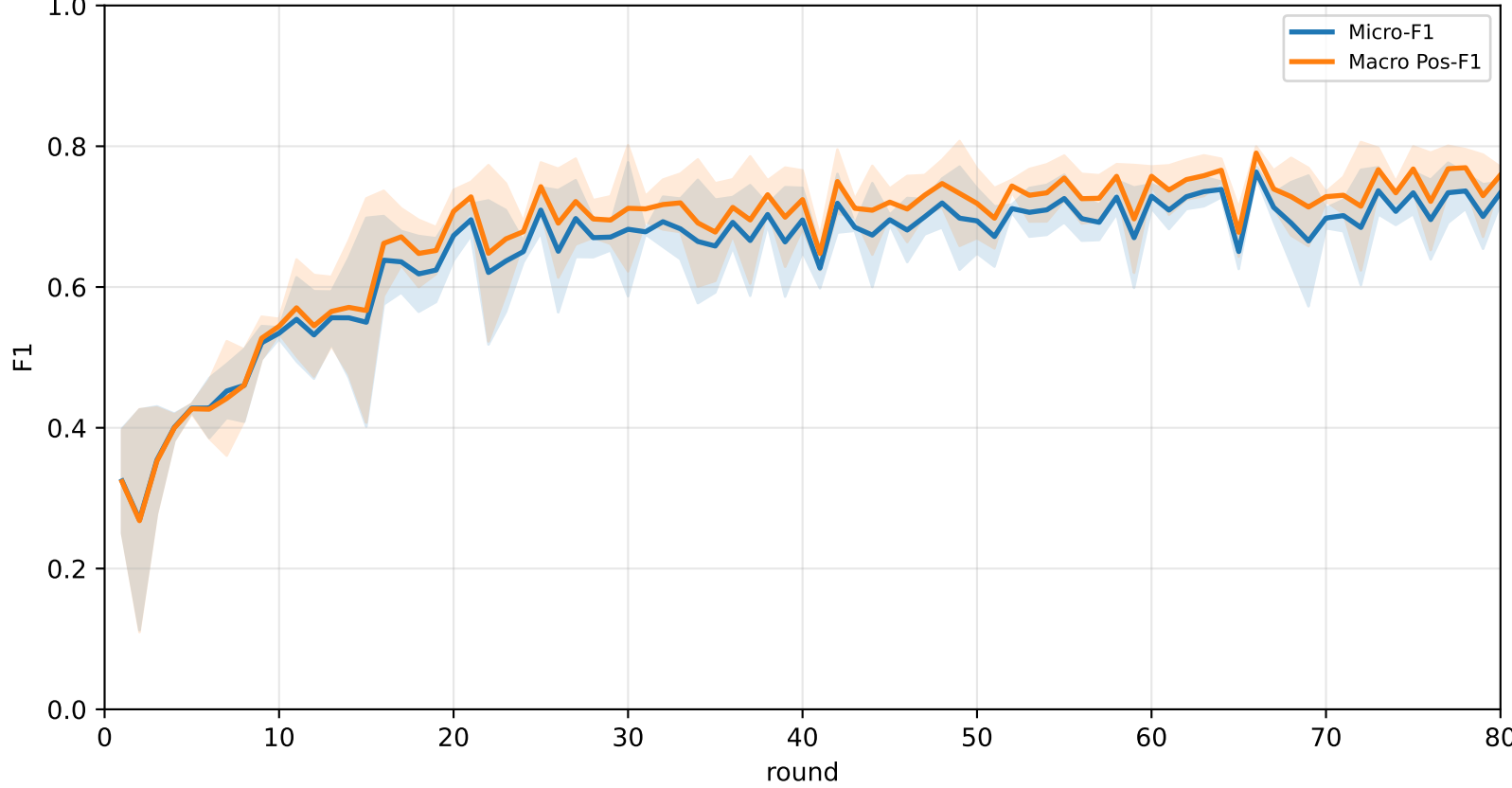
Validation loss: baselines vs selected APPLE-QOracle and end-of-round mismatch (rounds=80)



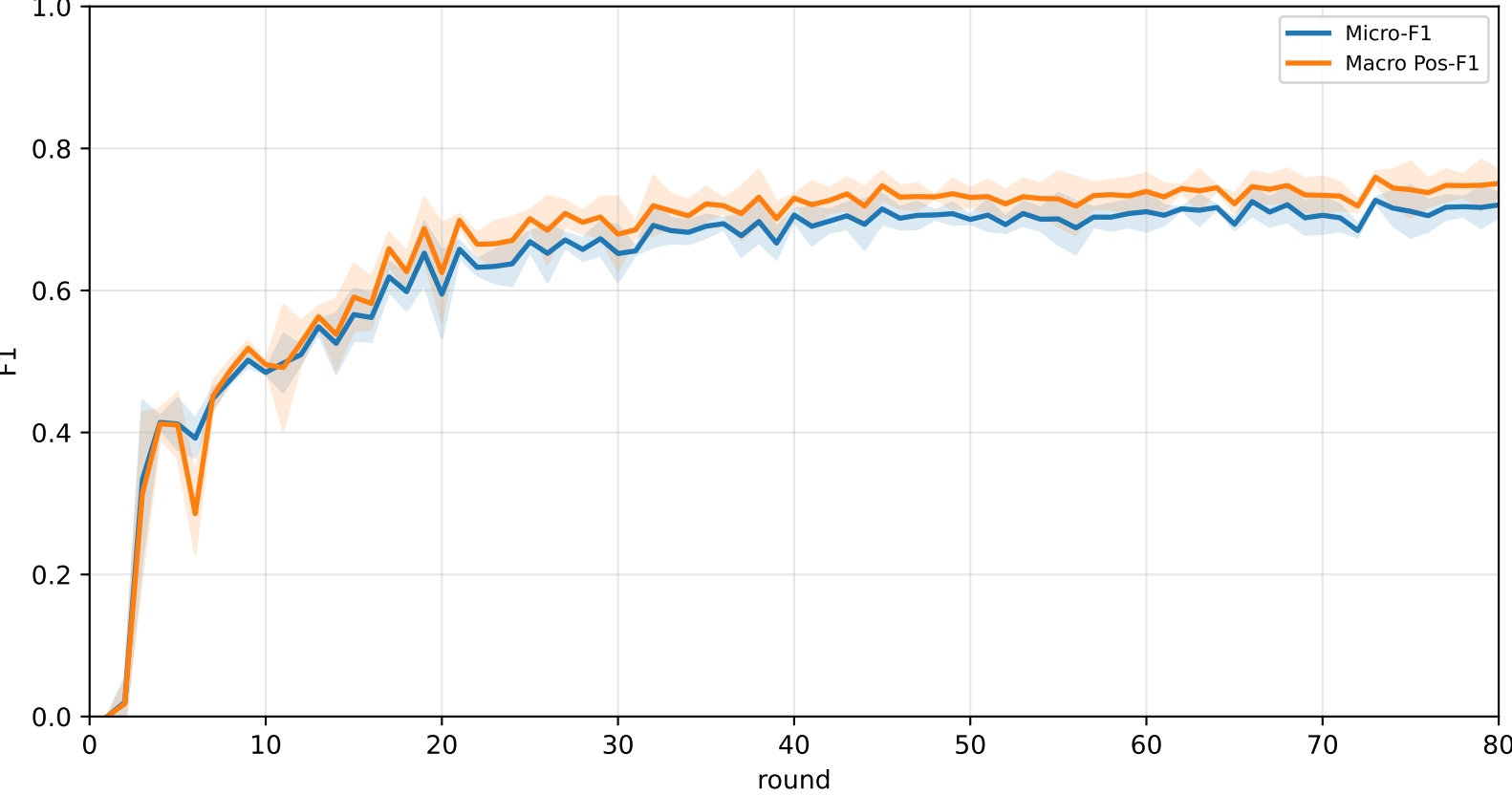
Pooled baseline micro / macro F1



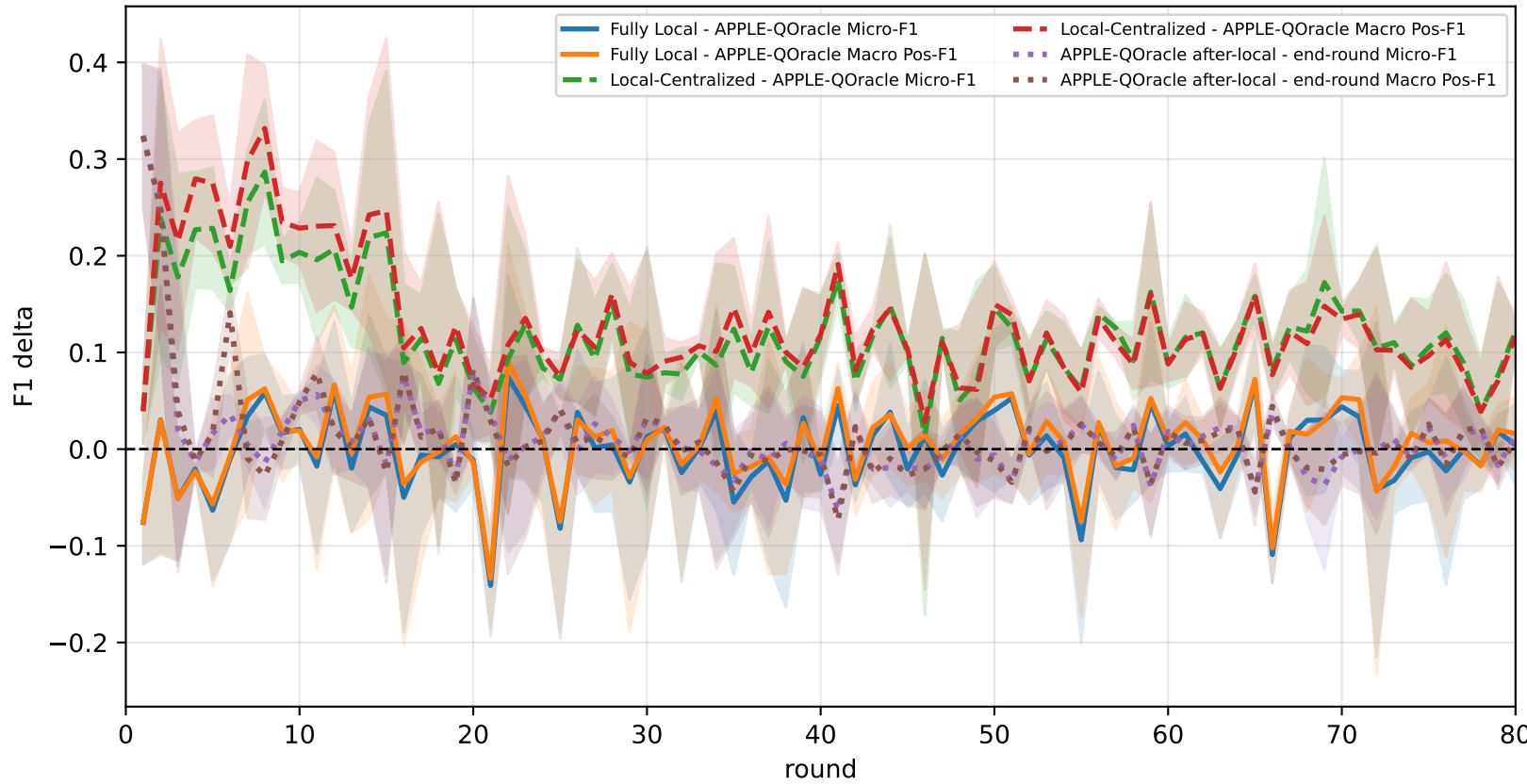
Pooled APPLE-QOracle micro / macro F1 (after-local selected model)



Pooled APPLE-QOracle micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QOracle mismatch gap

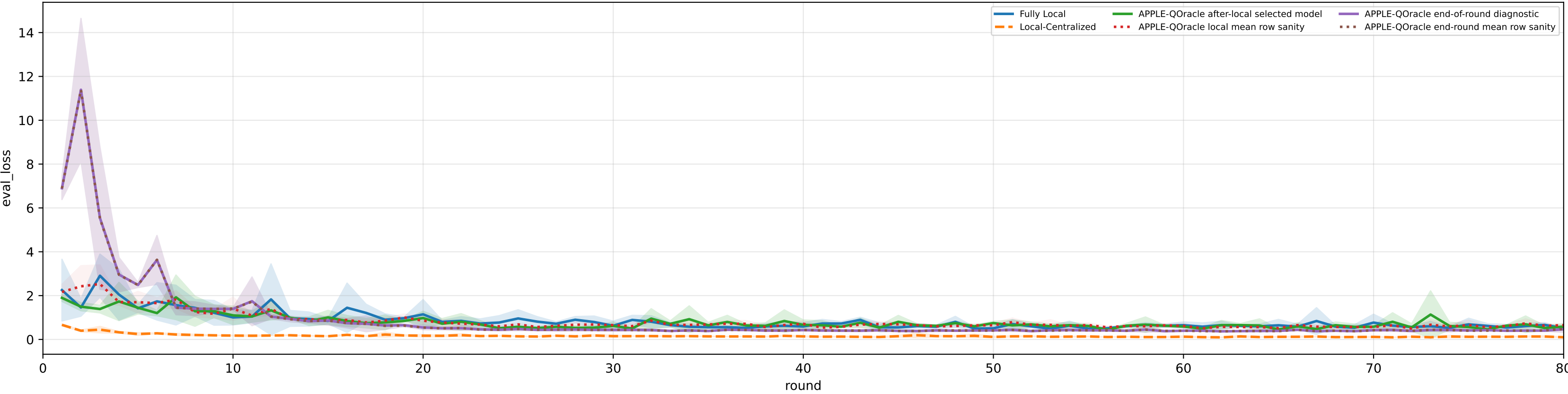


family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

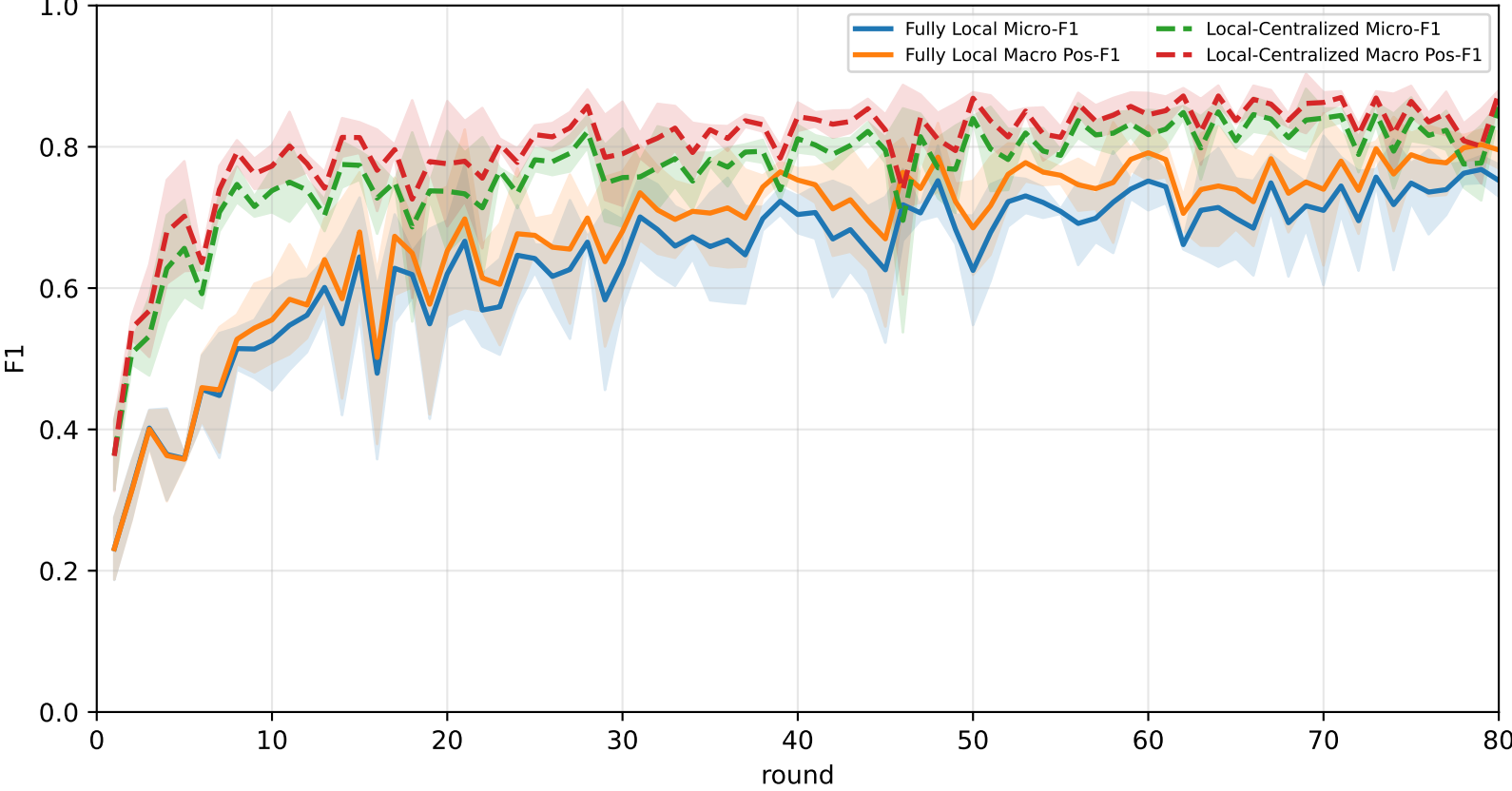
Best test performance (Fully Local vs Local-Centralized vs APPLE-QOracle, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	<u>70.5% ± 1.9%</u>	<u>74.2% ± 2.2%</u>	87.8% ± 6.8%	<u>86.4% ± 2.4%</u>	<u>74.0% ± 2.0%</u>	<u>69.7% ± 1.8%</u>	53.3% ± 1.7%
Local-Centralized	85.0% ± 0.7%	87.0% ± 0.6%	98.0% ± 1.1%	94.5% ± 1.4%	89.7% ± 1.8%	80.3% ± 0.5%	72.5% ± 1.3%
APPLE-QOracle	70.4% ± 2.9%	73.3% ± 3.1%	<u>88.2% ± 4.9%</u>	82.0% ± 4.1%	73.2% ± 2.3%	66.3% ± 5.5%	<u>56.6% ± 2.2%</u>

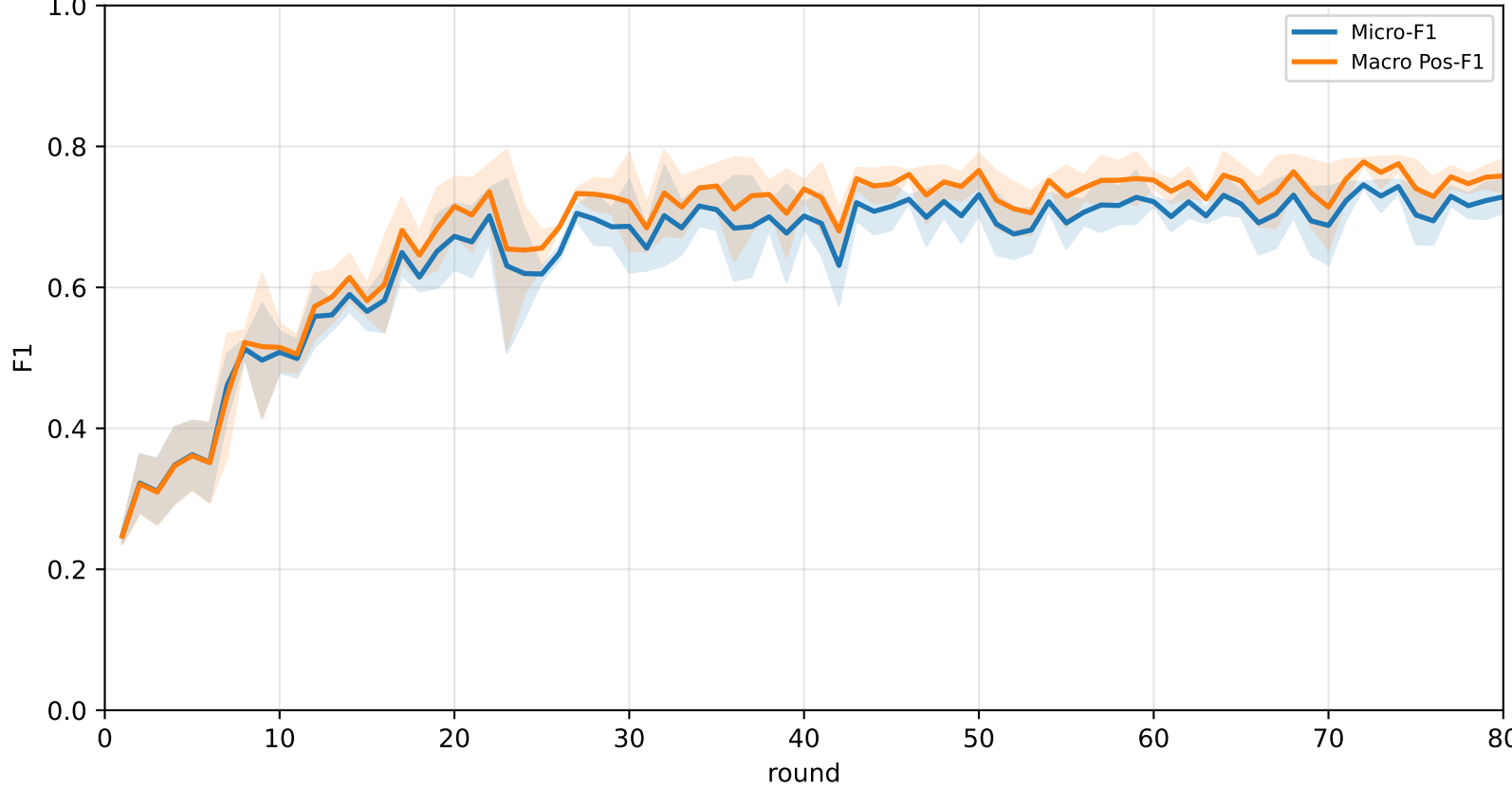
Validation loss: baselines vs selected APPLE-QOracle and end-of-round mismatch (rounds=80)



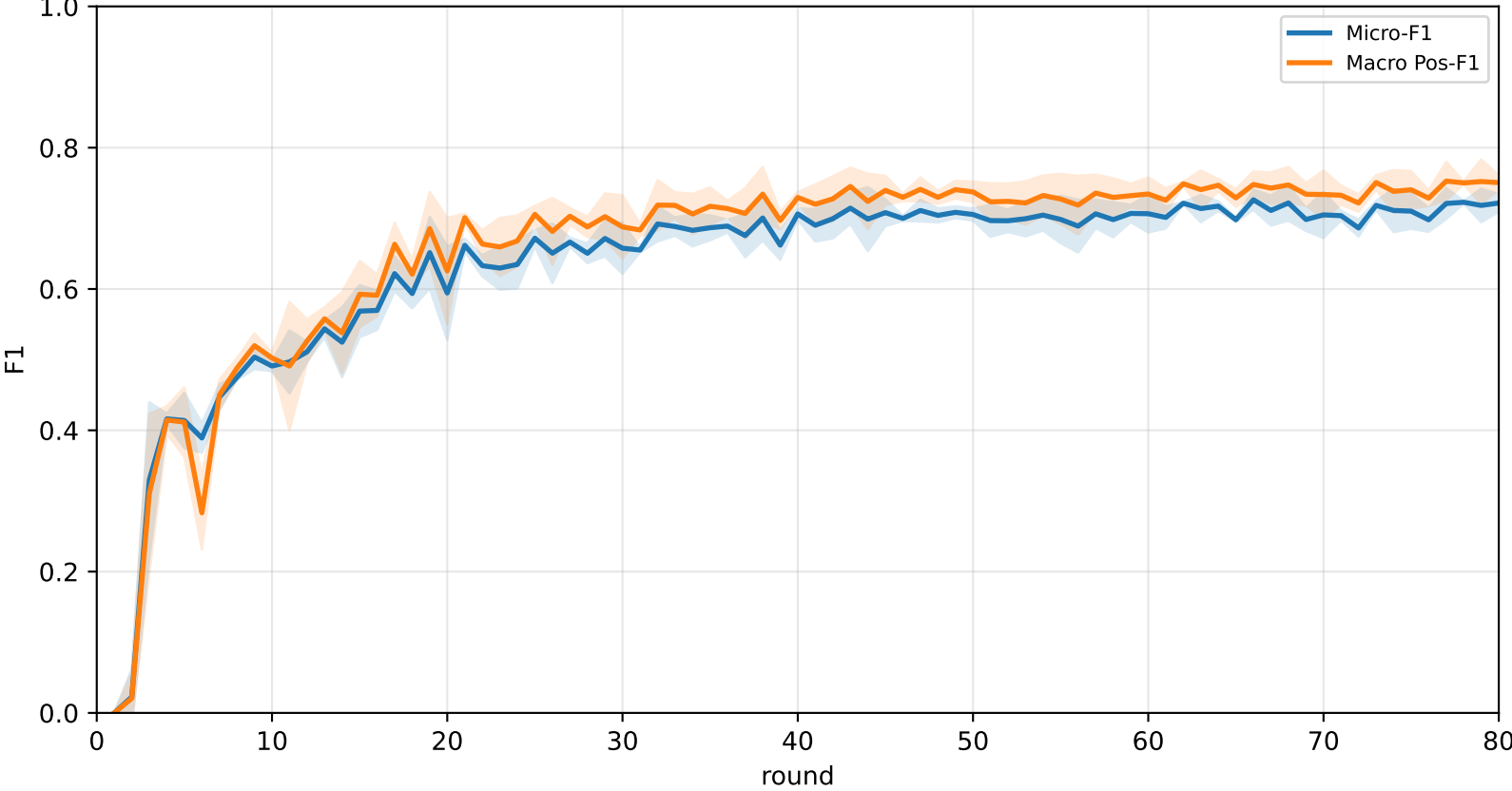
Pooled baseline micro / macro F1



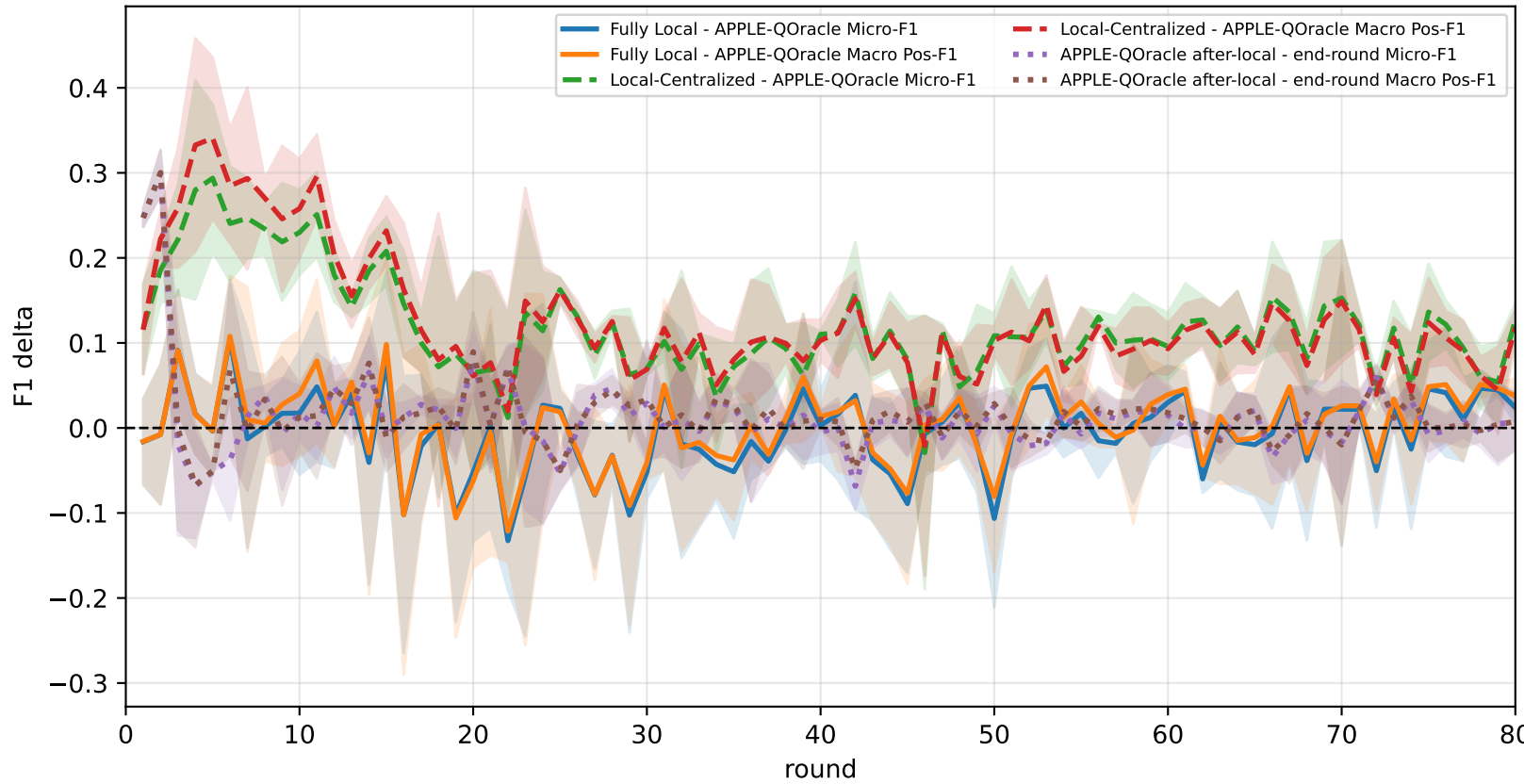
Pooled APPLE-QOracle micro / macro F1 (after-local selected model)



Pooled APPLE-QOracle micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QOracle mismatch gap

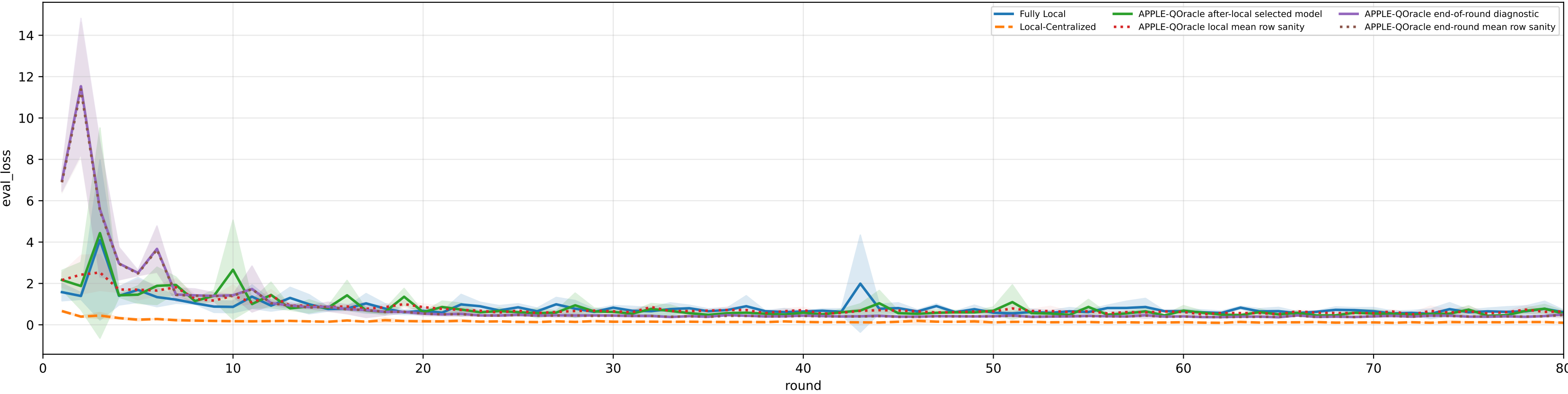


family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

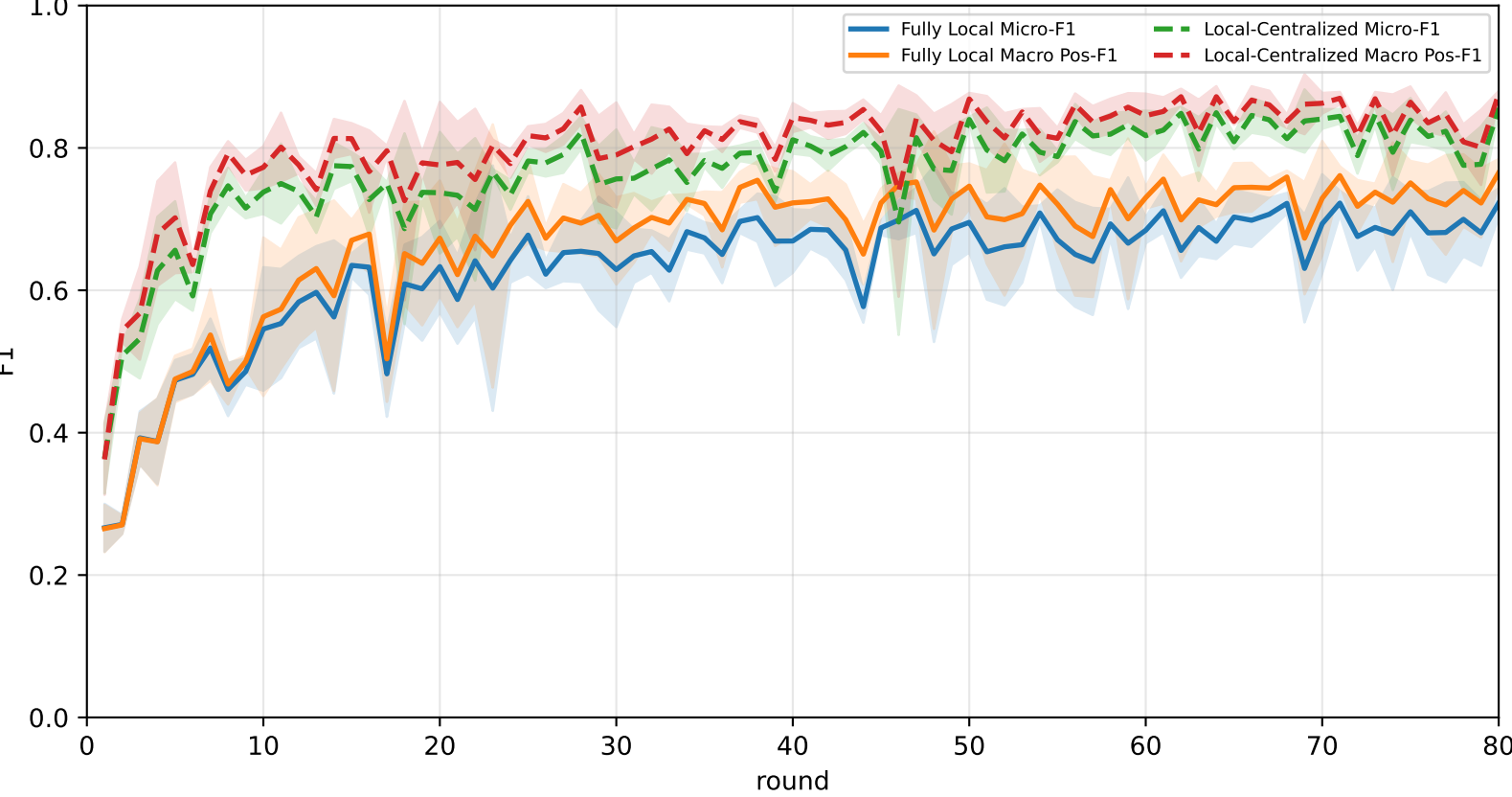
Best test performance (Fully Local vs Local-Centralized vs APPLE-QOracle, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	64.5% ± 3.5%	69.6% ± 3.3%	<u>87.3% ± 2.7%</u>	<u>84.4% ± 3.4%</u>	62.5% ± 7.2%	<u>67.0% ± 5.0%</u>	46.8% ± 1.7%
Local-Centralized	85.0% ± 0.7%	87.0% ± 0.6%	98.0% ± 1.1%	94.5% ± 1.4%	89.7% ± 1.8%	80.3% ± 0.5%	72.5% ± 1.3%
APPLE-QOracle	<u>67.8% ± 2.4%</u>	<u>71.1% ± 3.3%</u>	83.6% ± 10.1%	80.8% ± 2.8%	<u>72.9% ± 5.6%</u>	64.9% ± 2.6%	<u>53.2% ± 1.6%</u>

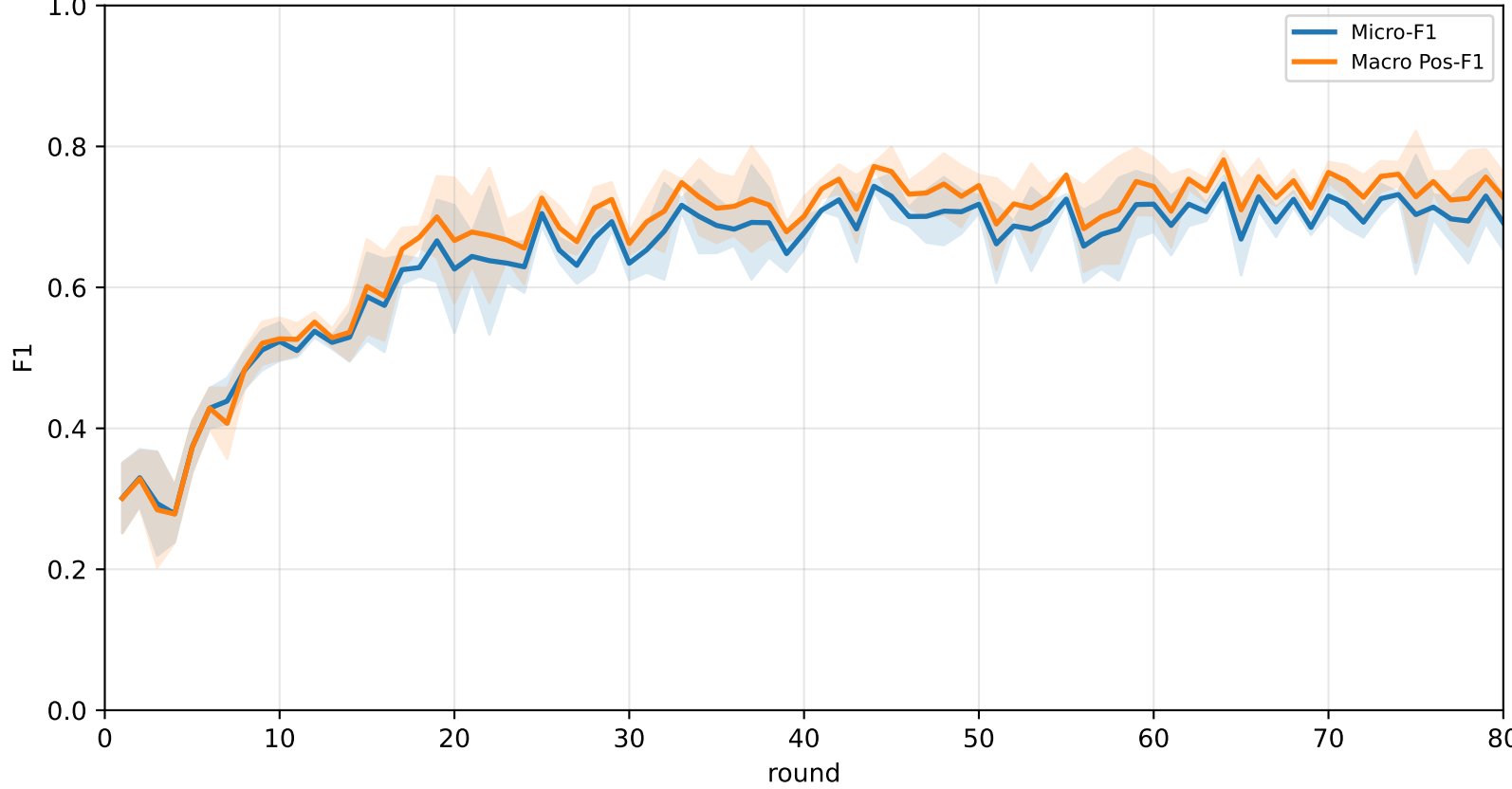
Validation loss: baselines vs selected APPLE-QOracle and end-of-round mismatch (rounds=80)



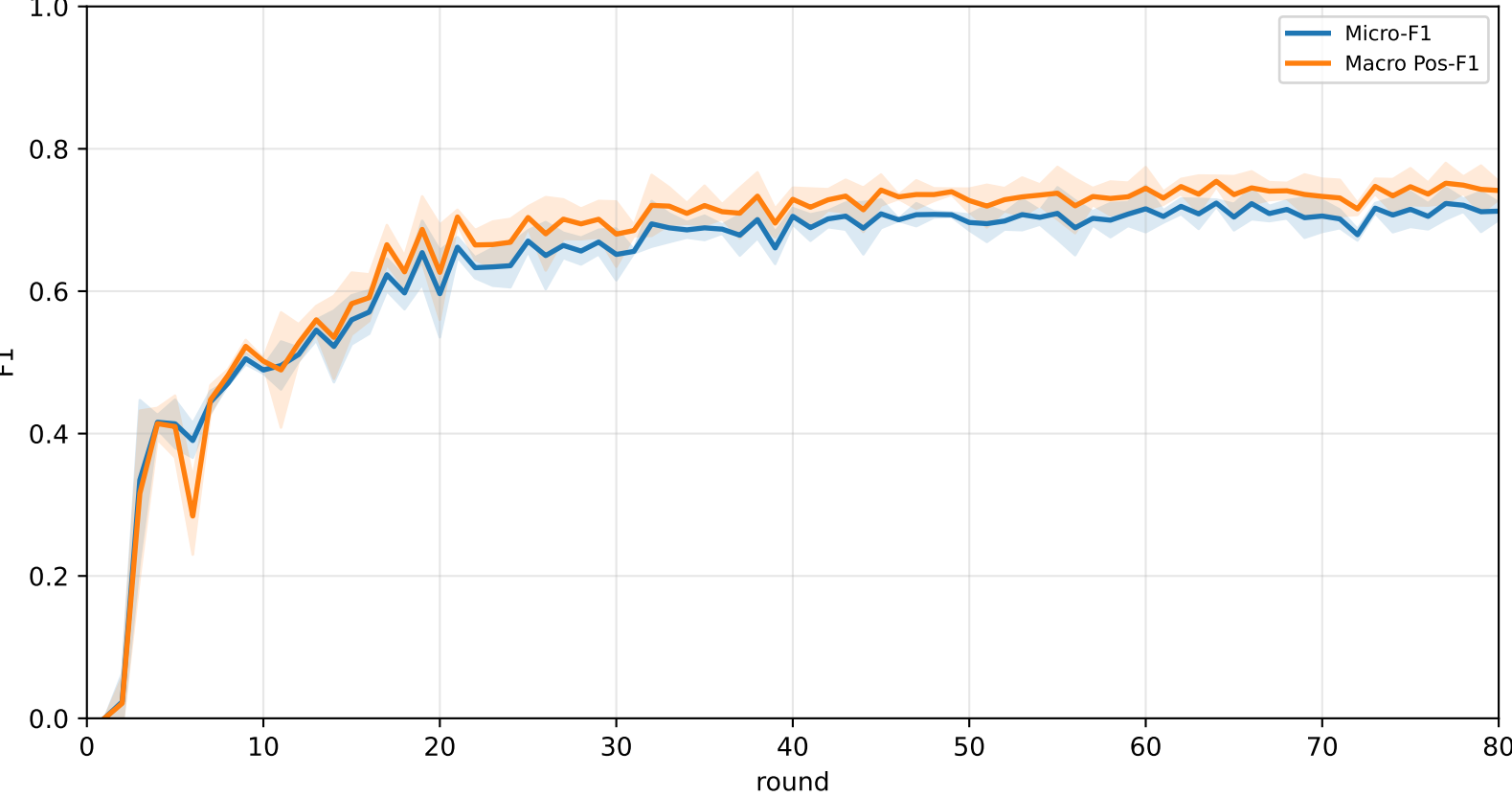
Pooled baseline micro / macro F1



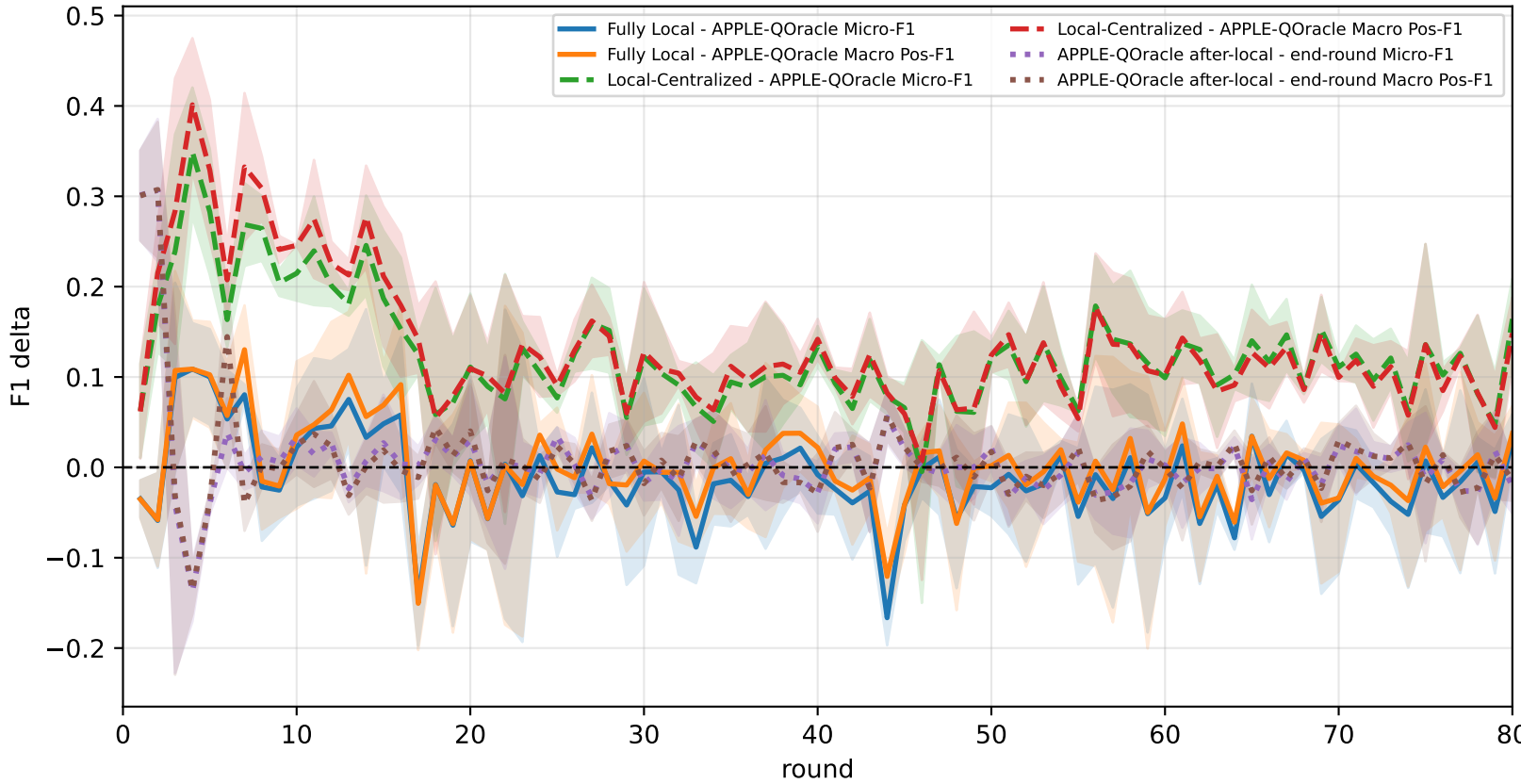
Pooled APPLE-QOracle micro / macro F1 (after-local selected model)



Pooled APPLE-QOracle micro / macro F1 (end-of-round diagnostic only)



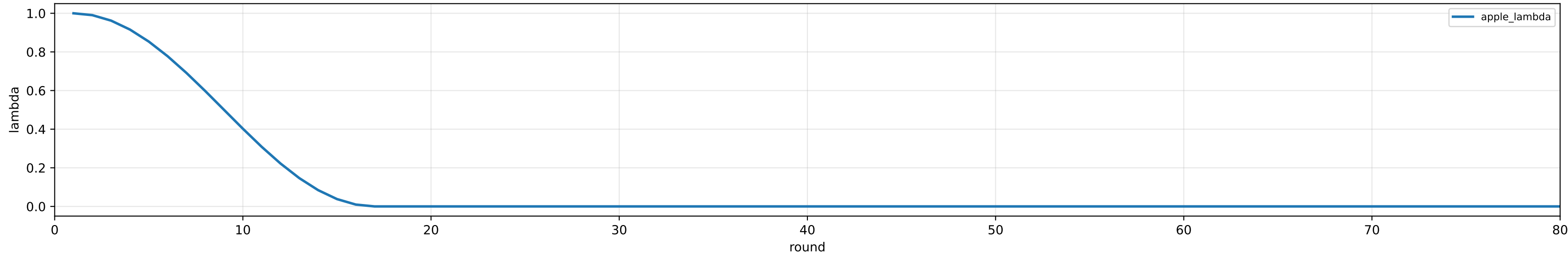
Delta diagnostics: baseline gaps and APPLE-QOracle mismatch gap



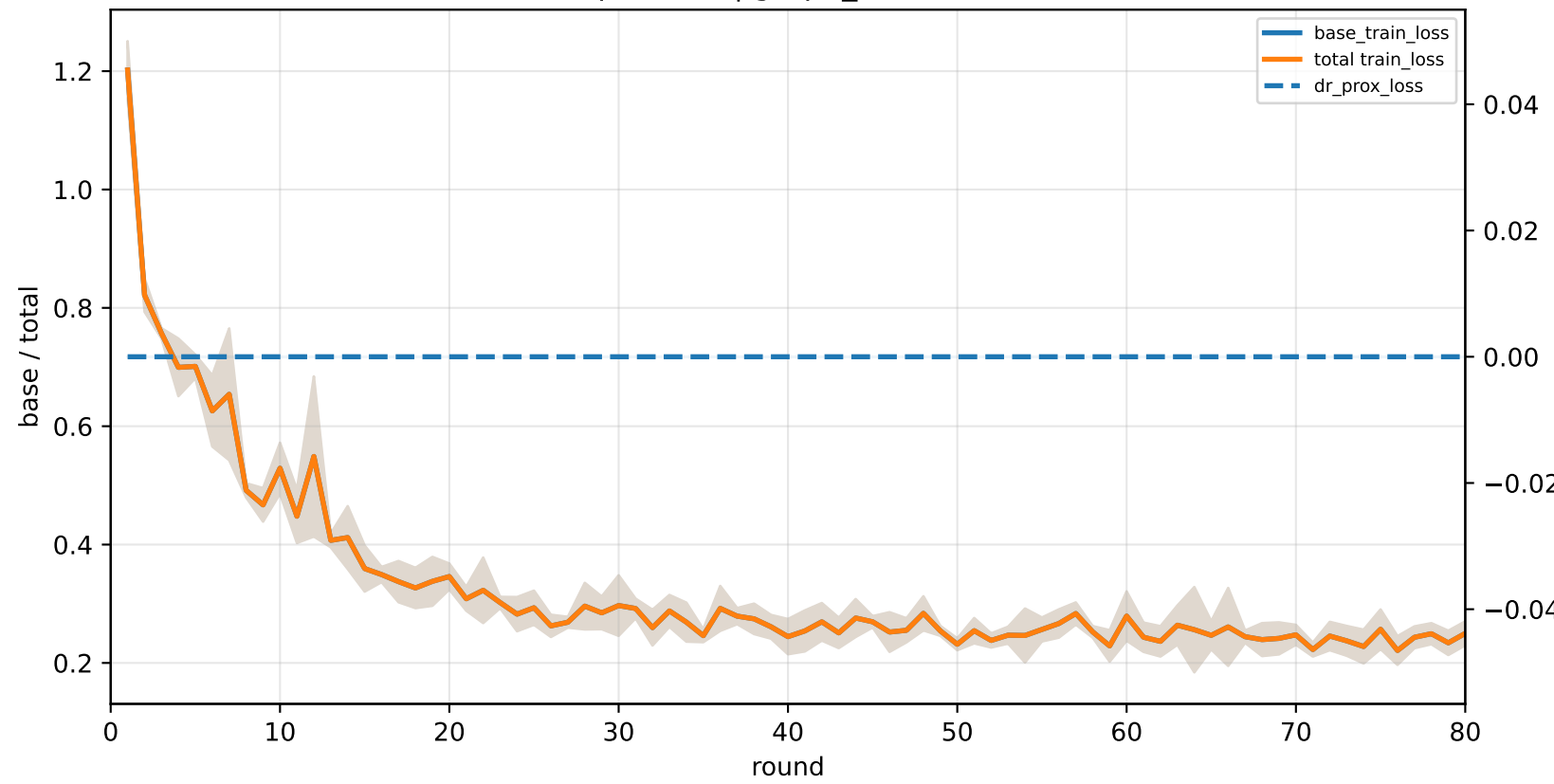
APPLE-QOracle training-loss decomposition | iid heterogeneity

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

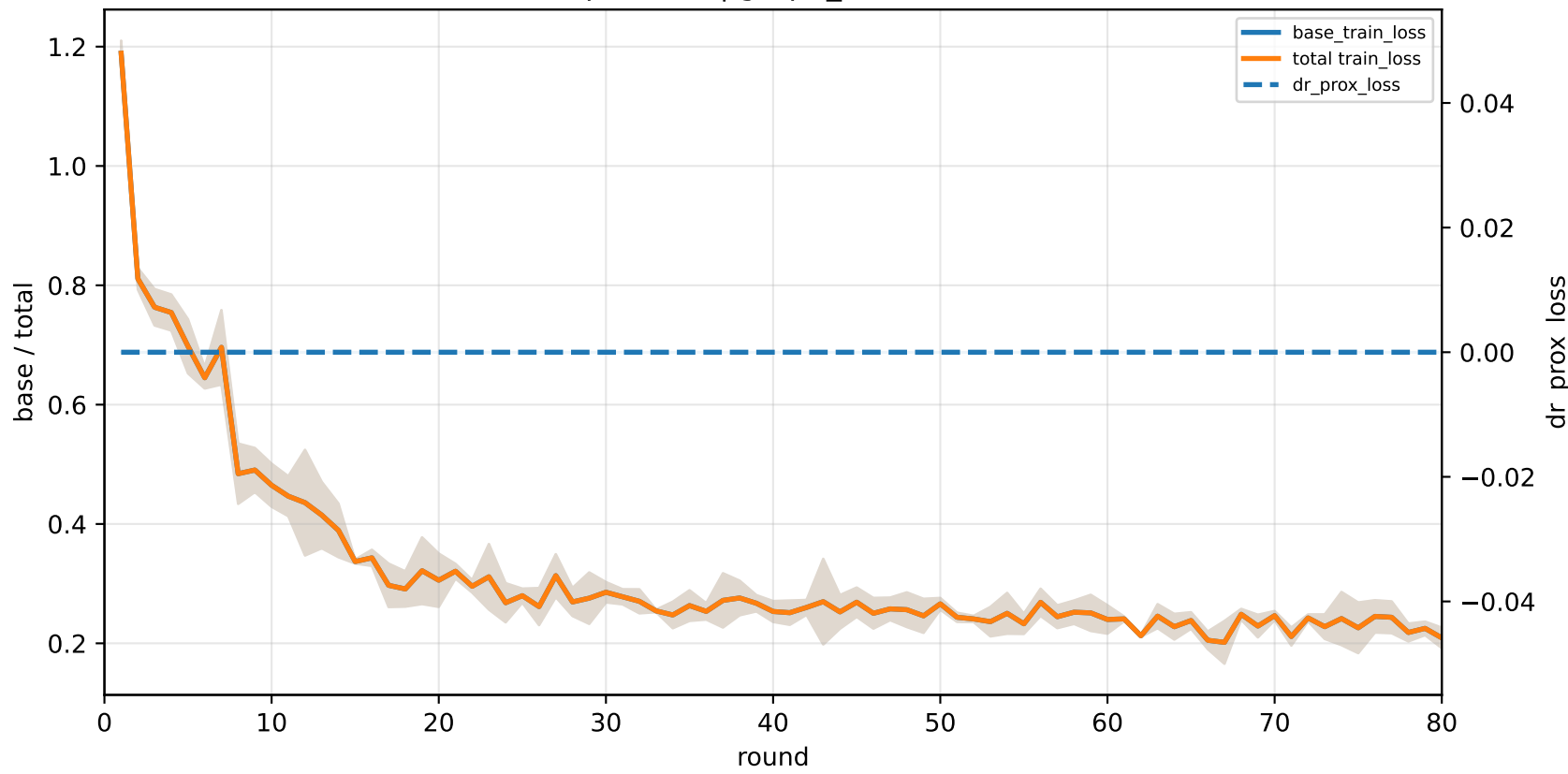
APPLE lambda scheduler | apple_mu=0



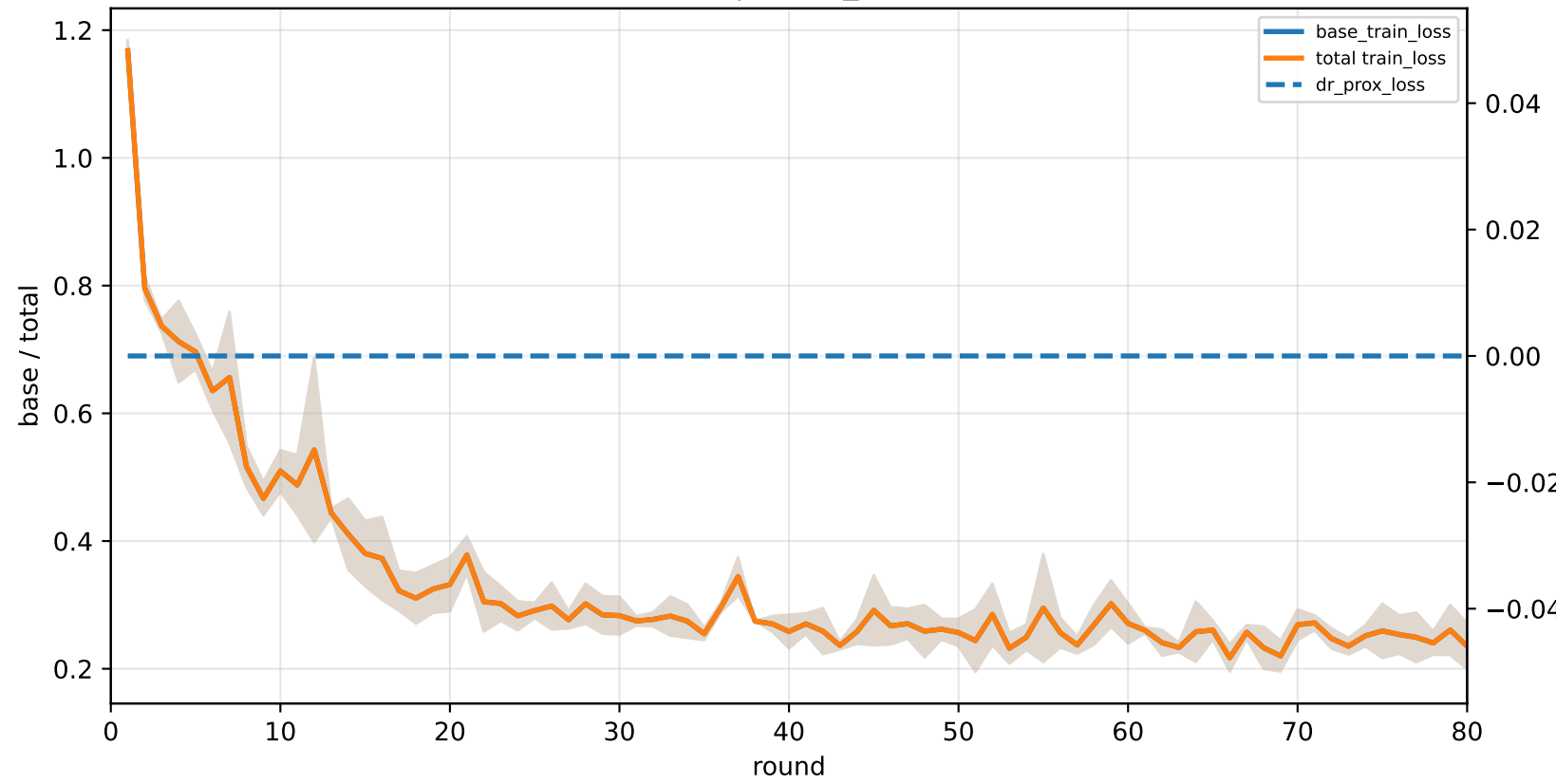
C2-specialist | graph_id=3000000



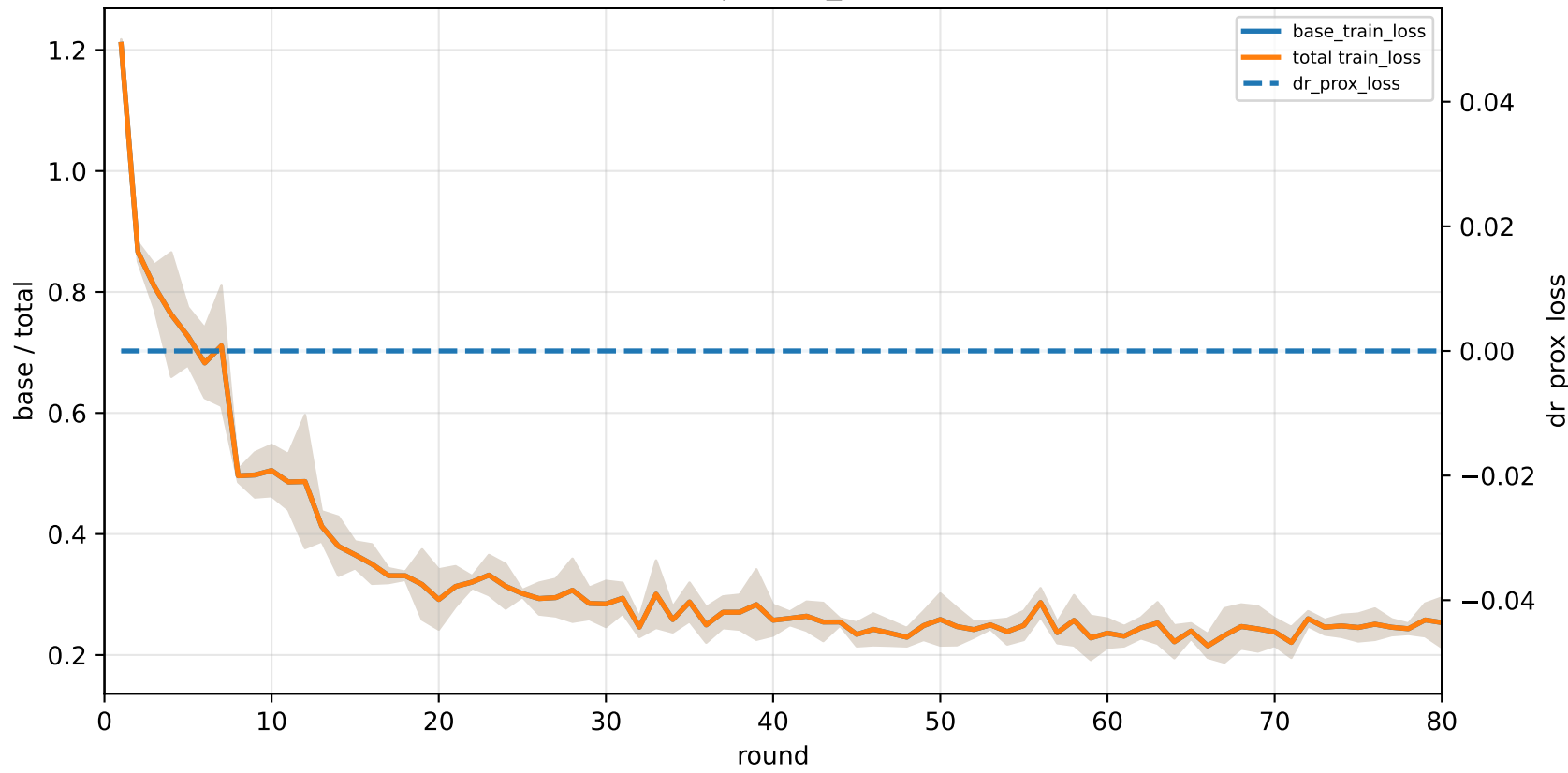
C3-specialist | graph_id=3000001



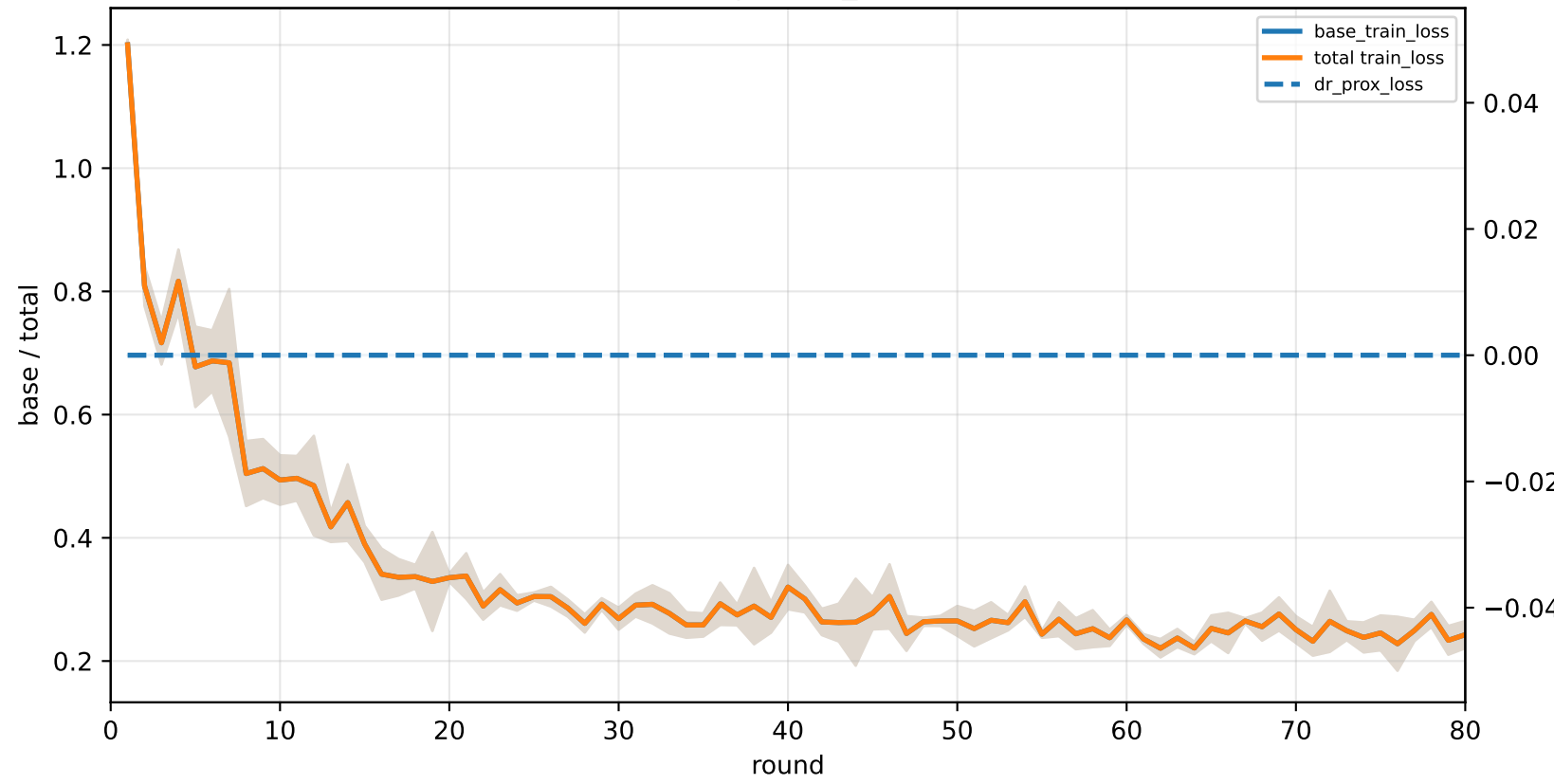
C4-specialist | graph_id=3000002



C5-specialist | graph_id=3000003

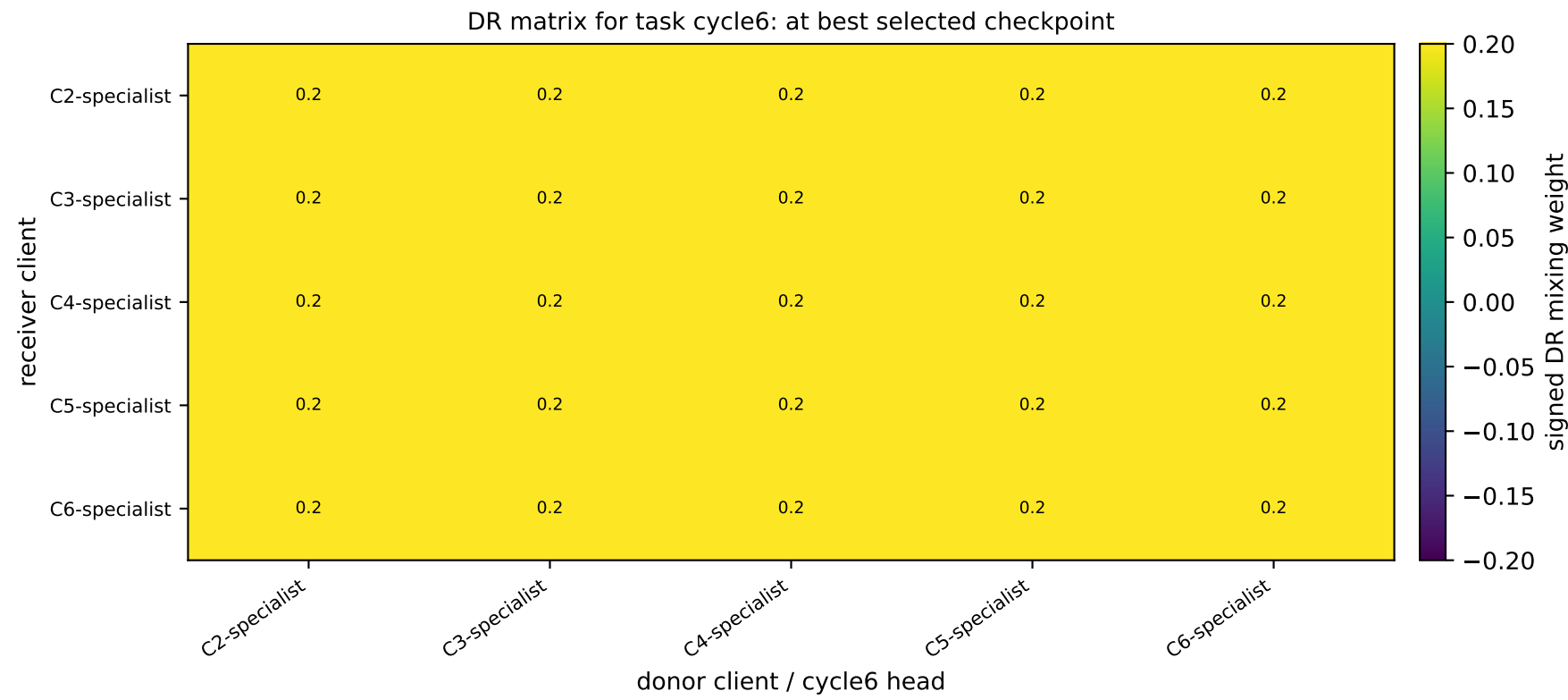
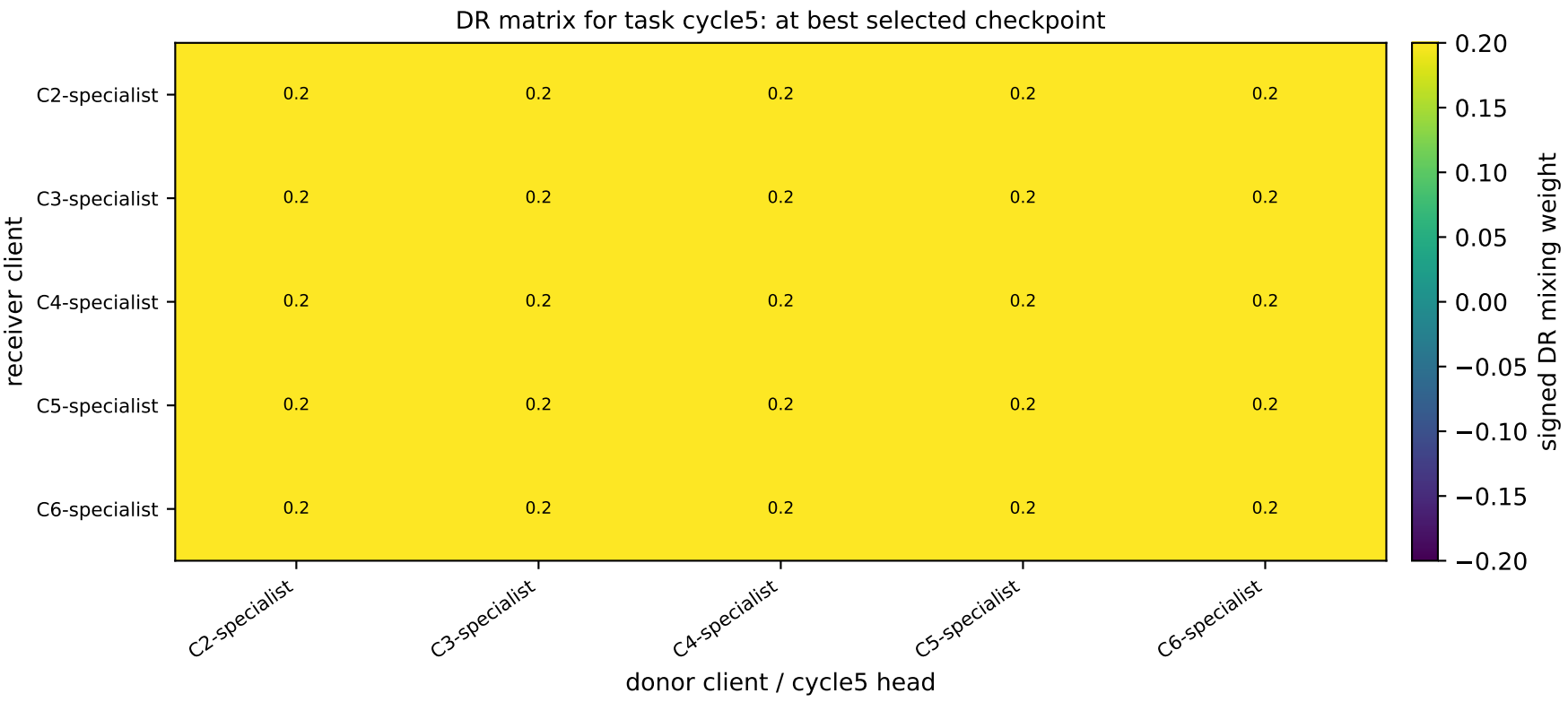
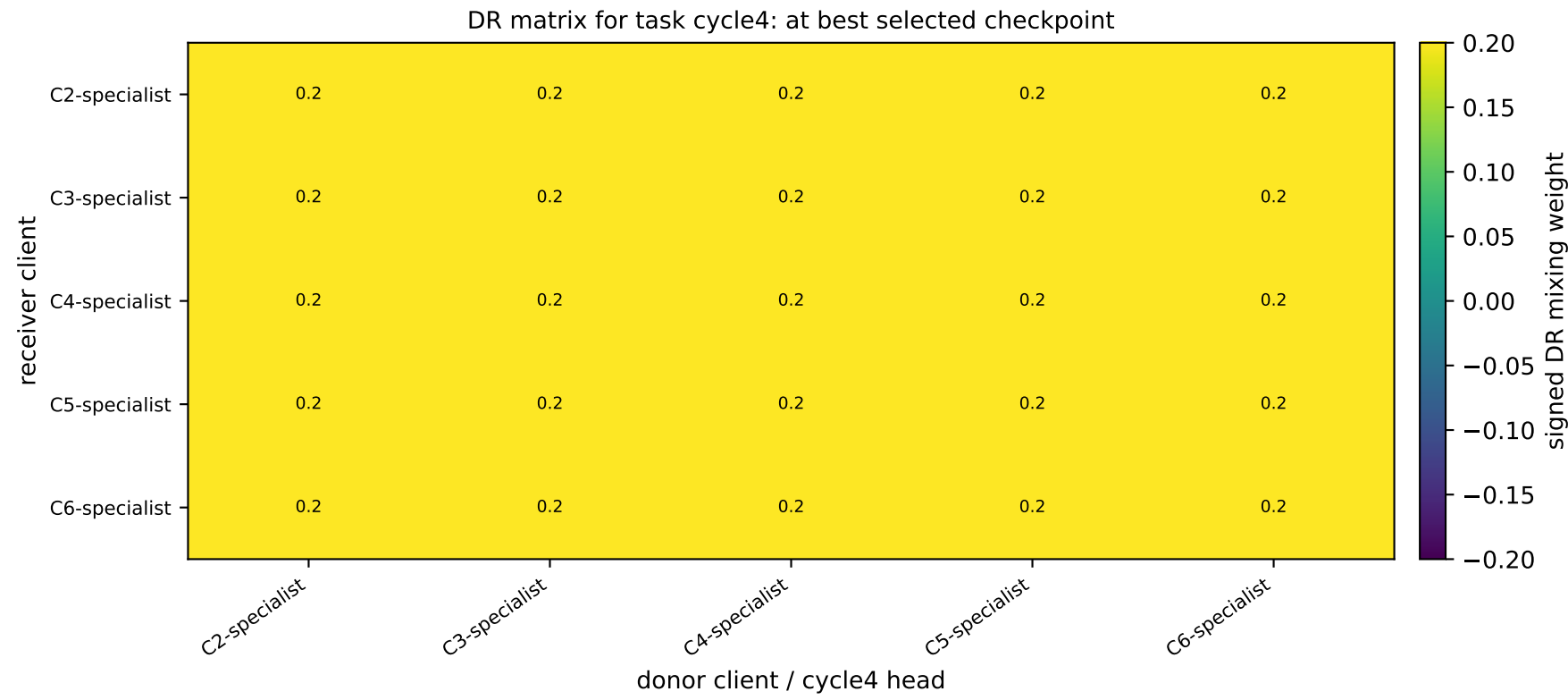
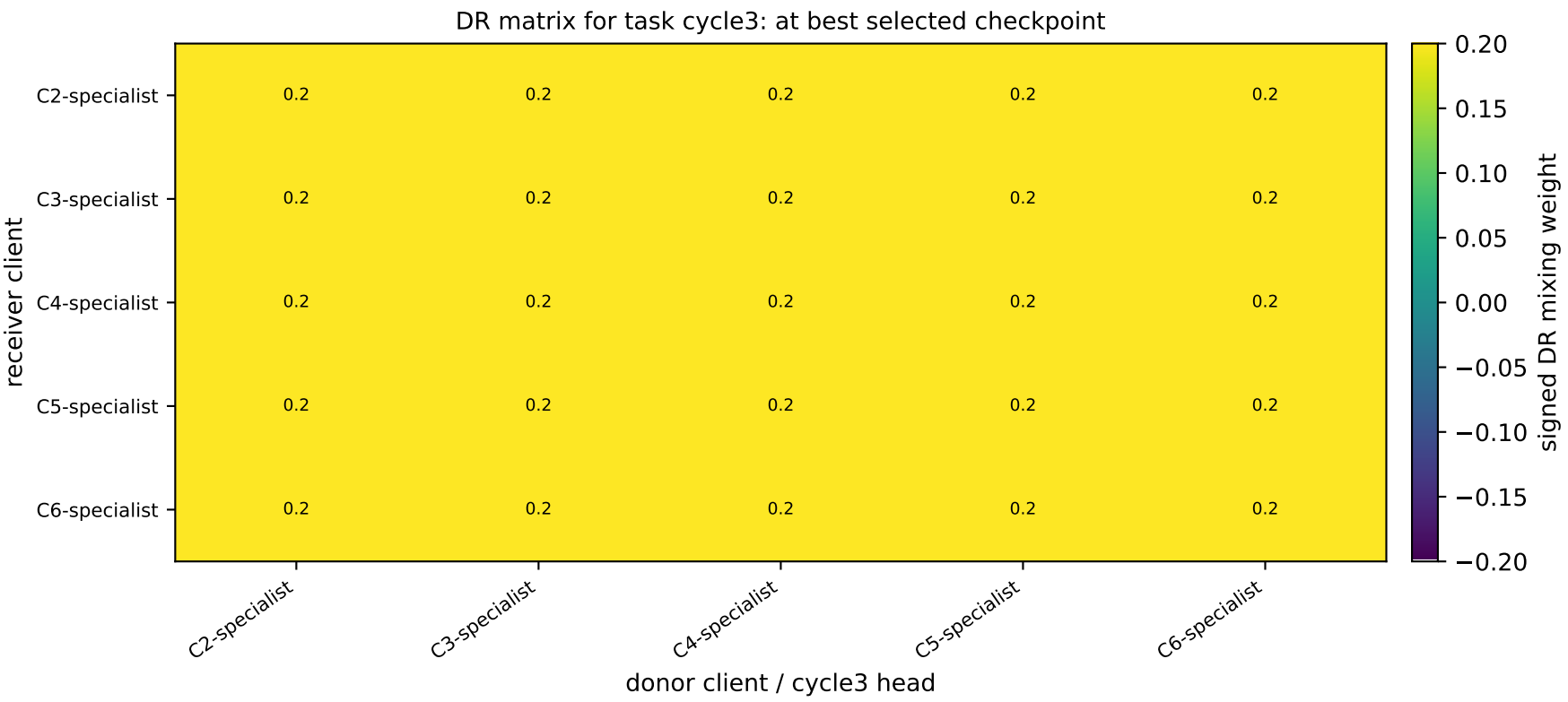
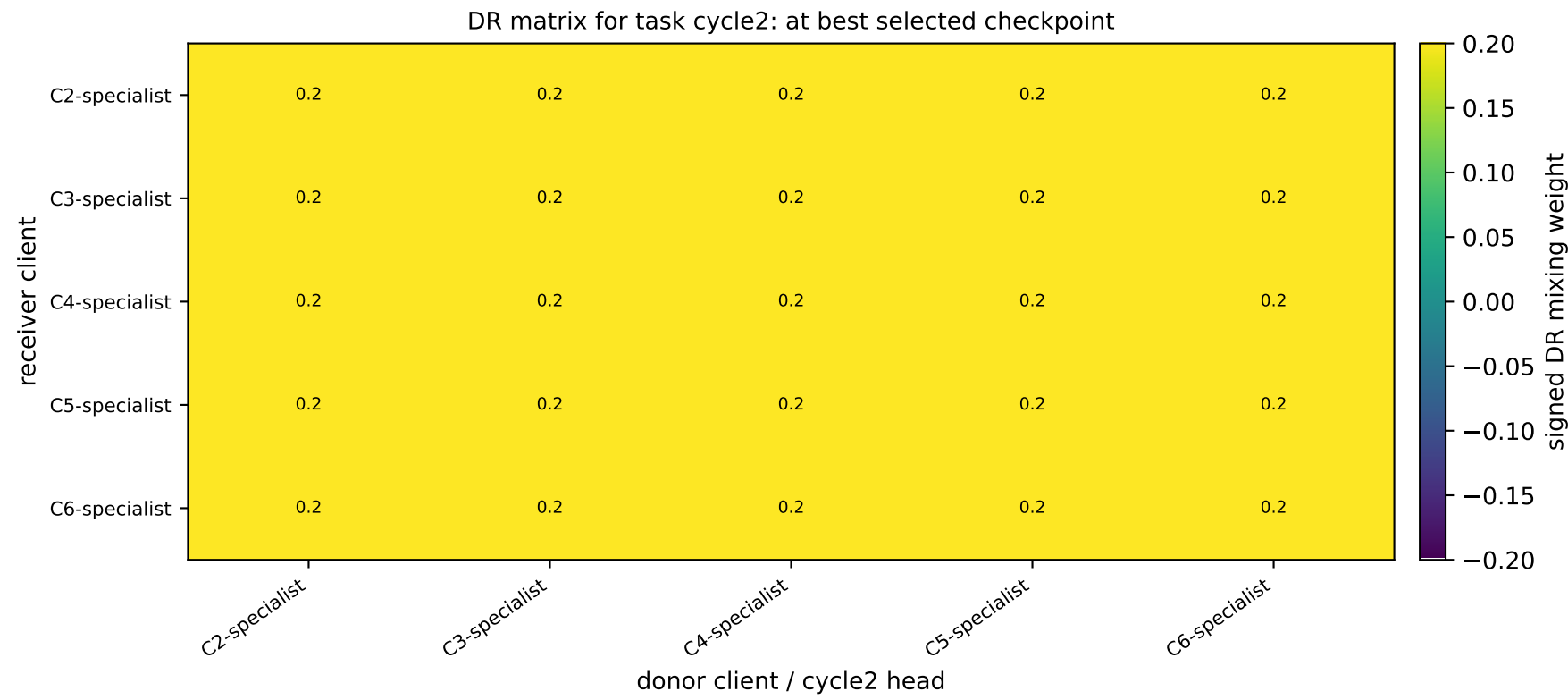


C6-specialist | graph_id=3000004



APPLE-QOracle task-wise DR matrices: receiver clients × donor task-heads | lid

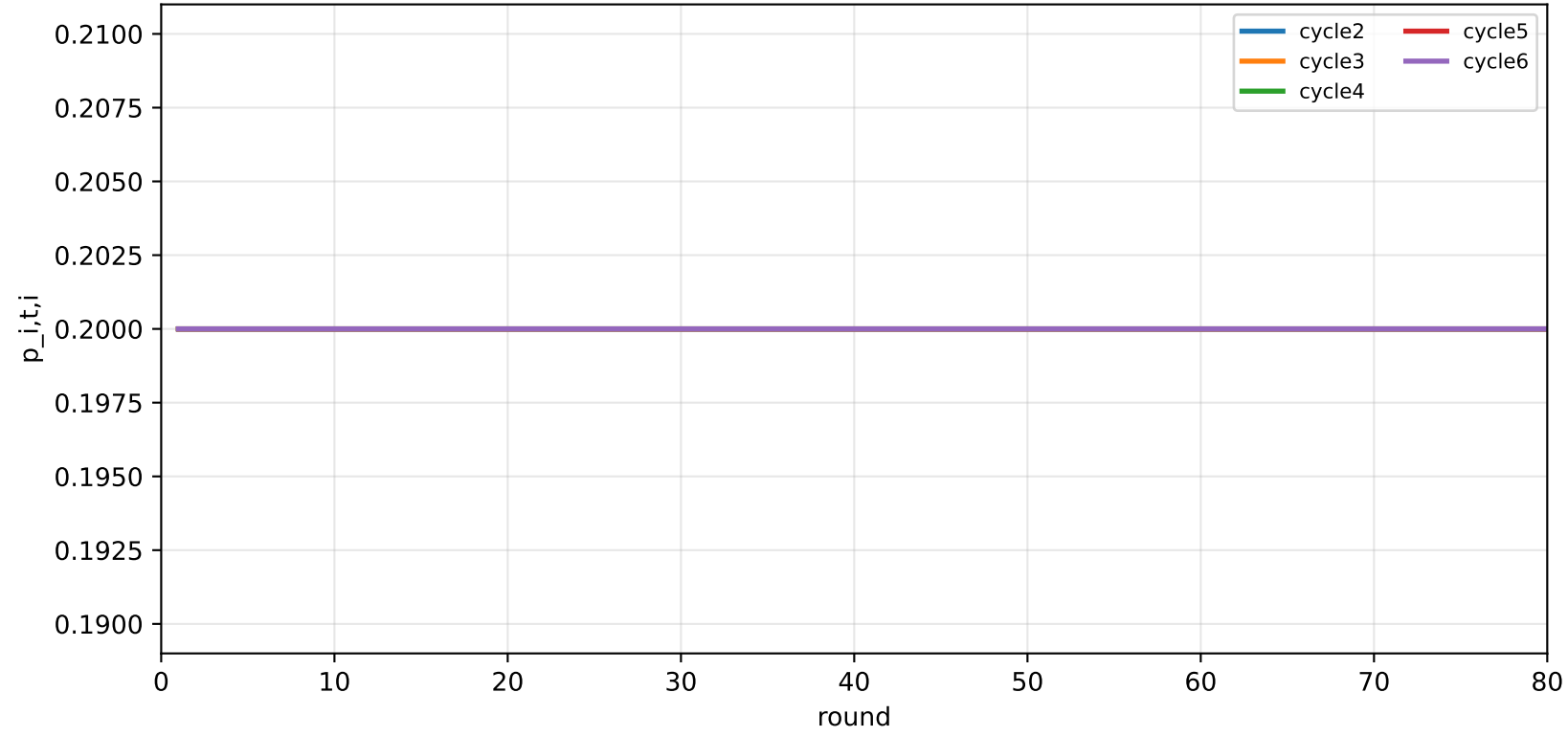
family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000



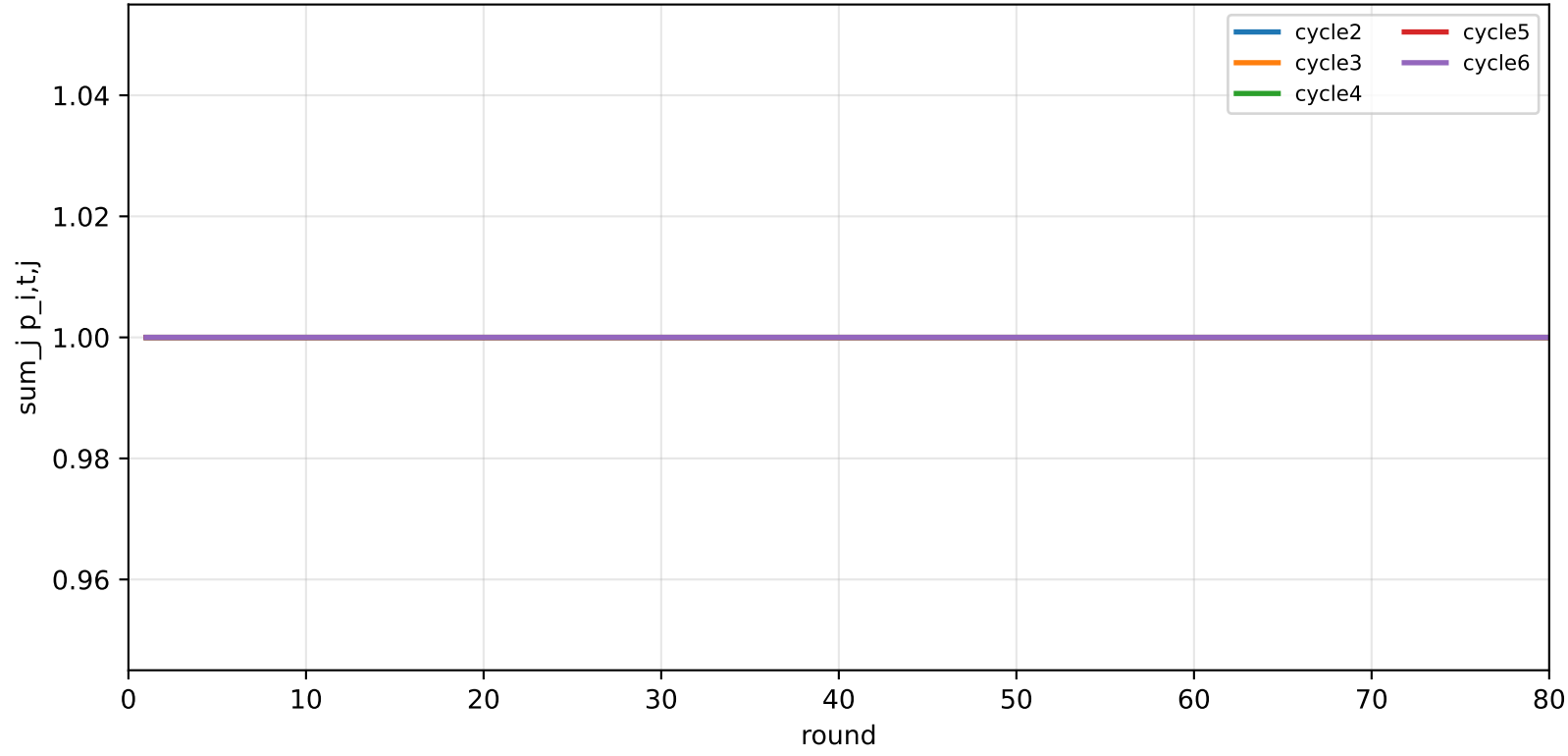
APPLE-QOracle task-wise DR dynamics | lid

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

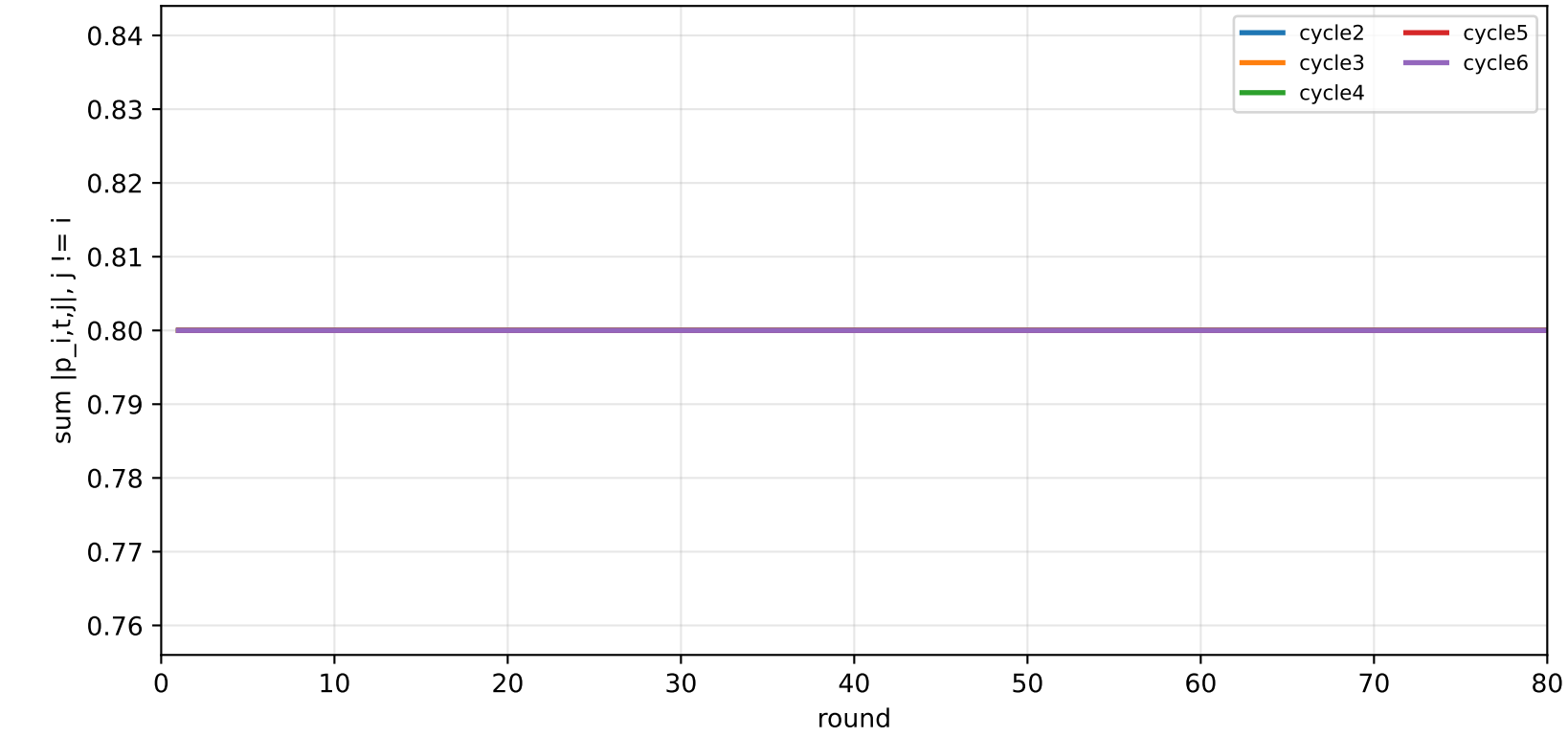
Task-head DR self-weight by task



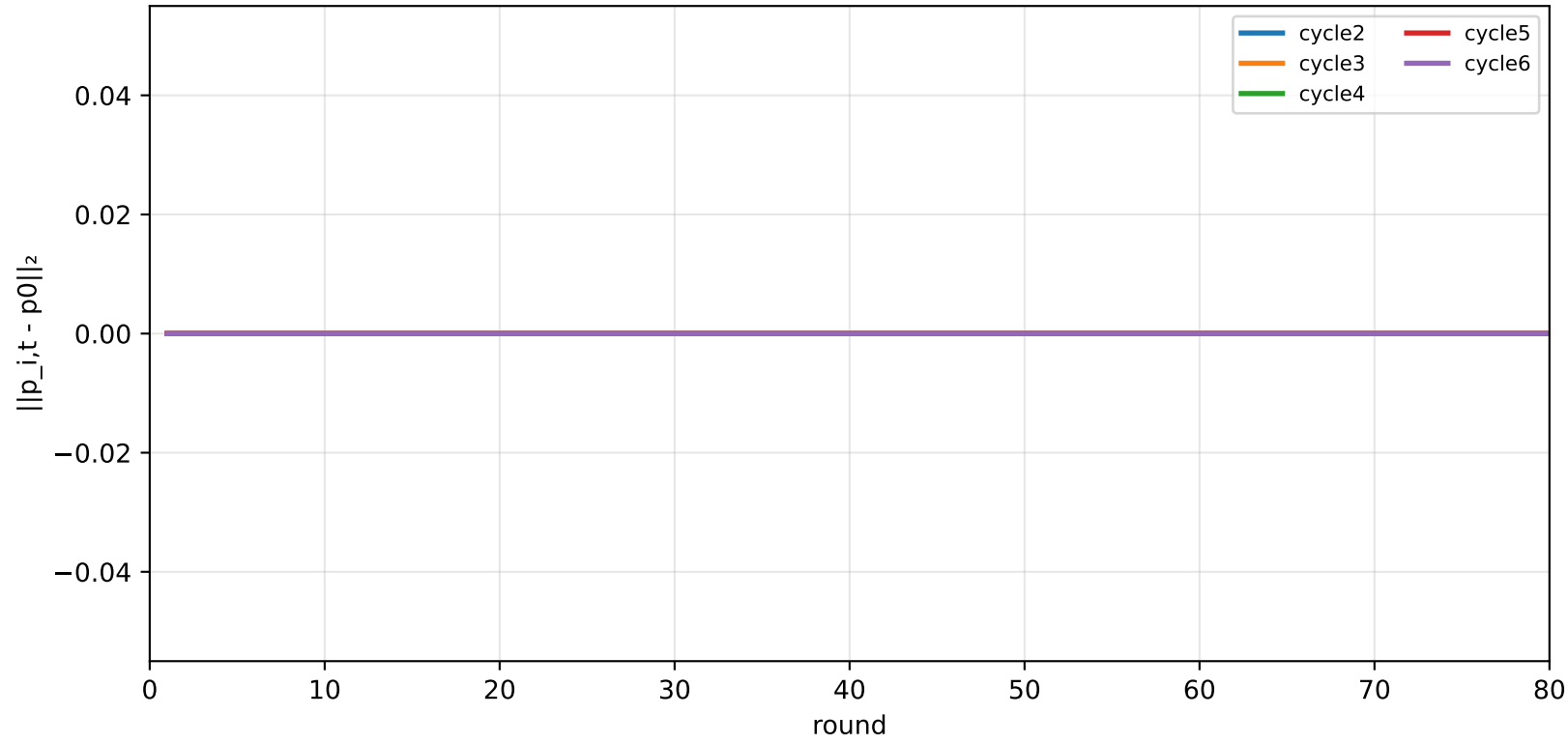
Task-head DR row sum by task



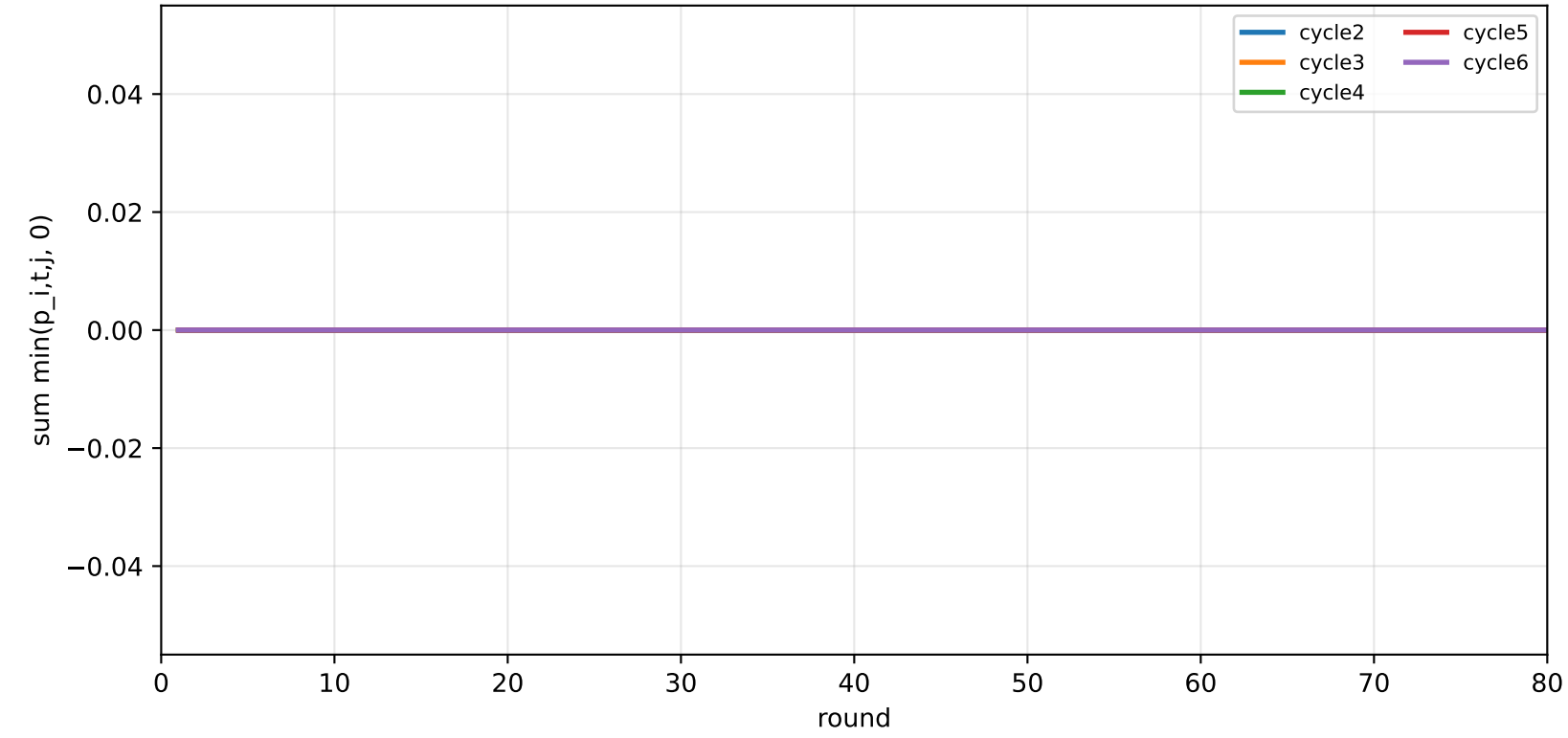
Task-head off-diagonal absolute mass by task



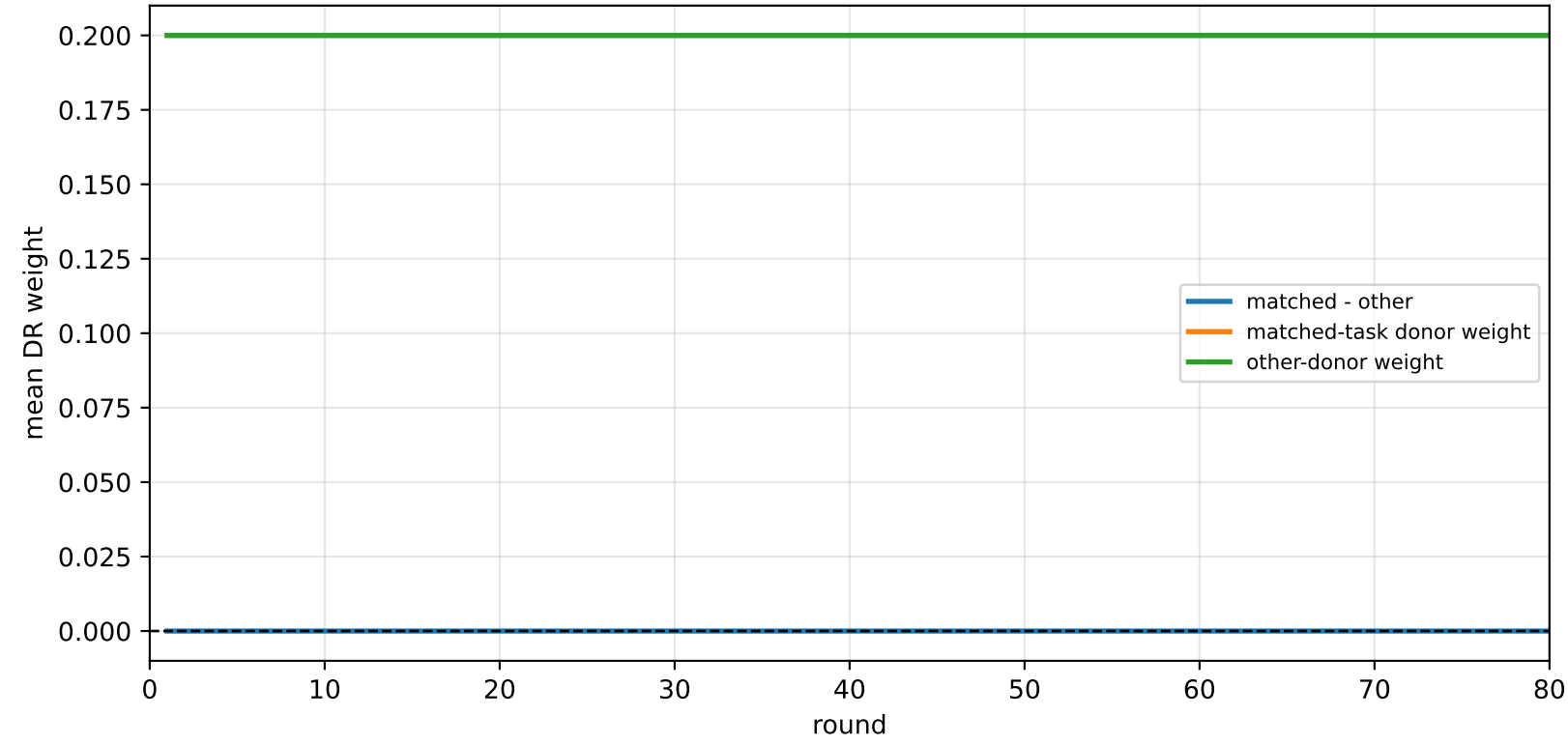
Task-head DR movement away from p0 by task



Task-head off-diagonal negative mass by task



Task-head DR alignment



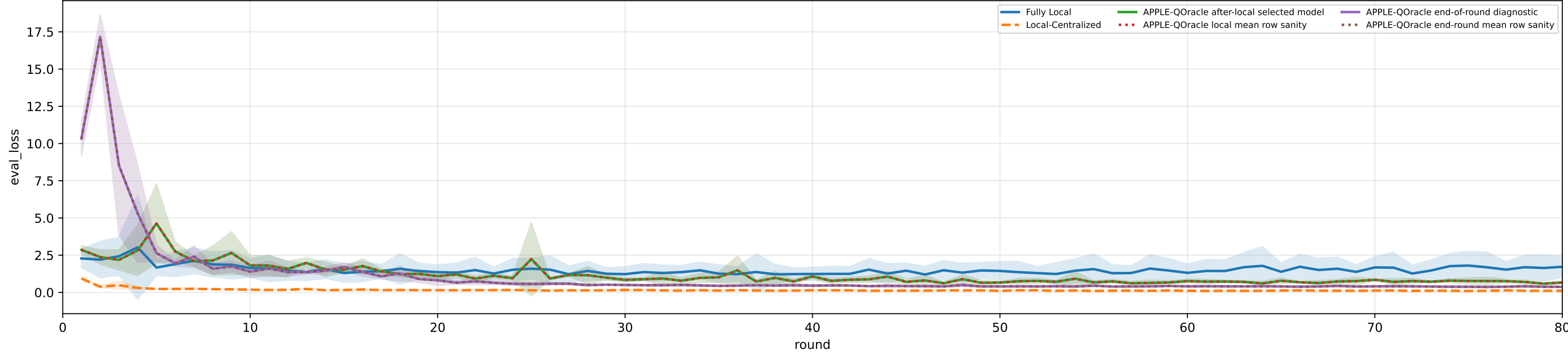
Controlled q-label heterogeneity: Mid | APPLE-QOracle diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

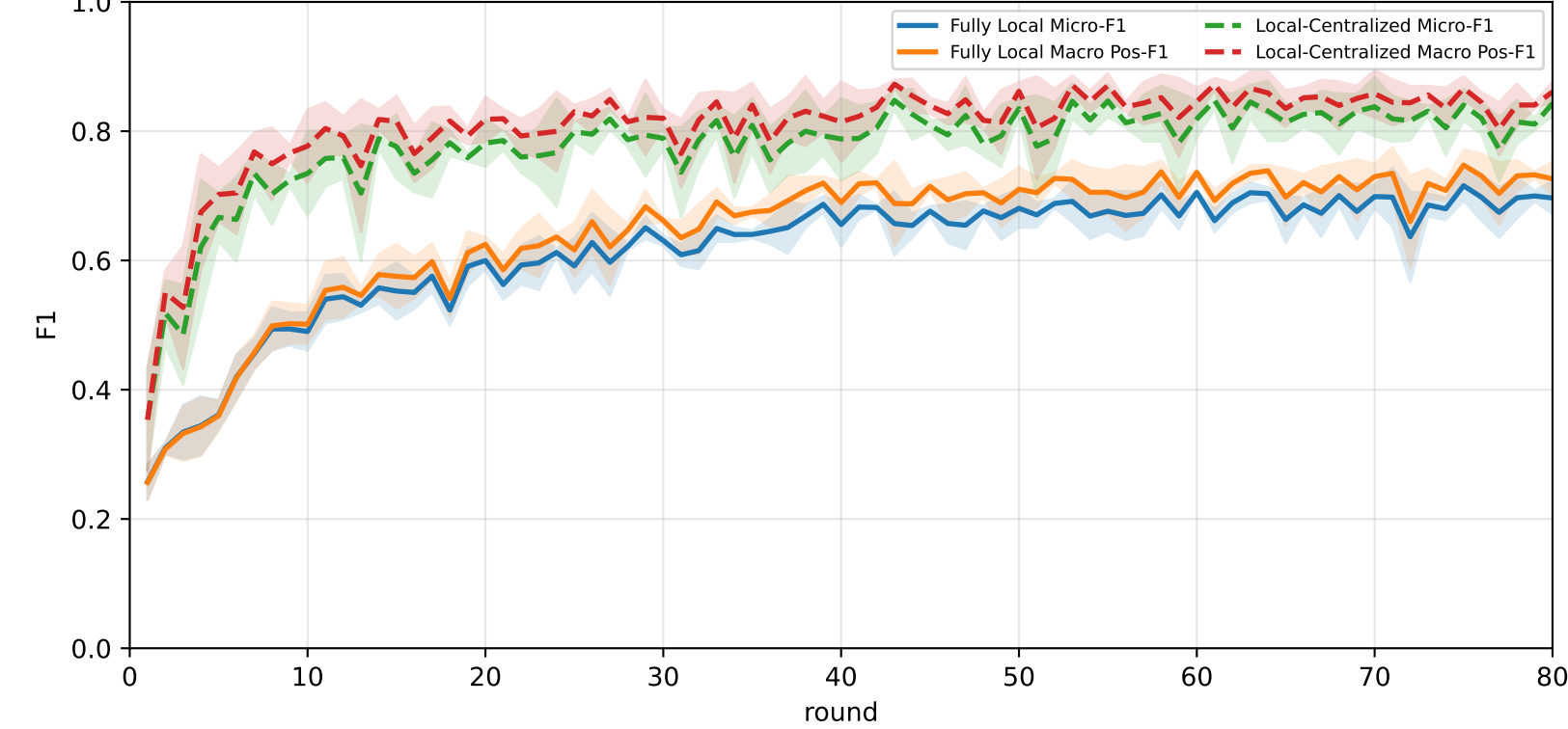
Best test performance (Fully Local vs Local-Centralized vs APPLE-QOracle, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	61.6% ± 4.0%	66.0% ± 4.3%	81.6% ± 7.0%	78.2% ± 5.1%	64.0% ± 3.6%	58.6% ± 5.2%	44.4% ± 3.0%
Local-Centralized	84.3% ± 1.6%	86.3% ± 1.4%	96.7% ± 1.1%	94.7% ± 0.6%	87.4% ± 4.2%	80.9% ± 2.2%	71.8% ± 3.9%
APPLE-QOracle	<u>70.5% ± 1.3%</u>	<u>73.1% ± 1.0%</u>	<u>82.0% ± 1.3%</u>	<u>83.7% ± 2.0%</u>	<u>71.0% ± 0.8%</u>	<u>69.8% ± 1.4%</u>	<u>57.0% ± 3.0%</u>

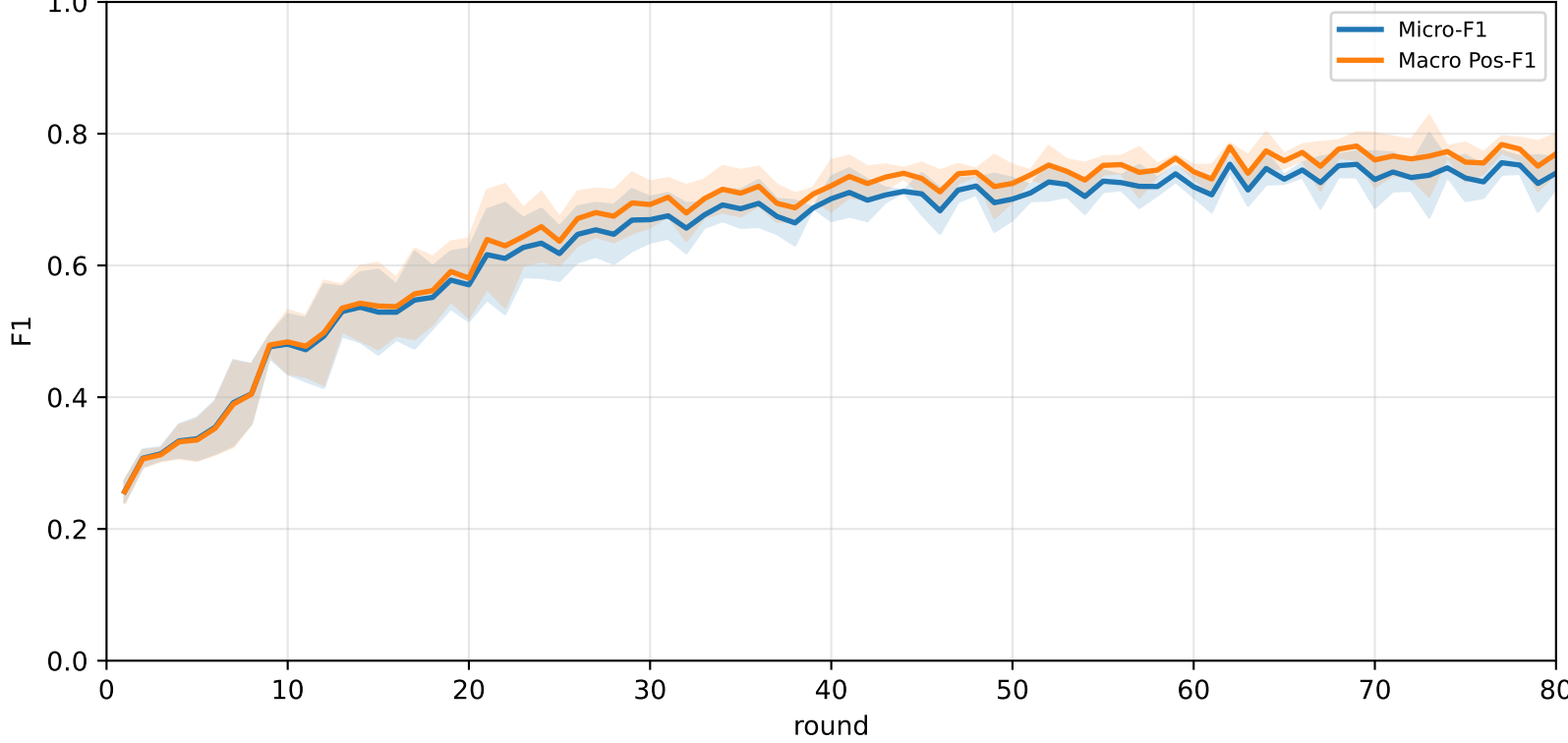
Validation loss: baselines vs selected APPLE-QOracle and end-of-round mismatch (rounds=80)



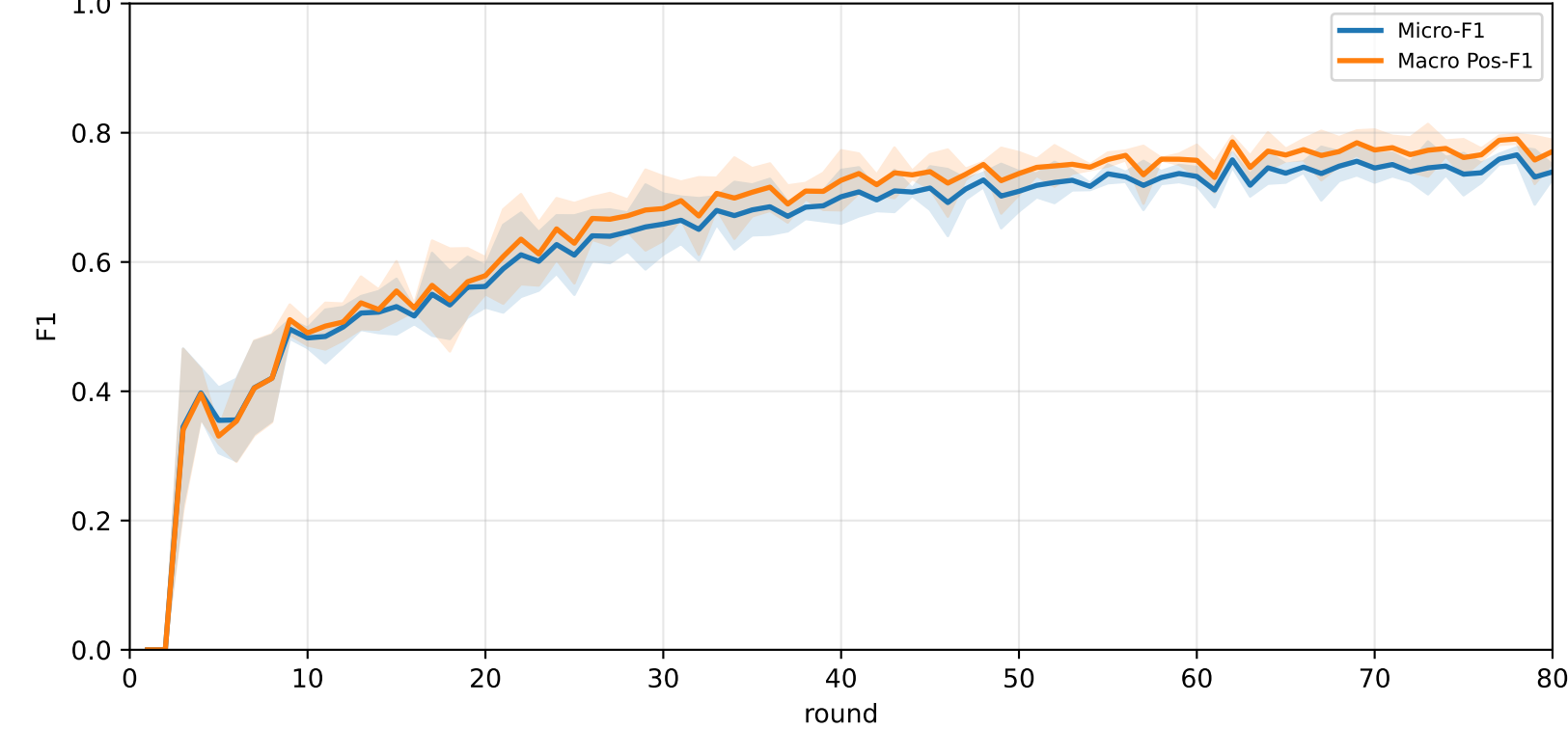
Pooled baseline micro / macro F1



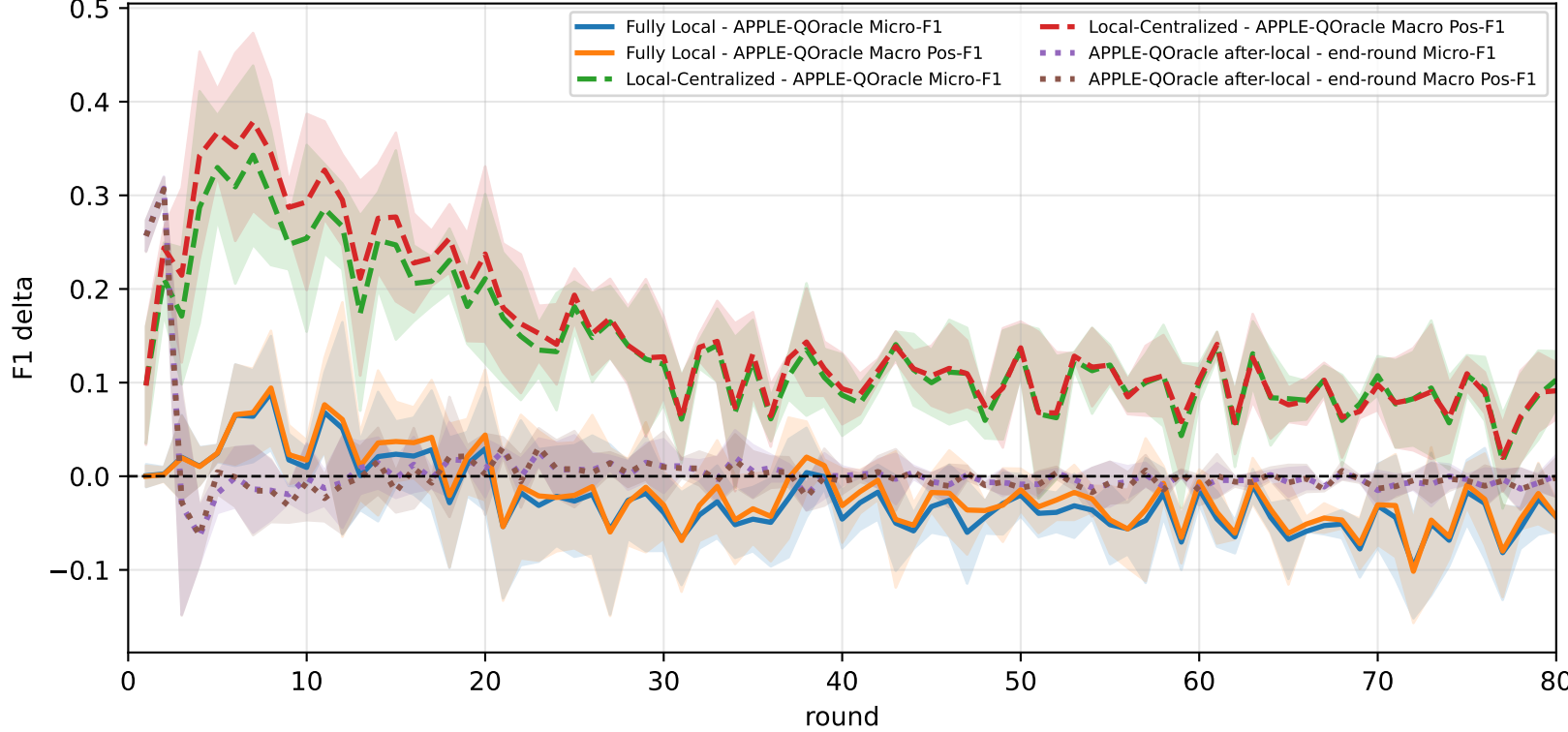
Pooled APPLE-QOracle micro / macro F1 (after-local selected model)



Pooled APPLE-QOracle micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QOracle mismatch gap

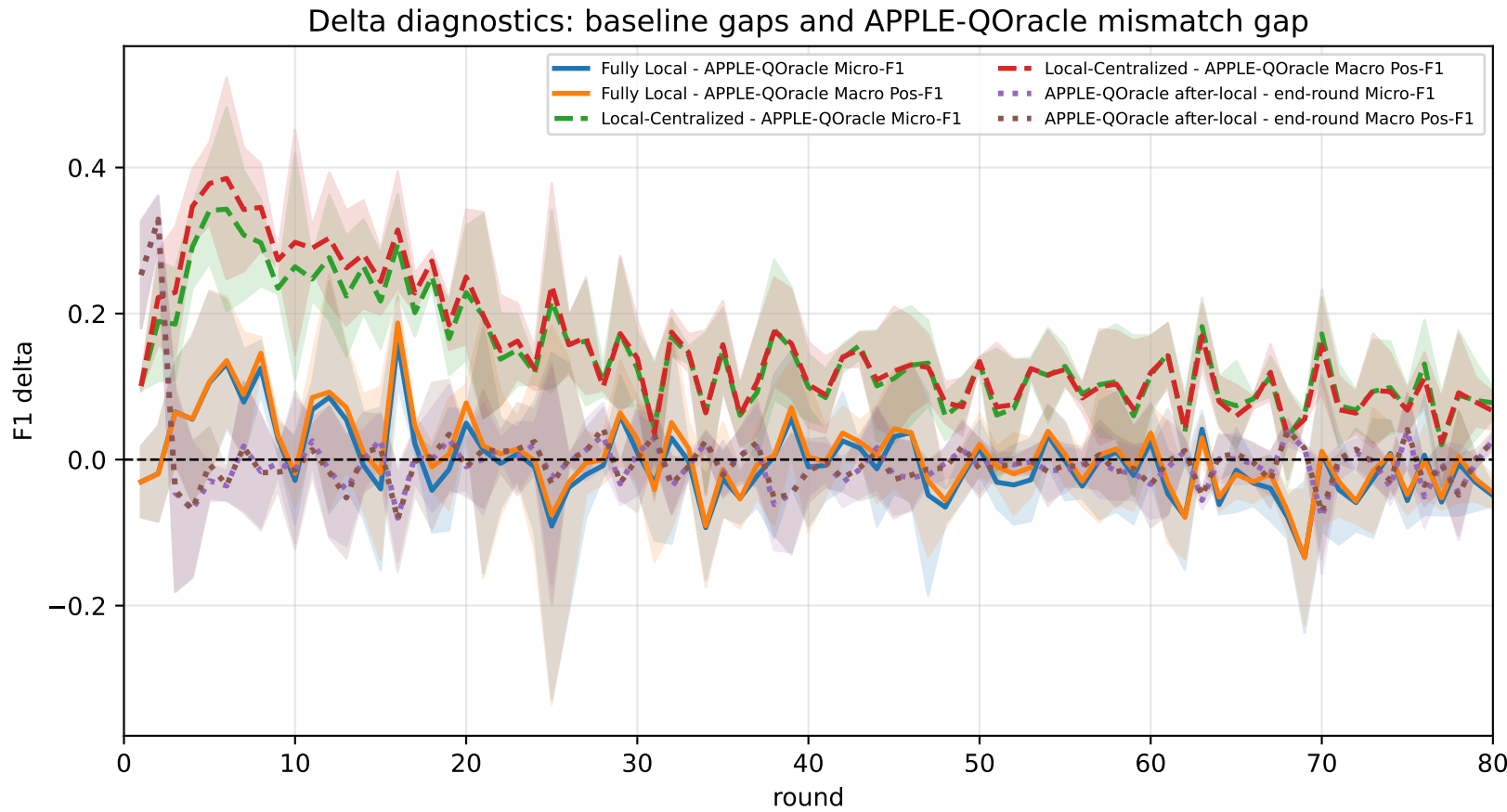
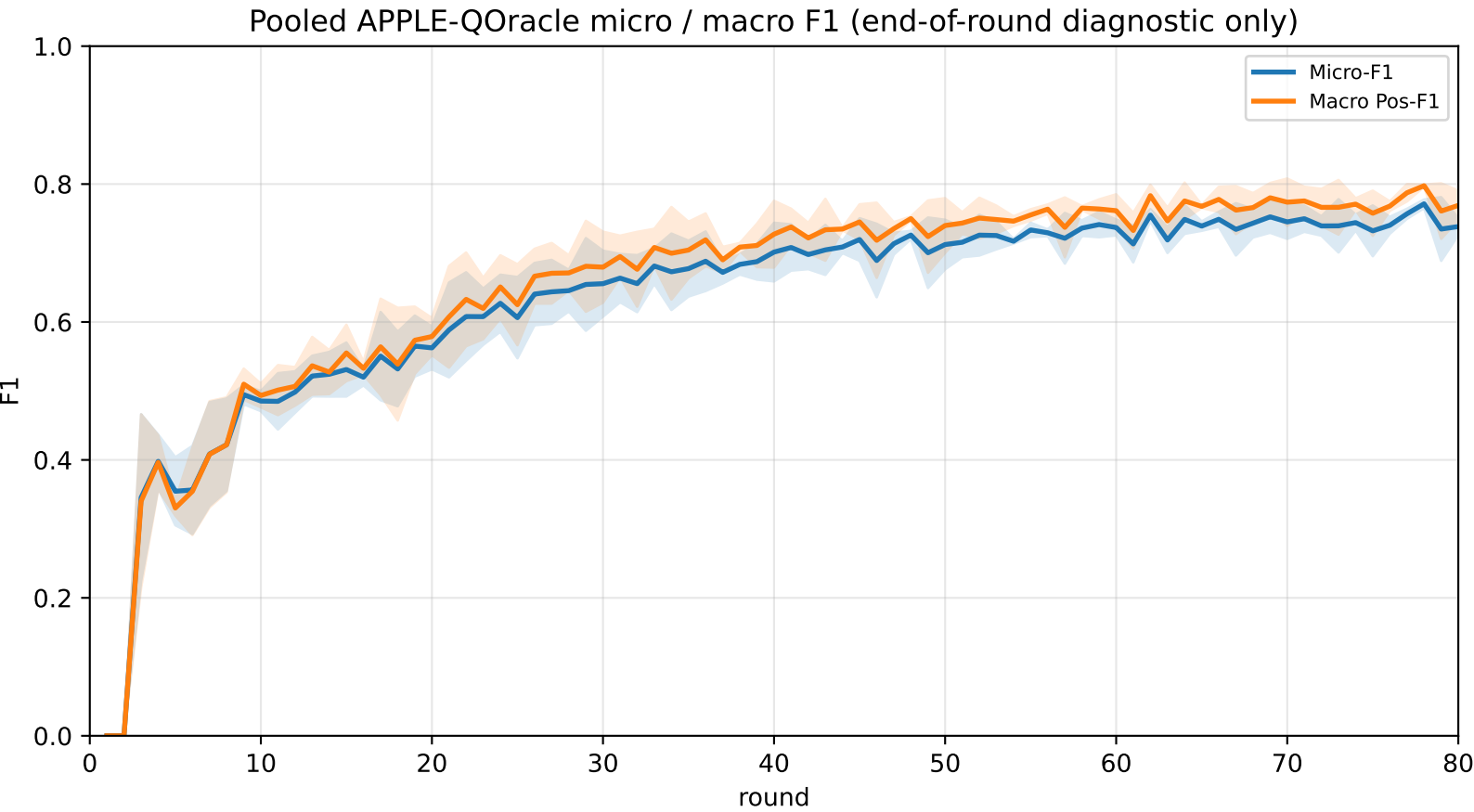
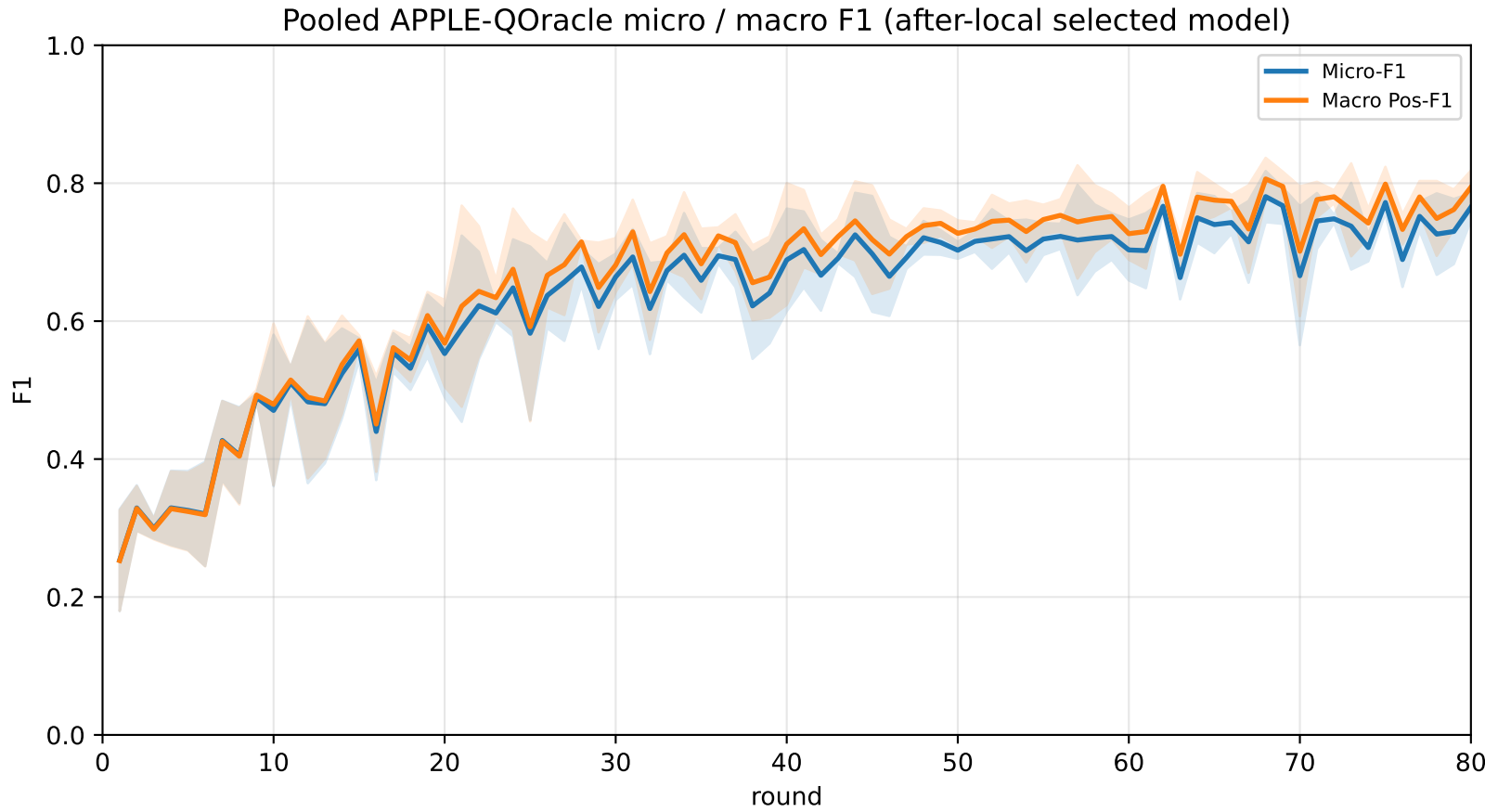
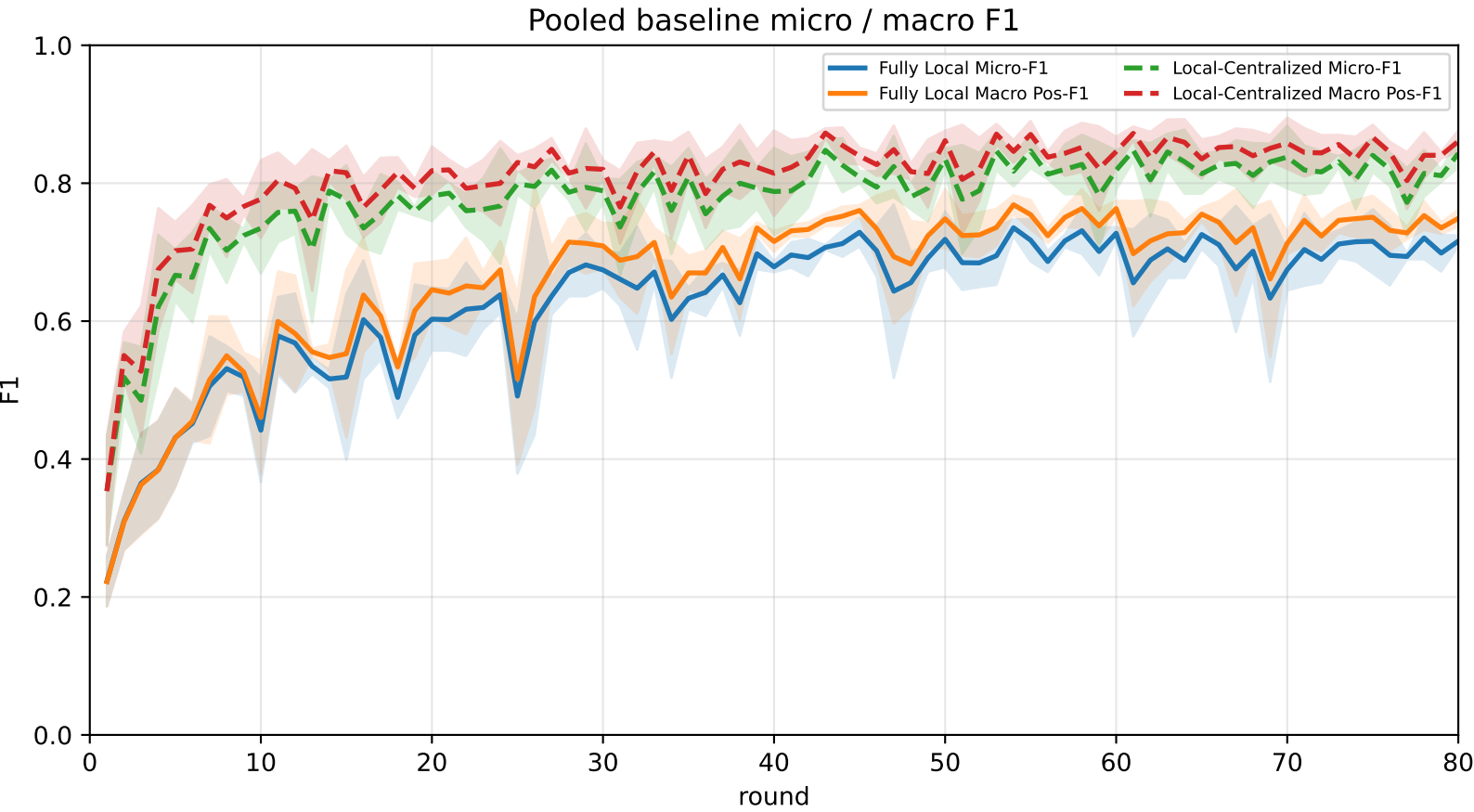
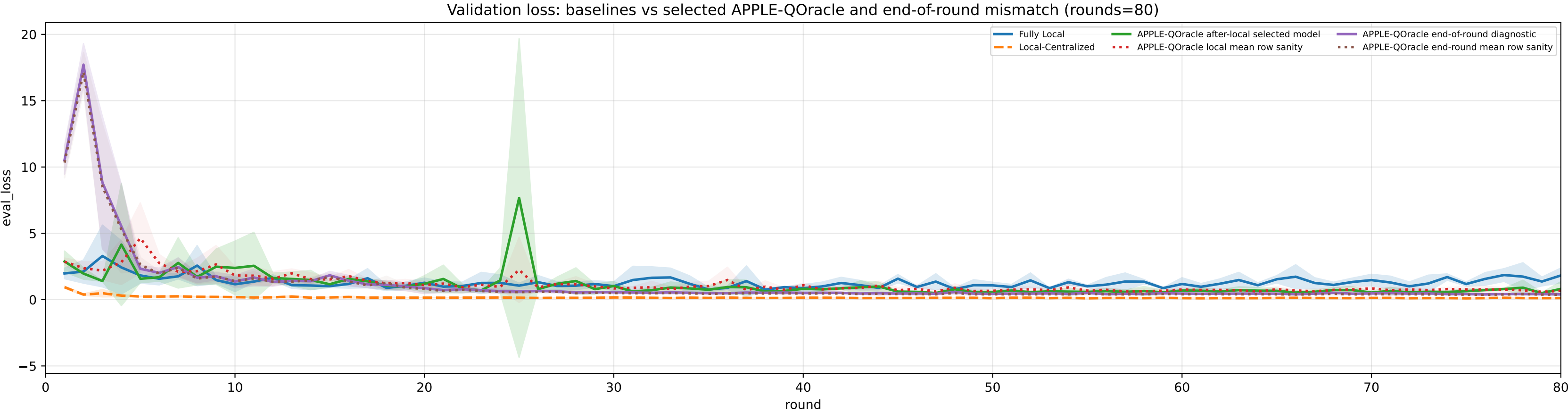


Controlled q-label heterogeneity: Mid | APPLE-QOracle diagnostics | family=q_task_cycle2 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% | heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

Best test performance (Fully Local vs Local-Centralized vs APPLE-QOracle, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	63.8% ± 5.4%	68.0% ± 5.8%	85.6% ± 8.0%	80.6% ± 4.3%	65.4% ± 6.2%	59.3% ± 9.8%	49.2% ± 2.7%
Local-Centralized	84.3% ± 1.6%	86.3% ± 1.4%	96.7% ± 1.1%	94.7% ± 0.6%	87.4% ± 4.2%	80.9% ± 2.2%	71.8% ± 3.9%
APPLE-QOracle	70.2% ± 3.0%	72.9% ± 2.5%	86.2% ± 1.8%	82.1% ± 3.8%	71.6% ± 2.4%	66.8% ± 2.9%	57.6% ± 5.4%

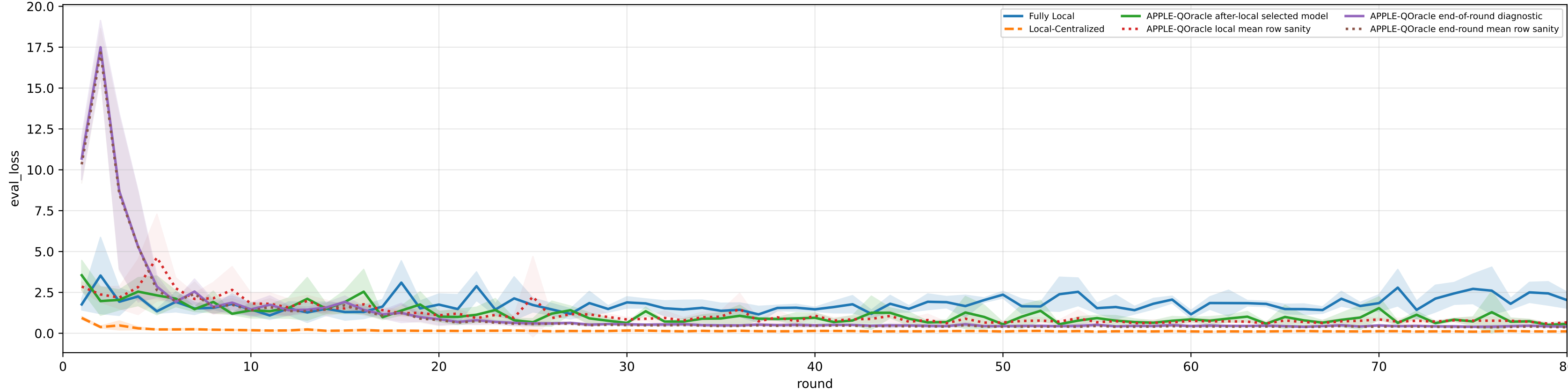


family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% | heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

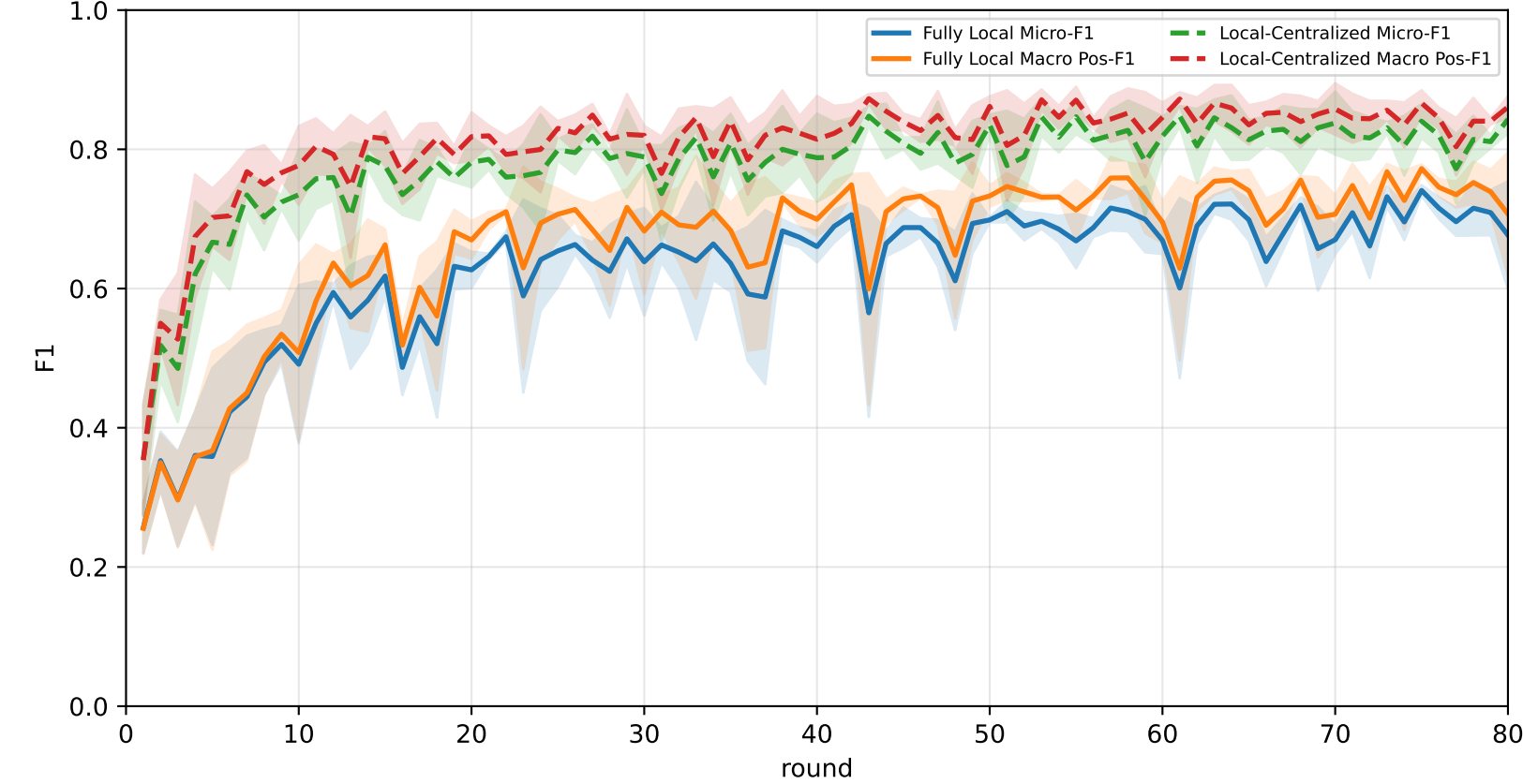
Best test performance (Fully Local vs Local-Centralized vs APPLE-QOracle, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	60.1% ± 11.6%	64.2% ± 10.9%	75.3% ± 14.5%	77.9% ± 12.0%	63.8% ± 6.1%	60.7% ± 11.5%	43.1% ± 11.1%
Local-Centralized	84.3% ± 1.6%	86.3% ± 1.4%	96.7% ± 1.1%	94.7% ± 0.6%	87.4% ± 4.2%	80.9% ± 2.2%	71.8% ± 3.9%
APPLE-QOracle	<u>70.4% ± 3.6%</u>	<u>72.7% ± 2.8%</u>	<u>75.8% ± 7.5%</u>	<u>87.2% ± 3.7%</u>	<u>74.4% ± 4.5%</u>	<u>68.8% ± 4.4%</u>	<u>57.2% ± 5.5%</u>

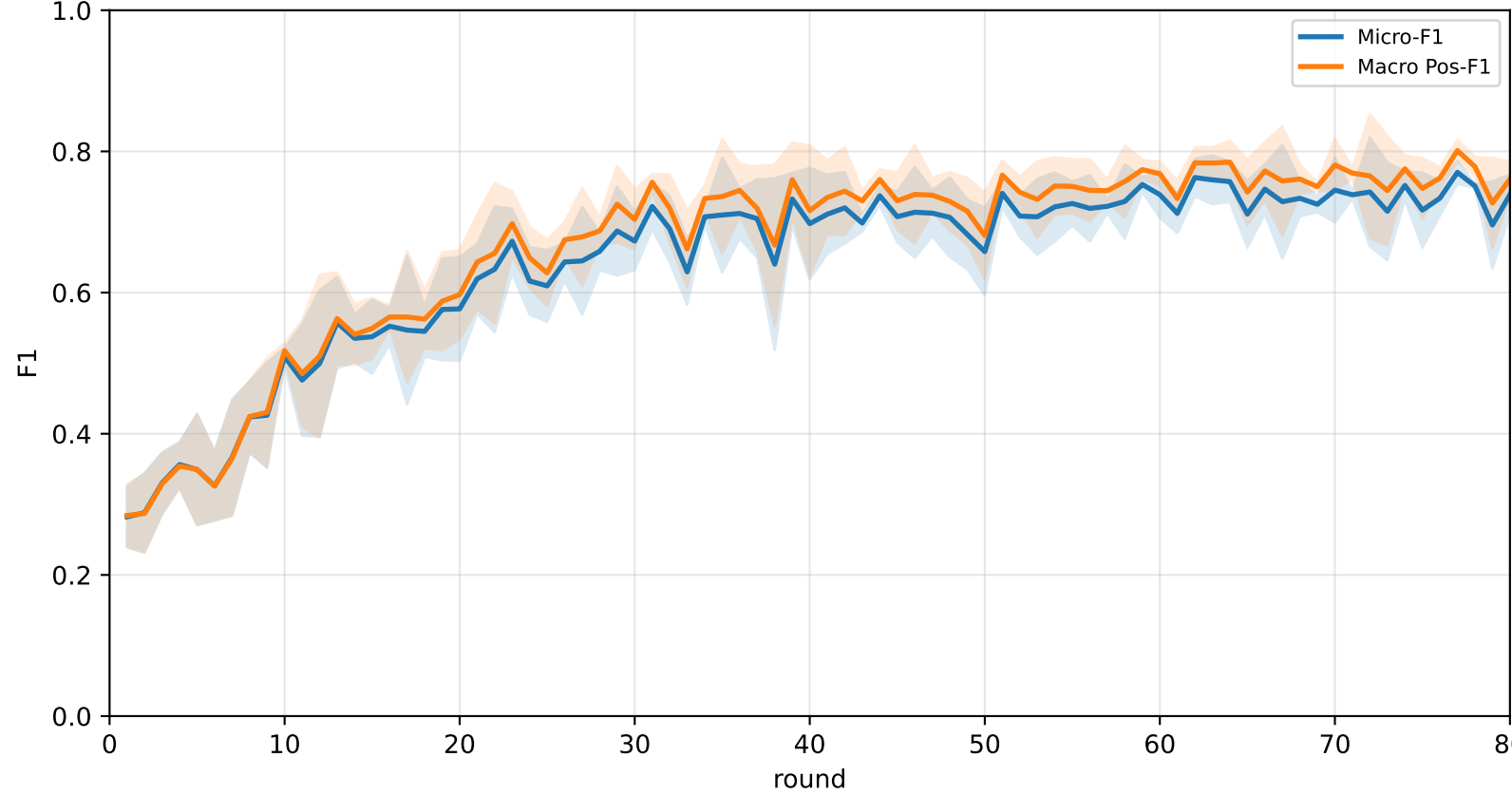
Validation loss: baselines vs selected APPLE-QOracle and end-of-round mismatch (rounds=80)



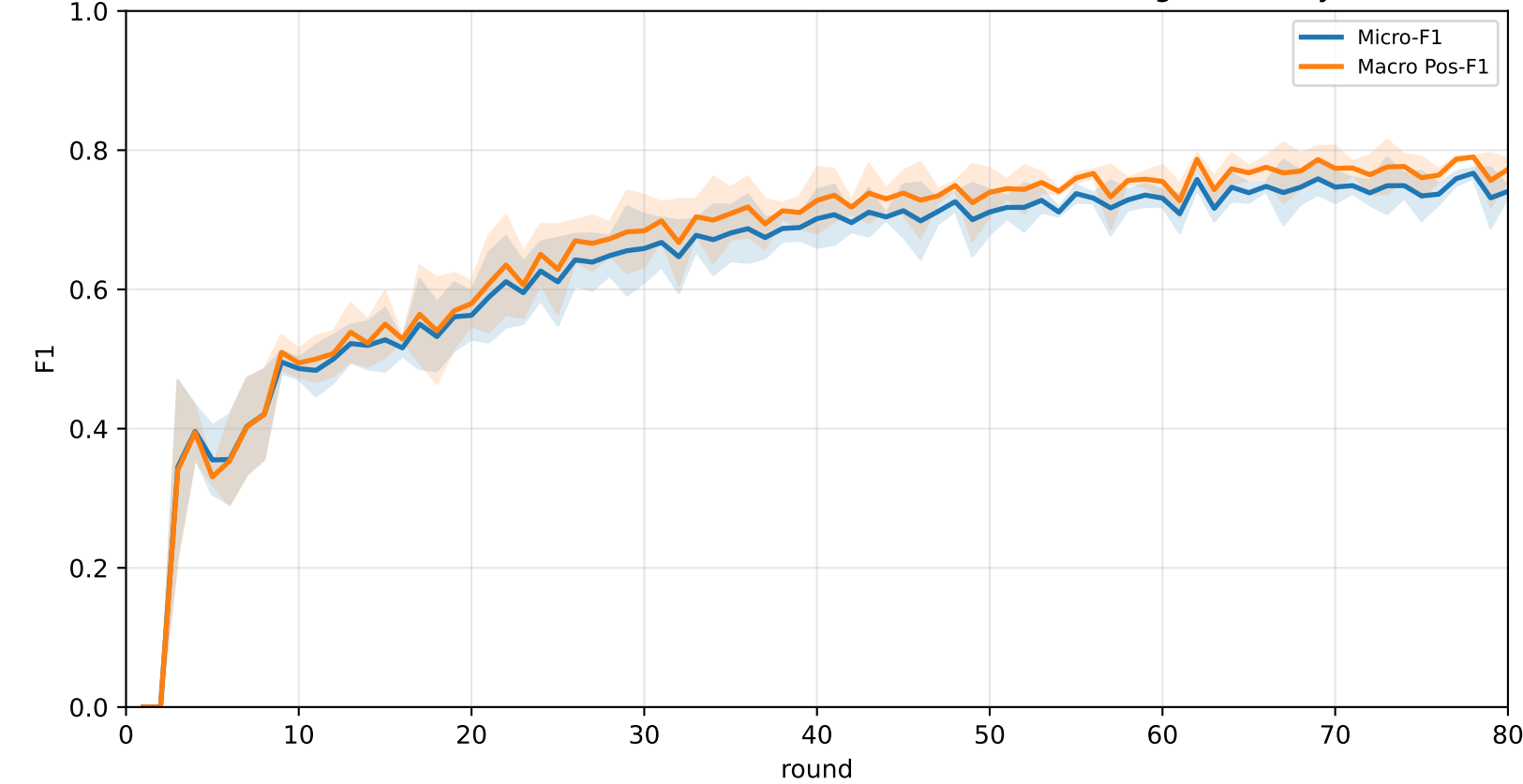
Pooled baseline micro / macro F1



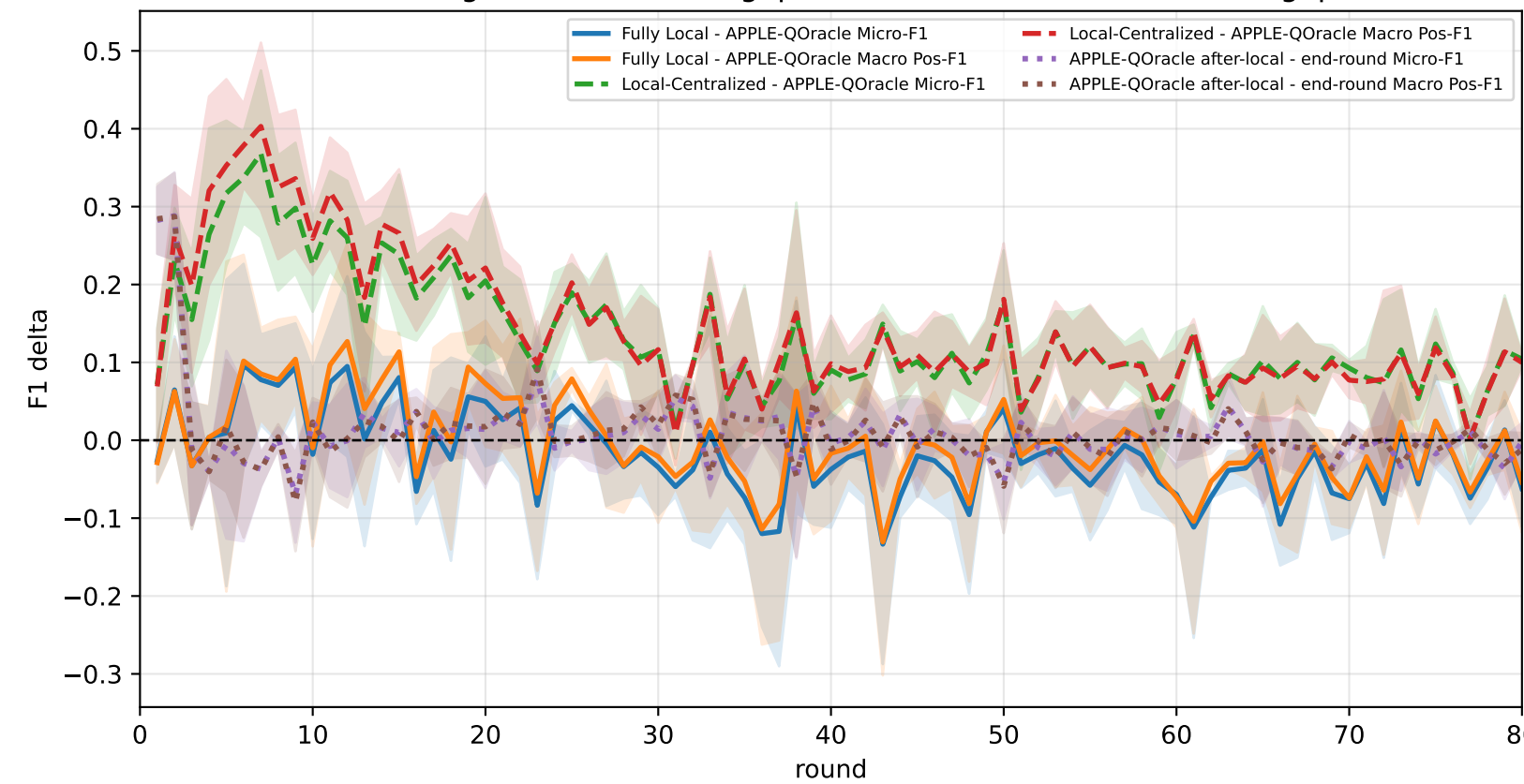
Pooled APPLE-QOracle micro / macro F1 (after-local selected model)



Pooled APPLE-QOracle micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QOracle mismatch gap

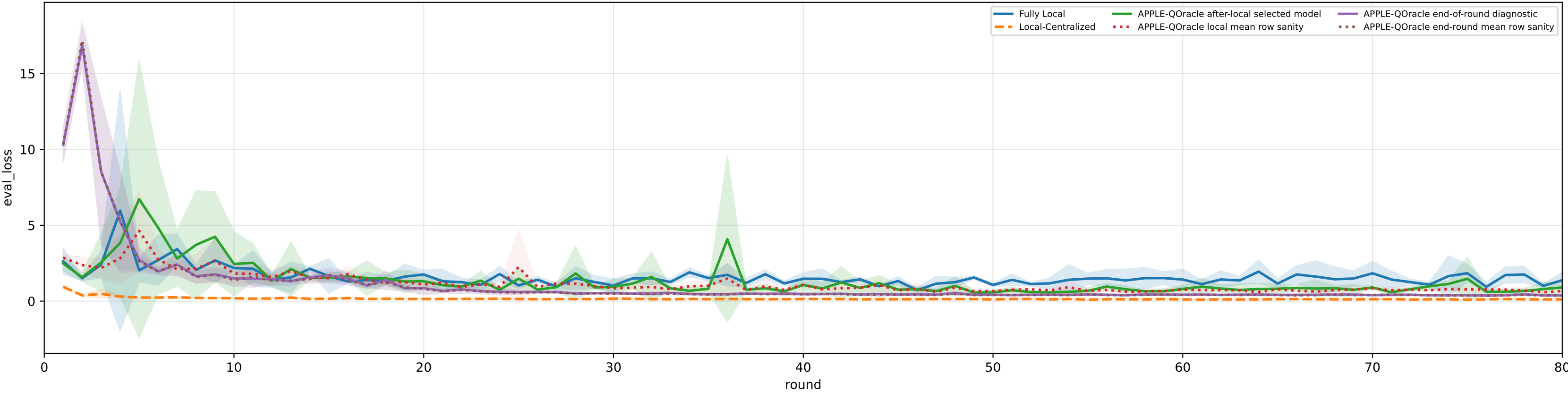


family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

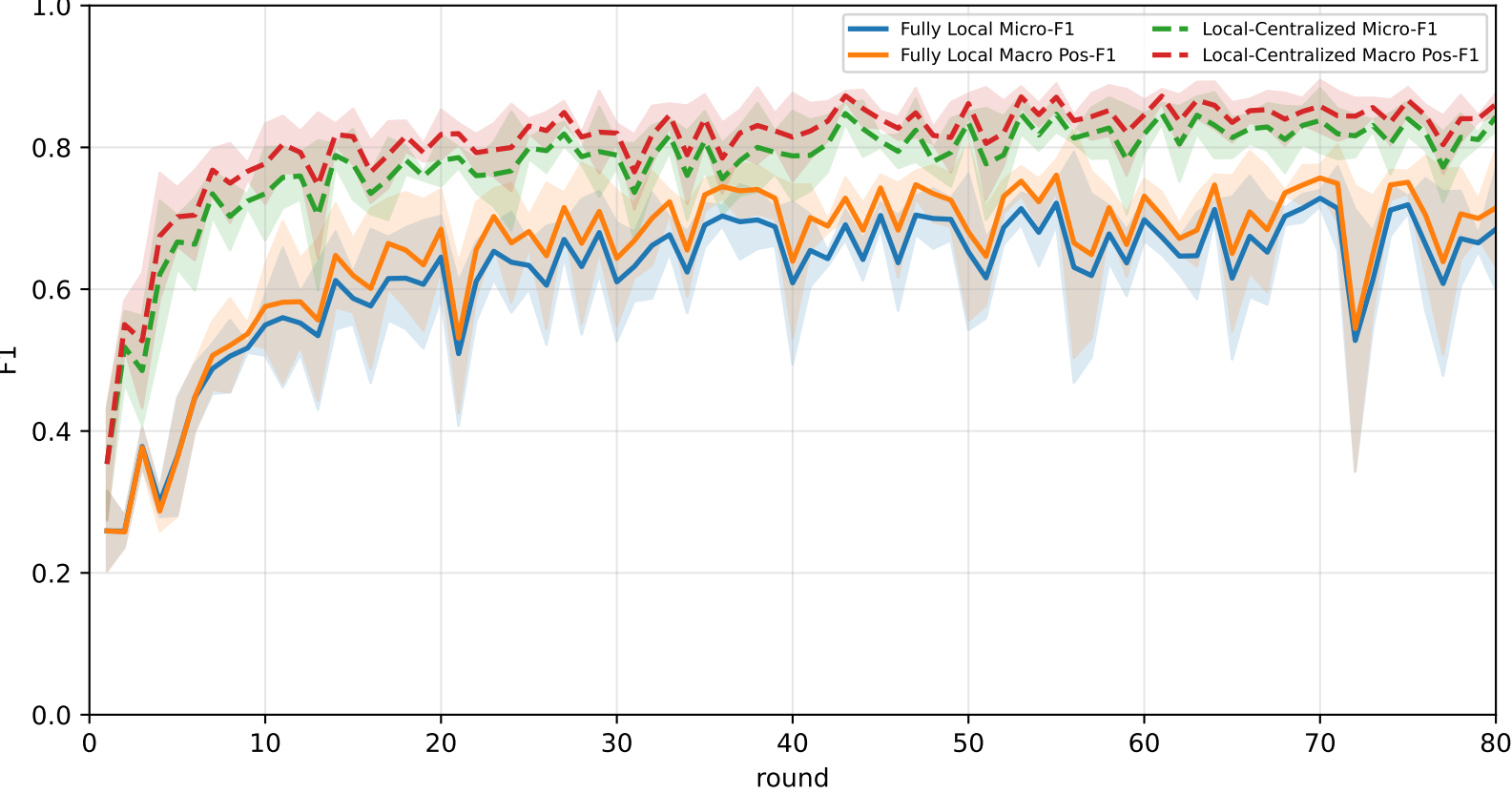
Best test performance (Fully Local vs Local-Centralized vs APPLE-QOracle, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	58.7% ± 5.5%	63.8% ± 5.1%	82.7% ± 2.7%	76.5% ± 5.8%	63.3% ± 5.5%	54.2% ± 7.8%	42.1% ± 3.8%
Local-Centralized	84.3% ± 1.6%	86.3% ± 1.4%	96.7% ± 1.1%	94.7% ± 0.6%	87.4% ± 4.2%	80.9% ± 2.2%	71.8% ± 3.9%
APPLE-QOracle	<u>70.8% ± 1.0%</u>	<u>74.0% ± 0.5%</u>	<u>90.9% ± 2.2%</u>	<u>82.9% ± 1.5%</u>	<u>70.5% ± 2.1%</u>	<u>68.9% ± 2.6%</u>	<u>56.7% ± 2.0%</u>

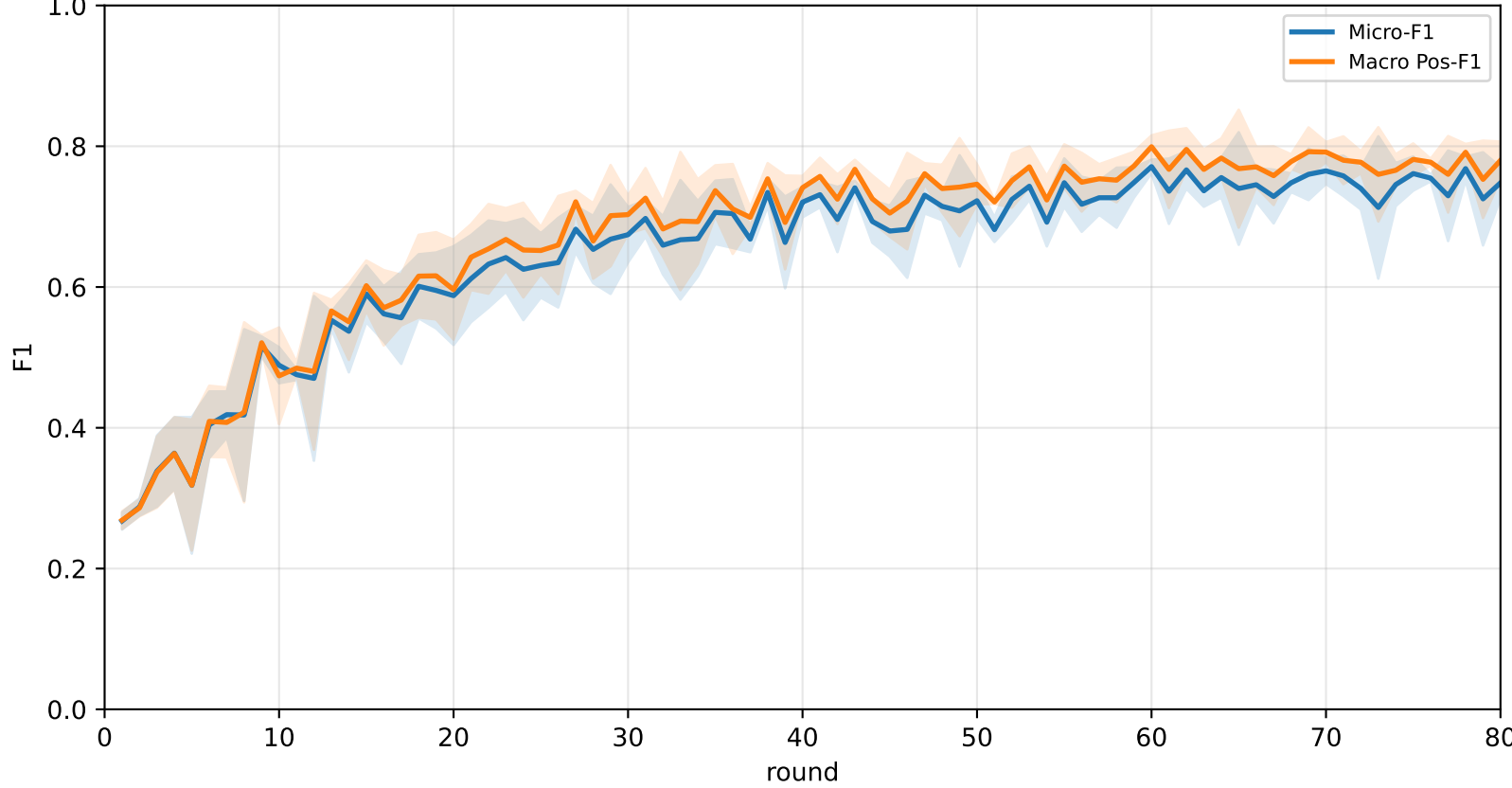
Validation loss: baselines vs selected APPLE-QOracle and end-of-round mismatch (rounds=80)



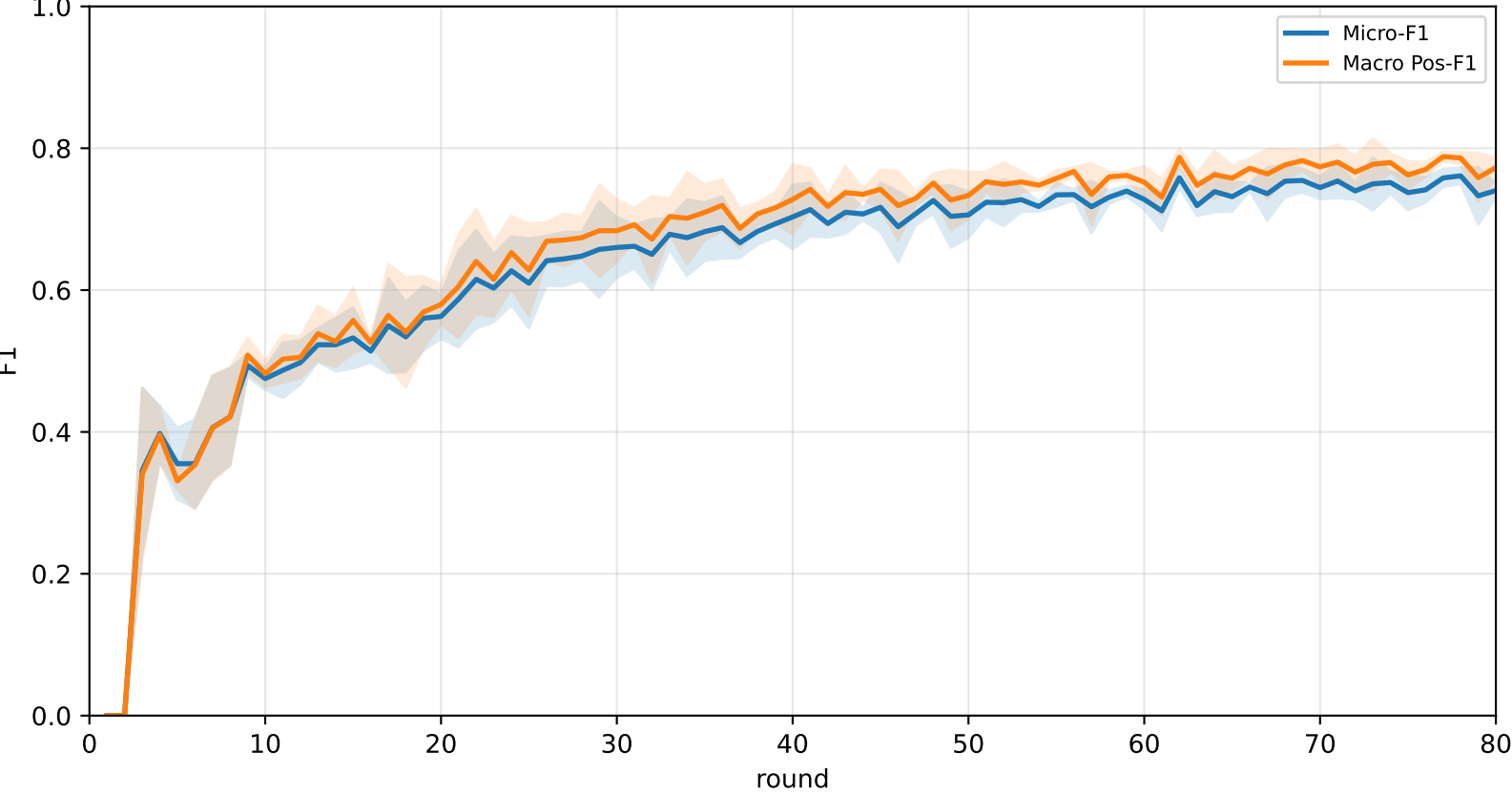
Pooled baseline micro / macro F1



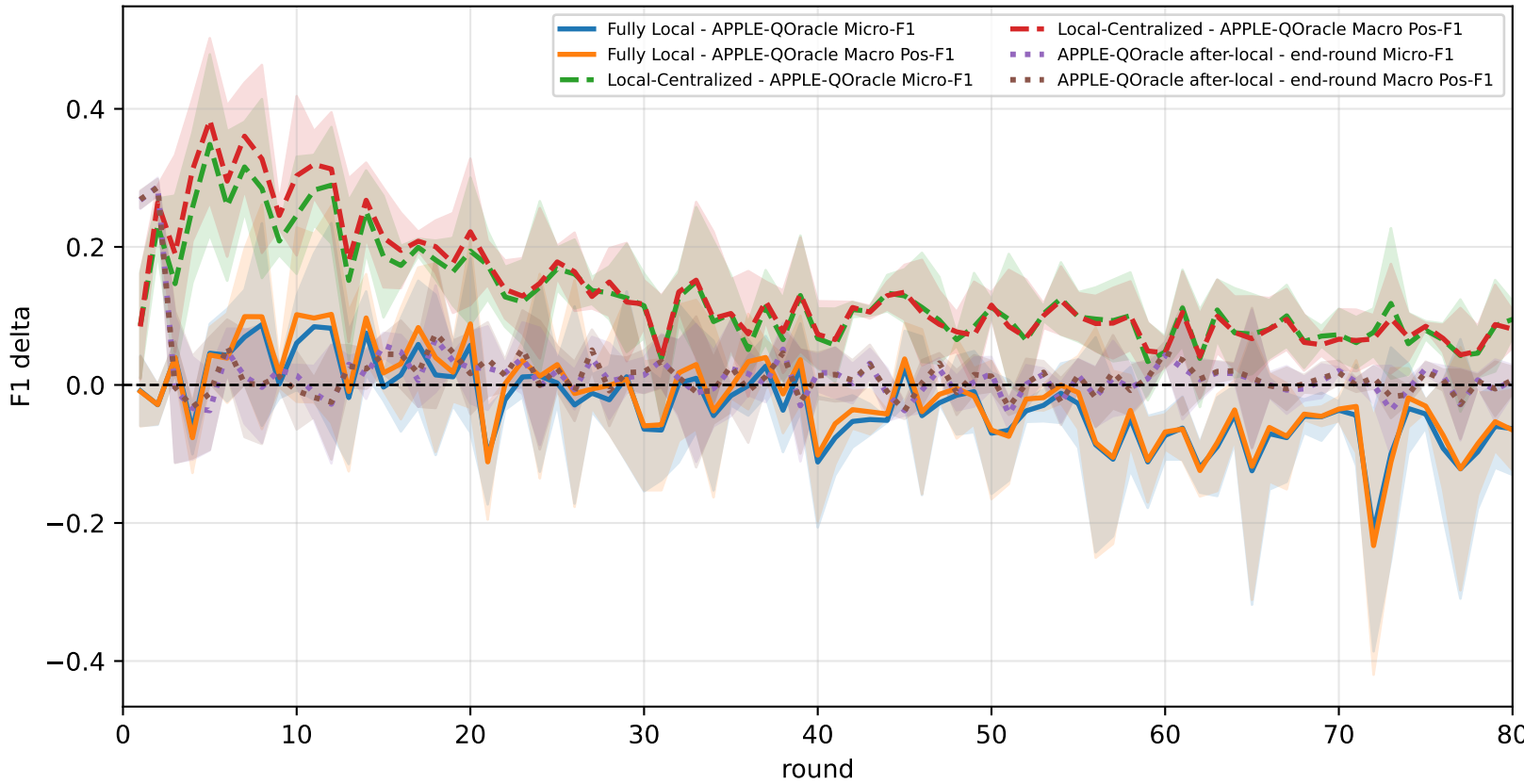
Pooled APPLE-QOracle micro / macro F1 (after-local selected model)



Pooled APPLE-QOracle micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QOracle mismatch gap

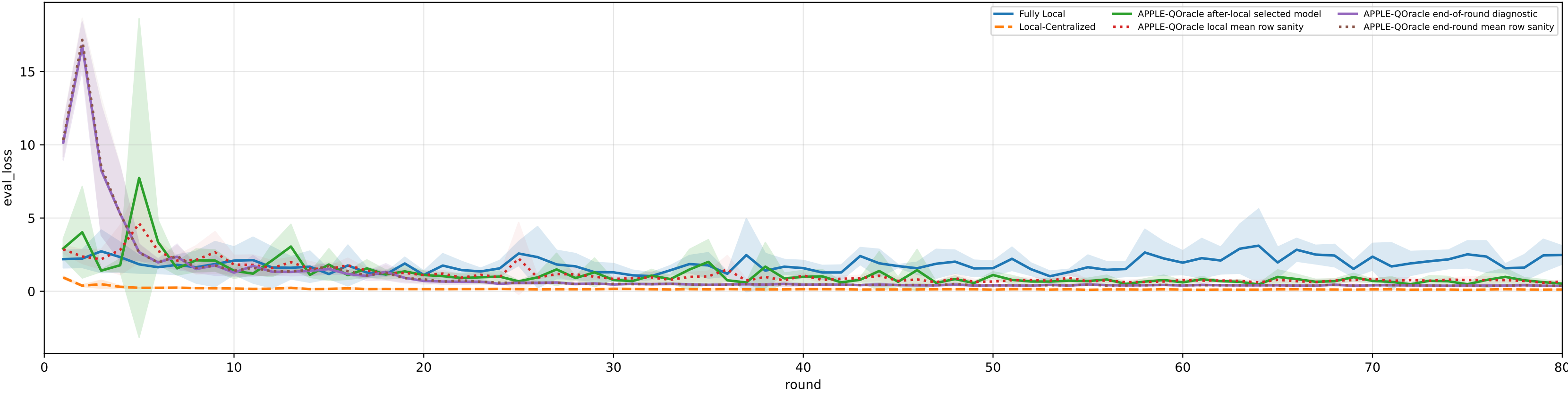


family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% | heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

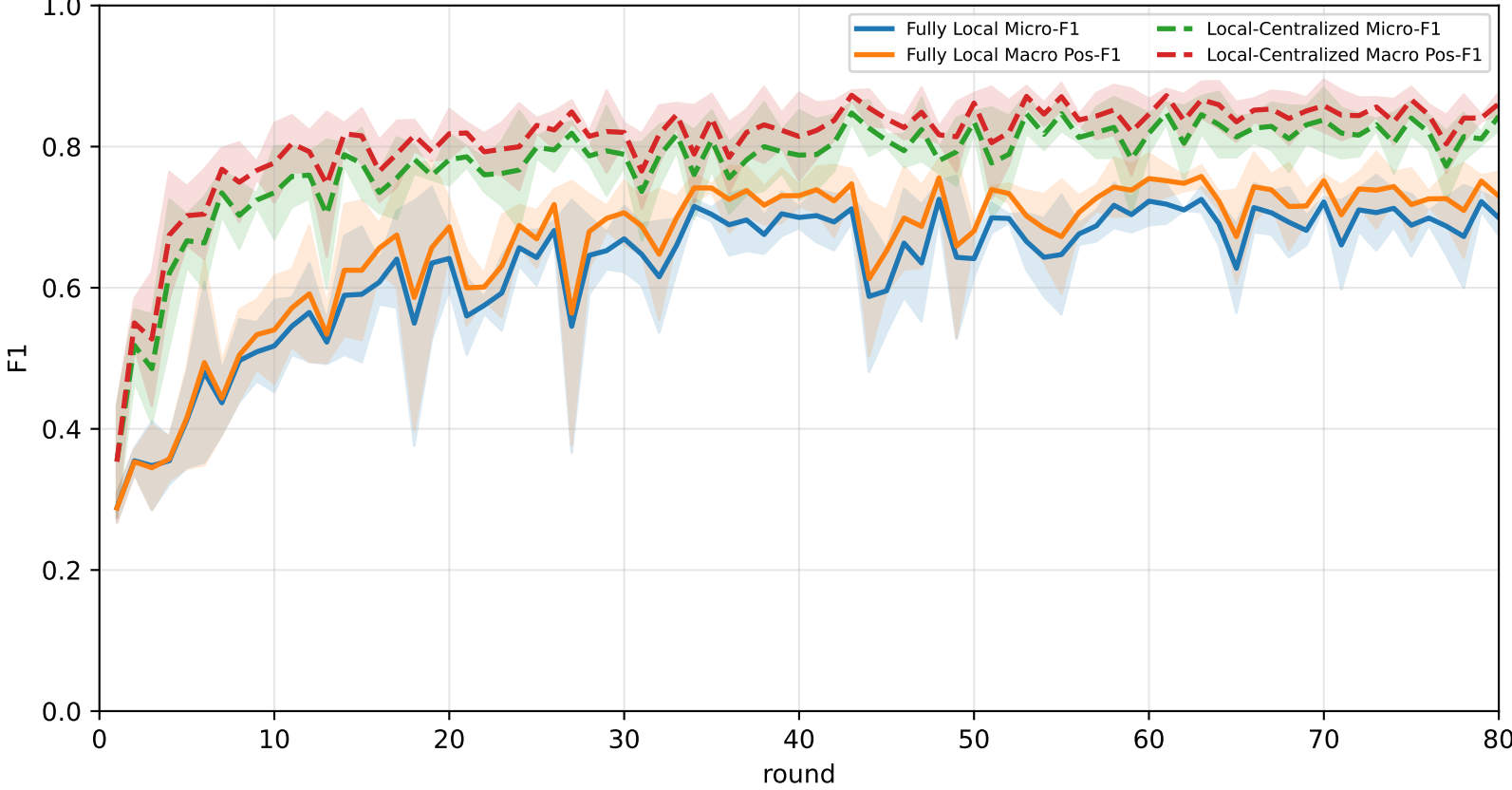
Best test performance (Fully Local vs Local-Centralized vs APPLE-QOracle, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	61.2% ± 0.9%	66.3% ± 1.4%	<u>86.3% ± 4.7%</u>	79.8% ± 5.8%	63.6% ± 1.5%	58.1% ± 1.4%	43.8% ± 3.4%
Local-Centralized	84.3% ± 1.6%	86.3% ± 1.4%	96.7% ± 1.1%	94.7% ± 0.6%	87.4% ± 4.2%	80.9% ± 2.2%	71.8% ± 3.9%
APPLE-QOracle	<u>70.9% ± 2.3%</u>	<u>72.6% ± 3.1%</u>	76.7% ± 7.9%	<u>82.2% ± 4.9%</u>	<u>74.0% ± 5.0%</u>	<u>71.4% ± 4.1%</u>	<u>58.8% ± 1.4%</u>

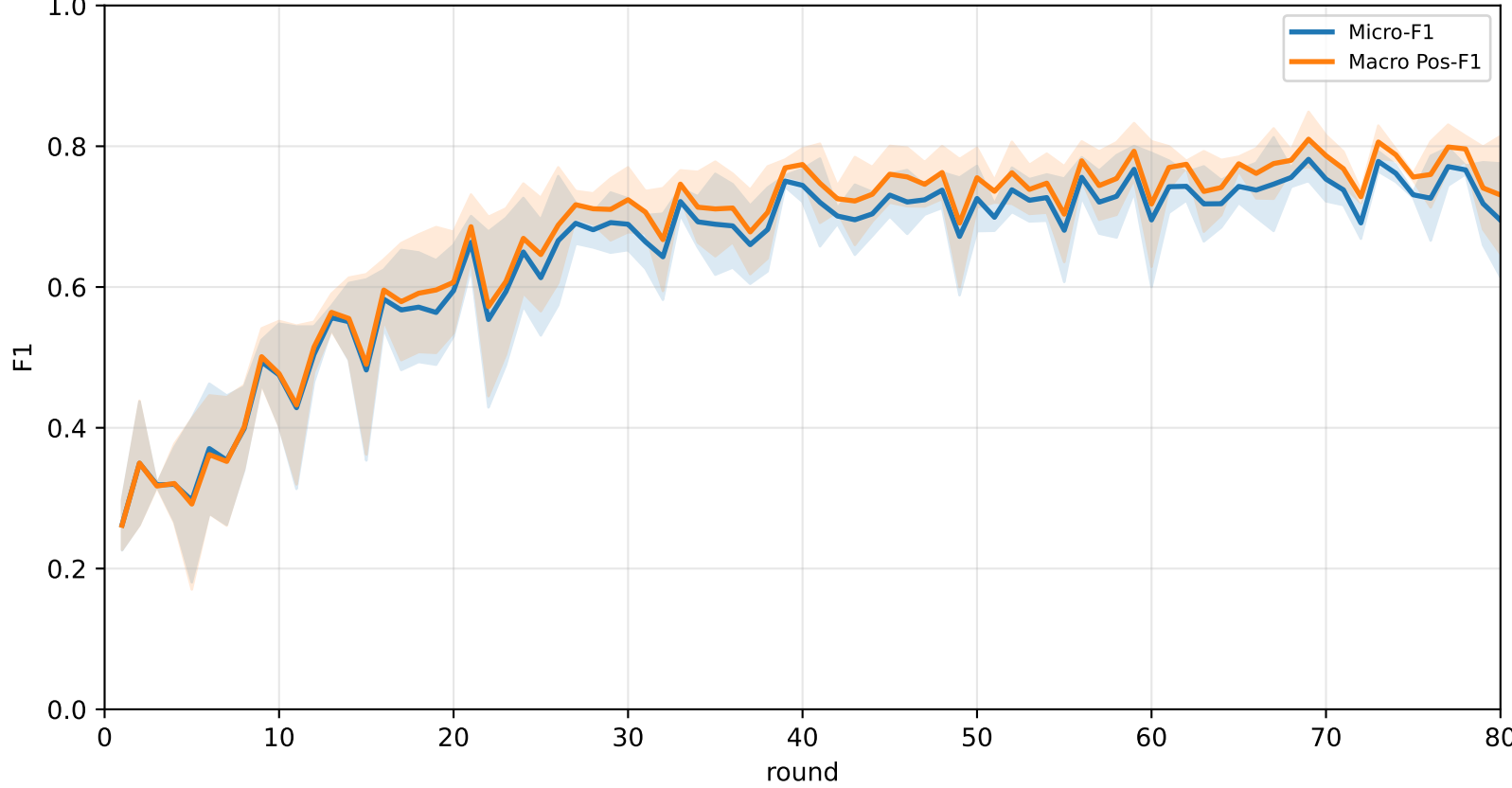
Validation loss: baselines vs selected APPLE-QOracle and end-of-round mismatch (rounds=80)



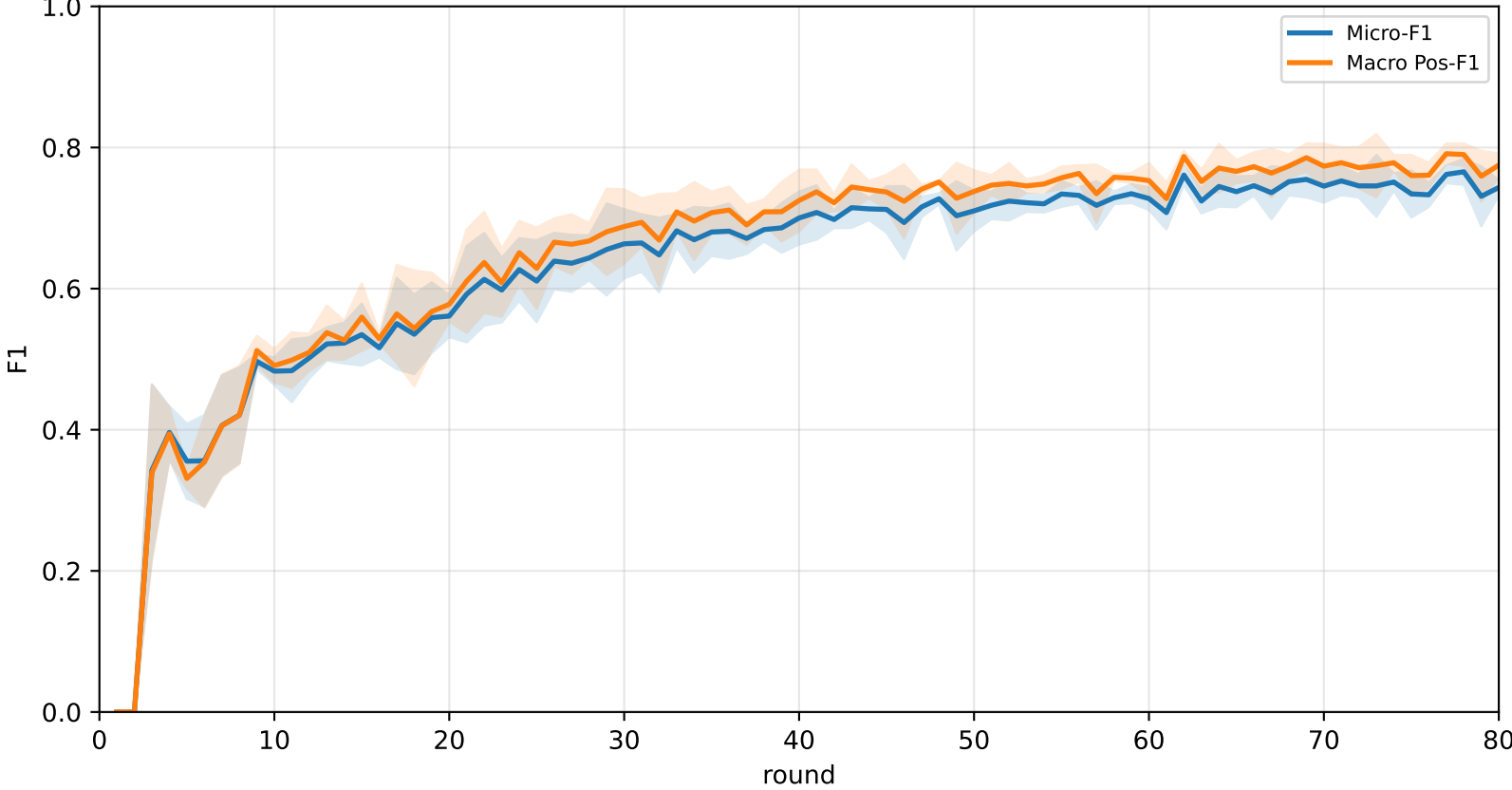
Pooled baseline micro / macro F1



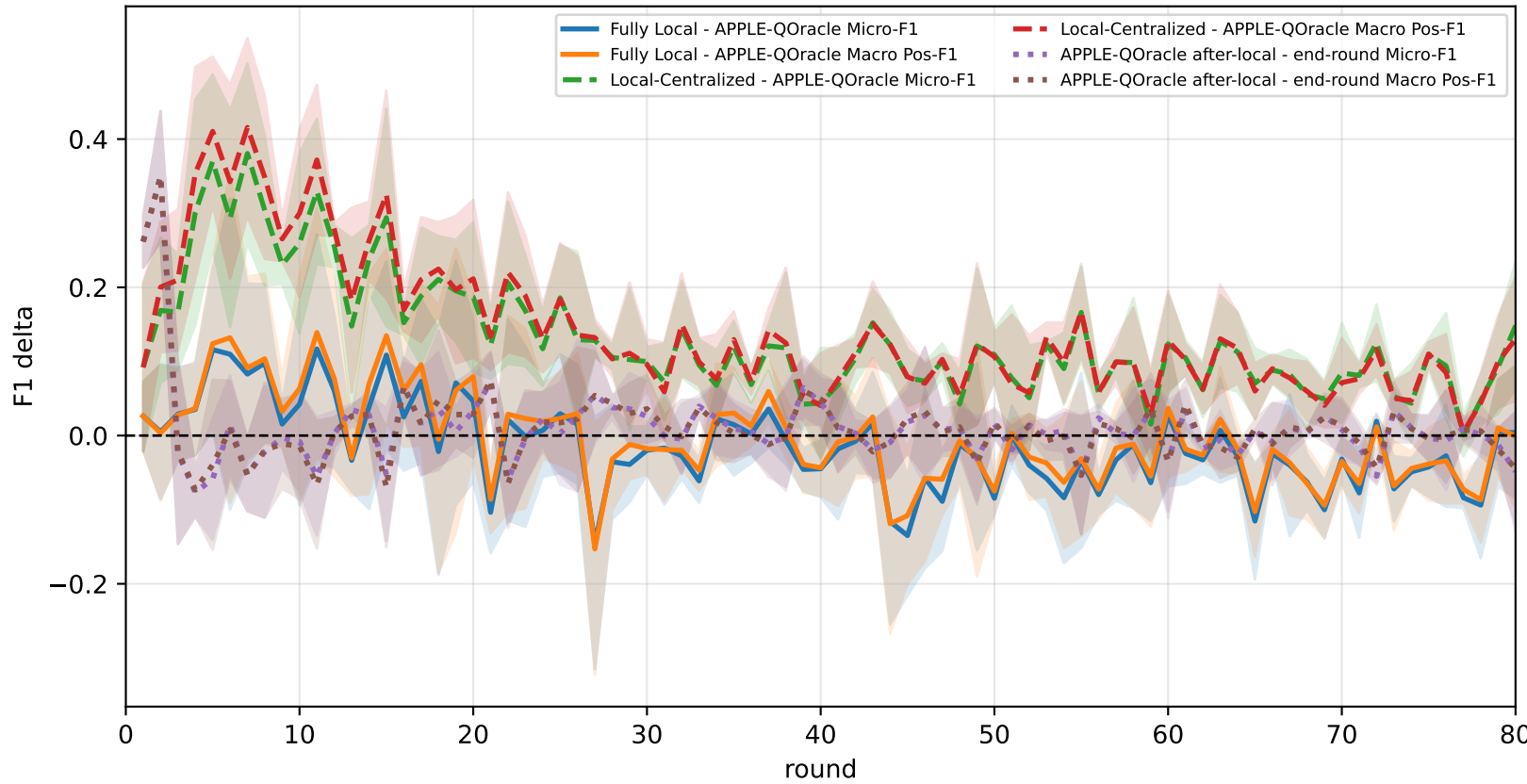
Pooled APPLE-QOracle micro / macro F1 (after-local selected model)



Pooled APPLE-QOracle micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QOracle mismatch gap



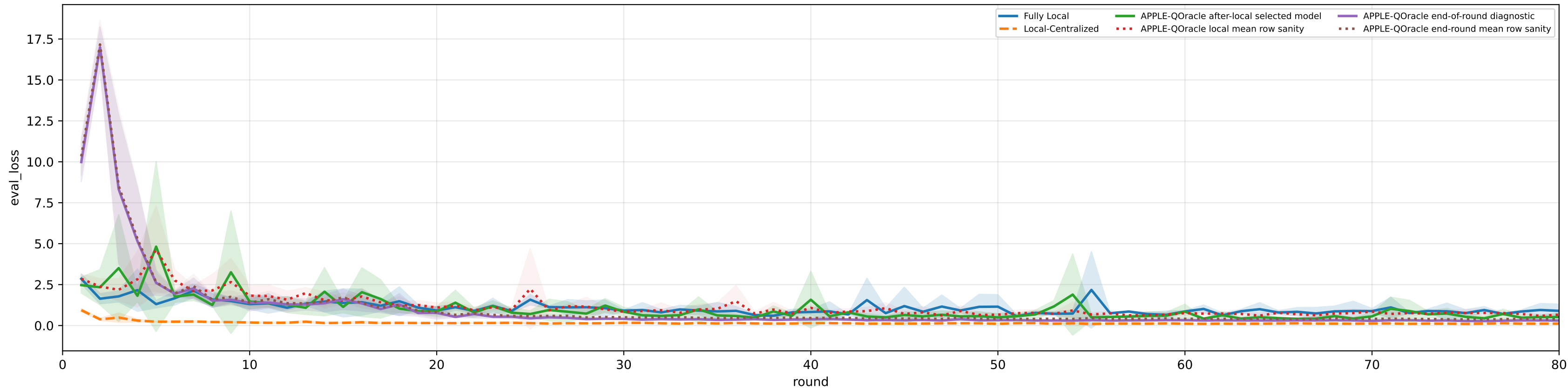
Controlled q-label heterogeneity: Mid | APPLE-QOracle diagnostics | family=q_task_cycle6 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

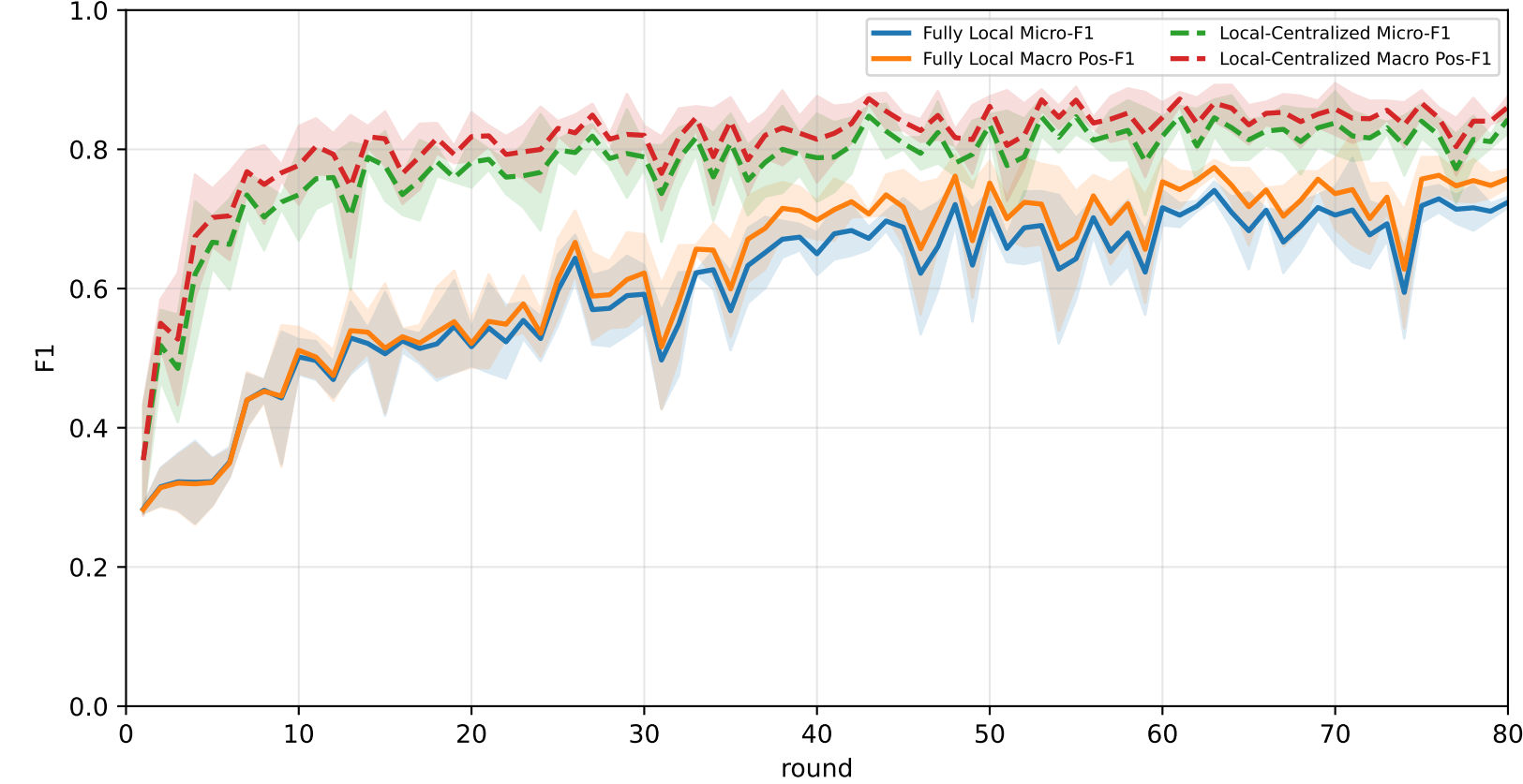
Best test performance (Fully Local vs Local-Centralized vs APPLE-QOracle, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	64.1% ± 1.8%	67.8% ± 2.4%	81.8% ± 8.2%	78.3% ± 6.8%	64.9% ± 1.4%	64.9% ± 4.4%	49.0% ± 3.2%
Local-Centralized	84.3% ± 1.6%	86.3% ± 1.4%	96.7% ± 1.1%	94.7% ± 0.6%	87.4% ± 4.2%	80.9% ± 2.2%	71.8% ± 3.9%
APPLE-QOracle	70.0% ± 1.6%	73.2% ± 1.6%	85.5% ± 9.8%	84.8% ± 3.2%	66.3% ± 4.3%	74.1% ± 1.7%	55.3% ± 2.6%

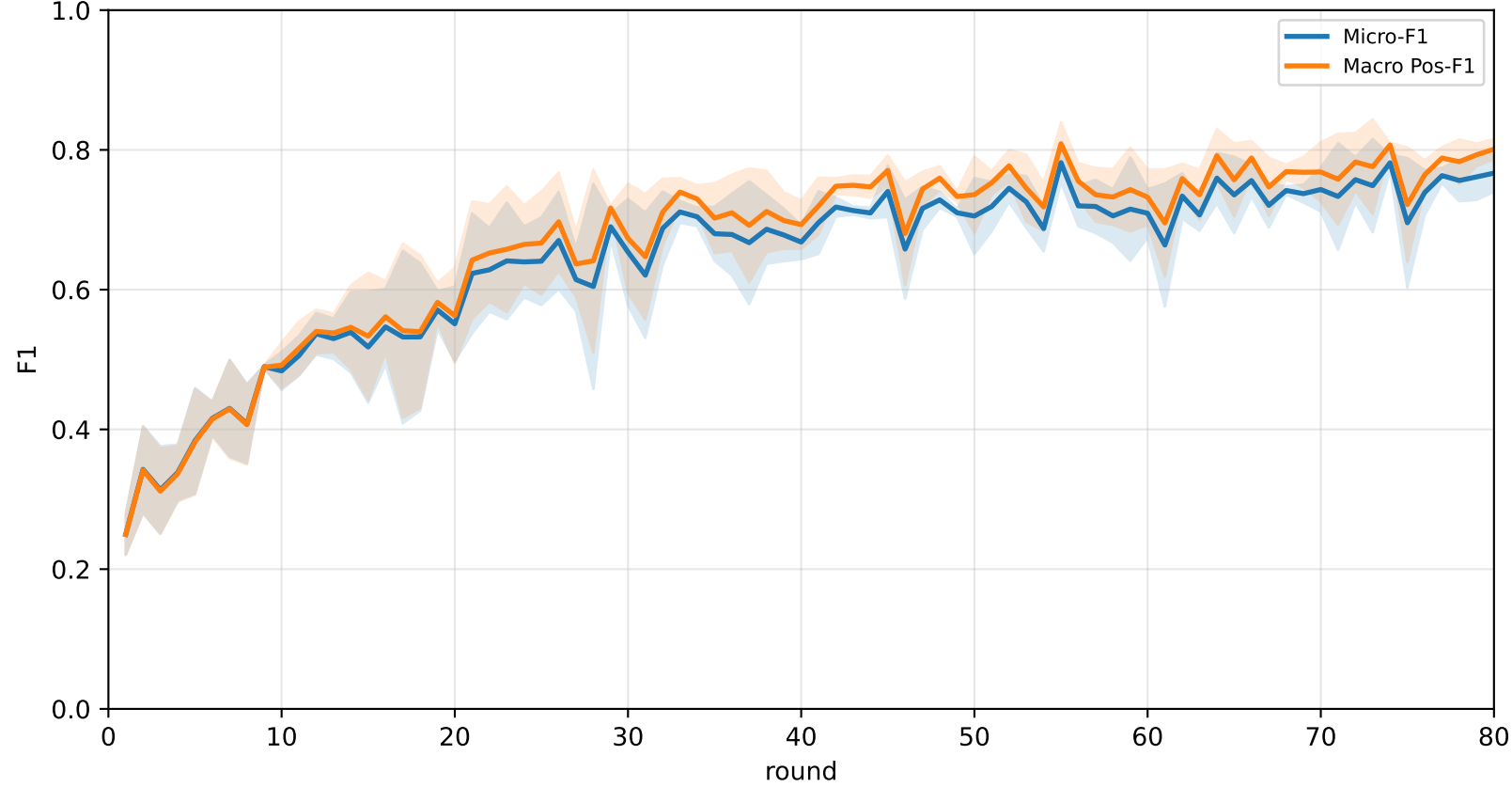
Validation loss: baselines vs selected APPLE-QOracle and end-of-round mismatch (rounds=80)



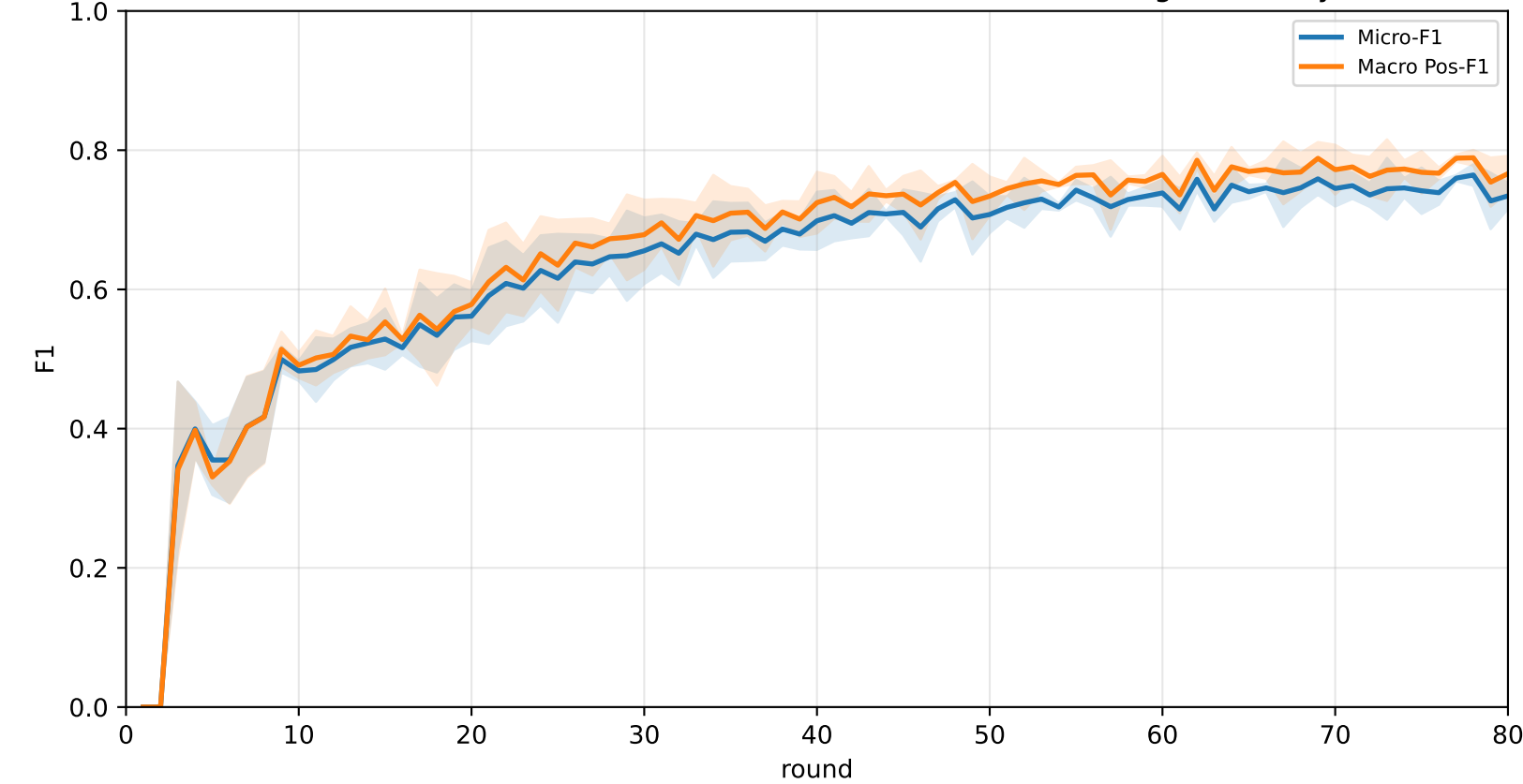
Pooled baseline micro / macro F1



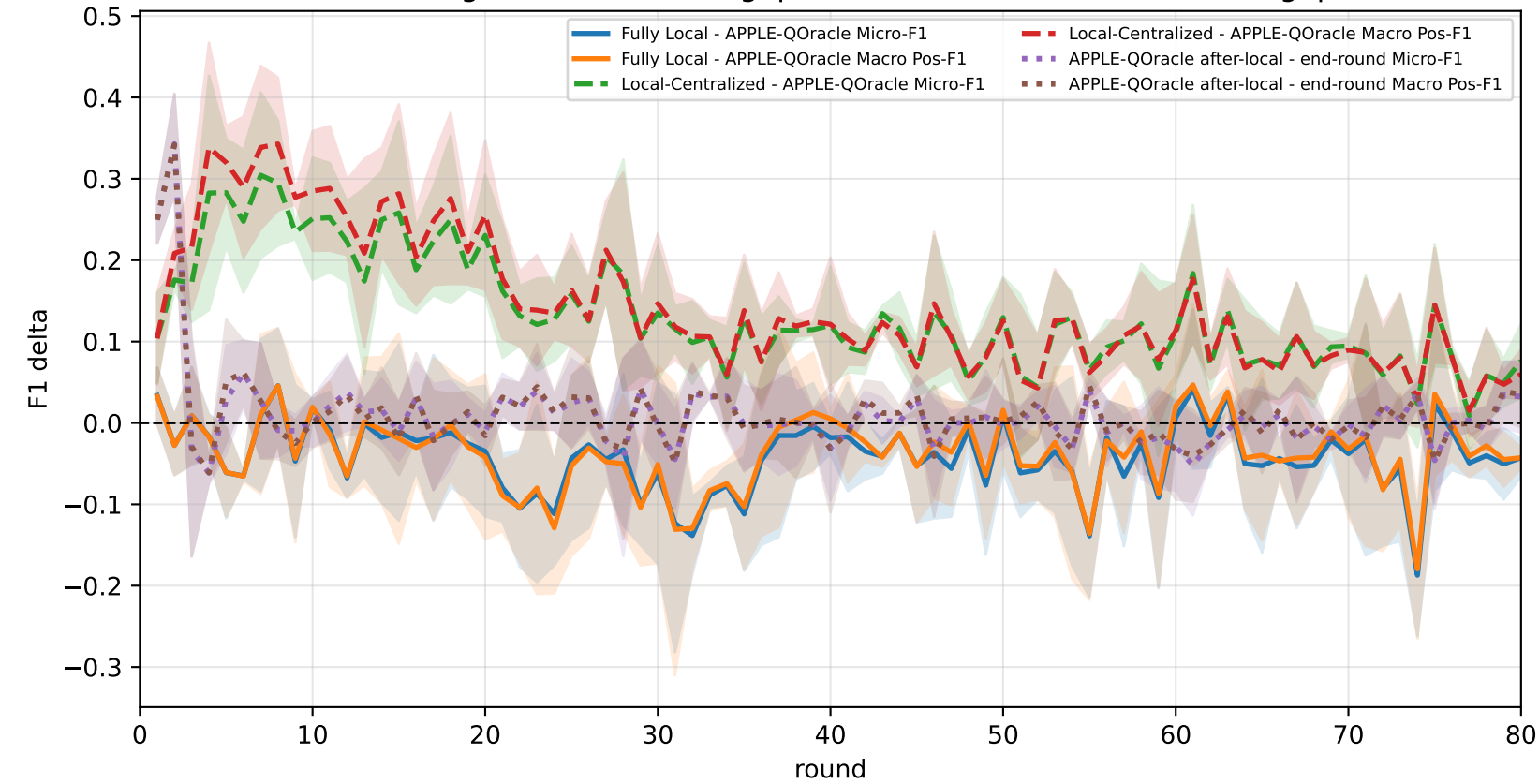
Pooled APPLE-QOracle micro / macro F1 (after-local selected model)



Pooled APPLE-QOracle micro / macro F1 (end-of-round diagnostic only)



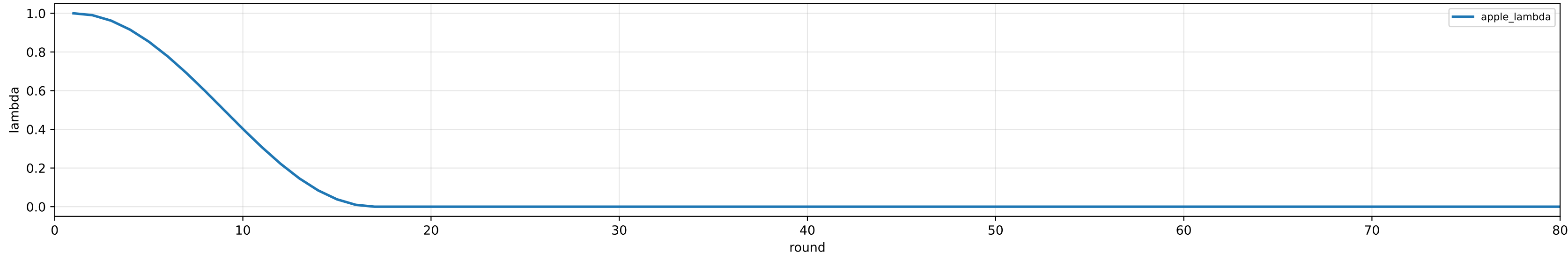
Delta diagnostics: baseline gaps and APPLE-QOracle mismatch gap



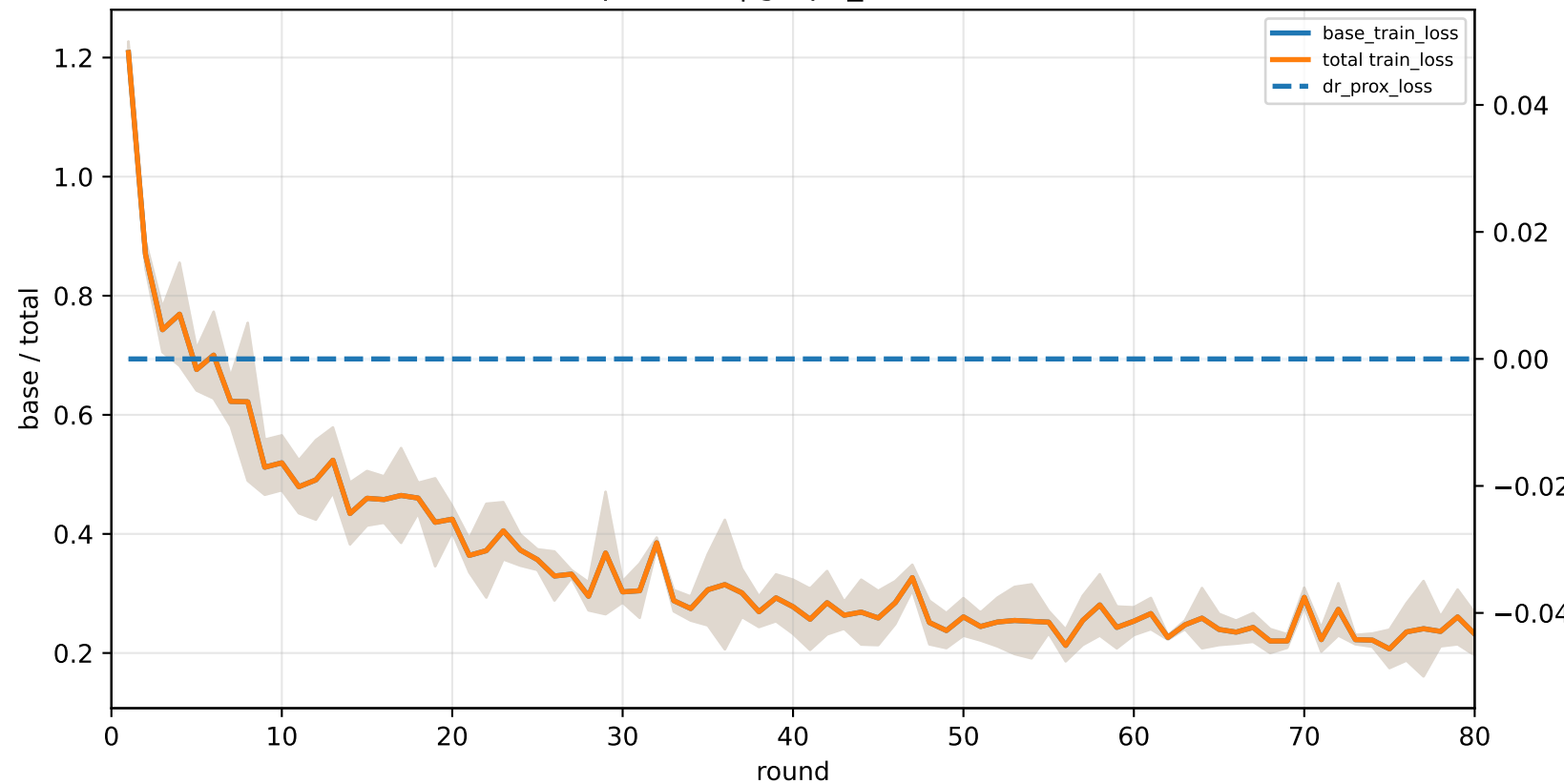
APPLE-QOracle training-loss decomposition | Mid heterogeneity

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

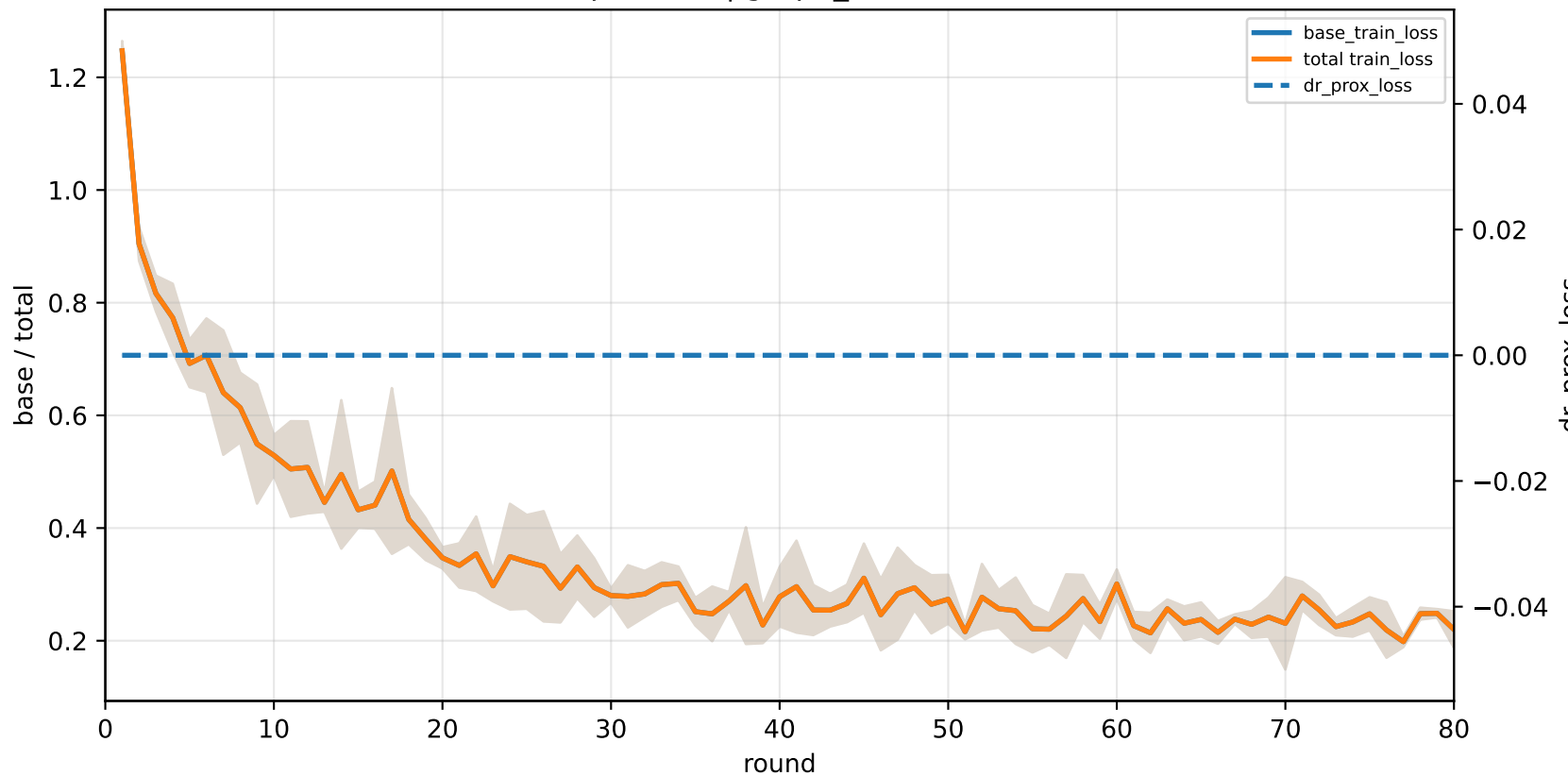
APPLE lambda scheduler | apple_mu=0



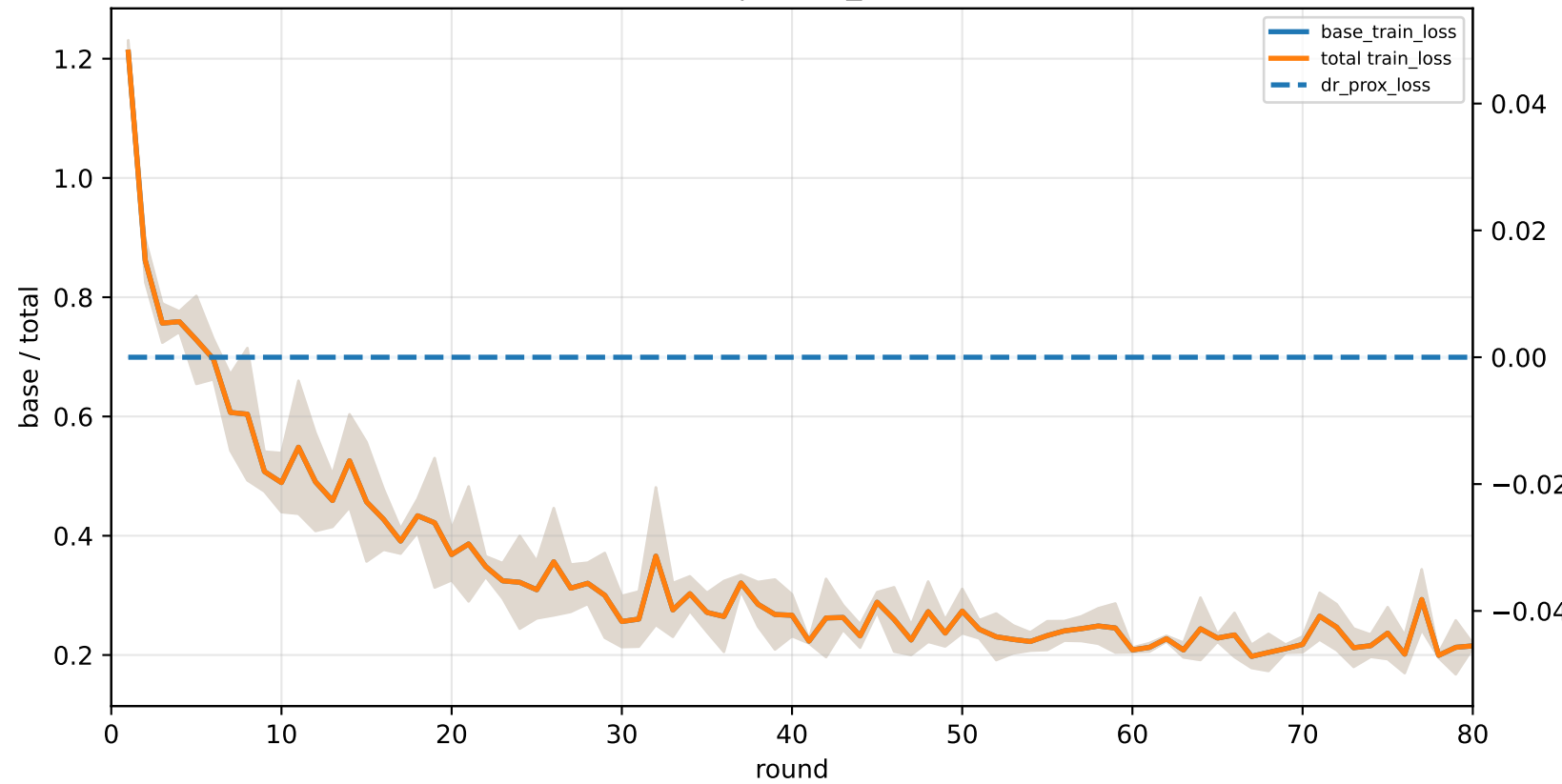
C2-specialist | graph_id=3000005



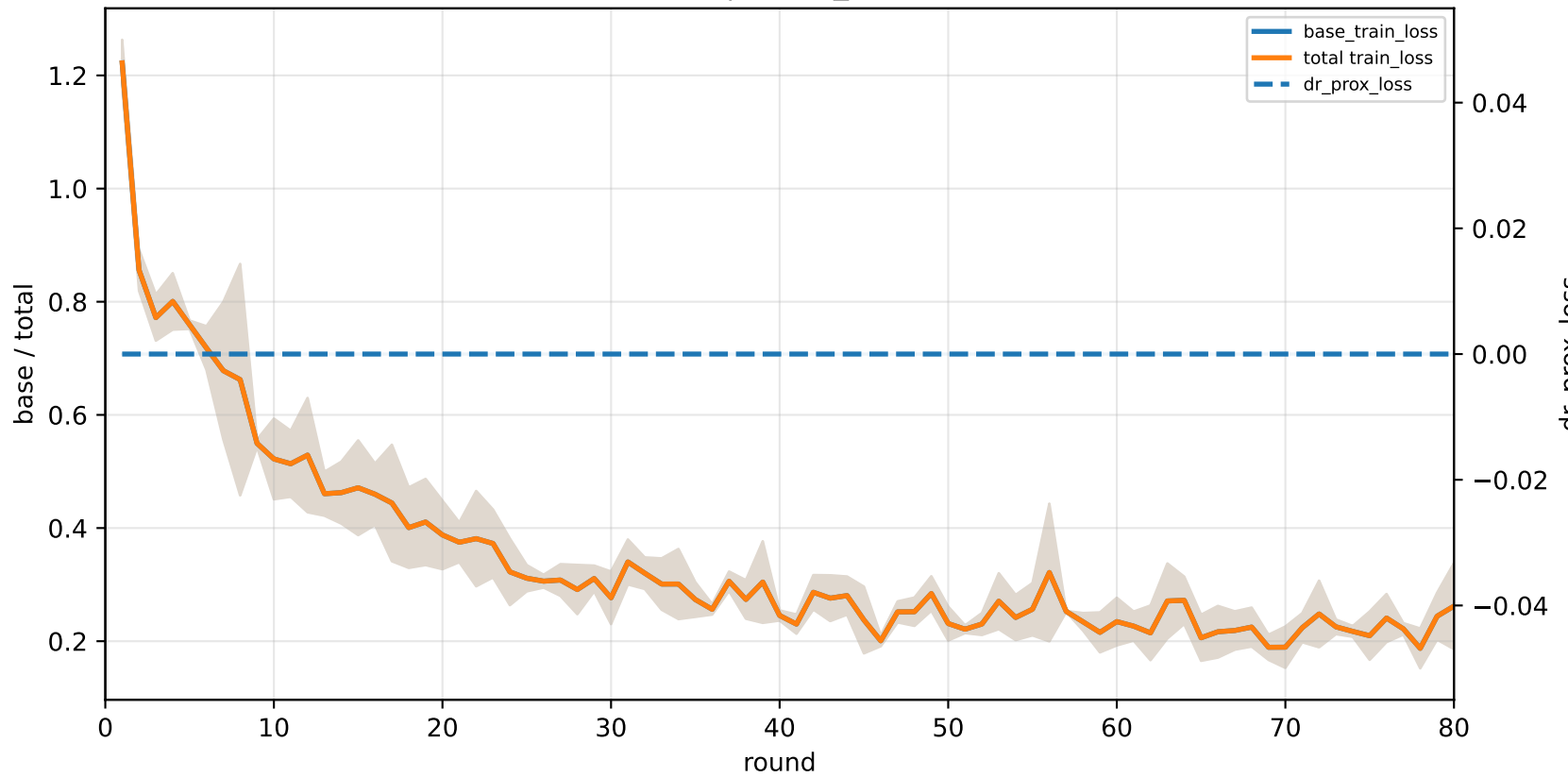
C3-specialist | graph_id=3000006



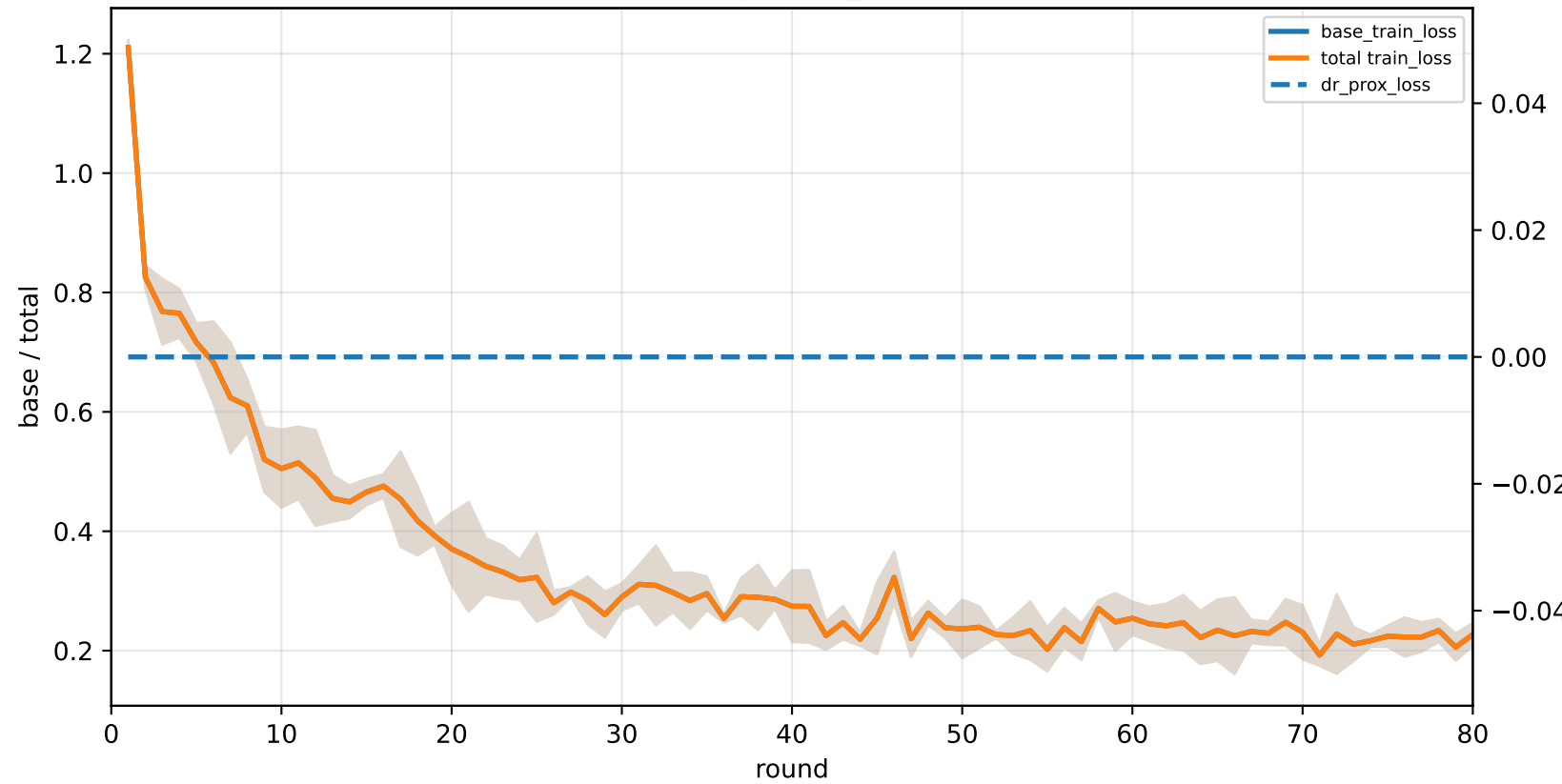
C4-specialist | graph_id=3000007



C5-specialist | graph_id=3000008

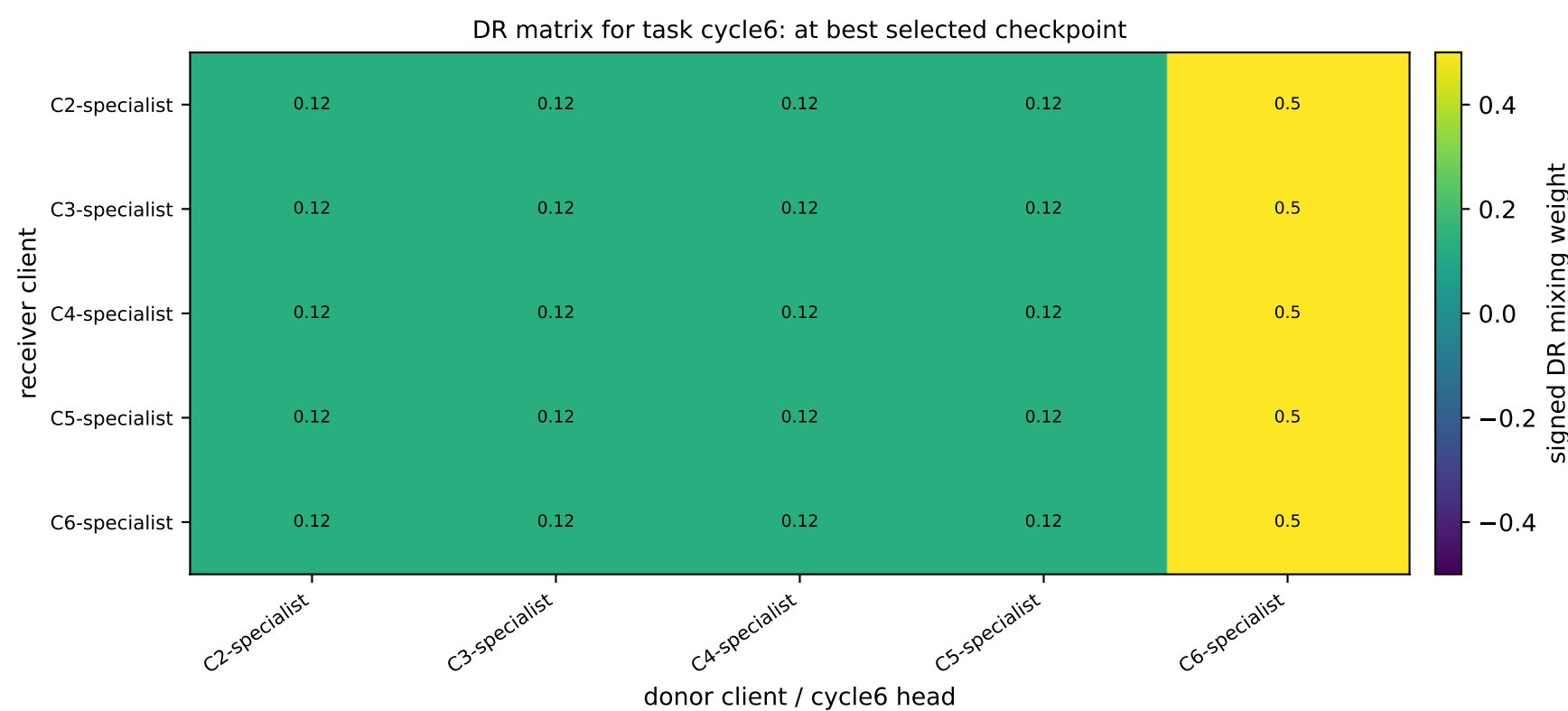
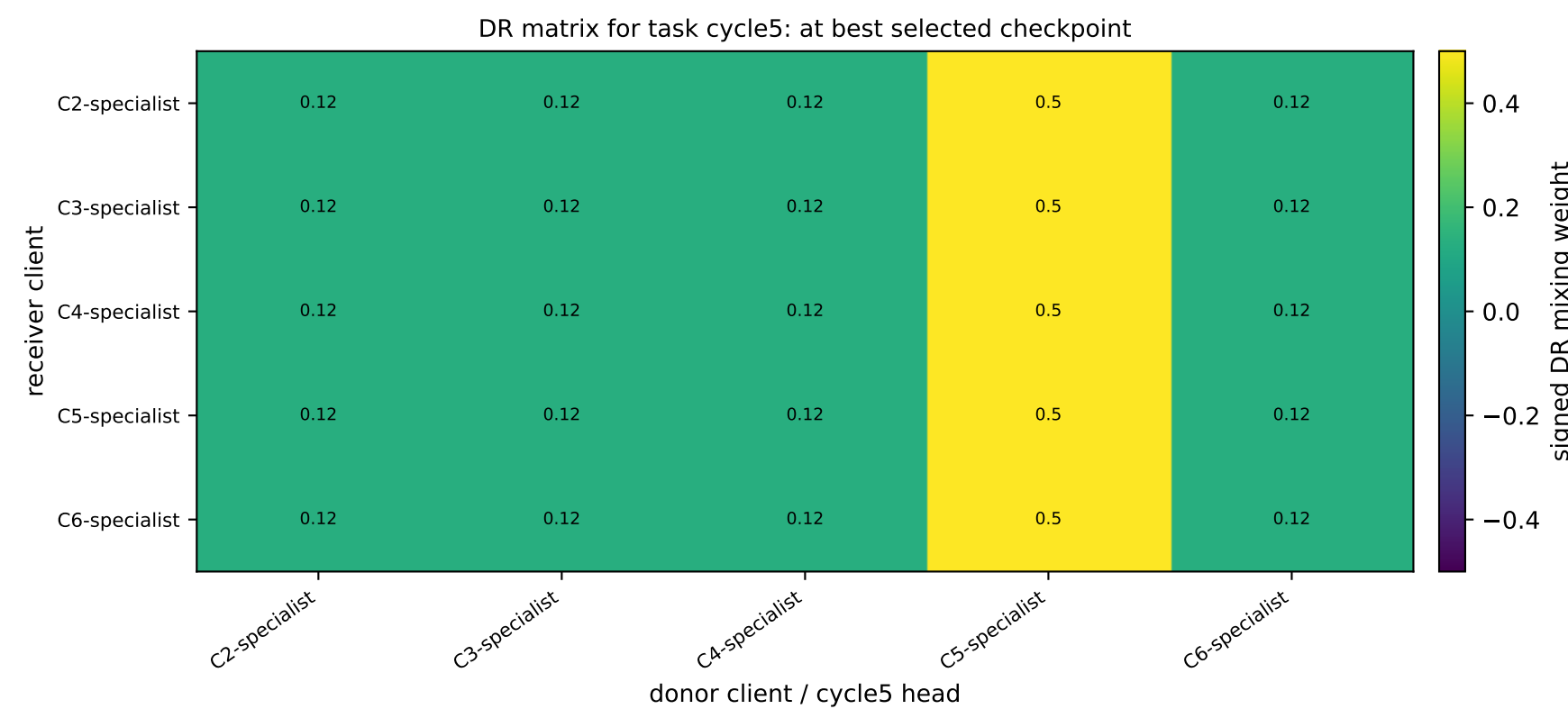
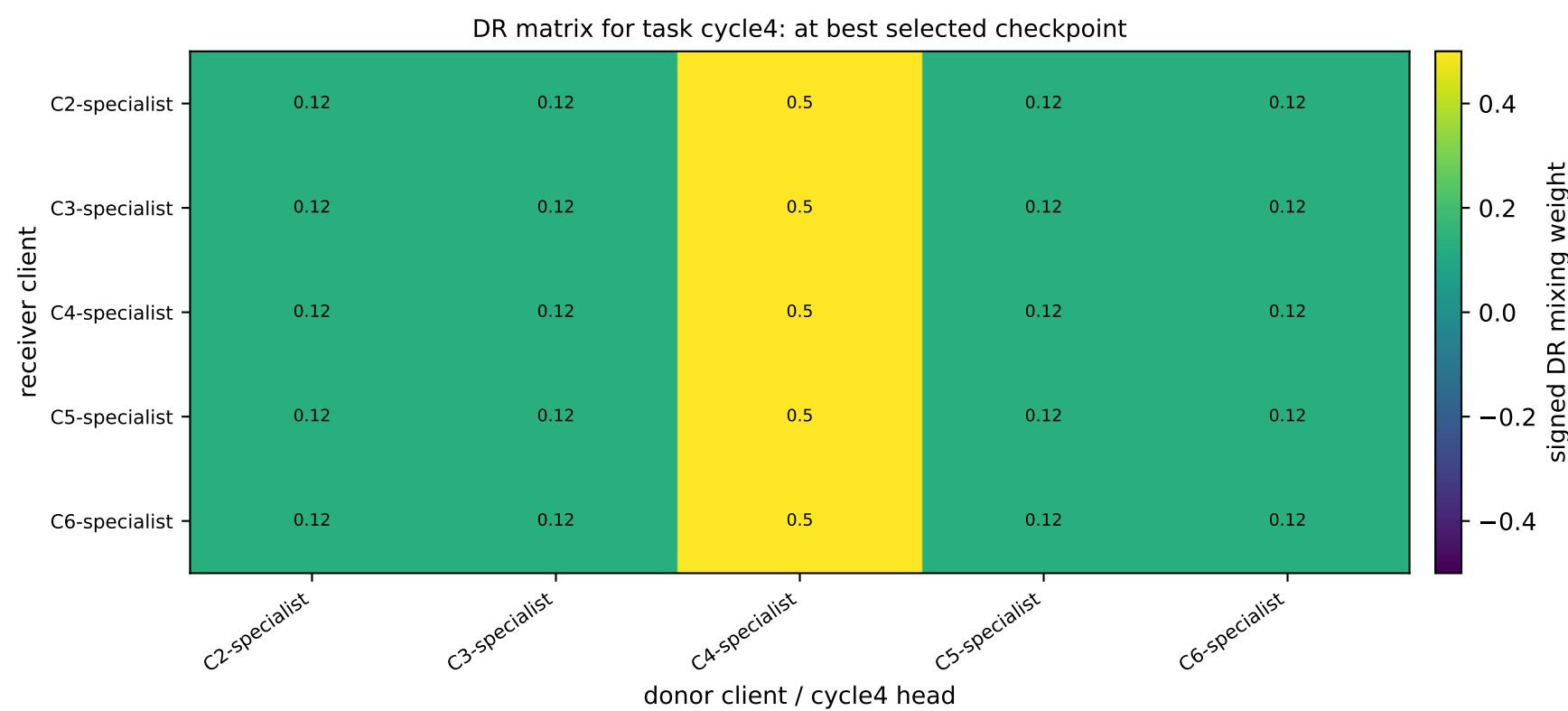
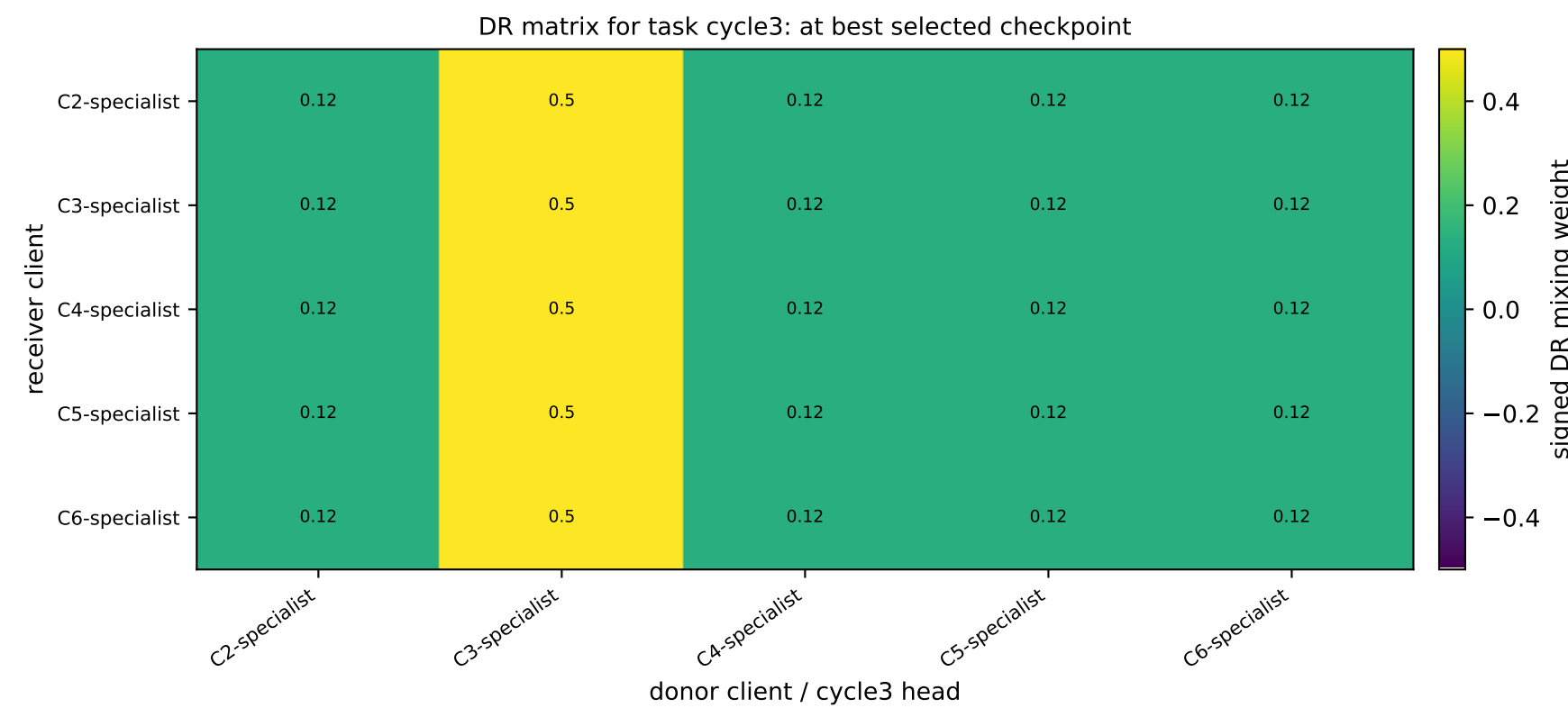
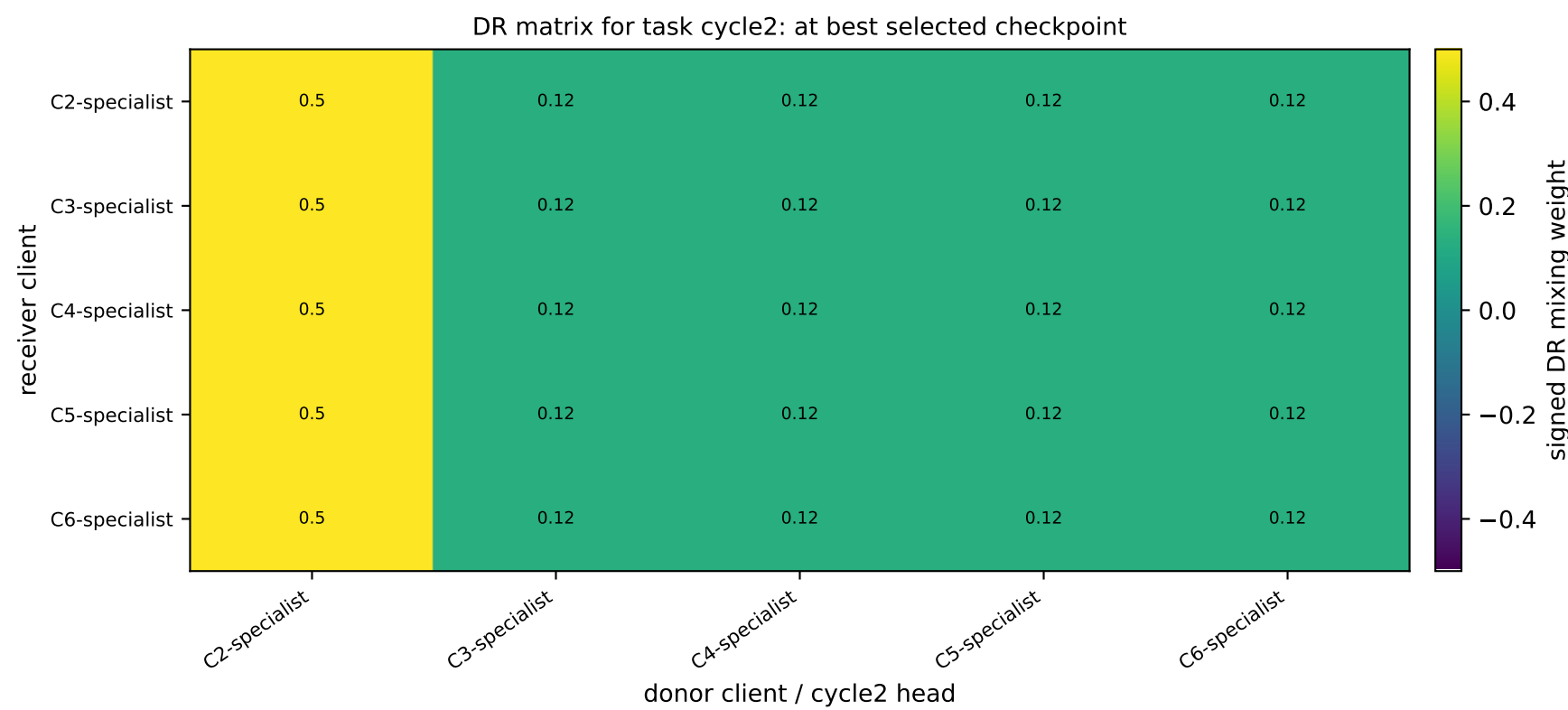


C6-specialist | graph_id=3000009



APPLE-QOracle task-wise DR matrices: receiver clients × donor task-heads | Mid

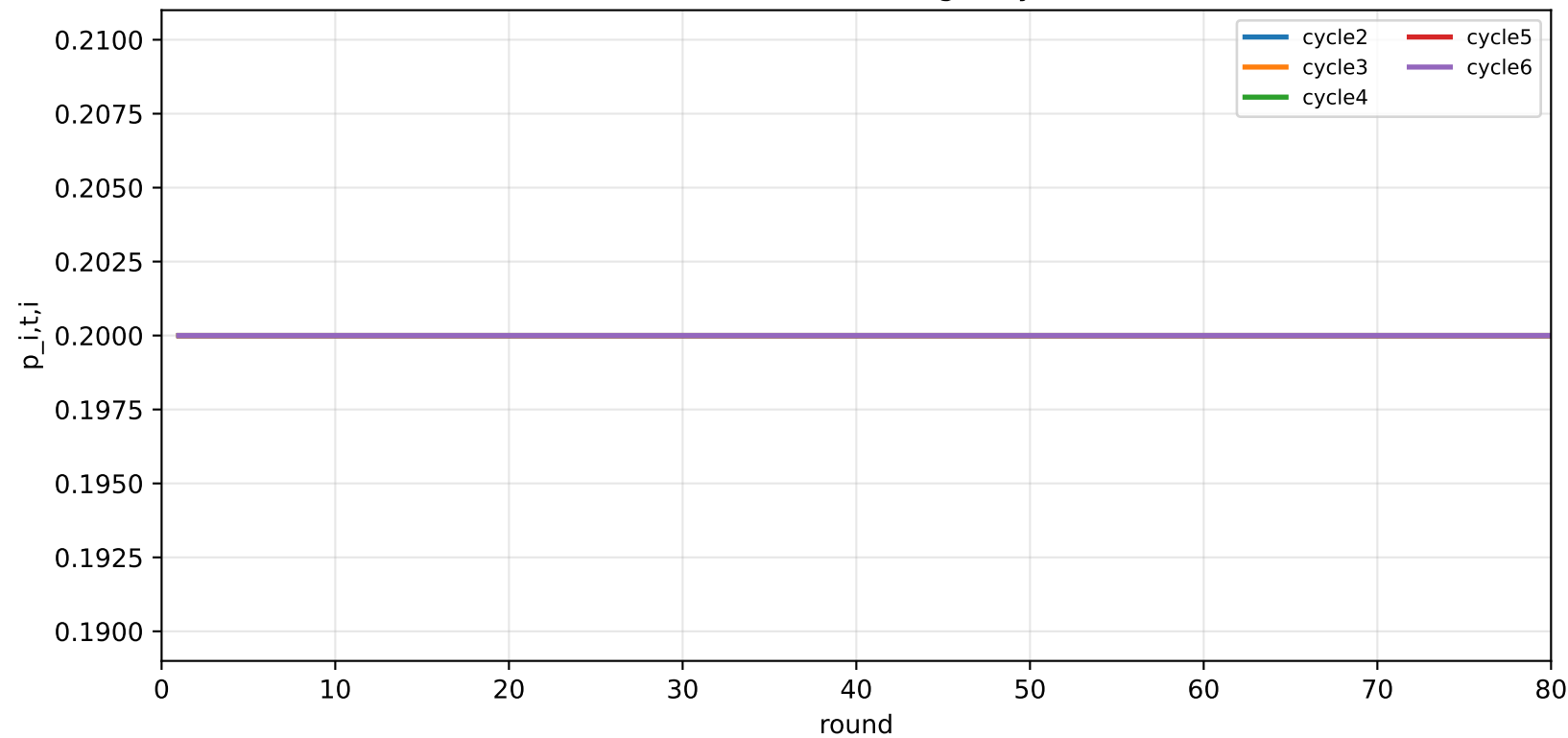
family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120



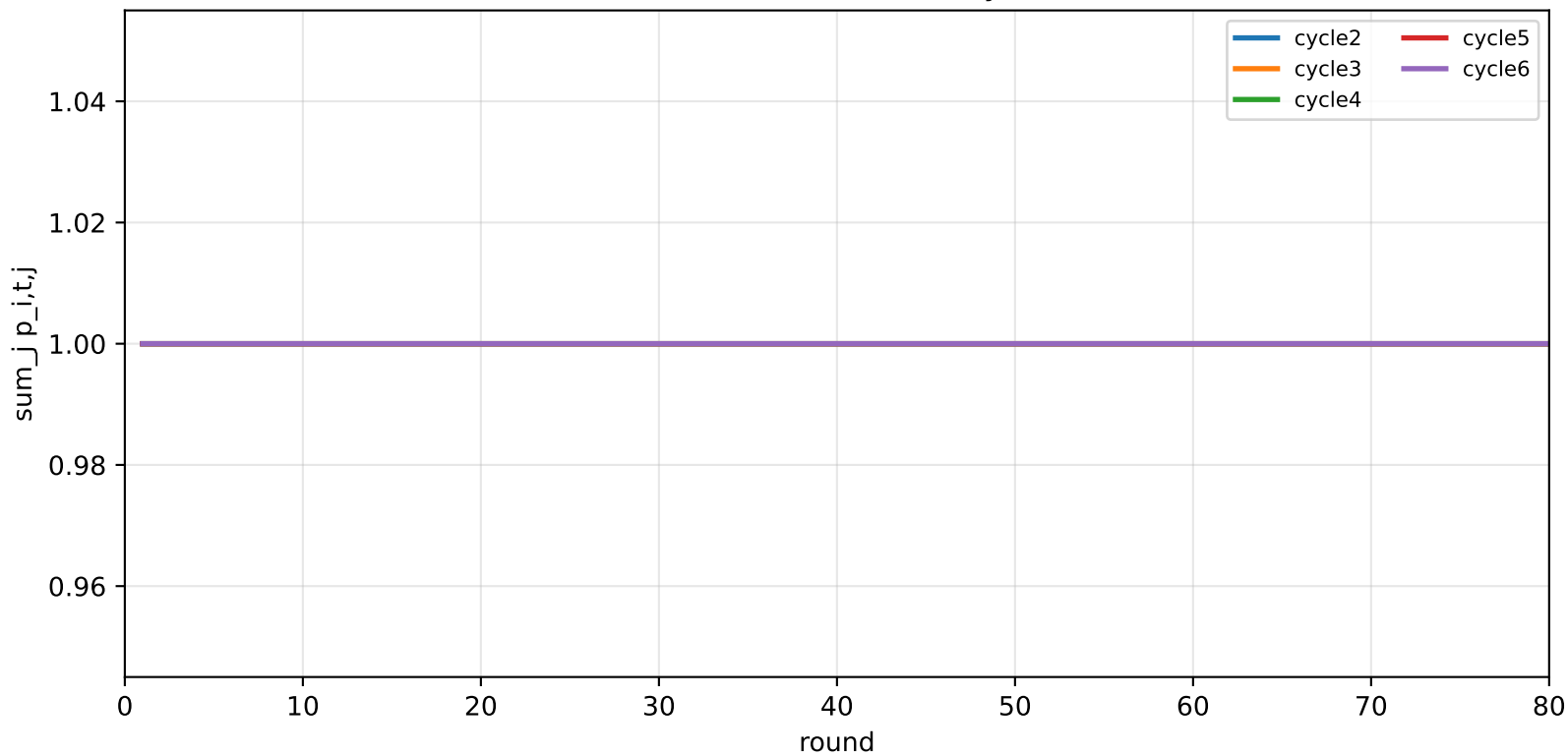
APPLE-QOracle task-wise DR dynamics | Mid

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

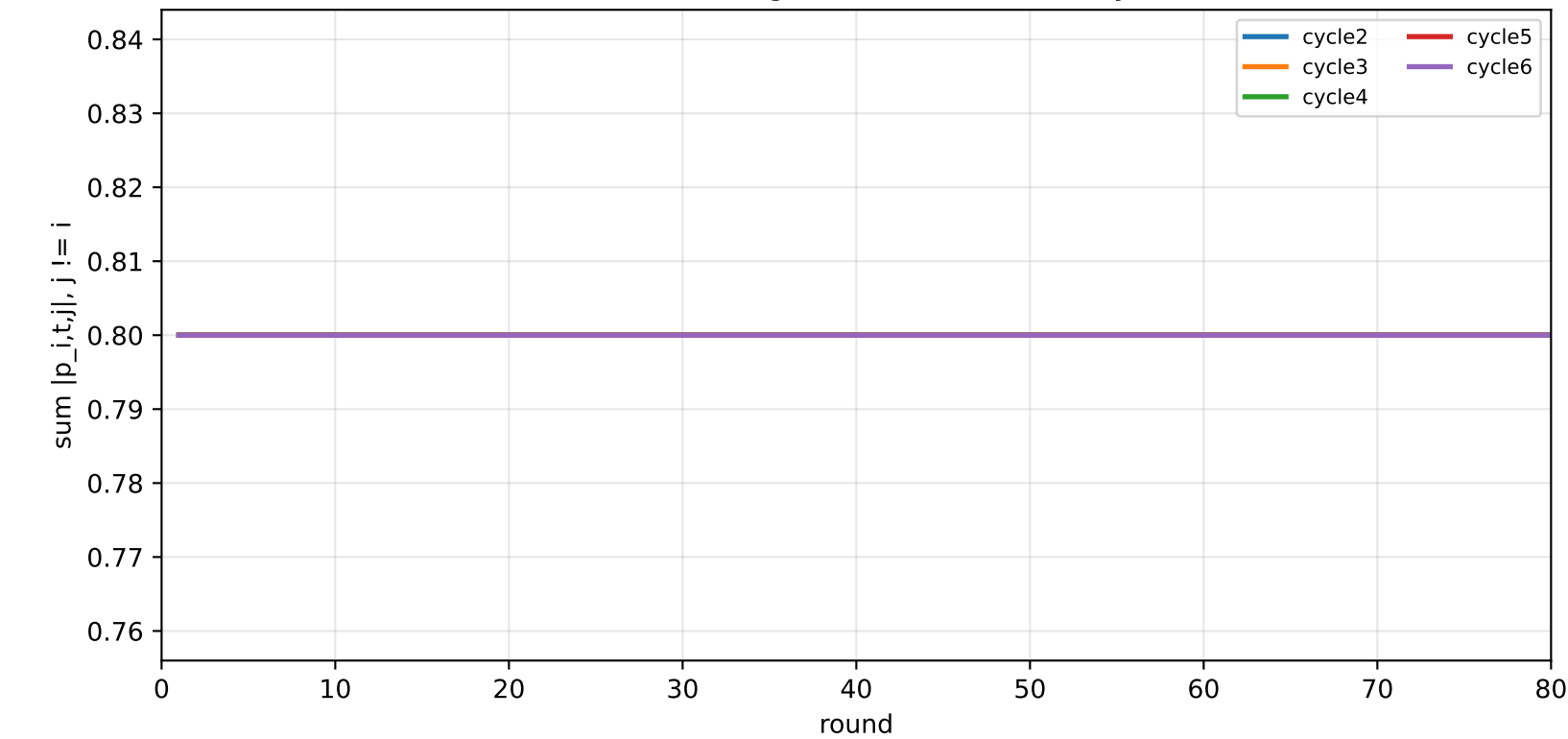
Task-head DR self-weight by task



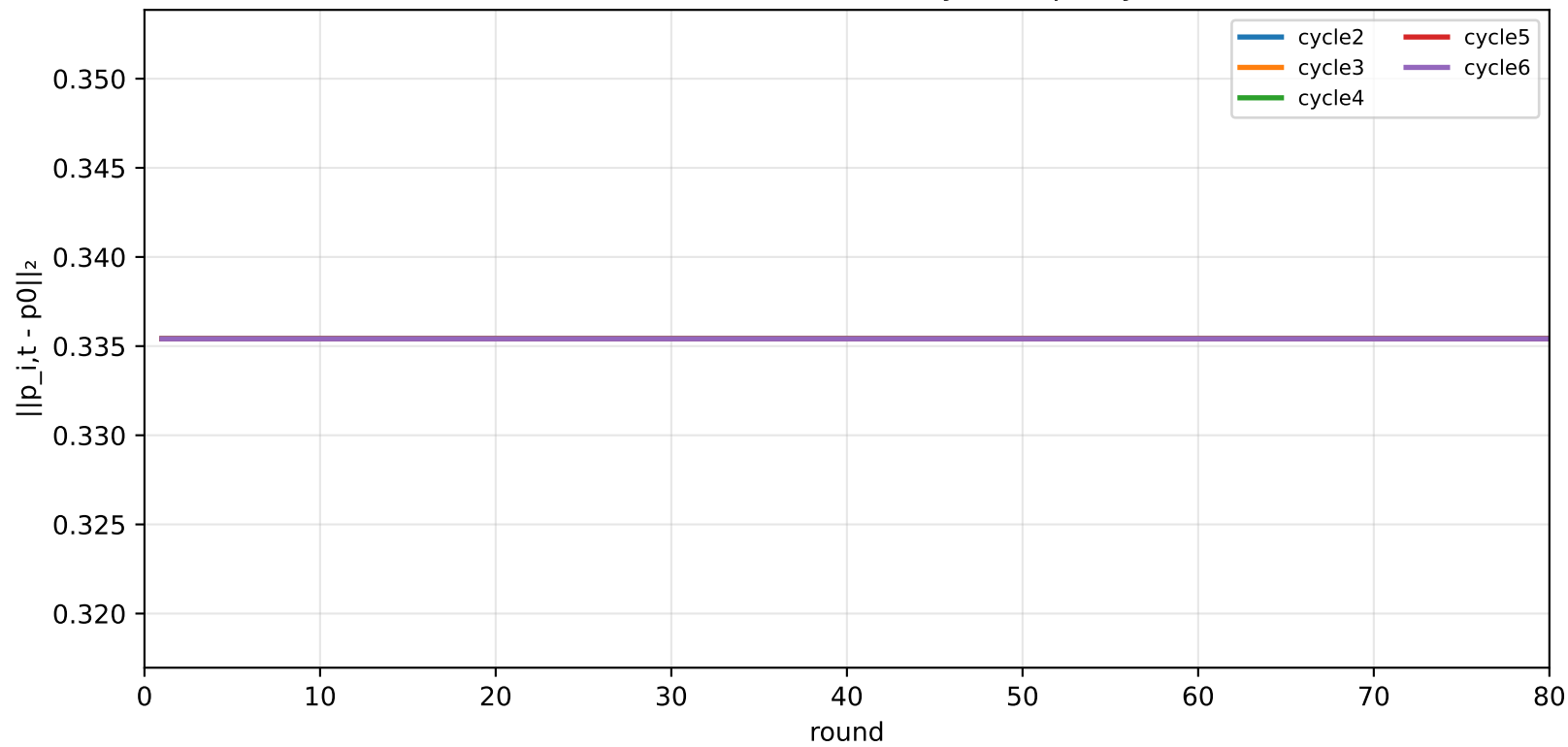
Task-head DR row sum by task



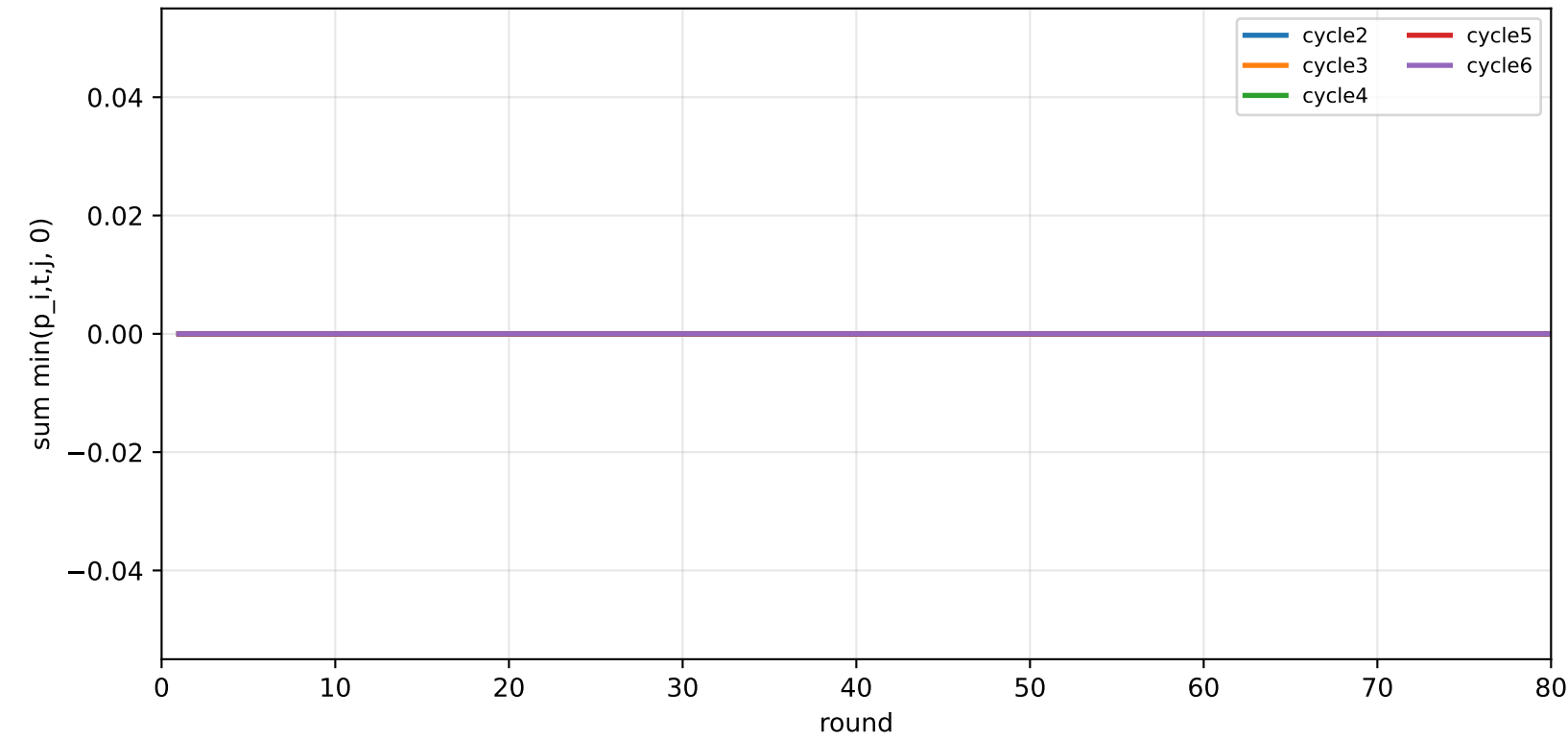
Task-head off-diagonal absolute mass by task



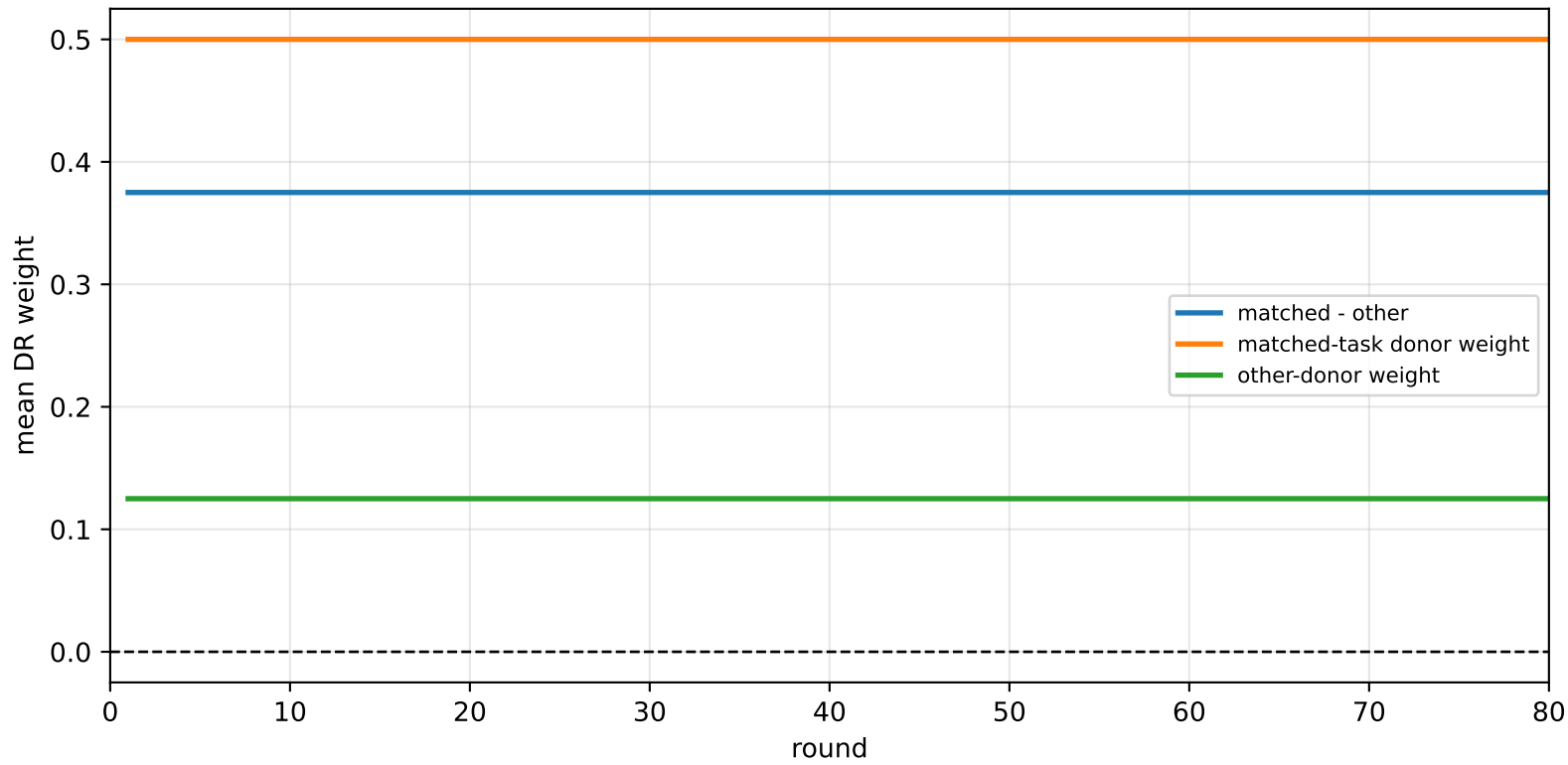
Task-head DR movement away from p0 by task



Task-head off-diagonal negative mass by task



Task-head DR alignment



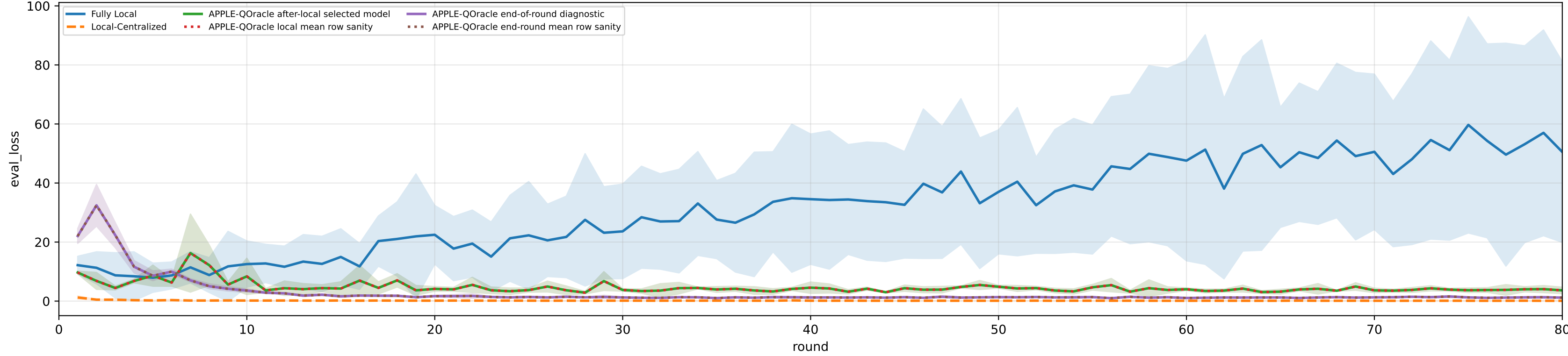
Controlled q-label heterogeneity | High | APPLE-QOracle diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

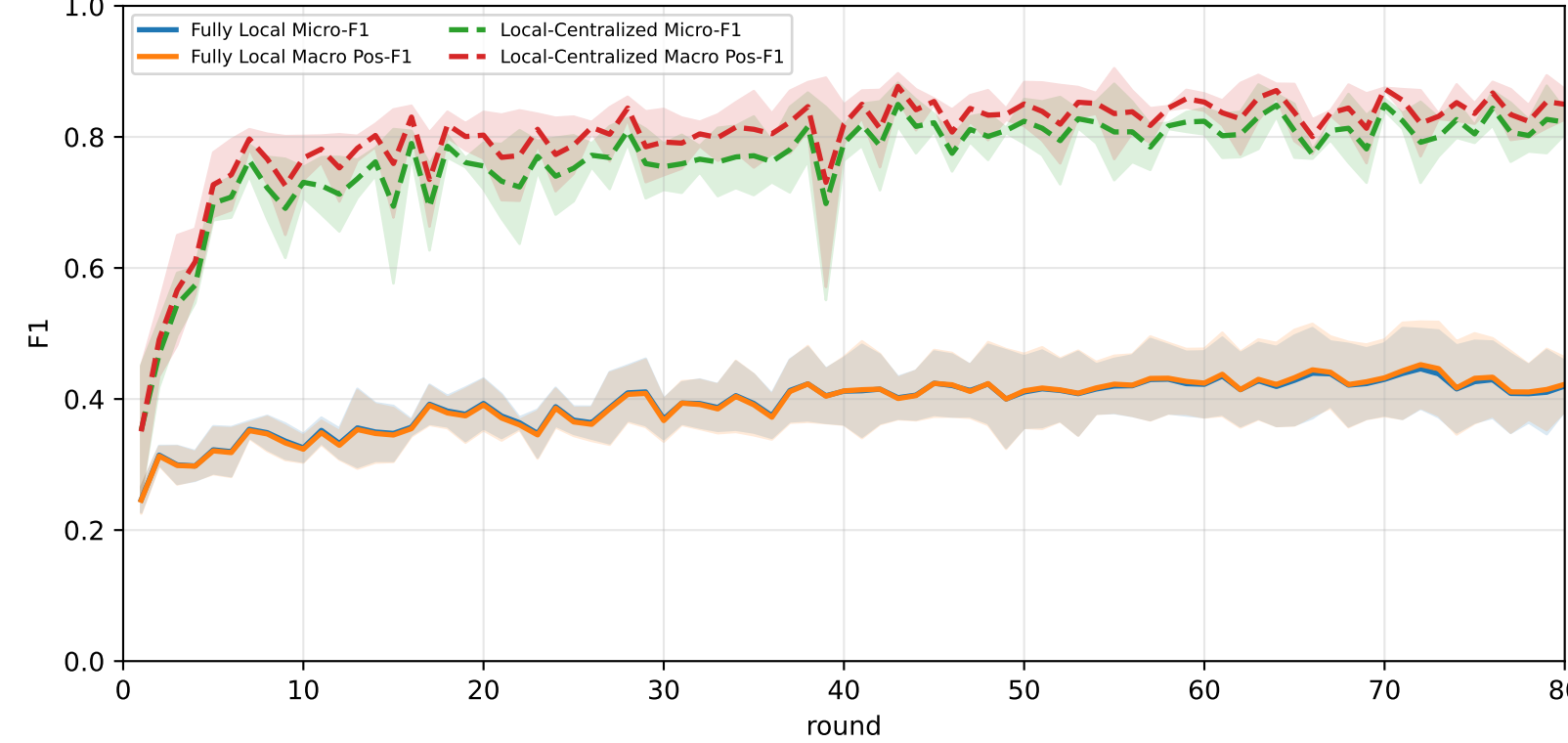
Best test performance (Fully Local vs Local-Centralized vs APPLE-QOracle, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	27.6% ± 1.8%	27.6% ± 1.8%	22.4% ± 1.5%	27.5% ± 2.1%	27.3% ± 1.2%	26.7% ± 1.2%	30.2% ± 1.5%
Local-Centralized	84.9% ± 1.3%	86.7% ± 0.8%	96.8% ± 1.6%	95.1% ± 1.4%	86.6% ± 2.5%	81.4% ± 1.3%	73.7% ± 5.8%
APPLE-QOracle	<u>58.5% ± 10.6%</u>	<u>60.6% ± 11.9%</u>	<u>68.5% ± 19.8%</u>	<u>68.3% ± 13.7%</u>	<u>58.9% ± 9.6%</u>	<u>57.0% ± 10.4%</u>	<u>47.9% ± 5.8%</u>

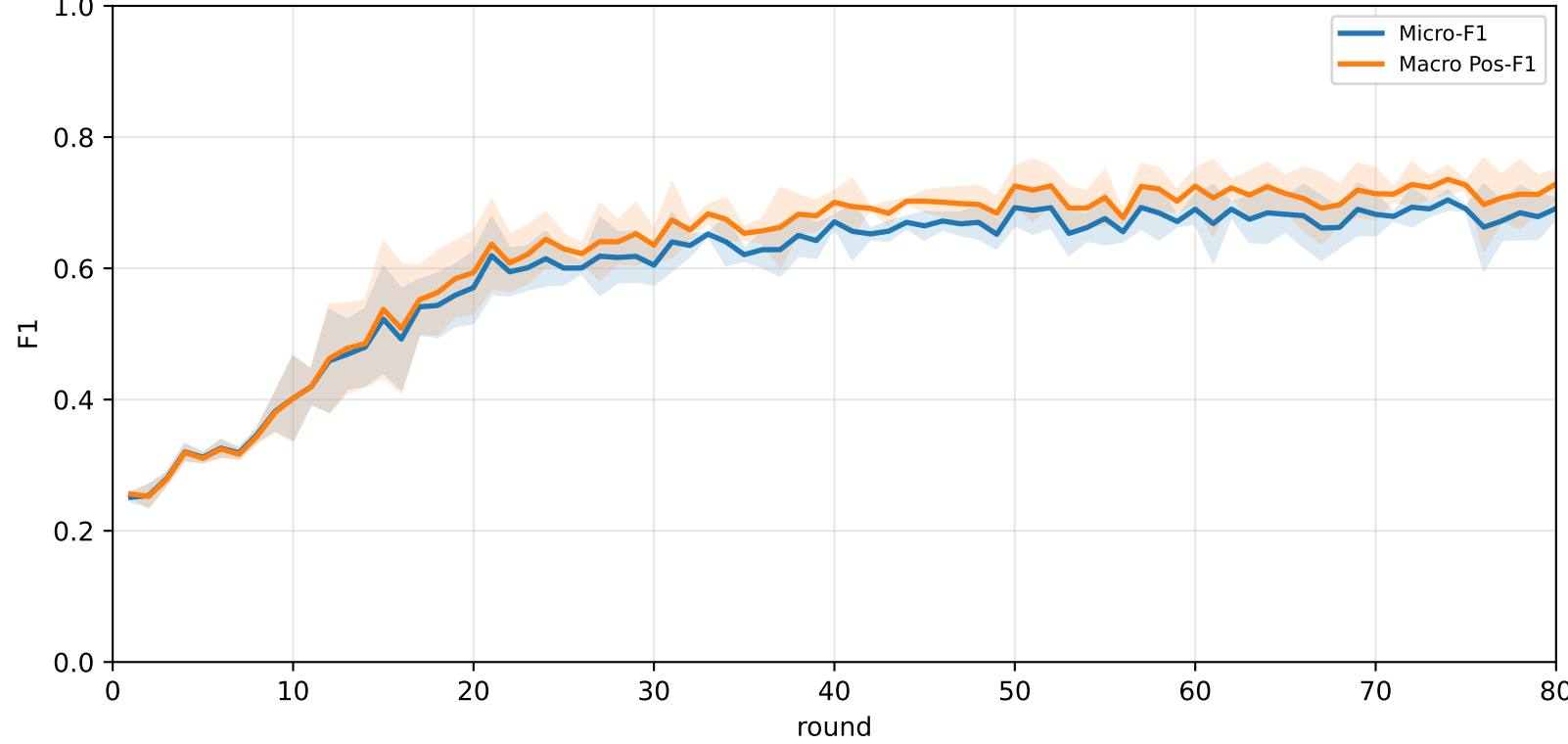
Validation loss: baselines vs selected APPLE-QOracle and end-of-round mismatch (rounds=80)



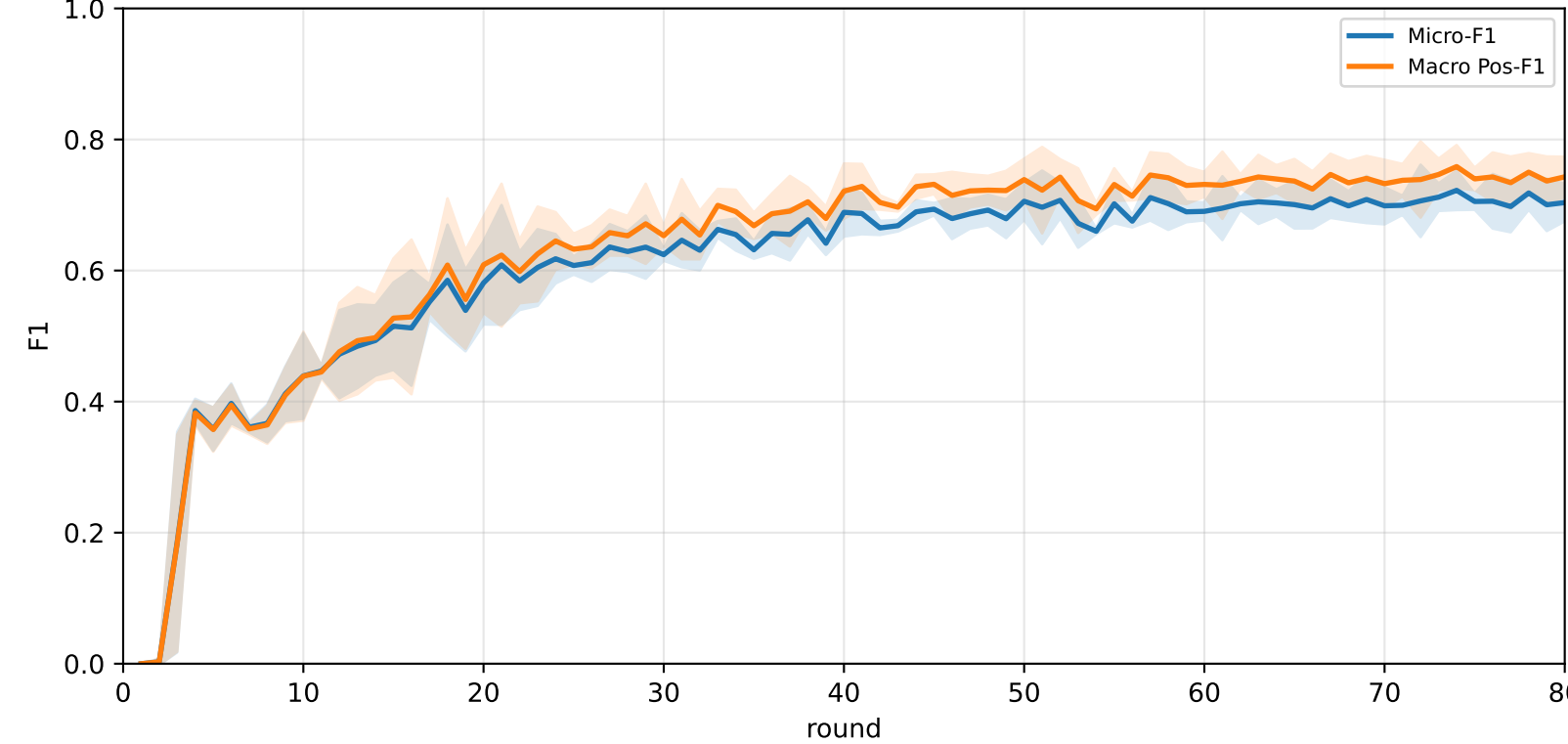
Pooled baseline micro / macro F1



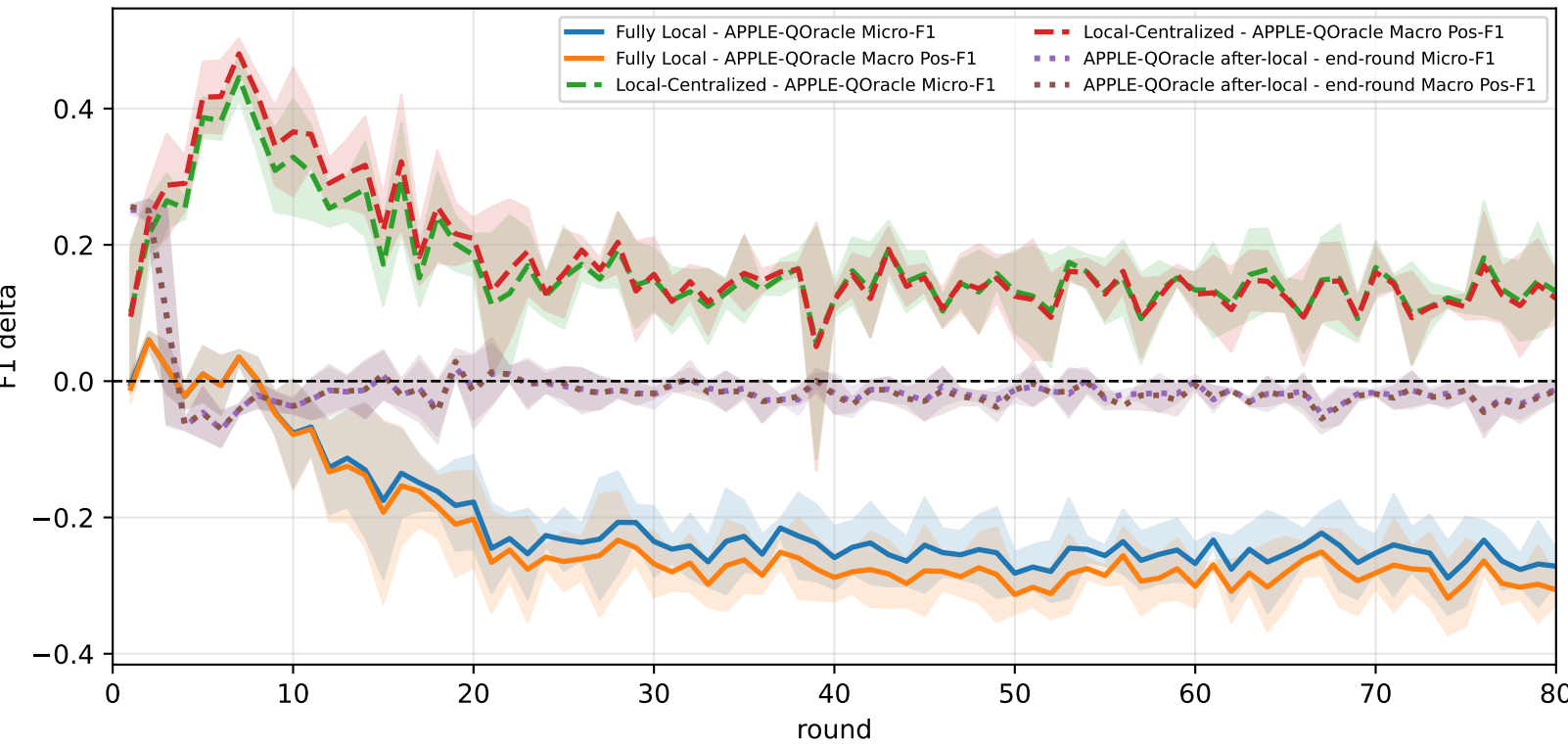
Pooled APPLE-QOracle micro / macro F1 (after-local selected model)



Pooled APPLE-QOracle micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QOracle mismatch gap

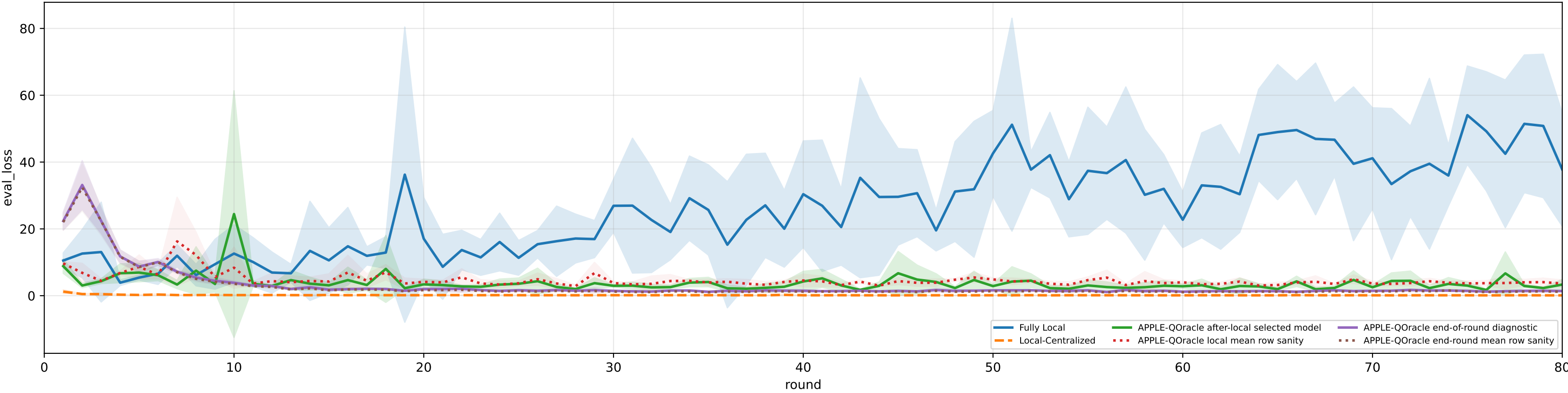


family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% | heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

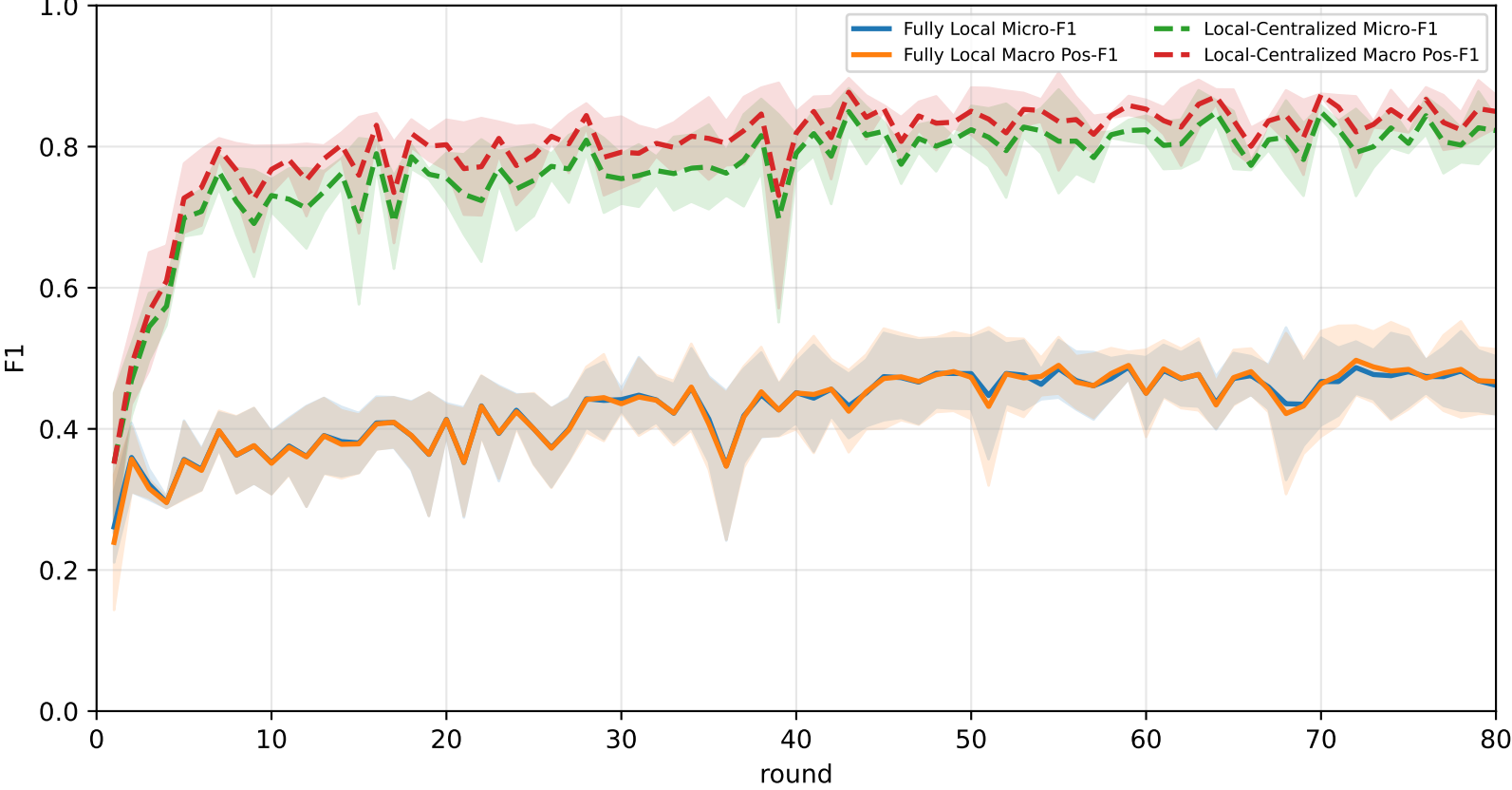
Best test performance (Fully Local vs Local-Centralized vs APPLE-QOracle, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	28.6% ± 0.7%	28.6% ± 0.7%	23.4% ± 1.7%	29.1% ± 1.2%	30.2% ± 0.8%	28.5% ± 1.4%	32.0% ± 1.3%
Local-Centralized	84.9% ± 1.3%	86.7% ± 0.8%	96.8% ± 1.6%	95.1% ± 1.4%	86.6% ± 2.5%	81.4% ± 1.3%	73.7% ± 5.8%
APPLE-QOracle	60.4% ± 13.9%	63.0% ± 15.5%	73.6% ± 24.4%	74.8% ± 18.6%	59.1% ± 12.2%	58.2% ± 14.7%	49.1% ± 7.8%

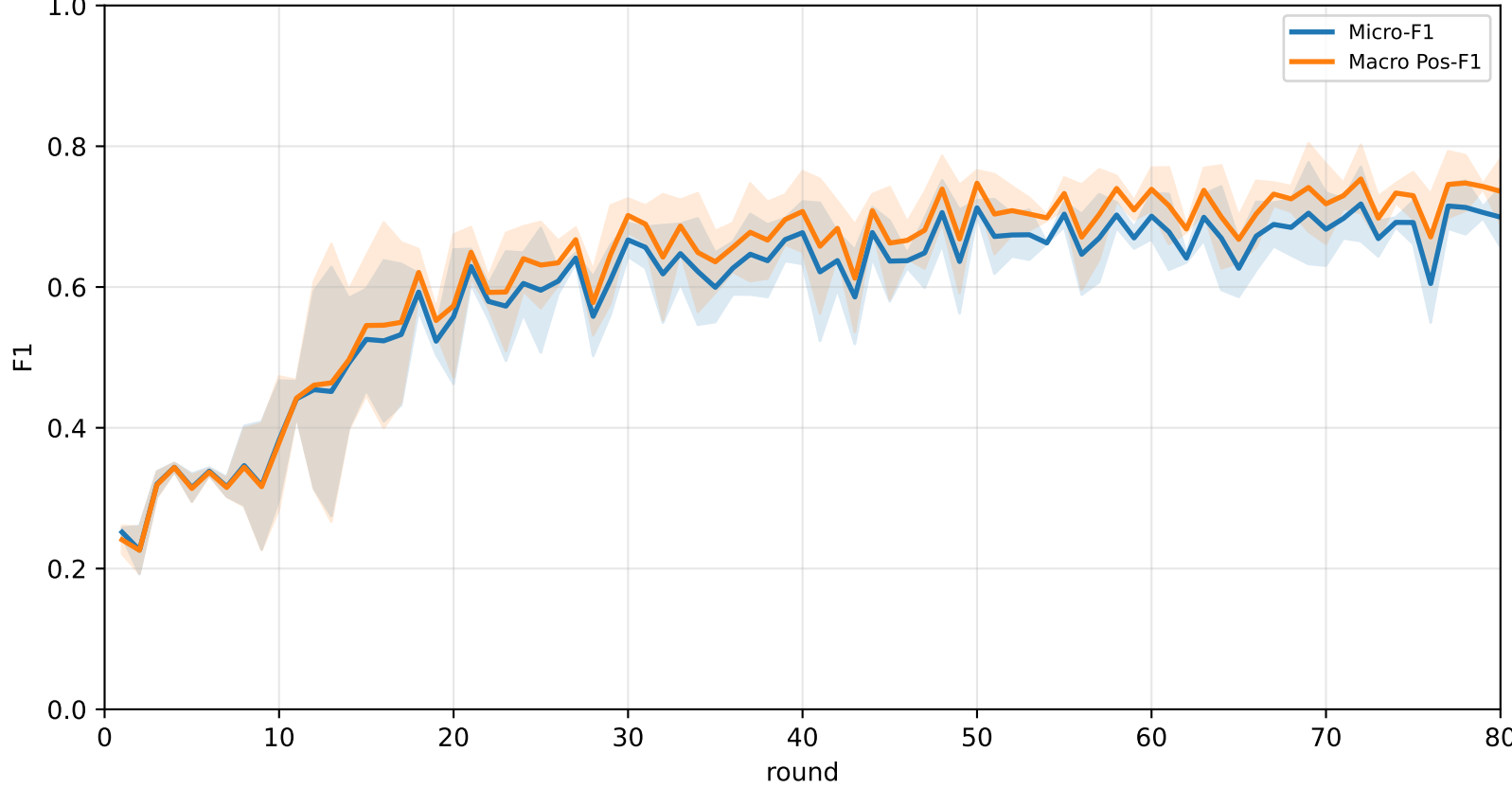
Validation loss: baselines vs selected APPLE-QOracle and end-of-round mismatch (rounds=80)



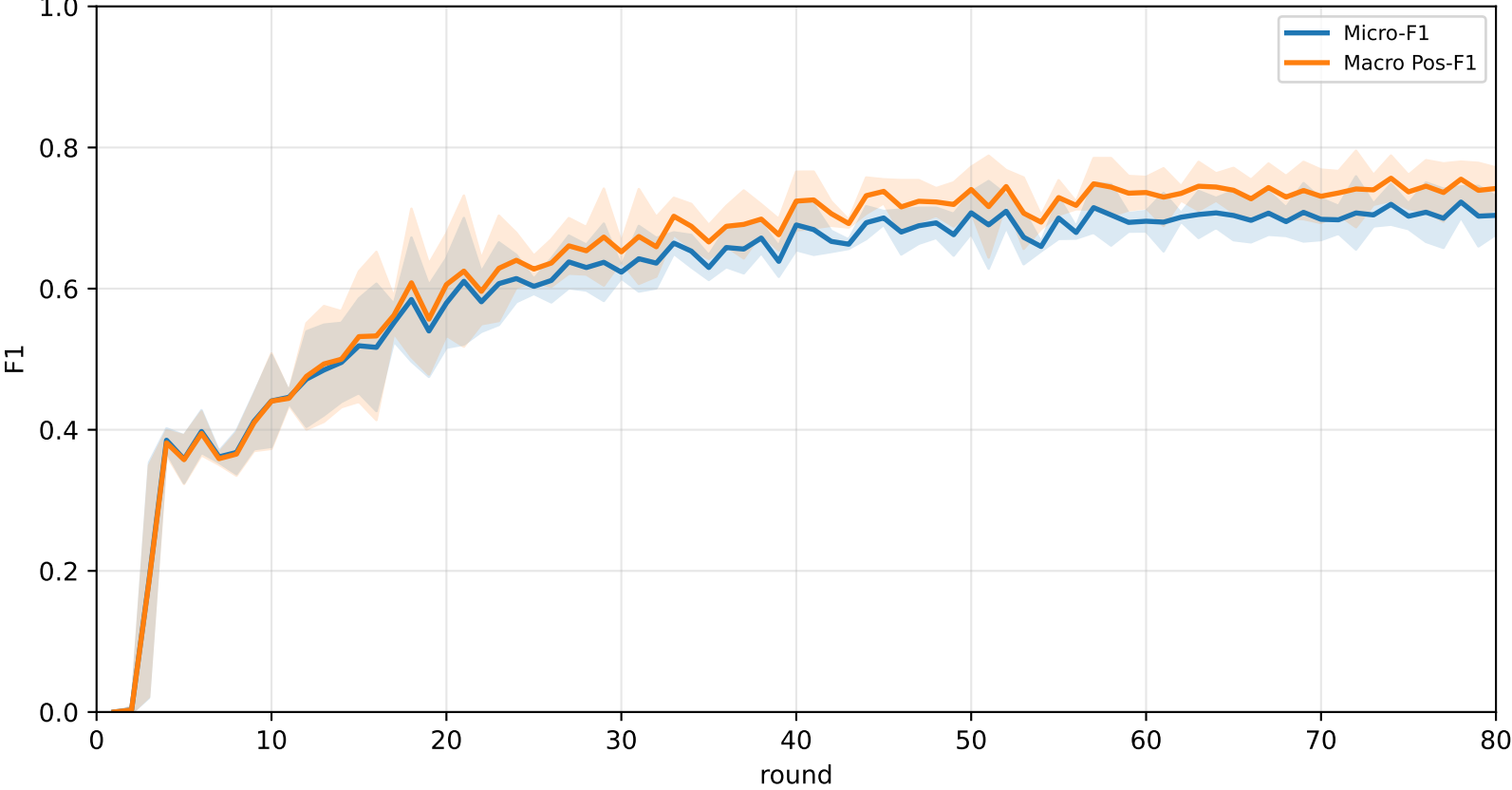
Pooled baseline micro / macro F1



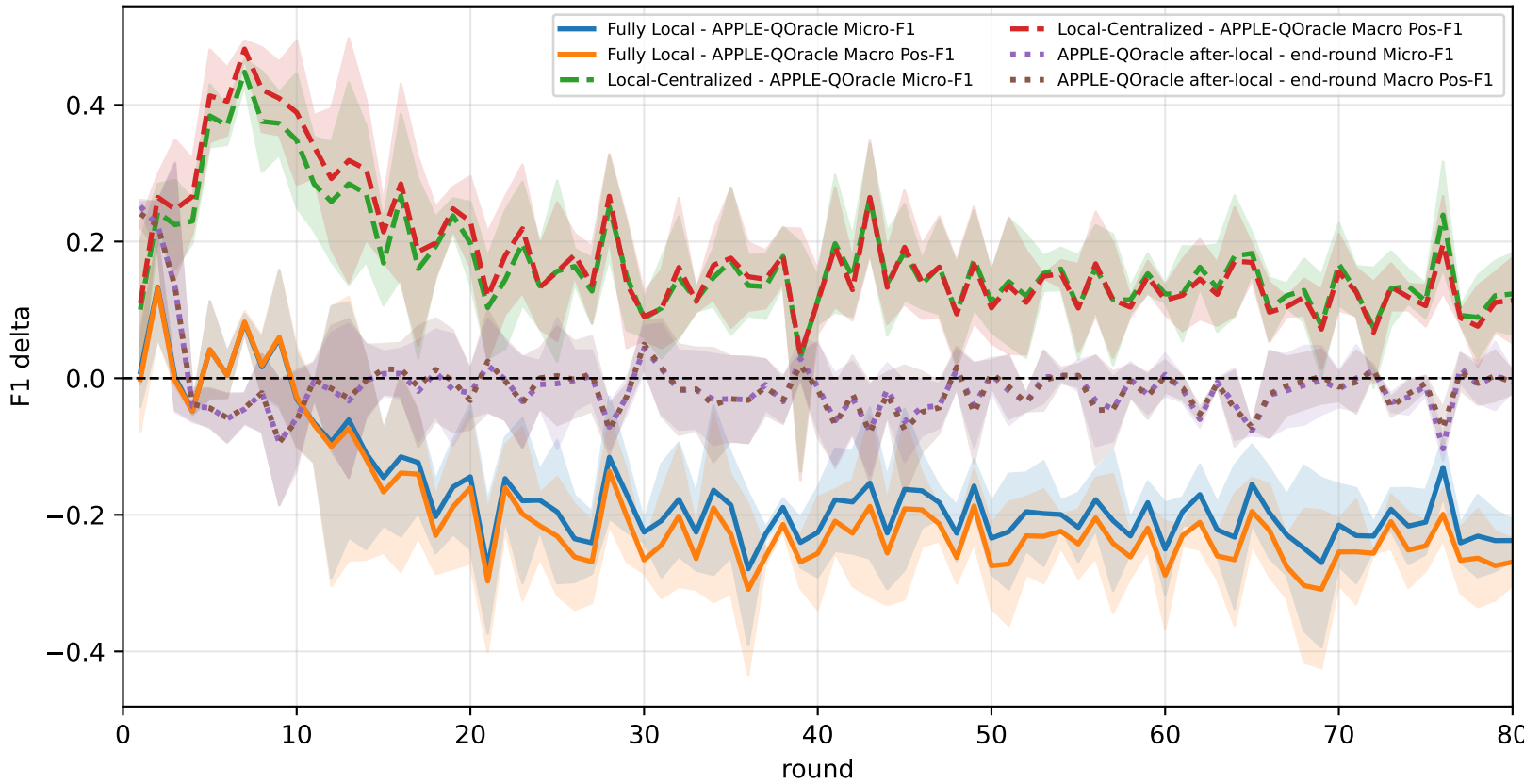
Pooled APPLE-QOracle micro / macro F1 (after-local selected model)



Pooled APPLE-QOracle micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QOracle mismatch gap

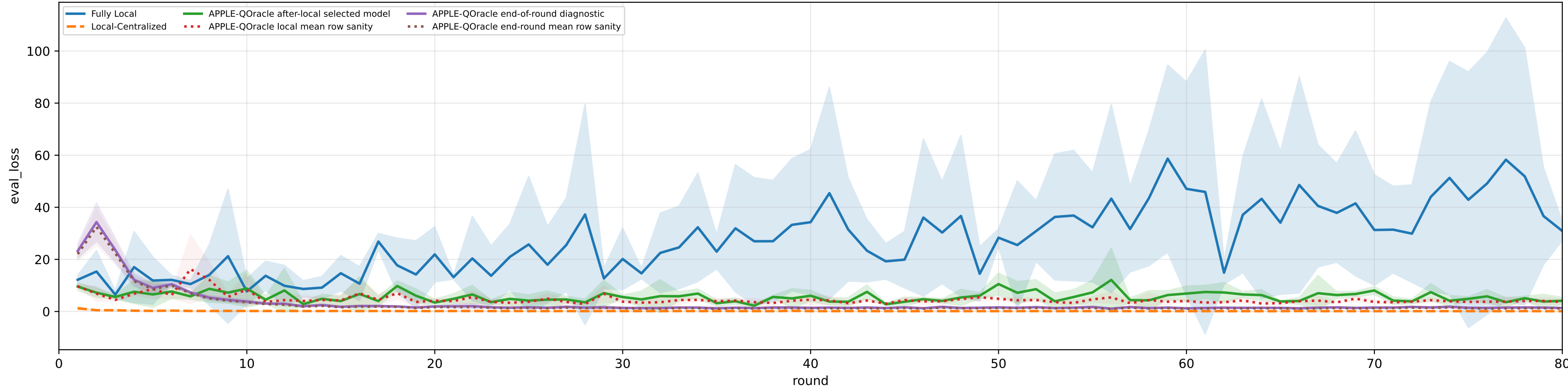


family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% | heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

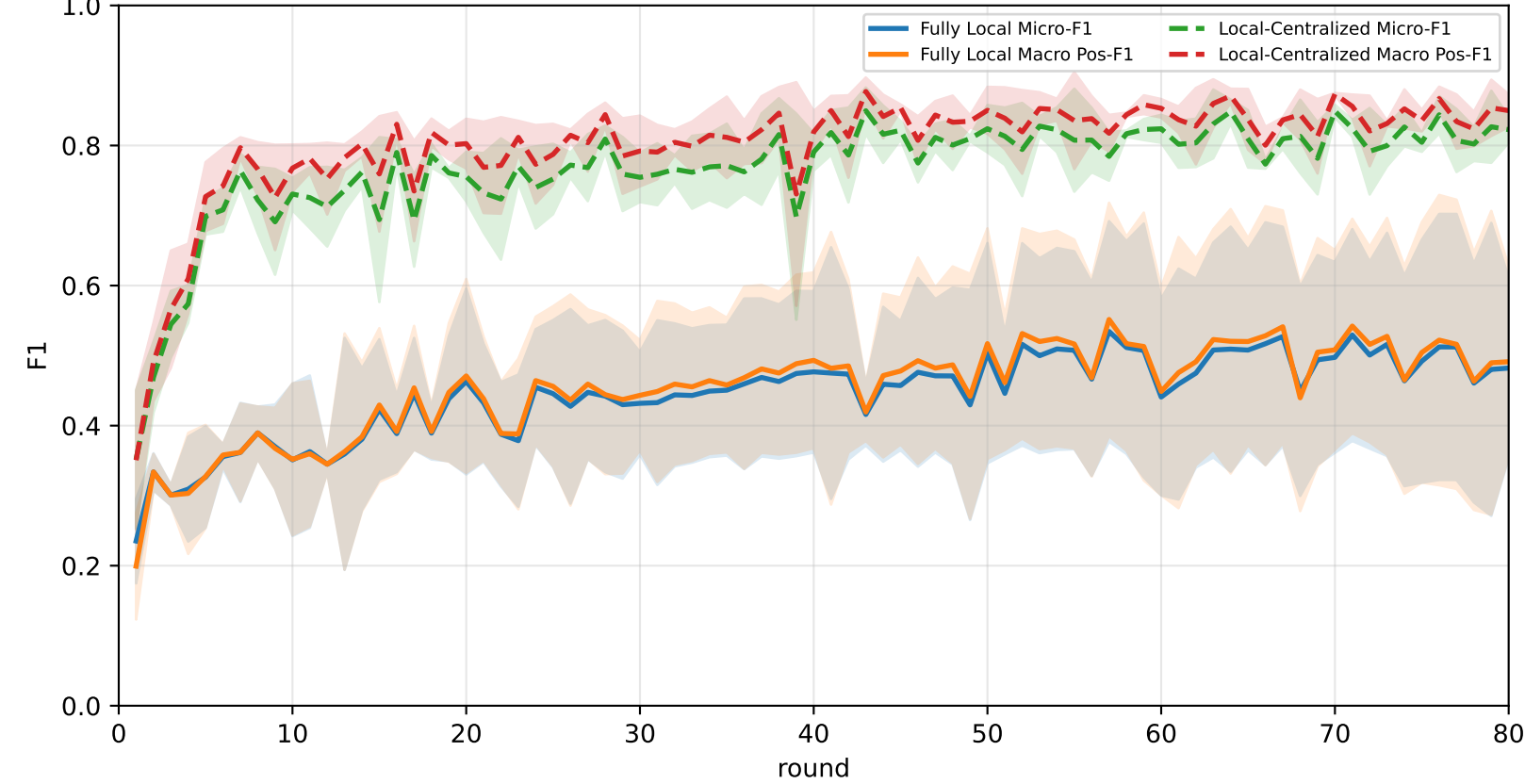
Best test performance (Fully Local vs Local-Centralized vs APPLE-QOracle, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	26.5% ± 4.7%	26.6% ± 4.7%	22.3% ± 4.2%	27.0% ± 7.0%	26.9% ± 5.3%	27.0% ± 4.1%	29.5% ± 3.2%
Local-Centralized	84.9% ± 1.3%	86.7% ± 0.8%	96.8% ± 1.6%	95.1% ± 1.4%	86.6% ± 2.5%	81.4% ± 1.3%	73.7% ± 5.8%
APPLE-QOracle	57.4% ± 8.5%	59.7% ± 10.0%	71.5% ± 21.5%	64.9% ± 13.4%	56.4% ± 6.6%	58.3% ± 6.6%	47.4% ± 4.1%

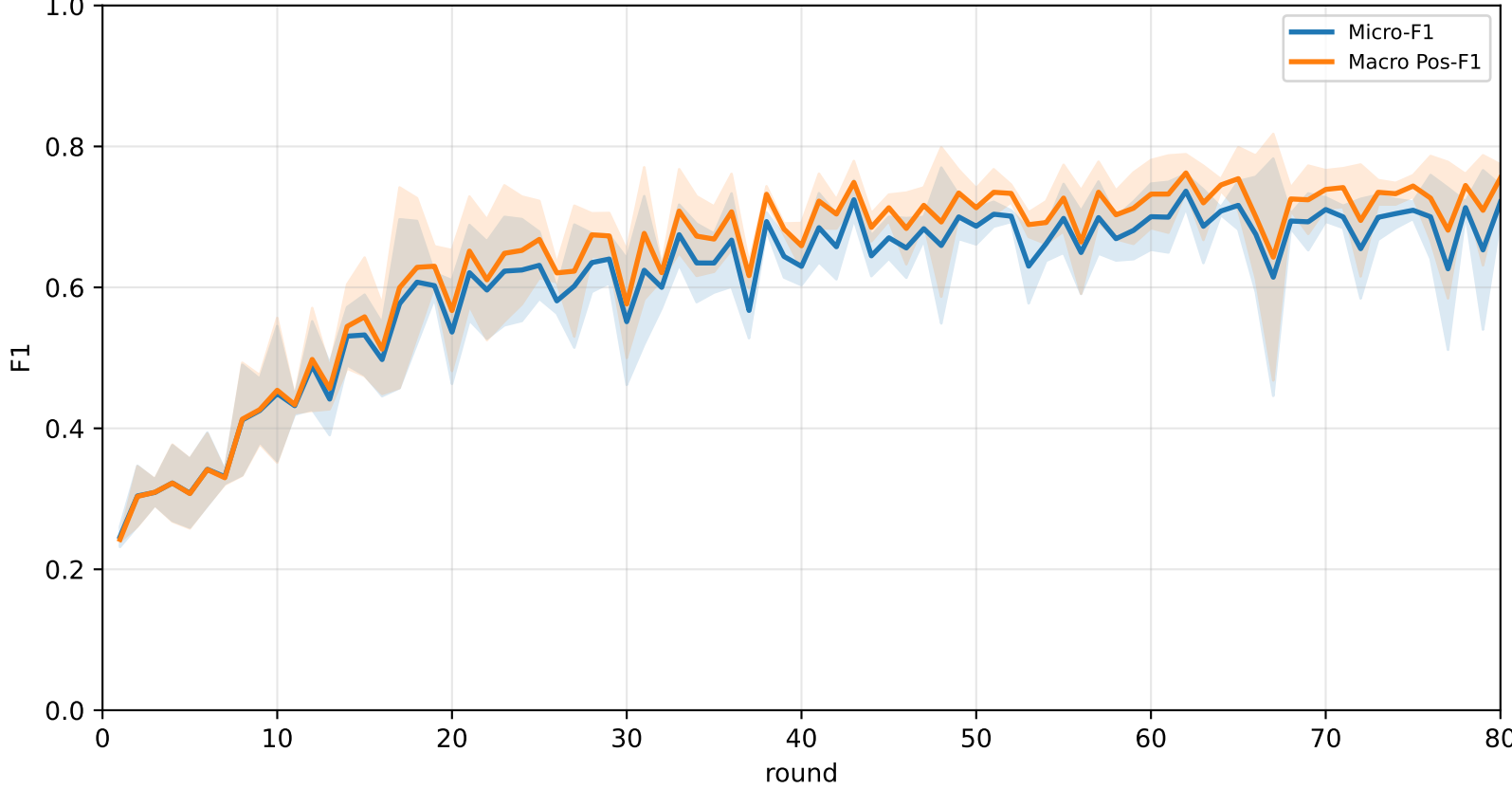
Validation loss: baselines vs selected APPLE-QOracle and end-of-round mismatch (rounds=80)



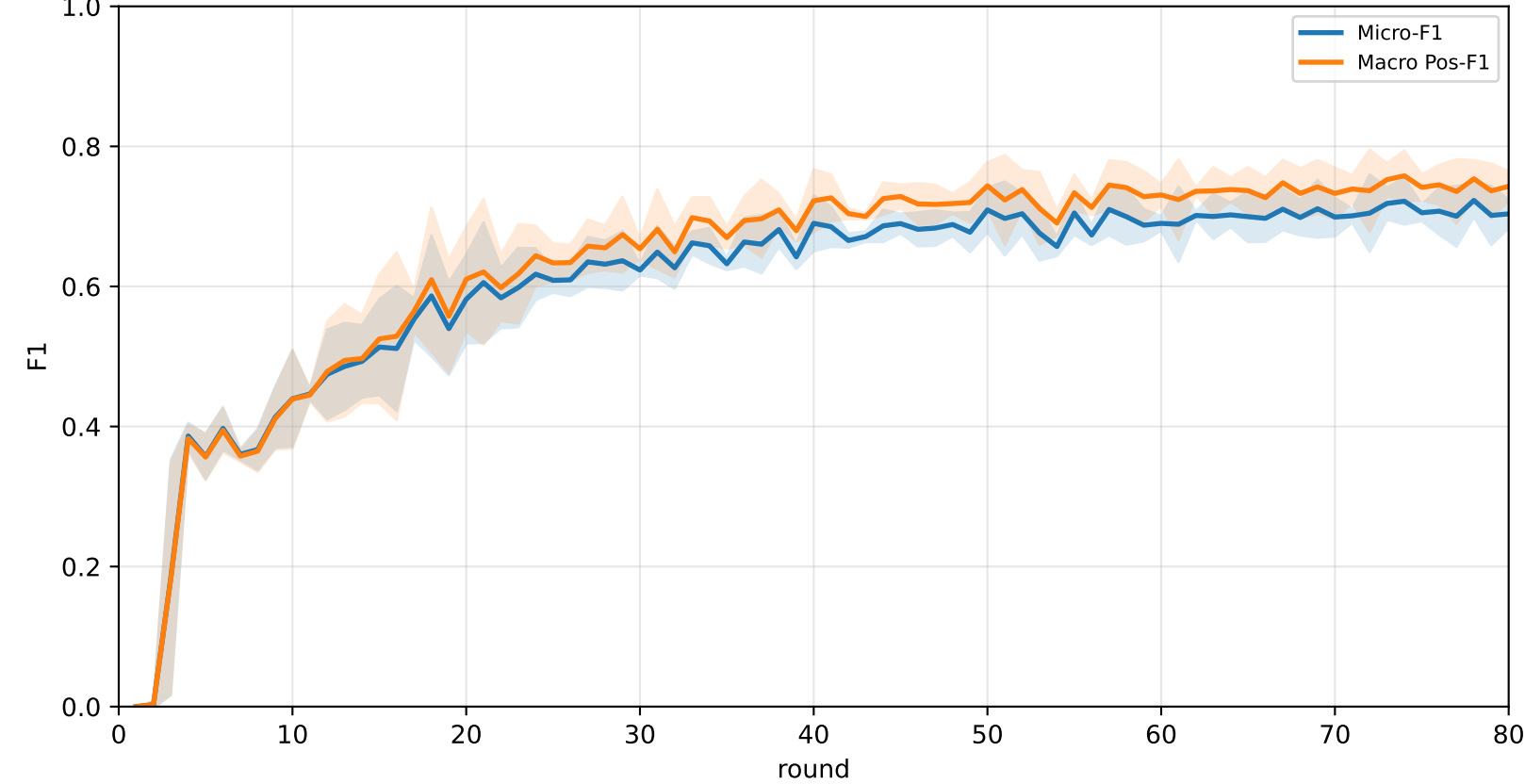
Pooled baseline micro / macro F1



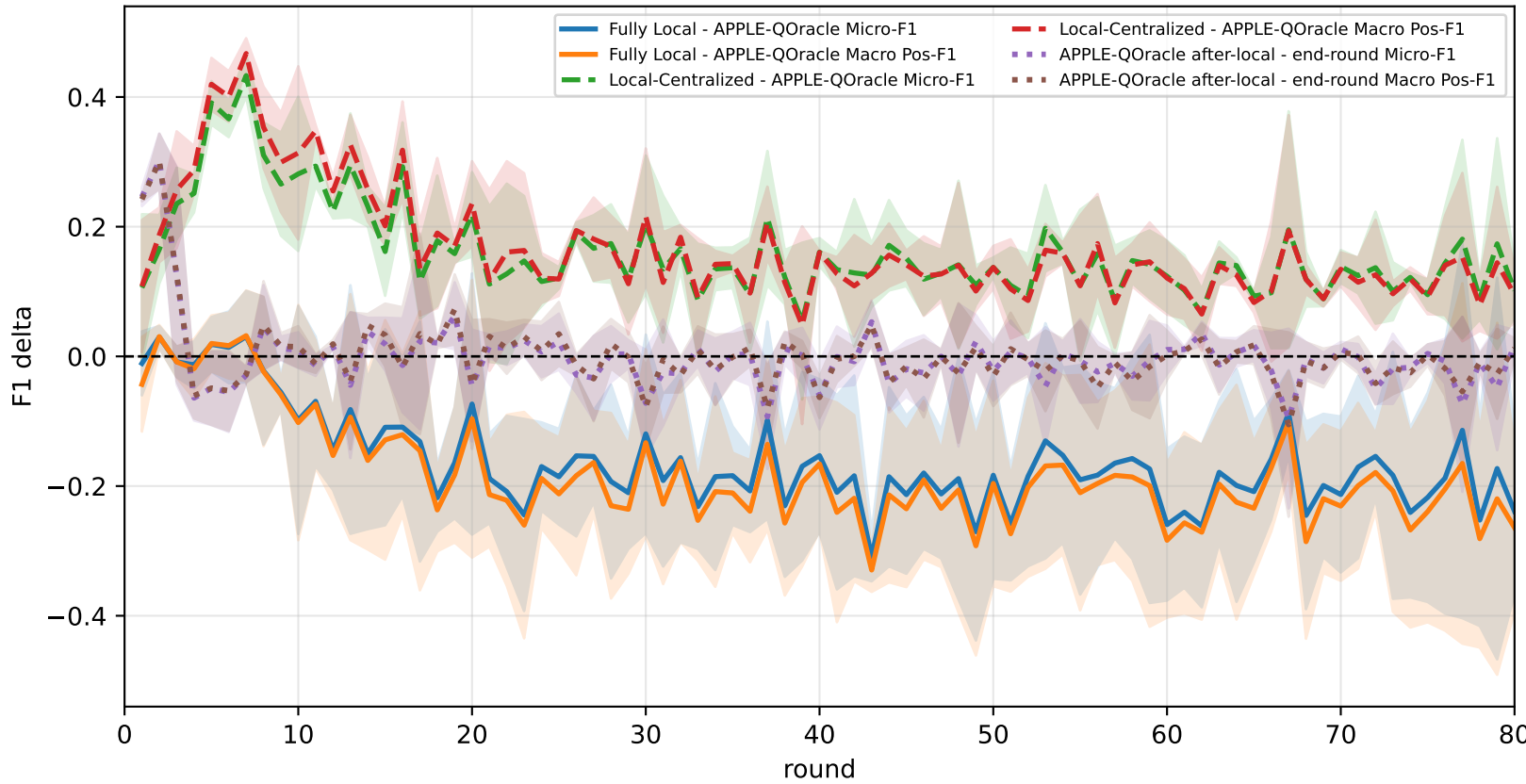
Pooled APPLE-QOracle micro / macro F1 (after-local selected model)



Pooled APPLE-QOracle micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QOracle mismatch gap

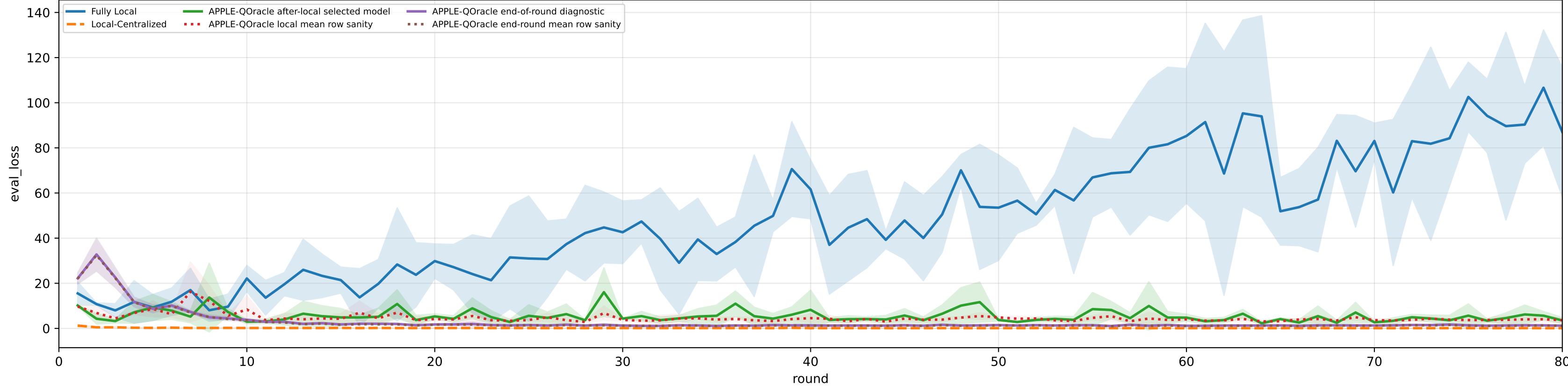


family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% | heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

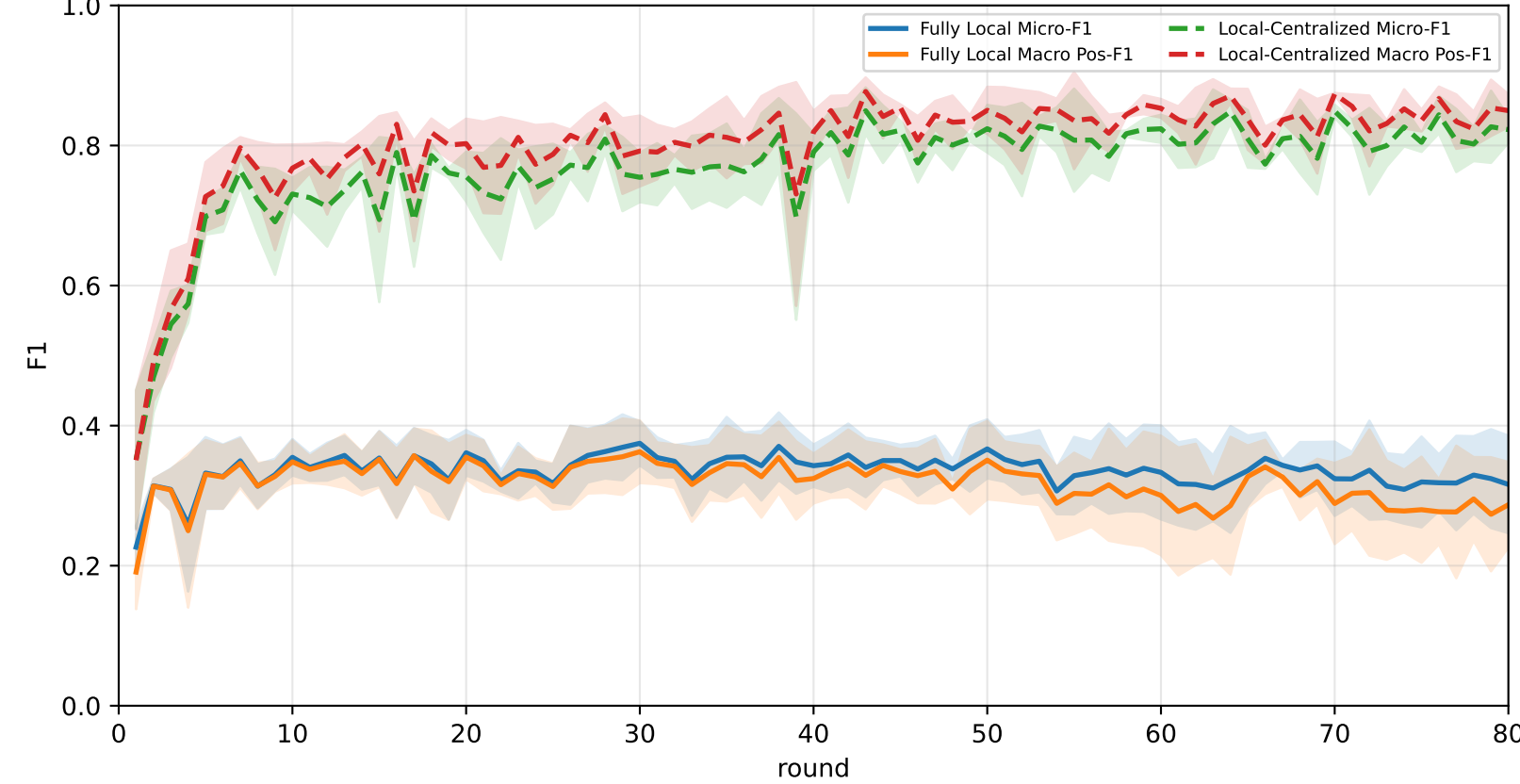
Best test performance (Fully Local vs Local-Centralized vs APPLE-QOracle, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	27.5% ± 1.5%	27.5% ± 1.4%	23.1% ± 1.8%	28.2% ± 0.7%	26.4% ± 1.8%	28.3% ± 1.7%	31.5% ± 2.1%
Local-Centralized	84.9% ± 1.3%	86.7% ± 0.8%	96.8% ± 1.6%	95.1% ± 1.4%	86.6% ± 2.5%	81.4% ± 1.3%	73.7% ± 5.8%
APPLE-QOracle	59.2% ± 10.6%	61.6% ± 11.9%	71.3% ± 20.9%	71.3% ± 15.1%	60.8% ± 9.3%	56.5% ± 12.1%	48.3% ± 5.2%

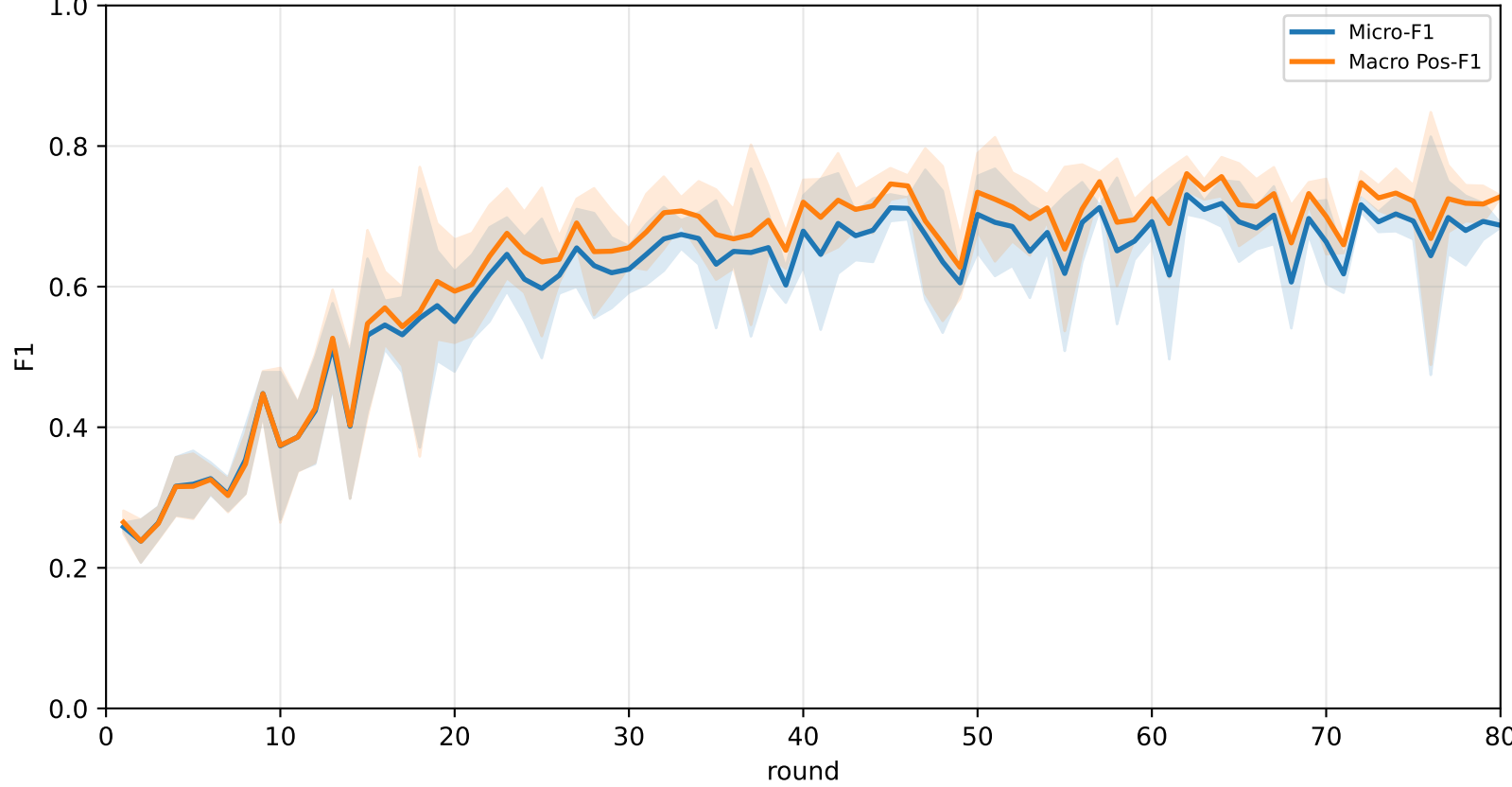
Validation loss: baselines vs selected APPLE-QOracle and end-of-round mismatch (rounds=80)



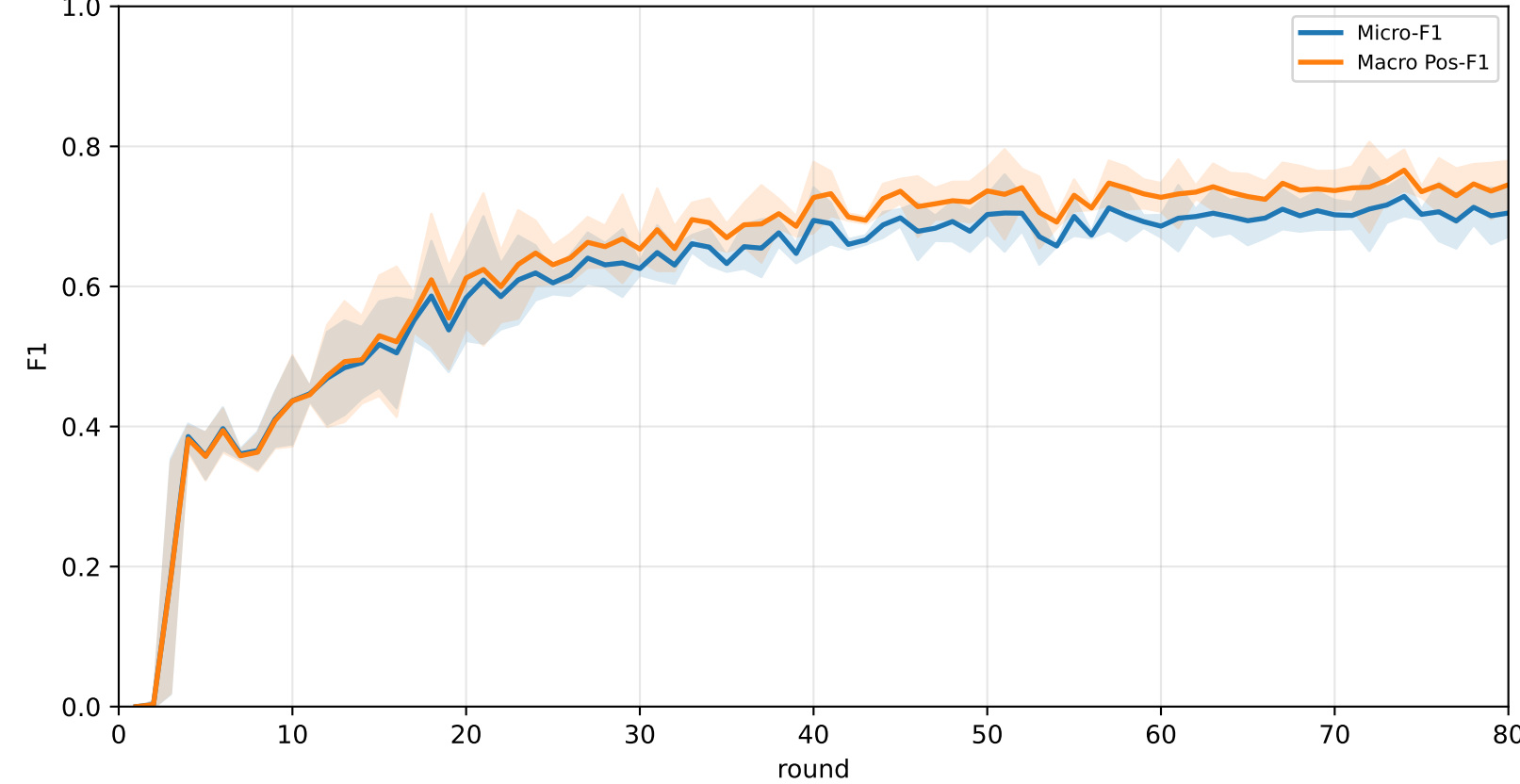
Pooled baseline micro / macro F1



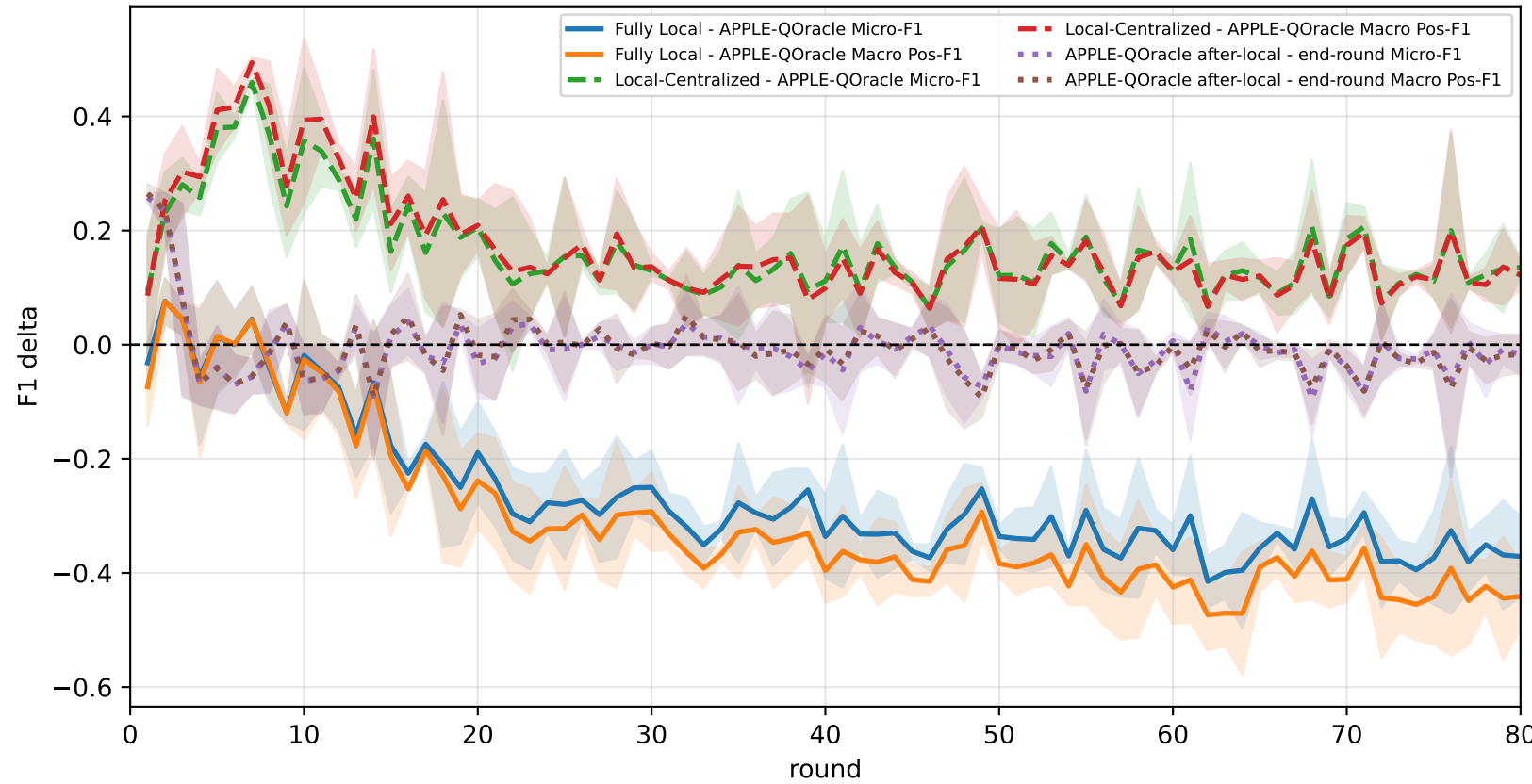
Pooled APPLE-QOracle micro / macro F1 (after-local selected model)



Pooled APPLE-QOracle micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QOracle mismatch gap

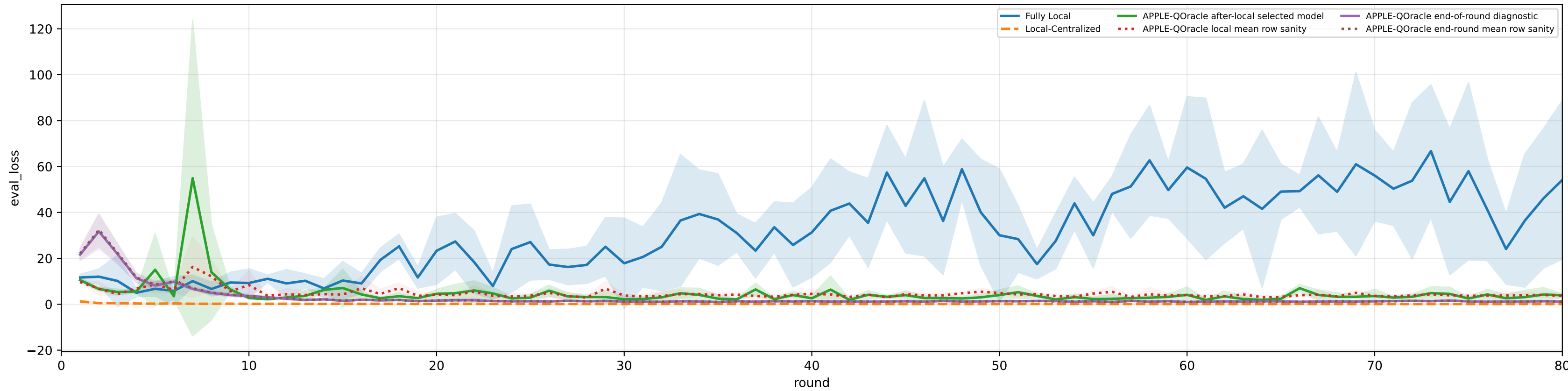


family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

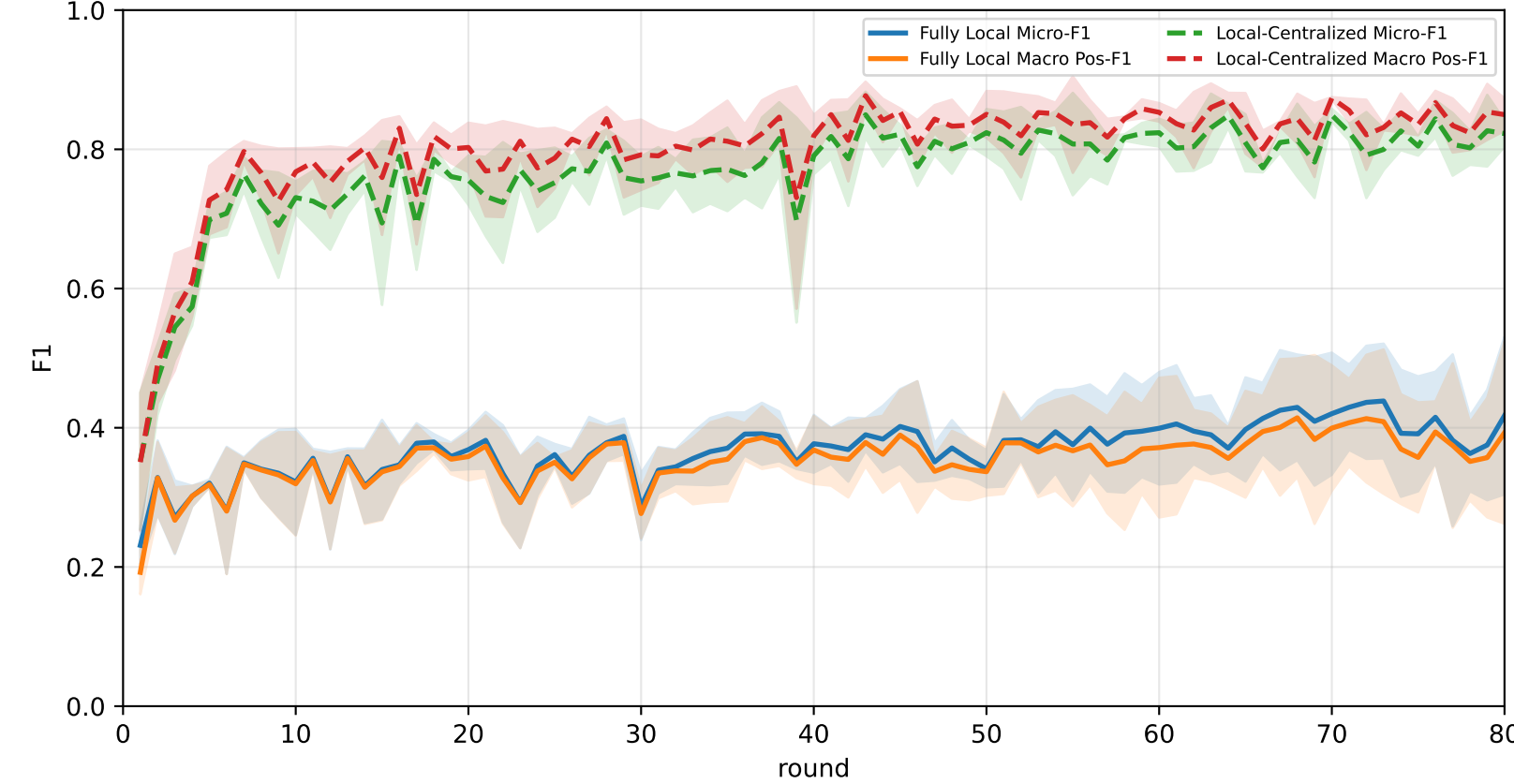
Best test performance (Fully Local vs Local-Centralized vs APPLE-QOracle, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	24.2% ± 1.1%	24.2% ± 1.1%	20.1% ± 0.7%	25.3% ± 1.1%	24.5% ± 1.6%	22.9% ± 1.3%	28.3% ± 1.1%
Local-Centralized	84.9% ± 1.3%	86.7% ± 0.8%	96.8% ± 1.6%	95.1% ± 1.4%	86.6% ± 2.5%	81.4% ± 1.3%	73.7% ± 5.8%
APPLE-QOracle	56.1% ± 12.0%	57.7% ± 12.4%	64.3% ± 18.5%	65.1% ± 11.9%	58.3% ± 10.5%	53.5% ± 12.2%	47.1% ± 10.5%

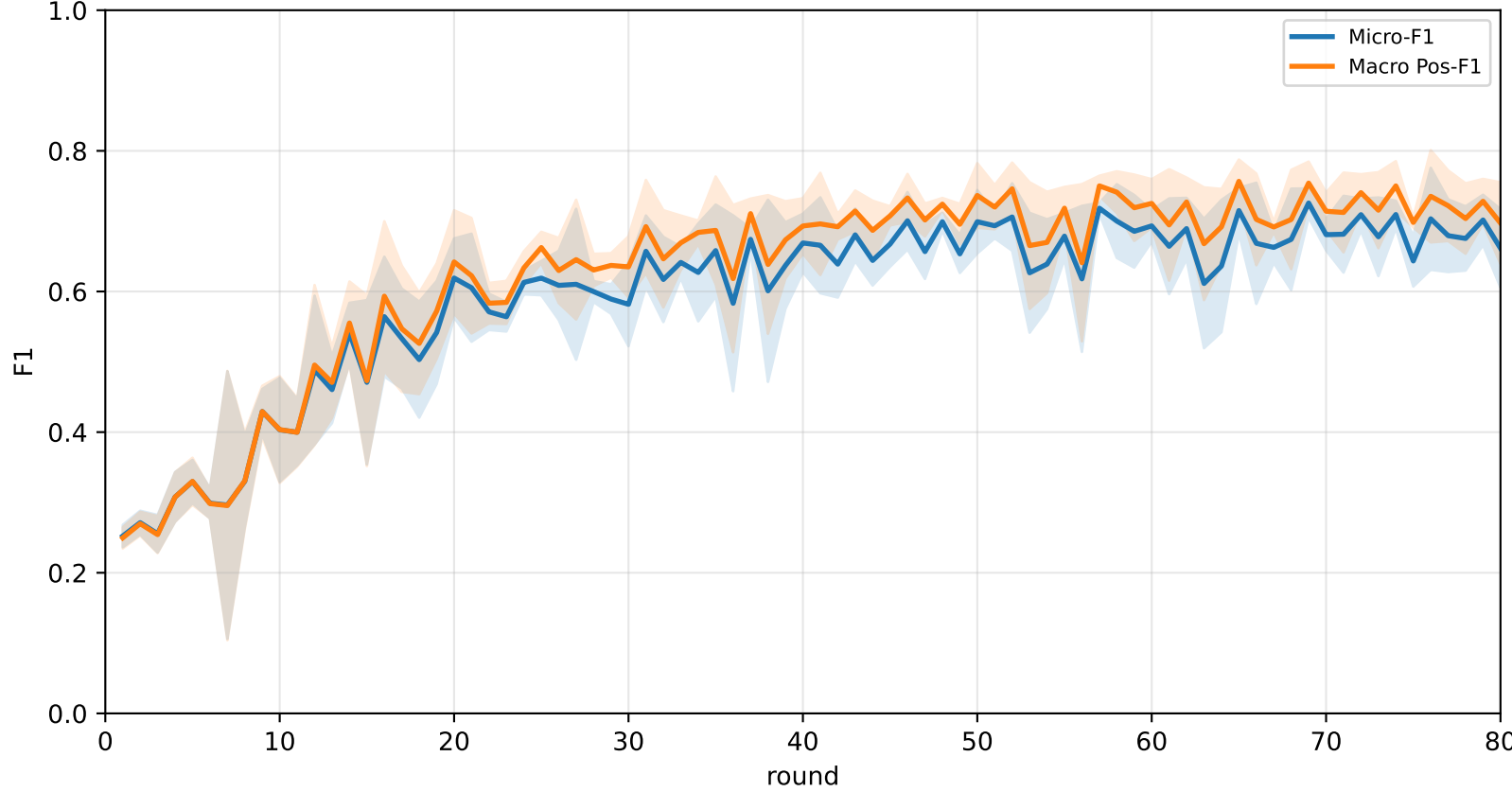
Validation loss: baselines vs selected APPLE-QOracle and end-of-round mismatch (rounds=80)



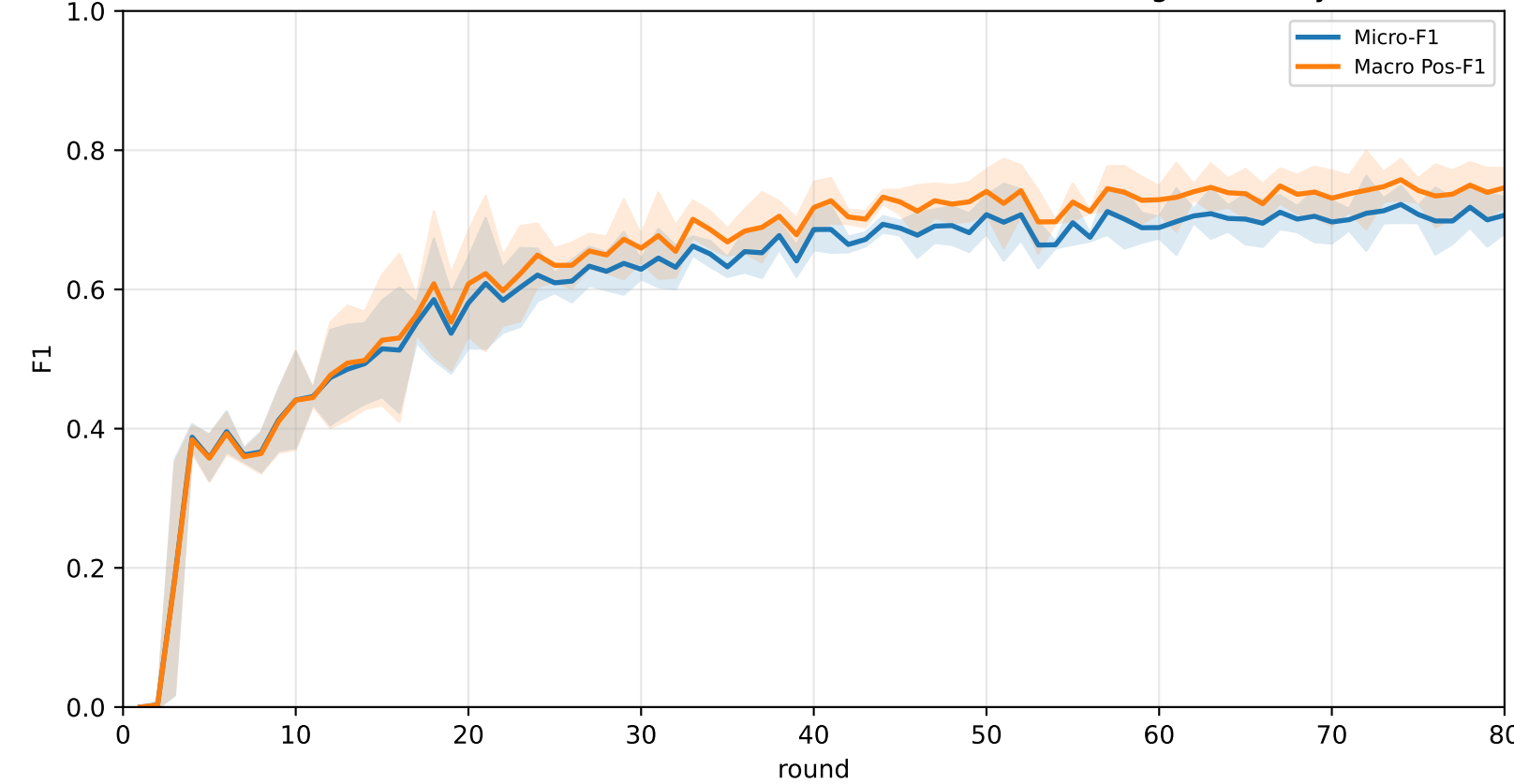
Pooled baseline micro / macro F1



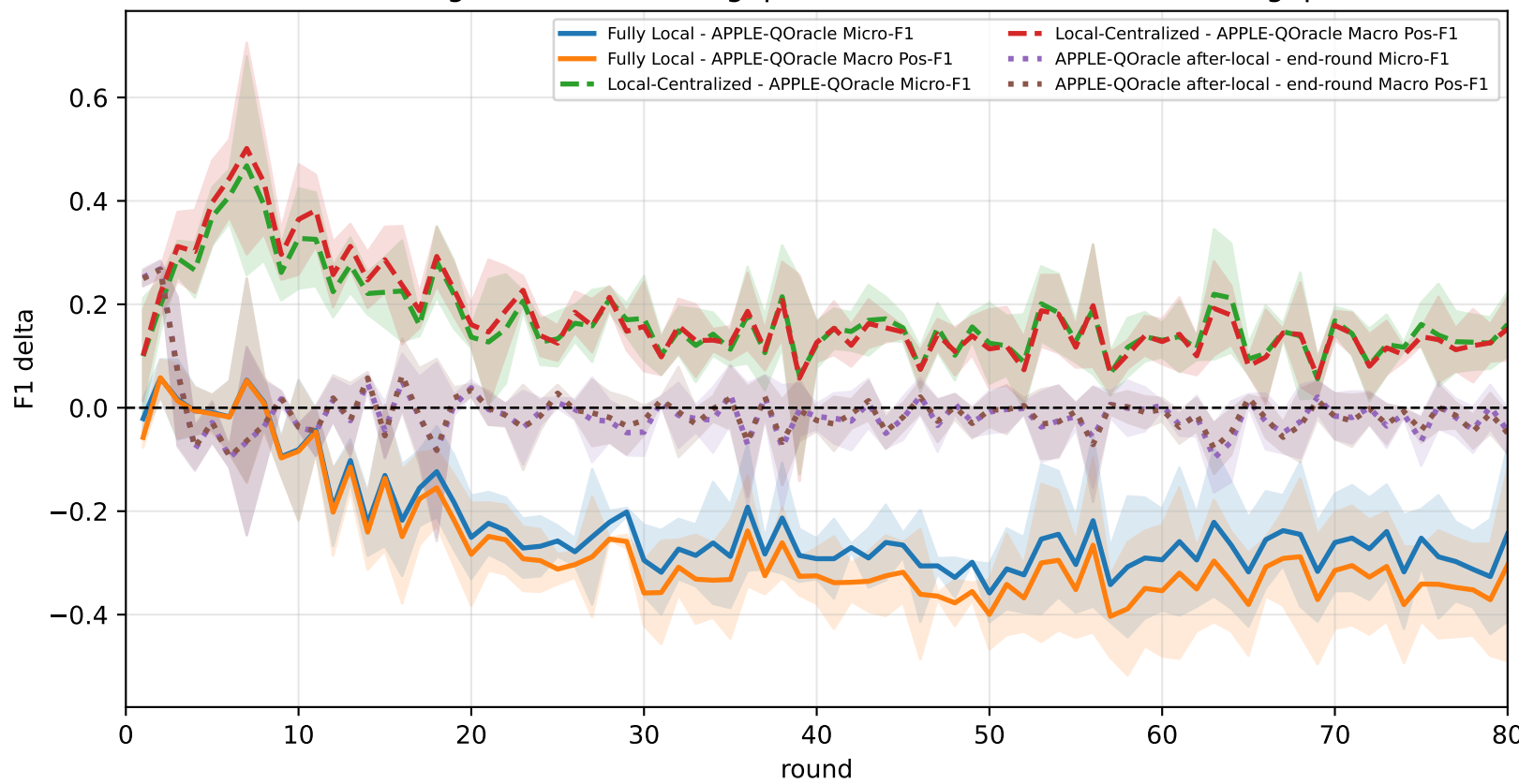
Pooled APPLE-QOracle micro / macro F1 (after-local selected model)



Pooled APPLE-QOracle micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QOracle mismatch gap

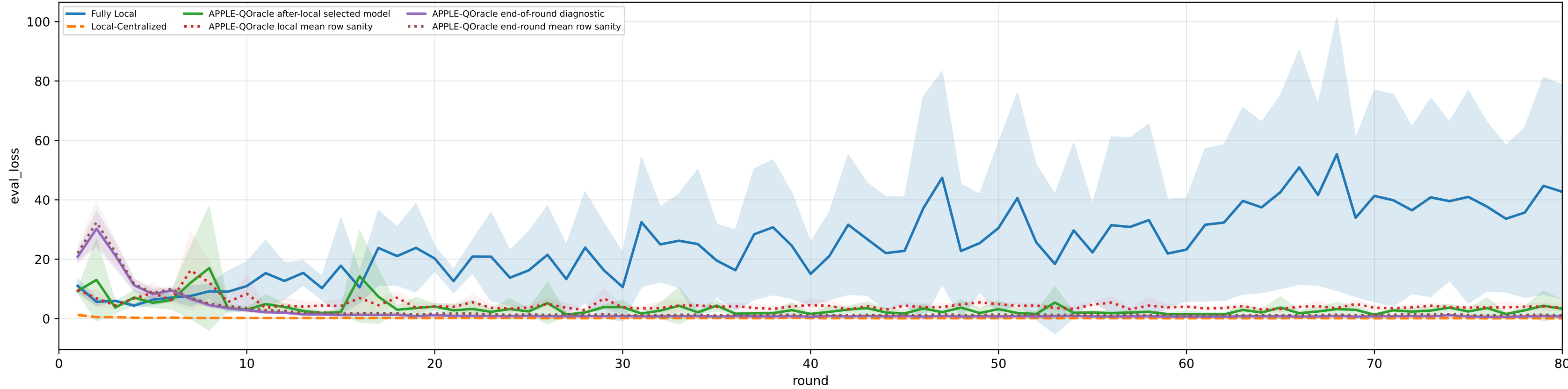


family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3% | heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

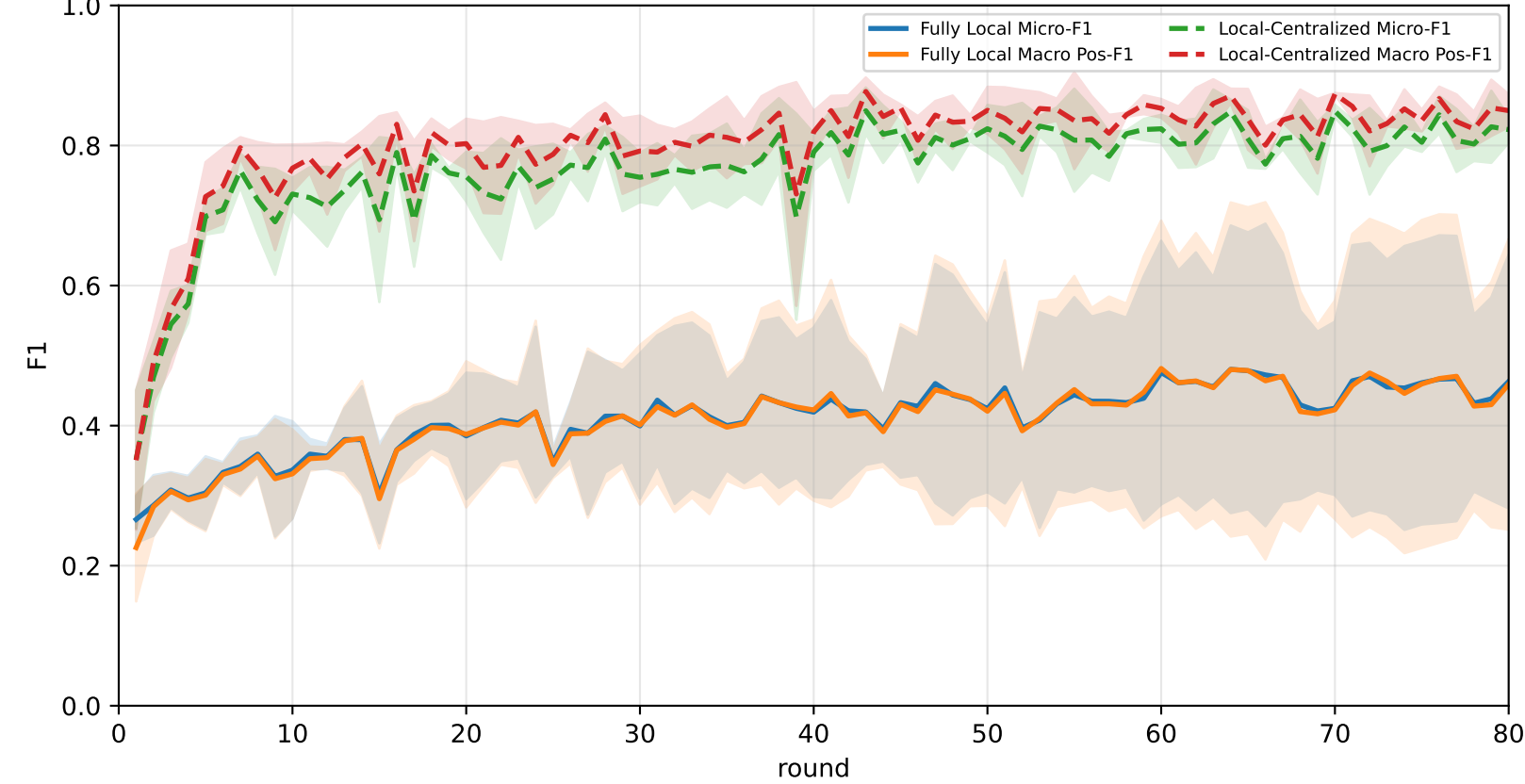
Best test performance (Fully Local vs Local-Centralized vs APPLE-QOracle, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	31.2% ± 8.8%	31.1% ± 8.9%	26.7% ± 8.1%	32.1% ± 9.5%	33.6% ± 11.2%	31.3% ± 9.2%	32.1% ± 6.8%
Local-Centralized	84.9% ± 1.3%	86.7% ± 0.8%	96.8% ± 1.6%	95.1% ± 1.4%	86.6% ± 2.5%	81.4% ± 1.3%	73.7% ± 5.8%
APPLE-QOracle	59.2% ± 10.7%	61.0% ± 12.0%	64.9% ± 18.7%	68.4% ± 15.4%	60.7% ± 10.9%	61.6% ± 11.3%	49.2% ± 5.4%

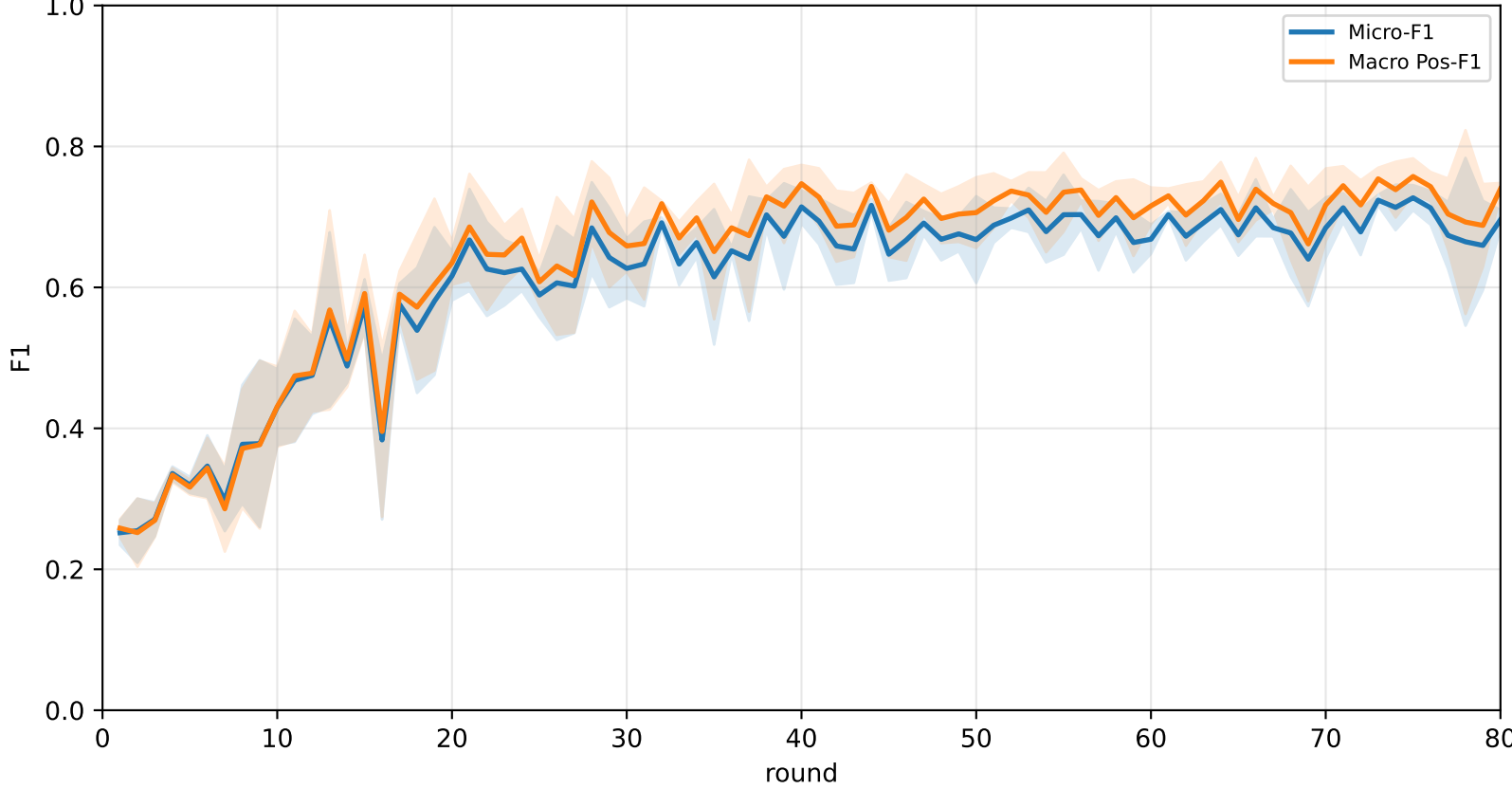
Validation loss: baselines vs selected APPLE-QOracle and end-of-round mismatch (rounds=80)



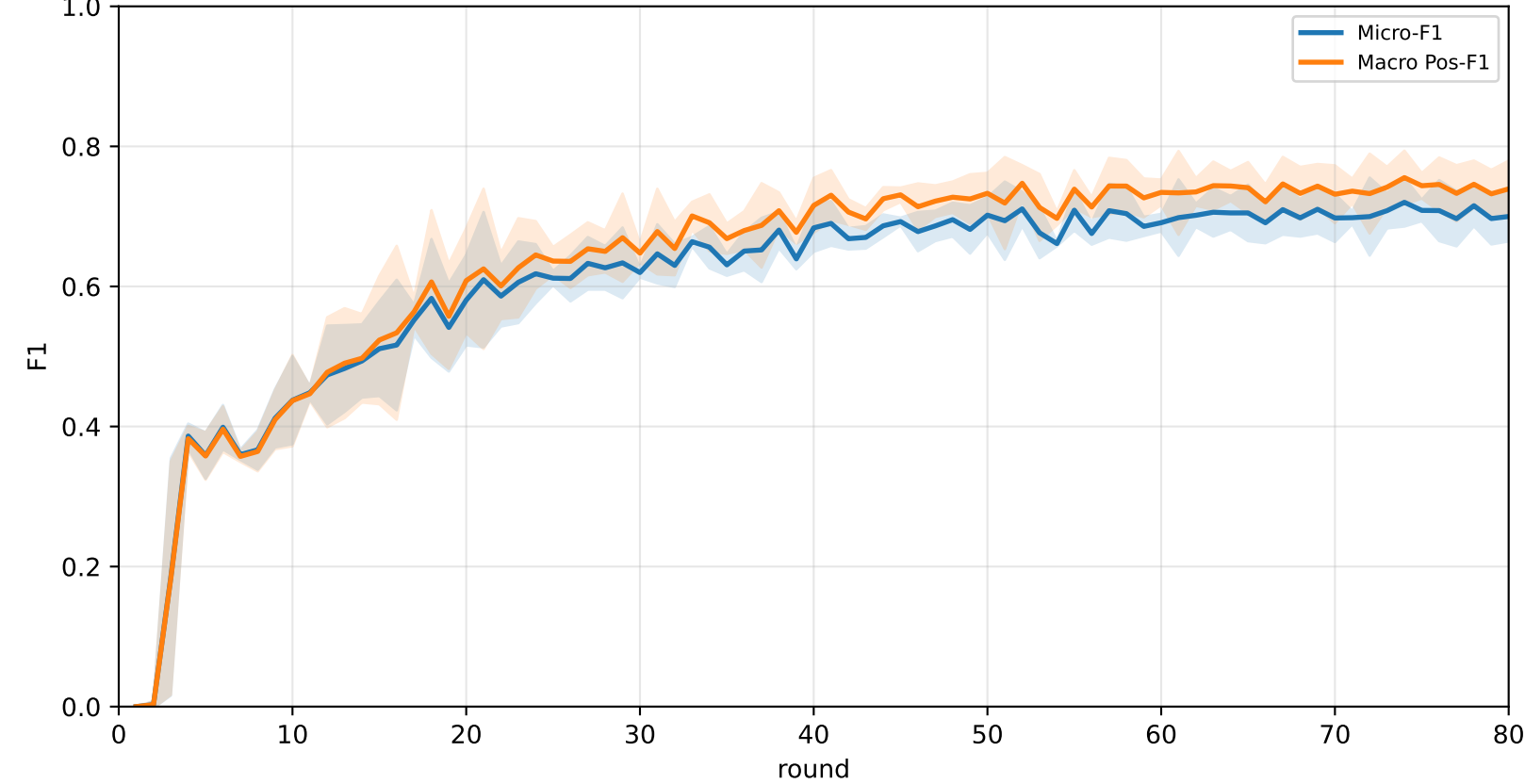
Pooled baseline micro / macro F1



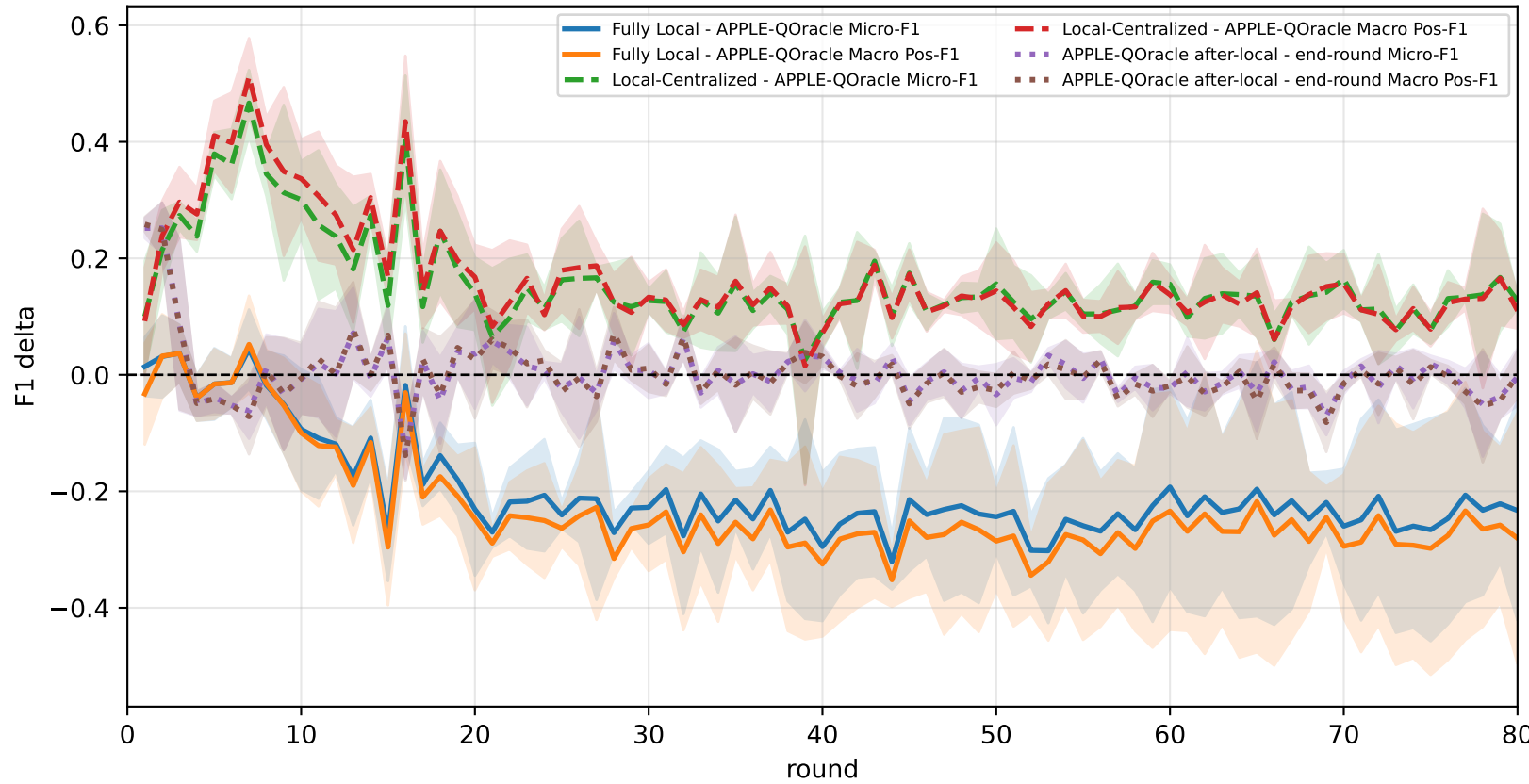
Pooled APPLE-QOracle micro / macro F1 (after-local selected model)



Pooled APPLE-QOracle micro / macro F1 (end-of-round diagnostic only)



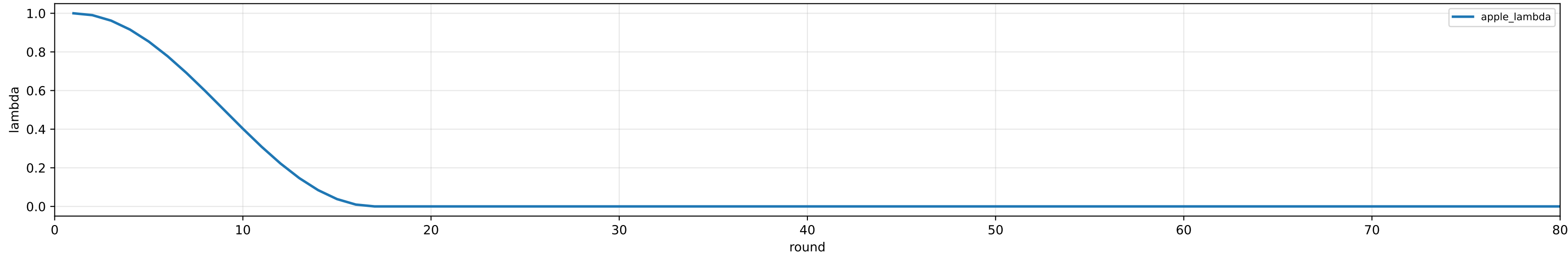
Delta diagnostics: baseline gaps and APPLE-QOracle mismatch gap



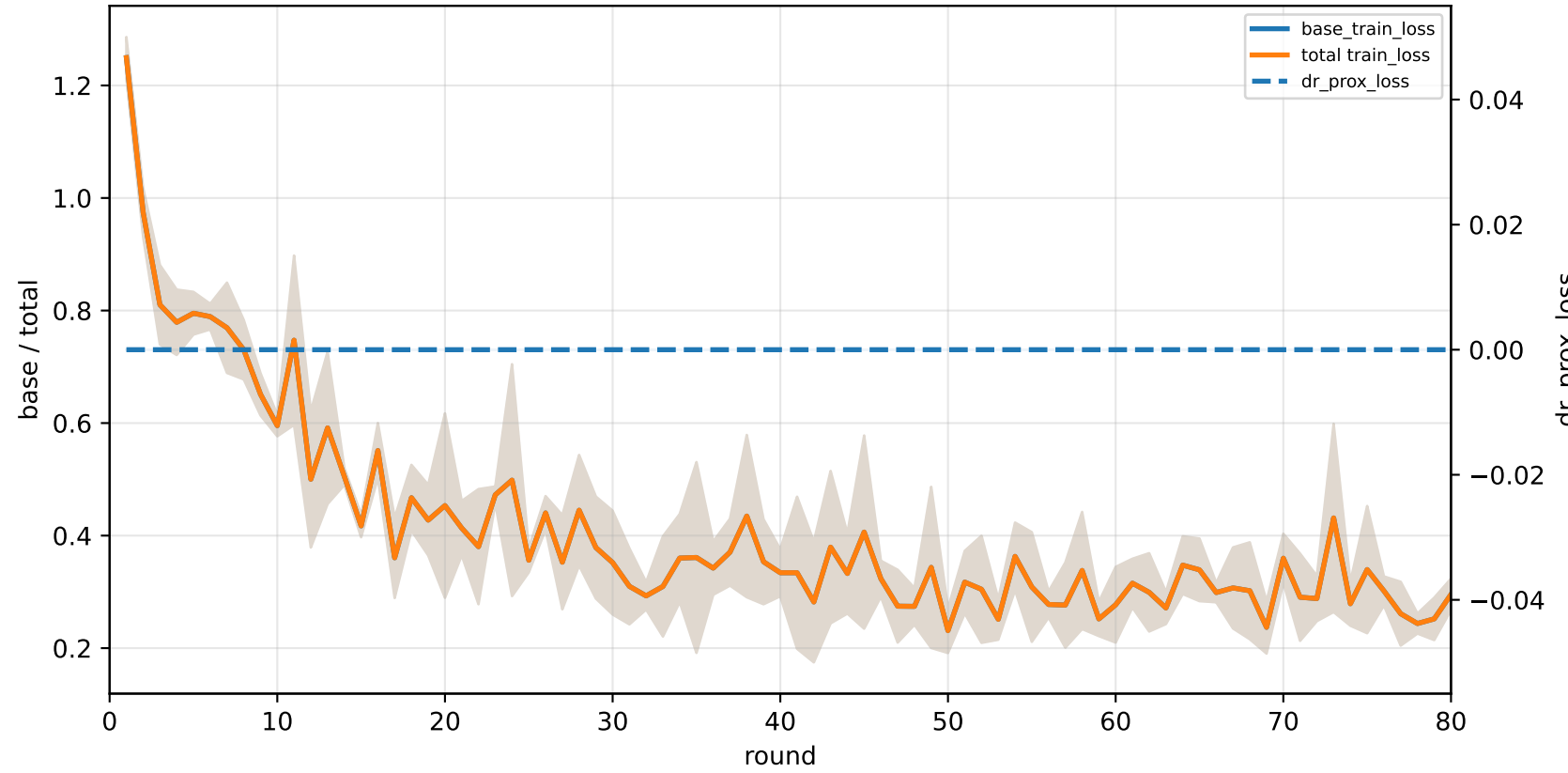
APPLE-QOracle training-loss decomposition | High heterogeneity

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

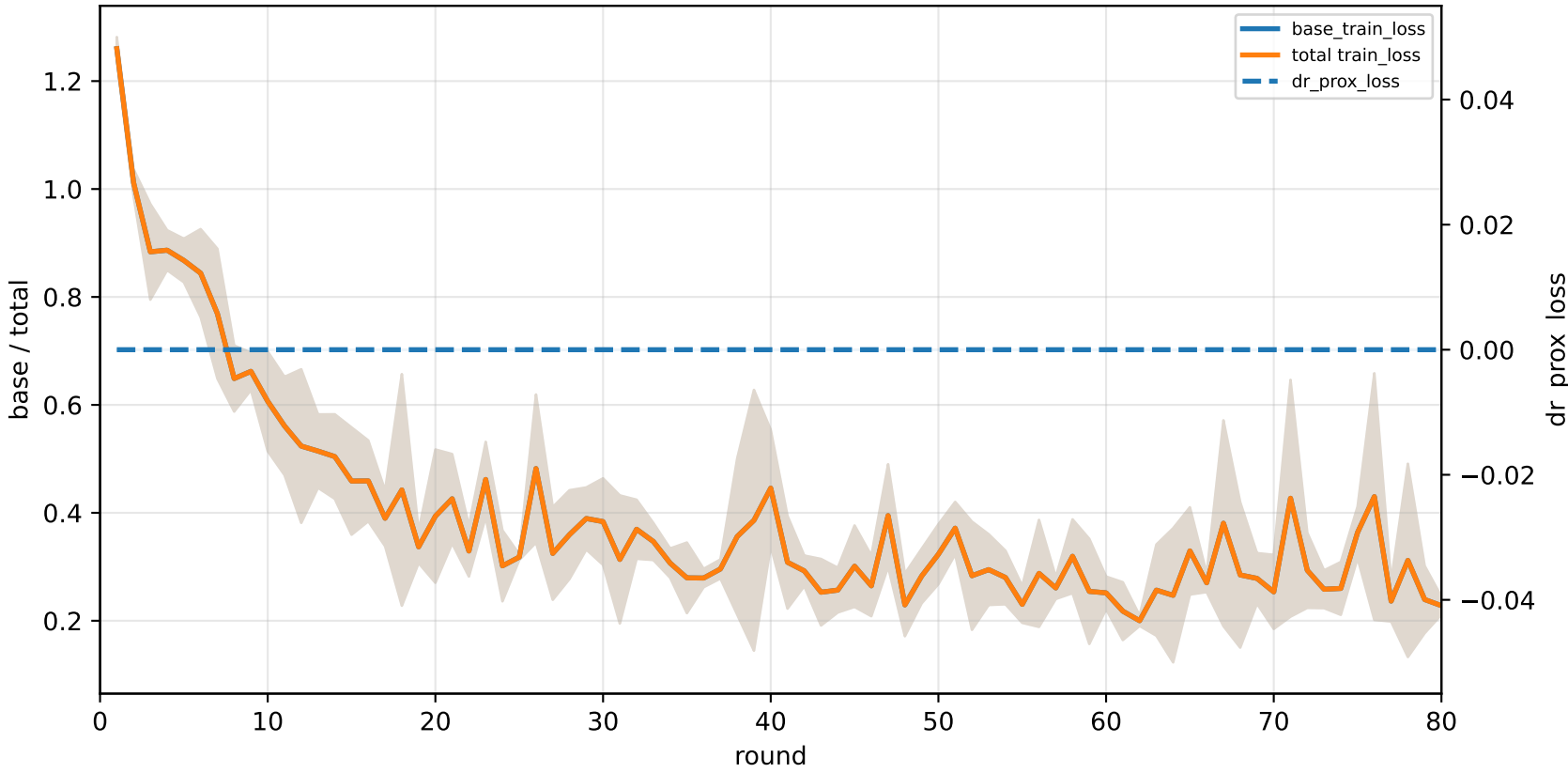
APPLE lambda scheduler | apple_mu=0



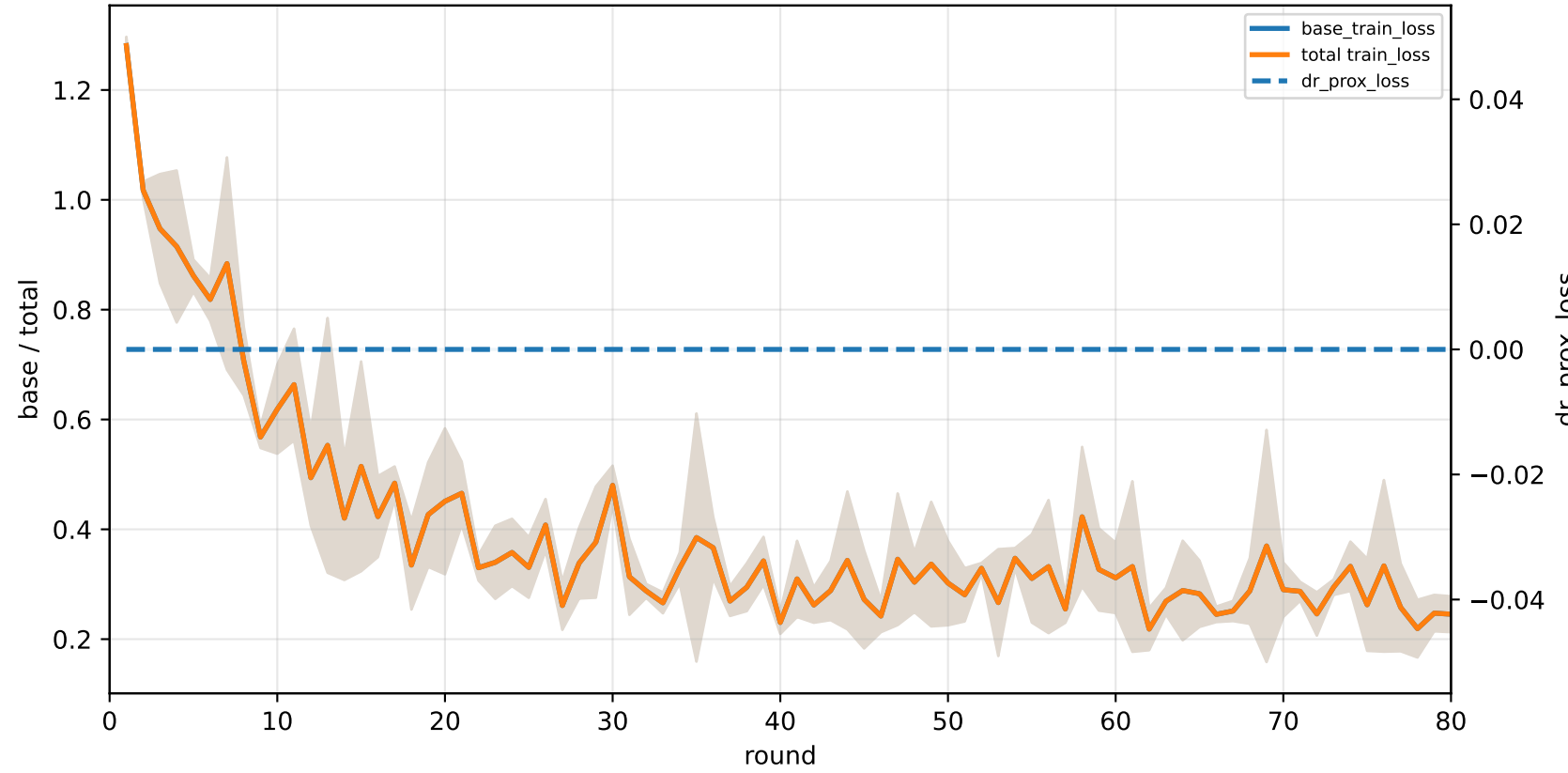
C2-specialist | graph_id=3000010



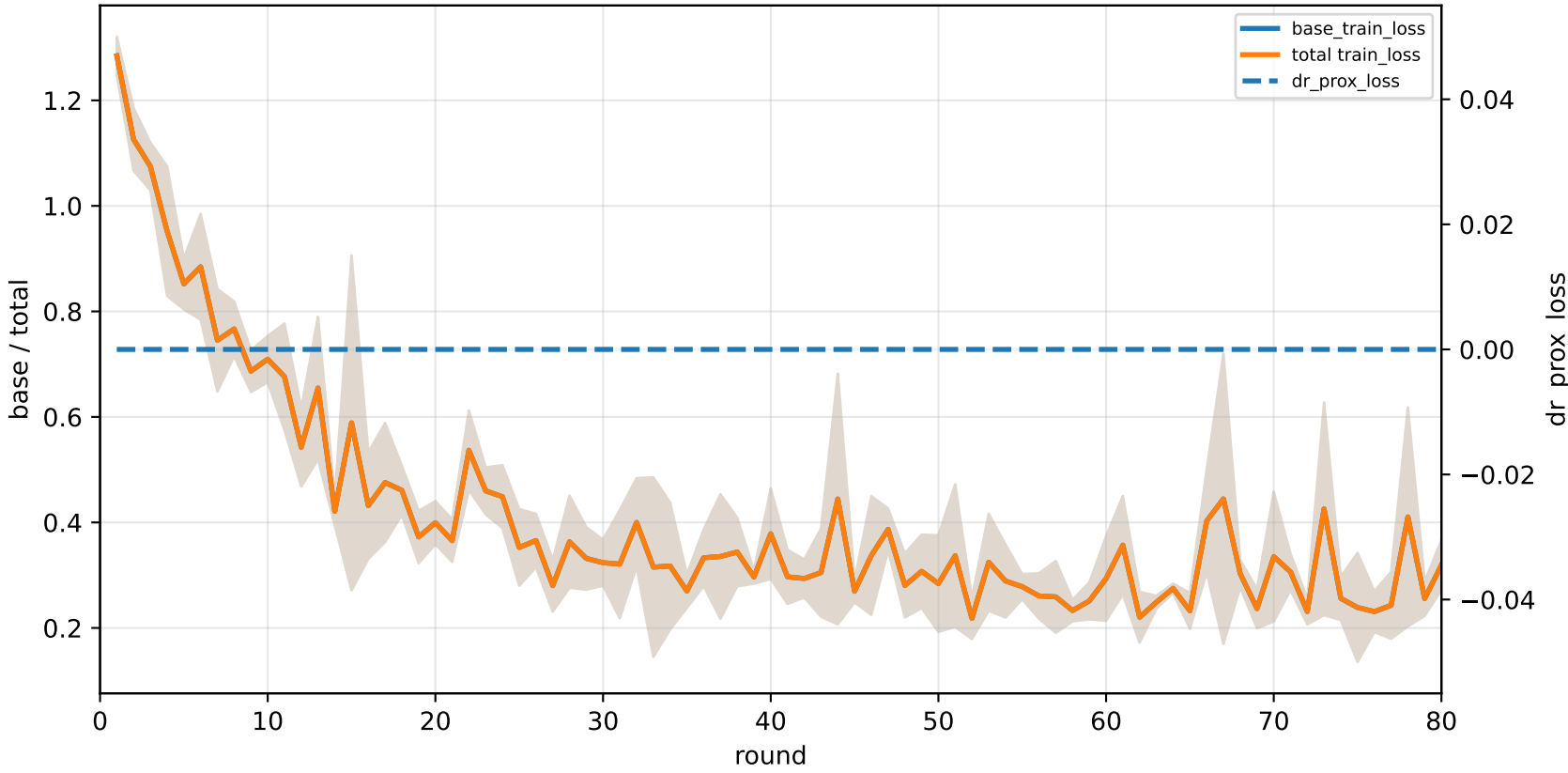
C3-specialist | graph_id=3000011



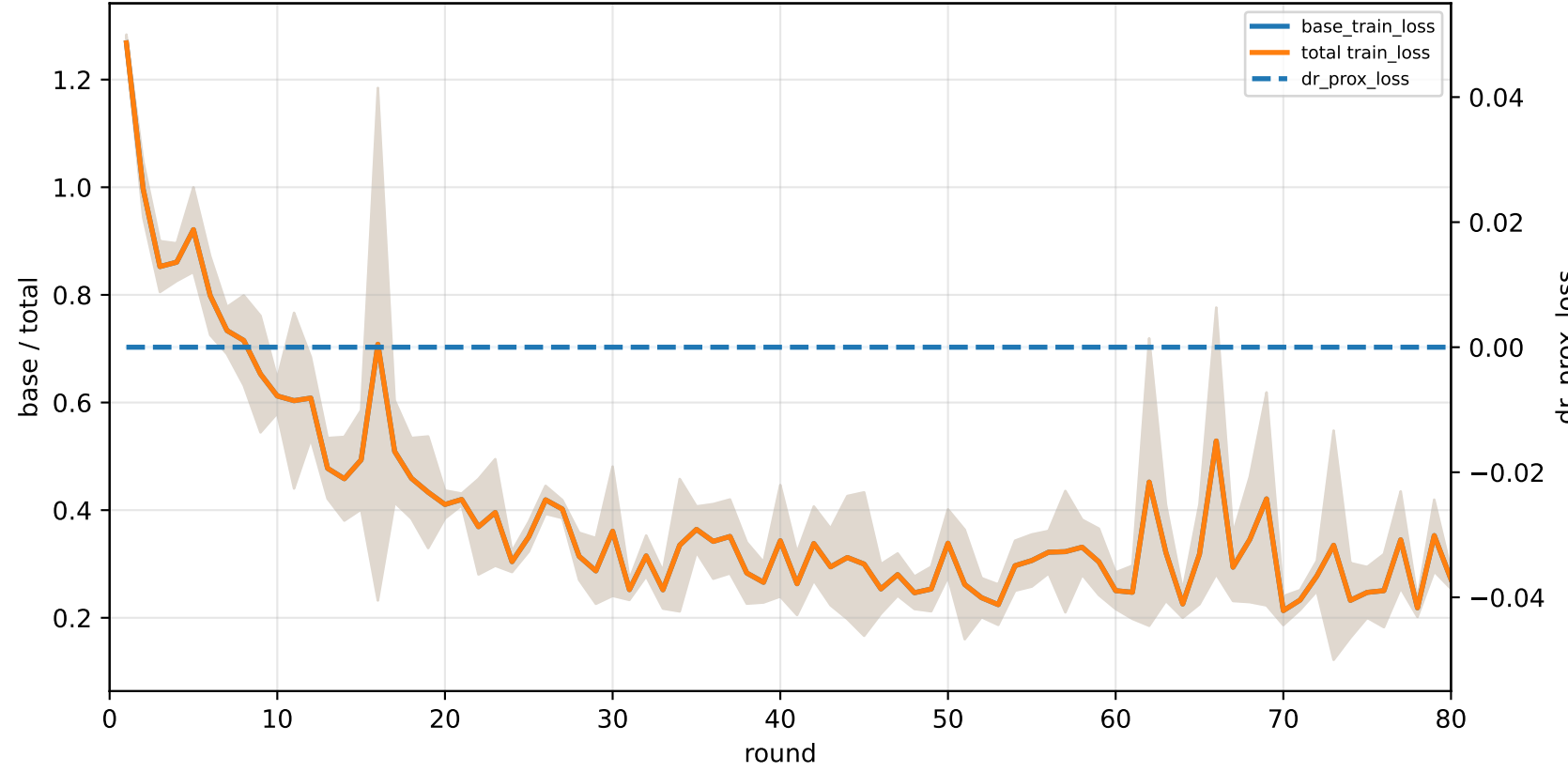
C4-specialist | graph_id=3000012



C5-specialist | graph_id=3000013

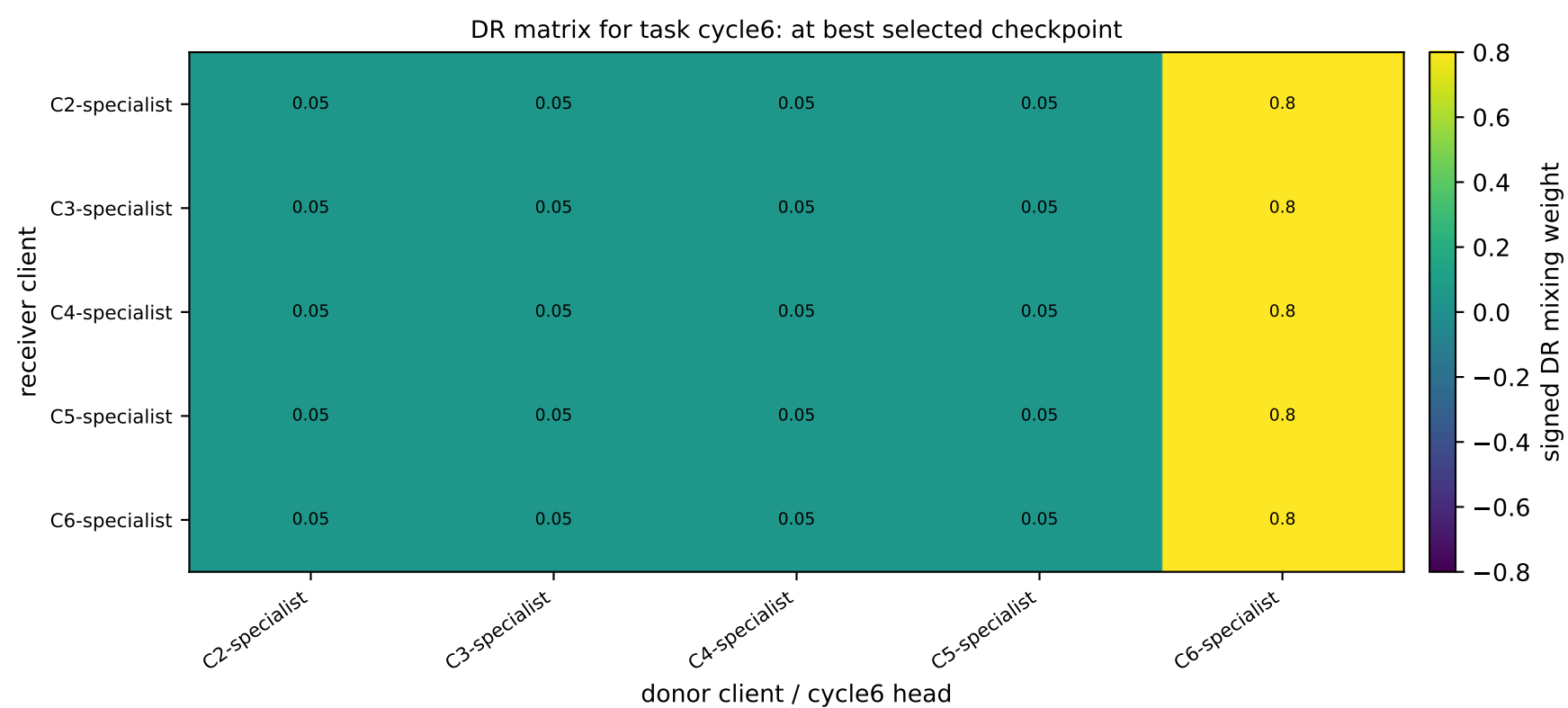
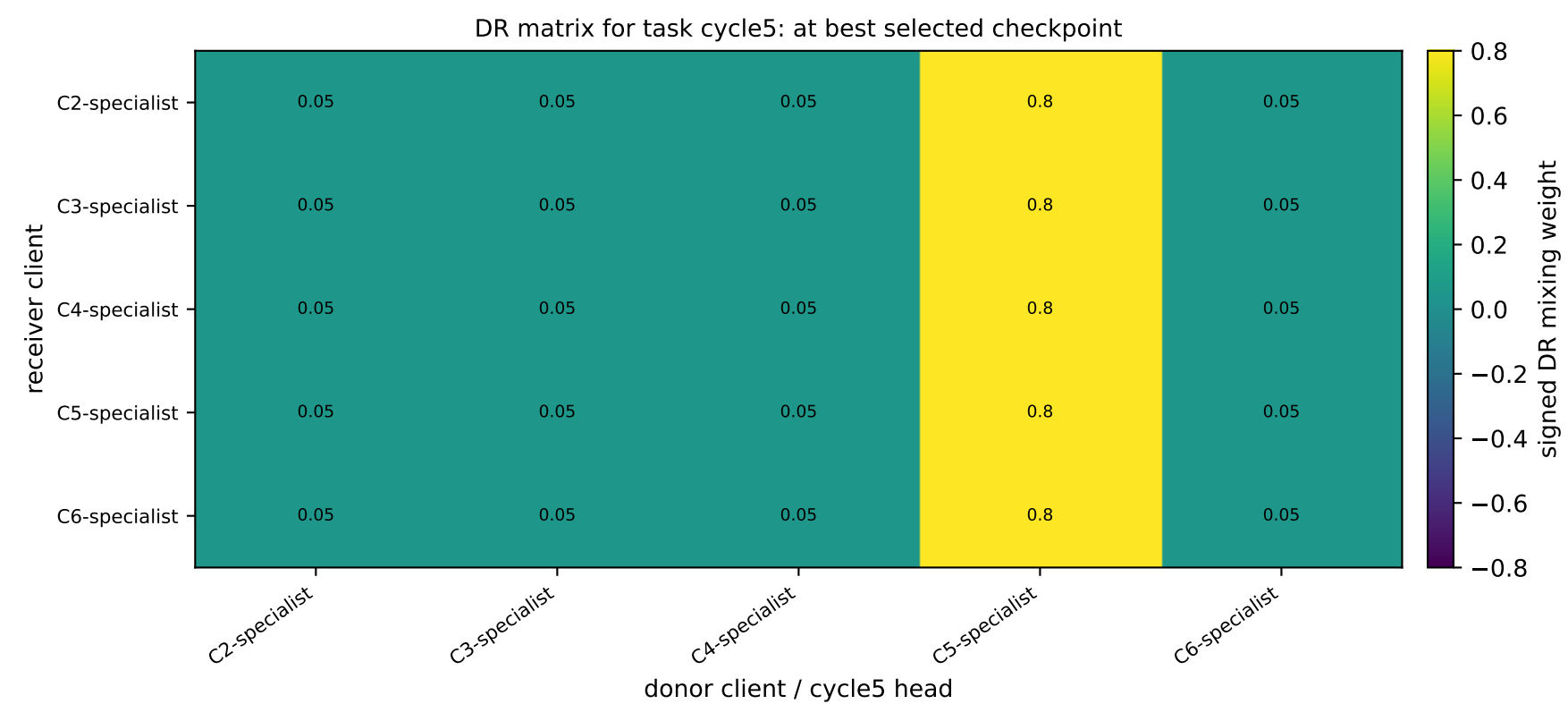
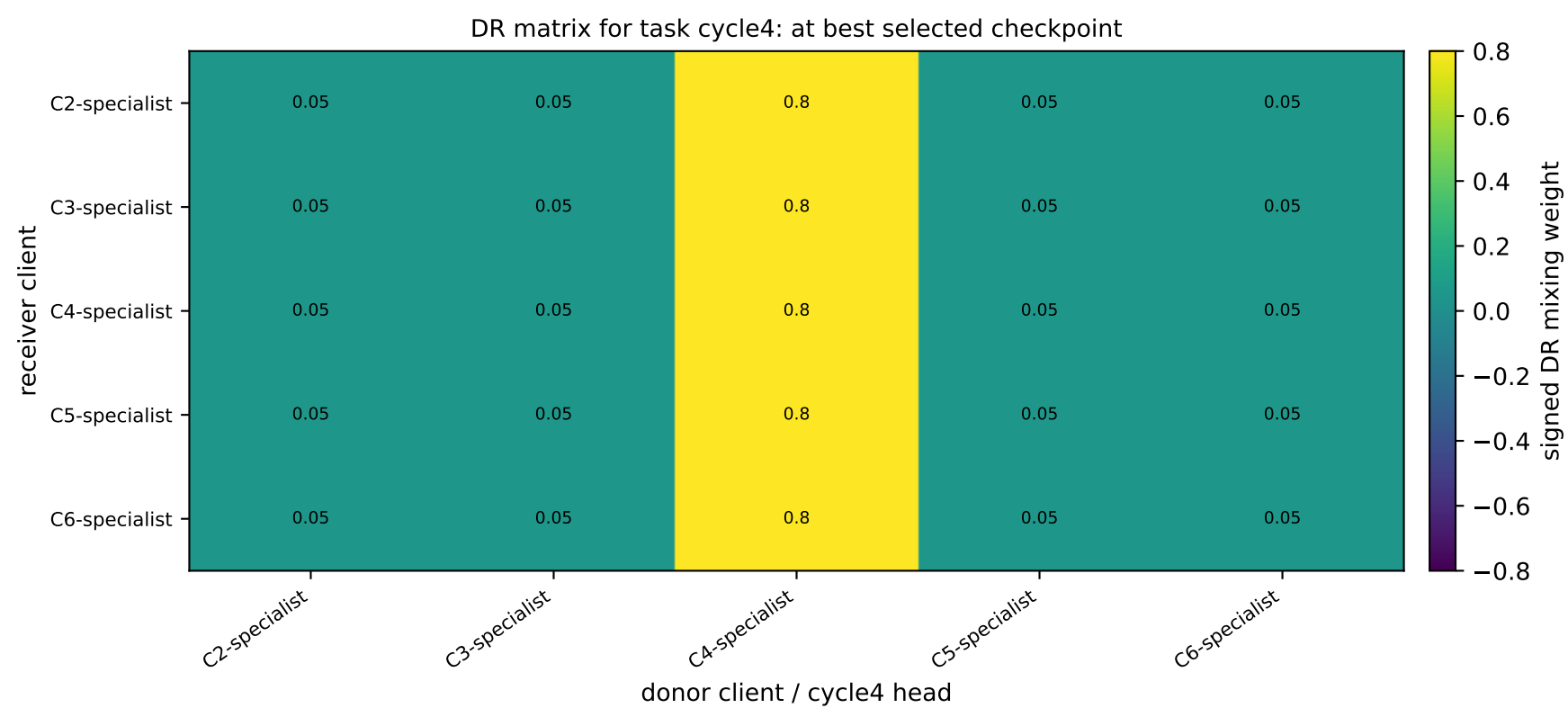
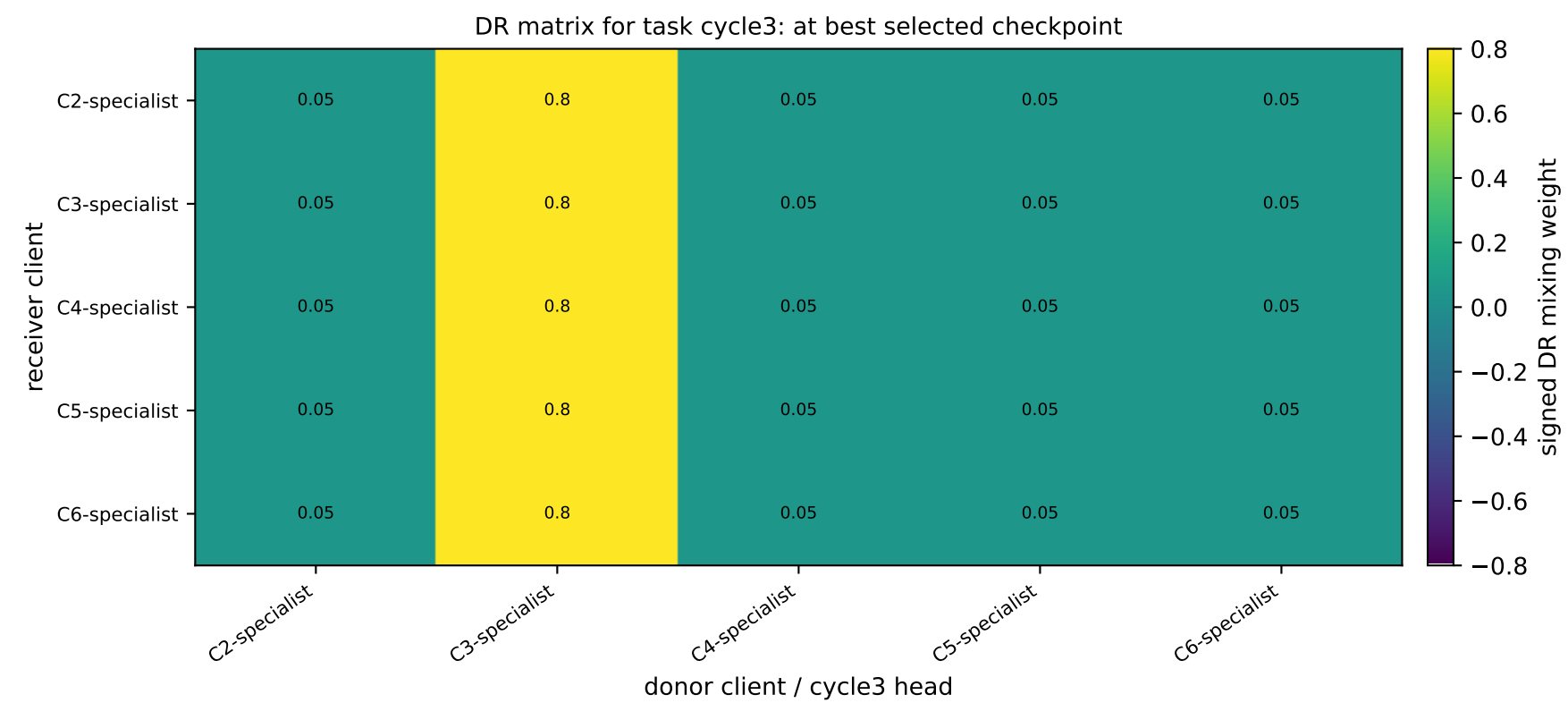
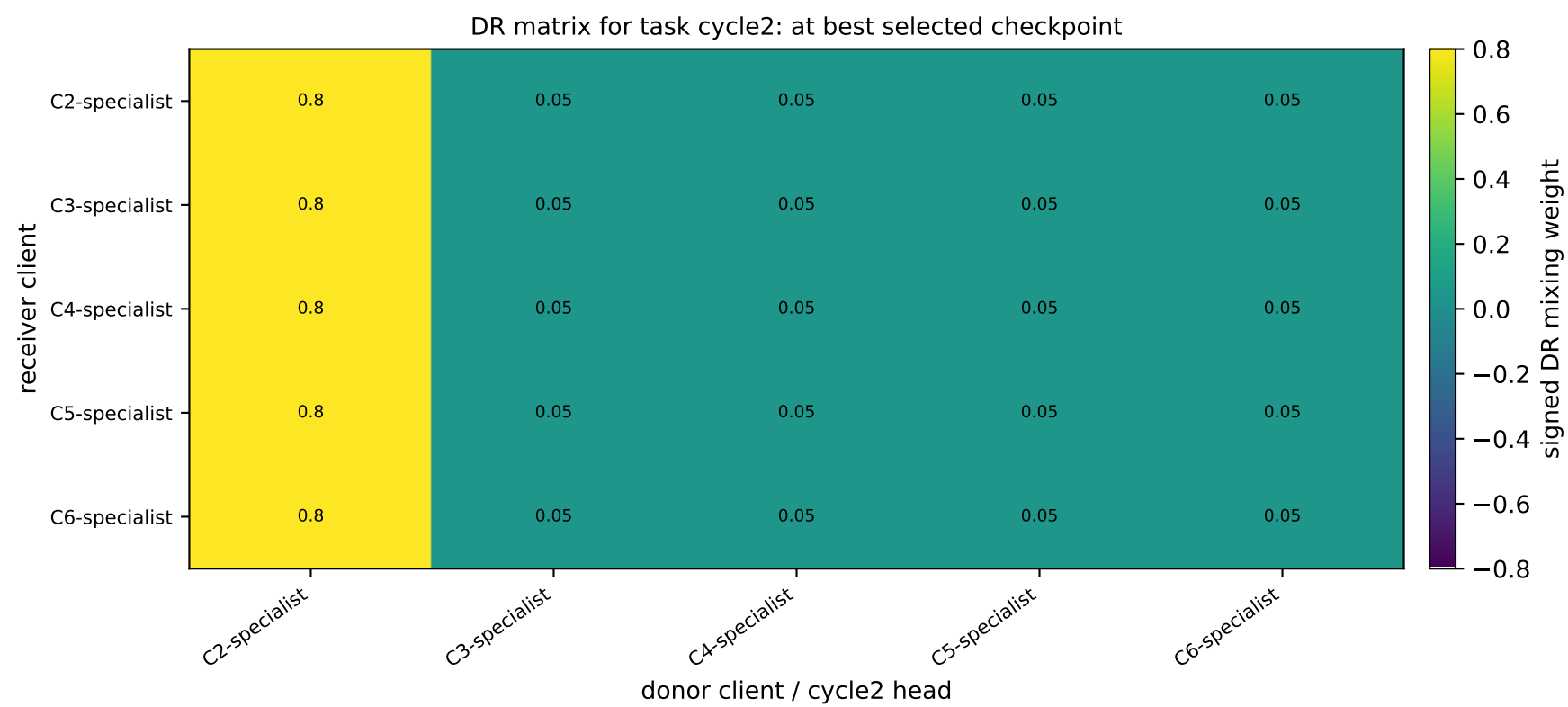


C6-specialist | graph_id=3000014



APPLE-QOracle task-wise DR matrices: receiver clients × donor task-heads | High

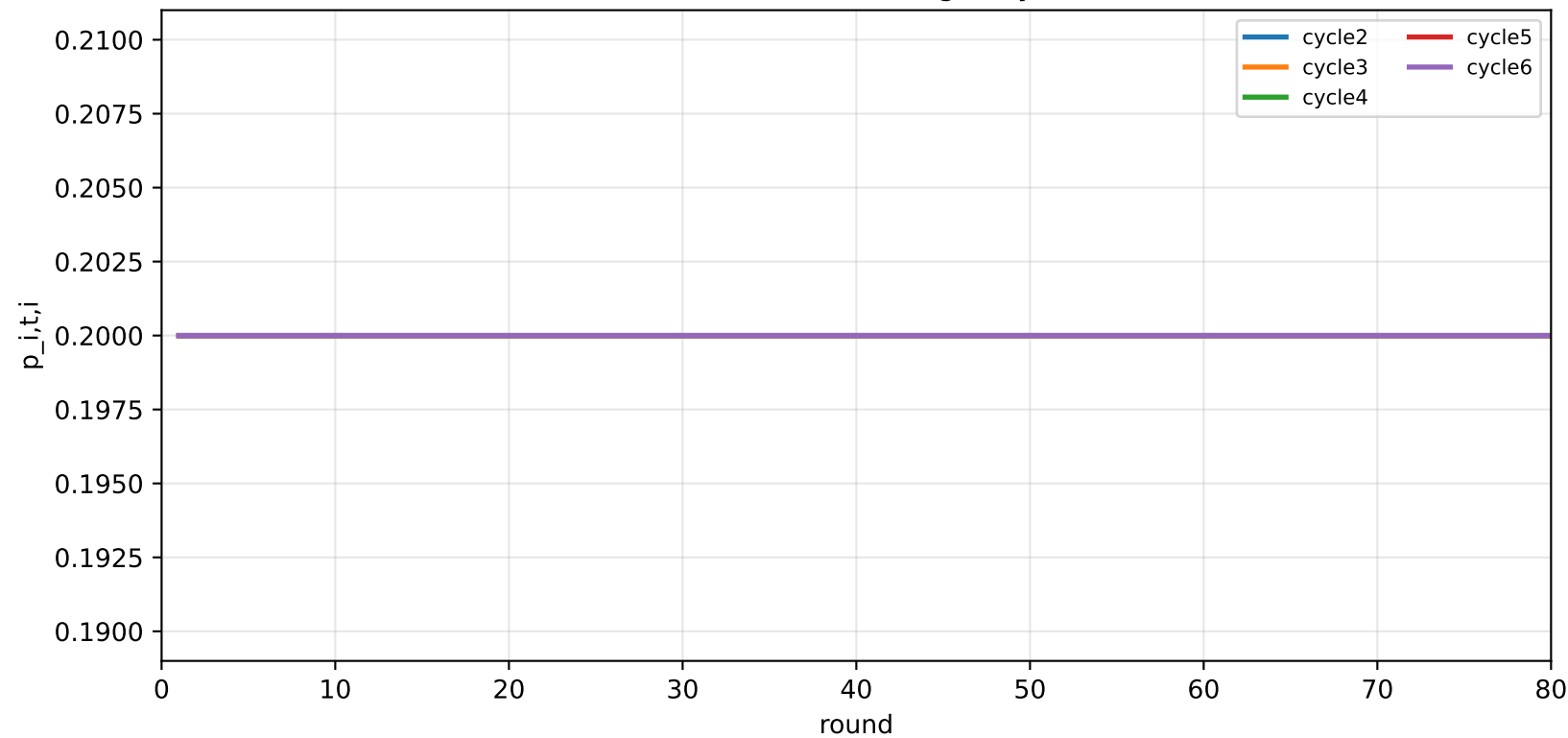
family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399



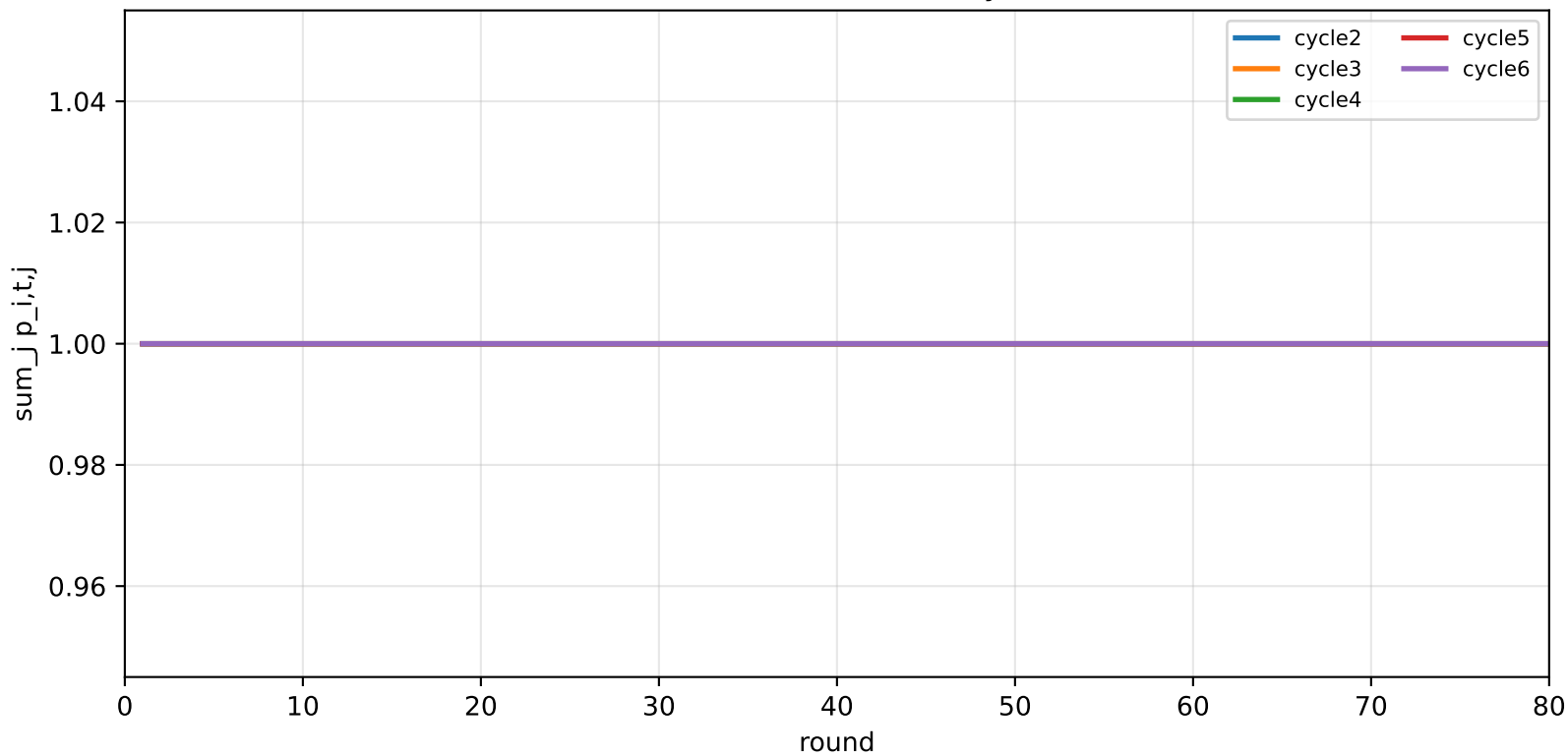
APPLE-QOracle task-wise DR dynamics | High

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

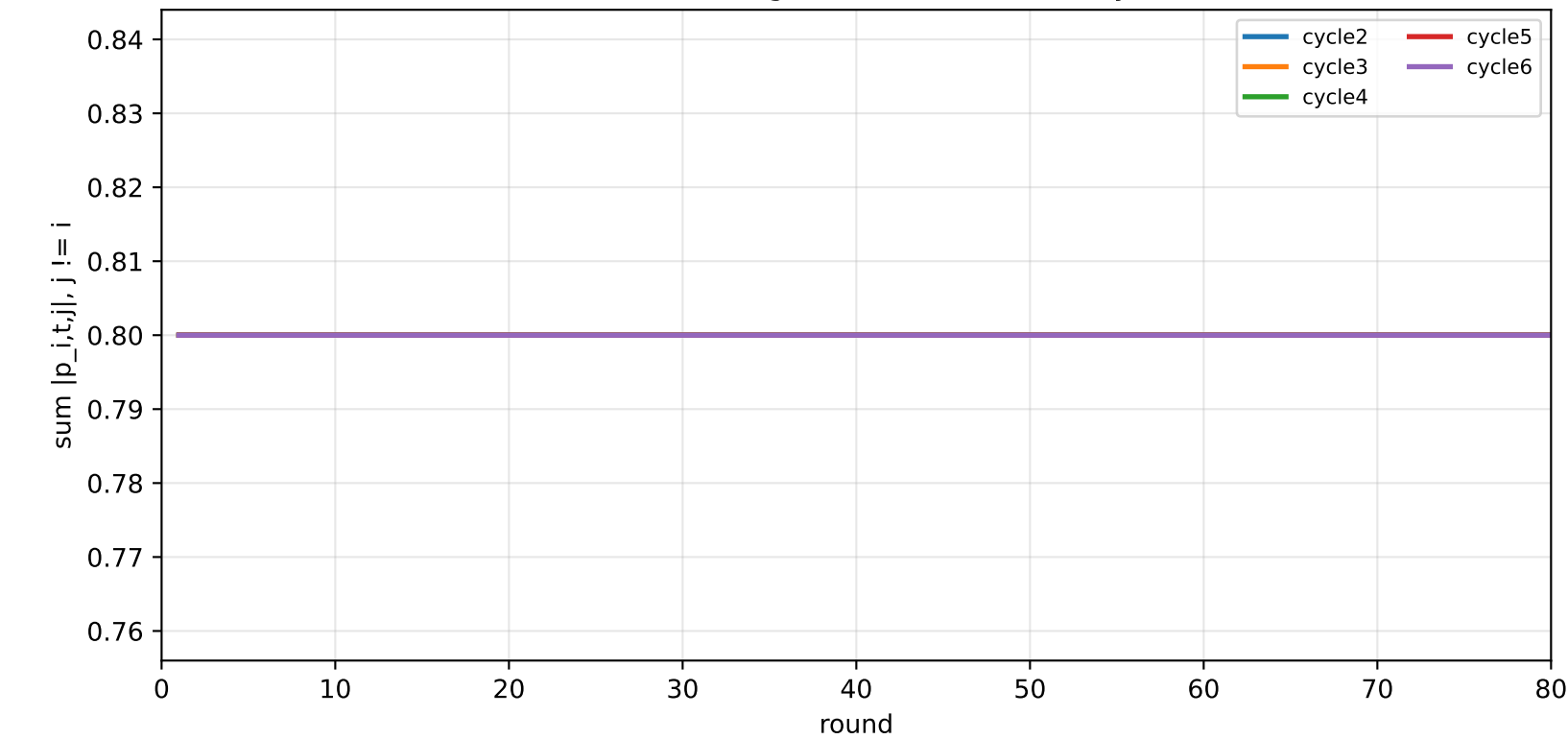
Task-head DR self-weight by task



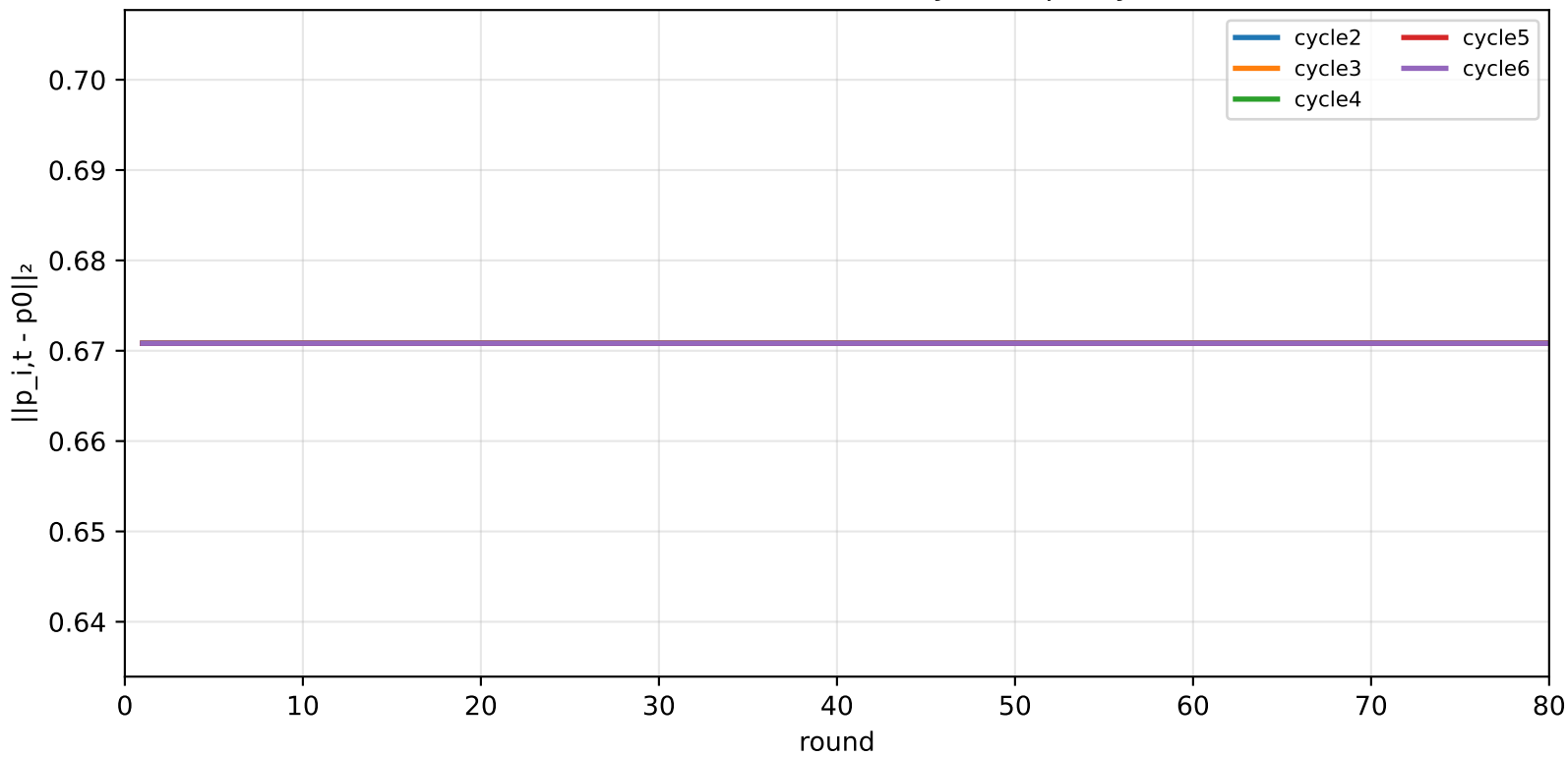
Task-head DR row sum by task



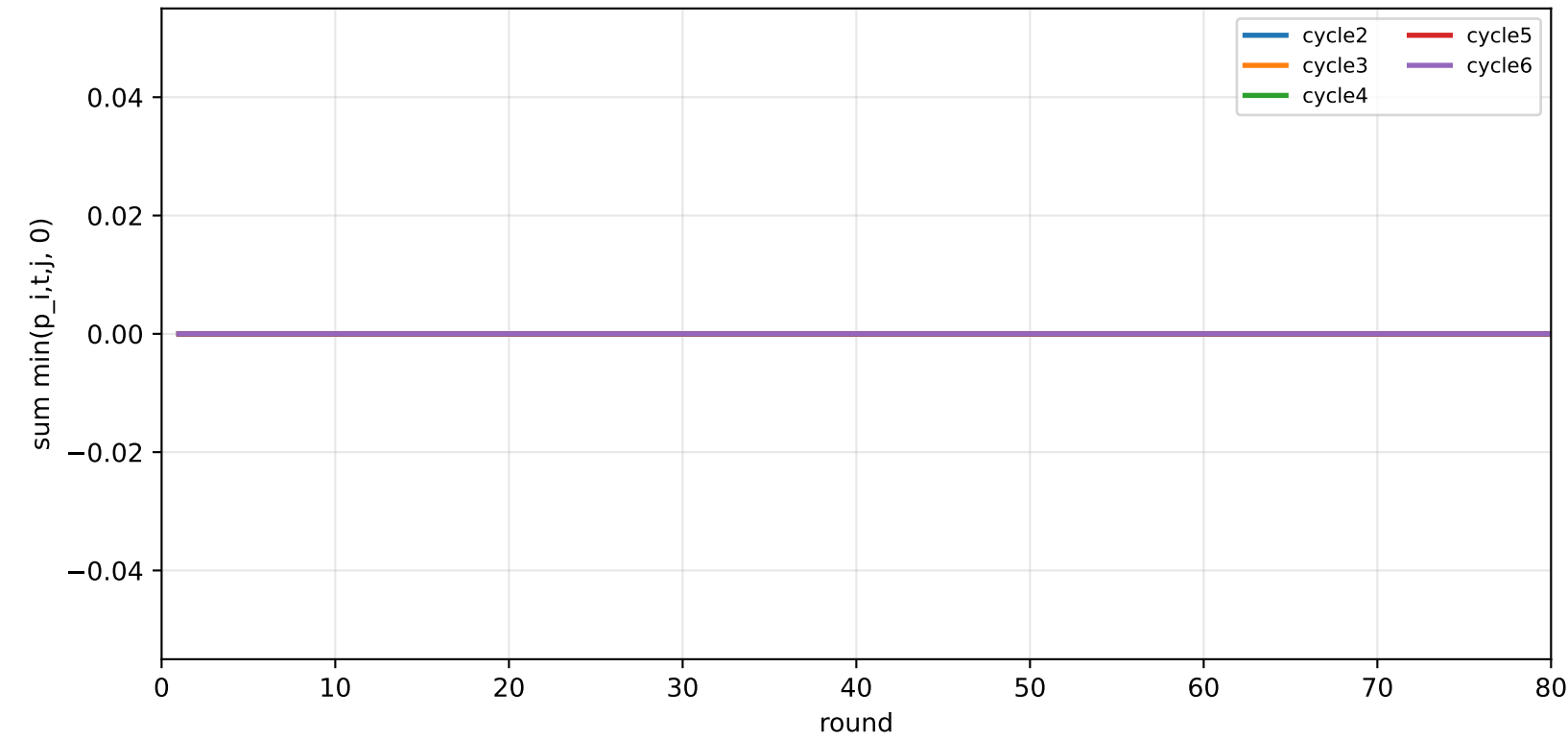
Task-head off-diagonal absolute mass by task



Task-head DR movement away from p0 by task



Task-head off-diagonal negative mass by task



Task-head DR alignment

